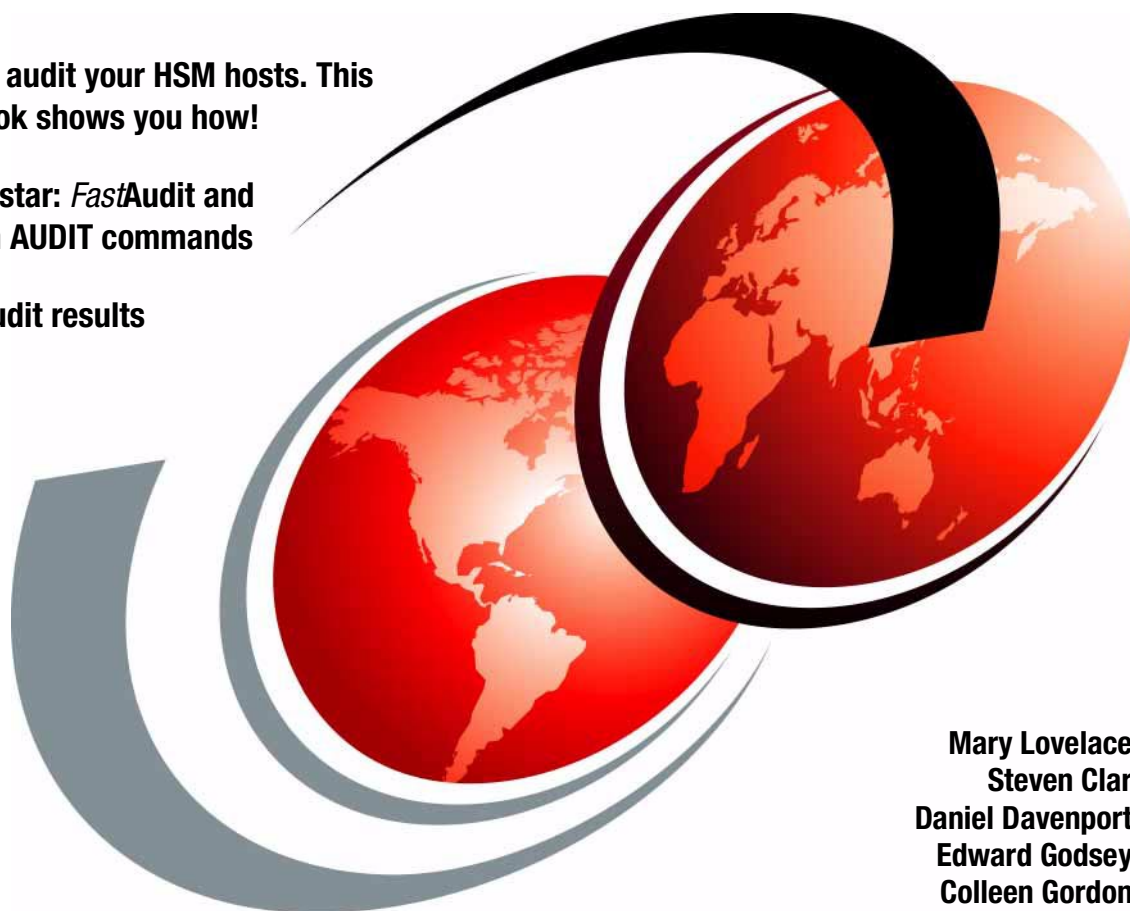


# DFSMShsm Audit and Mainstar: *FastAudit*/390

You should audit your HSM hosts. This IBM Redbook shows you how!

Learn Mainstar: *FastAudit* and DFSMSHsm AUDIT commands

Interpret audit results



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Daniel Davenport  
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Colleen Gordon





International Technical Support Organization

**DFSMSHsm Audit and Mainstar:***FastAudit/390*

May 2003

**Note:** Before using this information and the product it supports, read the information in “Notices” on page ix.

**First Edition (May 2003)**

This edition applies to HSM *FastAudit* (product number 5620-FHI), Tape Audit (5620-FHJ), and Catalog & DASD Audit (5620-FHK).

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# Preface

The DFSMSHsm™ control data sets, which are components of the DFSMSHsm product allow DFSMSHsm to manage your storage environment. The control data sets are virtual storage access method (VSAM) data sets, that contain the control information describing the status of the DFSMSHsm managed data sets and volumes.

The migration control data set (MCDS) stores information about data sets that are migrated to migration level 1 (ML1) DASD devices and migration level 2 (ML2) tape storage.

The backup control data set (BCDS) stores information about the backup copies of data sets that have been backed up by DFSMSHsm automatic functions or DFSMSHsm commands.

The offline control data set (OCDS) stores information about the data sets resident on tapes volumes used by DFSMSHsm.

Each control data set (CDS) requires regular maintenance (backup, reorganization) and auditing to ensure the integrity of the data managed by DFSMSHsm. The DFSMSHsm product provides the AUDIT command to detect, report and diagnose, and often to provide automated correction for discrepancies between the CDS, ICF user catalog, and DFSMSHsm-owned volumes.

Mainstar®: *FastAudit/390* family of products provides the same audit capabilities as with the DFSMSHsm audit command, but at faster speeds. Using the *FastAudit/390* products, the audits are comprehensive (perform multiple audits with one command), flexible, and easy to use. Audits of some, or all of the z/OS® or OS/390® metadata environments can now be done frequently, even daily, to identify and resolve errors before they lead to data loss or failed access.

Mainstar: *FastAudit/390* family of products detailed in this IBM® Redbook are:

- ▶ HSM *FastAudit* - Evaluates the Migration Control Data set (MCDS), Backup Control Data set (BCDS), and the Offline Control Data set (OCDS) to resolve structural and logical discrepancies that can prevent migrated data from being recalled, or backup data from being restored. Correlates entries with the ICF catalog environment to assure consistency. This function set duplicates most DFSMSHsm command audit functions, at many times the speed.
- ▶ Catalog & DASD Audit - Evaluates all master and user catalog data set entries, and reconciles against all DASD VTOC information on currently mounted volumes. Finds potential problems such as unused alias entries,

unconnected nodes, uncataloged data sets, and data sets cataloged to the wrong volumes. Catalog & DASD Audit also provides tools to automatically reconcile multiple master catalogs.

- **Tape Audit** - Correlates entries in the tape management catalog with ICF catalog status. Successful tape management depends sensitively on the correct correlation between tape management catalog information and ICF catalog information. This high performance audit makes this reconciliation fast and painless.

## The team that wrote this redbook

This redbook was produced by a team of specialists from around the world working at the International Technical Support Organization, San Jose Center.

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# Supporting metadata structures

This chapter provides an introduction to metadata structures and the common structures found in your environment. We discuss the Mainstar and IBM products that can be used to manage metadata structures.

## 1.1 DFSMS family of products

DFSMSHsm is a functional component of the DFSMS family of products providing facilities for managing your storage environment. DFSMSHsm uses policies defined to DFSMS to provide automated storage management, eliminating the need for batch jobs to perform backup, migration, or space reclamation functions. Using DFSMSHsm, data can be managed at the data set level, the device level, or the device pool level. DFSMSHsm manages data throughout its life cycle; from creation to expiration until the time its last backup is no longer needed.

DFSMSHsm is highly integrated with DFSMS. The five components of DFSMS and their functions are:

- ▶ DFSMSdfp™ – Provides storage, data, program, and device management functions through the storage management subsystem (SMS)
- ▶ DFSMSrmm™ – Provides tape management functions for removable media, such as tape cartridges and 3420 reels
- ▶ DFSMSdss™ – Provides data movement, copy, backup, and space management functions in a batch environment
- ▶ DFSMSOprt – Provides monitoring and reporting capabilities for multiple or single system environments
- ▶ DFSMSHsm – Provides backup, recovery, migration, and space management functions with optimum automation capabilities

## 1.2 DFSMSHsm CDSs are a metadata structure

The DFSMSHsm control data sets are metadata structures. Metadata is defined as data about data. DFSMSHsm control data sets function similar to ICF user catalogs for data that DFSMSHsm manages; therefore they are metadata structures providing access to your data.

As an important piece of the system catalog structure, the DFSMSHsm Migration Control Data Set (MCDS), Backup Control Data Set (BCDS), and Offline Control Data Set (OCDS) contains key information, critical for data access and recovery. In the course of normal operation of DFSMSHsm, the control data sets and ICF catalogs are subject to continual update activity; causing the control data sets to be particularly vulnerable to corruption from a variety of sources. These sources of corruption include un-coordinated IPLs, system failure, power hits, and forcing DFSMSHsm down with a cancel or force command.

Corruption in the control data sets results in an inability to access data managed by DFSMSHsm. This could result in valid user data becoming inaccessible and can be potentially disastrous to the data center.

Regular audits of the DFSMSHsm environment can dramatically reduce the danger of data loss or temporary loss of access to data managed by DFSMSHsm.

### 1.2.1 Supporting metadata structures in your environment

ICF catalogs, tape management databases, volume table of contents (VTOC), VSAM volume data sets (VVDS), and automated tape library (ATL) databases are also metadata structures in your environment. It is equally important to audit and maintain these structures to ensure access to your data.

#### **Mainstar: Catalog RecoveryPlus**

Mainstar: Catalog RecoveryPlus provides complete support for your catalogs, VTOC, and the VVDS structures in your environment, as detailed in the IBM Redbook *ICF Catalog Backup and Recovery, A Practical Guide*, SG24-5644.

Catalog RecoveryPlus functions include:

- ▶ A backup and restore facility for BCS and VVDS that results in better integrity and faster processing.
- ▶ A forward recovery facility for BCS and VVDS that uses SMF data, merged with the backup records, to update the BCS or VVDS from the date/time of the backup to the current time.
- ▶ A backup and restore facility for VSAM KSDS files that is specifically designed for situations where the index is damaged, and the records must be unloaded from the current file.
- ▶ An expanded DIAGNOSE facility that enables frequent health-checks of the BCS-VVDS-VTOC structure, with an audit-check and "fix" capability to resynchronize entries that are in error.
- ▶ A fast and safe MERGECAT facility that enables splitting or merging of BCS entries.
- ▶ A master catalog "alias" audit facility that compares master catalogs to determine if they contain consistent and appropriate alias entries, with "fix" capability to resynchronize the catalogs if necessary.
- ▶ A powerful SUPERCLIP facility that allows you to change the volser of online DASD volumes in a single command, without forcing you to unload and reload all data on that volume.

- ▶ An ALTER command that provides several BCS/VVDS modify facility, including volser changes to BCS records, BCS back-pointer correction in the VVDS, and SnapShot Volume support for updating all BCS entries to point to the new volser.
- ▶ A ZAP command that provides an easy-to-use AMASPZAP-like facility to delete, print, and patch BCS and VVDS record structures.
- ▶ An EXPLORE command that provides a powerful catalog search facility, with extensive dataset naming and file attribute filters to identify data sets of interest.
- ▶ The "simulate" feature on several commands that enables you to check critical catalog changes in advance, determining what effect the command will have when run.
- ▶ The "journaling" facility on critical commands that enables a "restart" or "backout" in the event of failure or interruption during command execution.

### **Mainstar: *FastAudit/390***

*FastAudit/390* is a complete metadata interrogation tool that helps keep your z/OS environment trouble-free with fast, accurate, and flexible audits. The selectable unit design allows you to choose only the groups of functionality that are meaningful to you. Comprehensive audit support is provided in three key areas:

- ▶ HSM *FastAudit* - Evaluates the Migration Control Data Set (MCDS), Backup Control Data Set (BCDS), and the Offline Control Data Set (OCDS) to resolve structural and logical discrepancies that can prevent migrated data from being recalled, or backup data from being restored. Correlates entries with the ICF catalog environment to assure consistency. This function set duplicates most DFSMSHsm command audit functions, at many times the speed.

- ▶ Catalog & DASD Audit - Evaluates all master and user catalog data set entries, and reconciles against all DASD VTOC information on currently-mounted volumes. It can be used to find such problems as unused alias entries, unconnected nodes, uncataloged data sets, and data sets cataloged to the wrong volumes. Catalog & DASD Audit also provides tools to automatically reconcile multiple master catalogs.

Catalog & DASD Audit uses DCOLLECT data to find discrepancies between the master catalog and user catalogs, reporting on GDG bases with no entries and auditing ICF catalogs and VTOCs. A variety of automated fixes are created to resolve the errors found.

- ▶ Tape Audit - Provides audit, reporting and error correction for IBM's Removable Media Manager (DFSMSRmm) and Computer Associate's CA-1 tape management product. It correlates entries in the tape management catalog with ICF catalog status. Tape Audit provides a means to reconcile the

tape management database with ICF catalogs and report discrepancies. This high performance audit makes this reconciliation fast and painless. Tape Audit is useful for identifying tapes that may be reclaimed by the data center. ICF catalogs and tape management databases (DFSMSrmm and CA-1) are compared and the discrepancies are reported. Automated fixes are created to resolve the errors found in the environment.

Additional features of *FastAudit/390* include:

- ▶ *FastAudit/390* offers audit functions for DFSMSHsm control data sets at faster speeds than standard DFSMSHsm audit and enhanced audit functions without the need to halt DFSMSHsm functions while the audit is executing.
- ▶ *FastAudit/390* HSM *FastAudit* is a complete replacement for DFSMSHsm command audit.
- ▶ *FastAudit/390* ISPF panels group audit errors into categories rather than manually parsing through pages and pages of output.
- ▶ Error message text provides a short, easy to understand explanation of the error for viewing online, coupled with a more detailed explanation and diagnostic tips provided in the product documentation.
- ▶ A variety of audit error fixes, provided for each error category, provides both diagnostic information and FIX capabilities based on the unique environmental requirements of each organization.
- ▶ The ISPF interface improves productivity and provides ease of use.
- ▶ Audit fixes are efficiently applied to all of the errors; eliminating redundant coding and submission of fix commands, reducing resource requirements to apply the fixes.
- ▶ The automated fixes take the guesswork out of FIXCDS, IDCAMS, and other command structures and syntax requirements, and gives the user complete control and confidence in the fixes they are applying.
- ▶ *FastAudit/390* takes the complexity out of running audits, analyzing the output, and applying the correct fix, increasing your productivity while increasing the reliability of your DFSMSHsm environment.





## DFSMSHsm control data sets

In this chapter, we include some common data set allocation techniques, making recommendations for allocating, performing backup and forward recovery, and sharing control data sets in a sysplex environment. Most of the recommendations we make are based on experience gathered over the past 20 years. We do not attempt to describe how to implement DFSMSHsm, code statements in DFSMSHsm PARMLIB, work with the DFSMSHsm Starter Set JCL, or provide information on the day-to-day maintenance of DFSMSHsm. Our intent for this chapter is to provide enough information about the control data sets, their structure, the records they contain, and backup and recovery requirements for proper maintenance of these data sets. Later in this book we discuss how the control data sets become corrupted over time and why frequent, comprehensive audits of these important metadata structures are critical to the proper operation of the DFSMSHsm environment.

## 2.1 Types of DFSMSHsm CDSs

The DFSMSHsm control data sets are VSAM KSDS files central to the operation of the product. The CDS data sets may reside on separate volumes, may be multi-volume, and may be multiclustered. They vary in size depending on the size of the environment that DFSMSHsm is managing. To use DFSMSHsm, only the migration control data set (MCDS) is required, a typical site utilizes all three control data sets.

### 2.1.1 Migration control data set (MCDS)

The MCDS contains records and pointers to data sets that have been migrated by DFSMSHsm and to the volumes where the data has been migrated to/from. In addition, it contains internal processing statistics, processing-unit-environment data, and several other types of records such as volume records and Migration Control Records (MCD). The information stored in the MCDS is critical to properly manage the data sets and volumes. This information is also useful to verify, monitor, and tune your storage environment.

Each hour, DFSMSHsm writes daily statistics records and volume statistics records to the MCDS. These statistics records can be used to report on DFSMSHsm activities in your environment. These two statistics records are also written to the system management facilities (SMF) if you specify the **SETSYS SMF** command in PARMLIB and assign an SMF ID.

An MCDS resident on a DASD device shared by two or more processors is defined as a multiple DFSMSHsm-host environment to DFSMSHsm. DFSMSHsm assumes that there is more than one host sharing the MCDS (and other control data sets) if the MCDS is on a shared DASD device or if you specify **CDSSHR=RLS** or **CDSSHR=YES**.

### 2.1.2 Backup control data set (BCDS)

The BCDS contains the records related to data set and volume backup. These backup activities include DFSMSHsm automatic backup, automatic dump and ABARS. DFSMSHsm requires this information to manage its backup data sets and volumes. The most common backup records in the BCDS are the MCB or B record and the MCC or C record. Every time a data set is backed up by DFSMSHsm automatic backup, a B record is created to manage the backup copy. Additional backup versions of the same data set cause a C record to be created. There can be up to one hundred C records for every B record in the BCDS.



### 2.1.3 Offline control data set (OCDS)

Every time DFSMSHsm writes a backup copy to a tape or migrates a data set to ML2 tape, the OCDS is updated with a TTOC or T record. This T record describes the contents of each migration or backup tape that DFSMSHsm currently owns. The OCDS can provide information about each migration and backup tape and about each data set residing on the tape. The T record is the only record in the OCDS.

## 2.2 Multi-cluster control data sets

Large storage environments, where DFSMSHsm is managing large amounts of data, require thousands of records to be written and stored in the control data sets. A large number of entries in the control data sets may also be a result of storing data in DFSMSHsm migration or backup for long periods of time. As the control data sets grow, they can require more space than the physical DASD device allow. When this occurs, the control data sets can be split and placed on multiple volumes to accommodate the larger size.

Both the MCDS and BCDS control data sets can be split up into as many as four separate files. IBM recommends the use of non-key-range multi-cluster control data sets.

To implement the use of non-key-range multi-cluster control data sets, remove the **KEYRANGE((...))** parameter from the **IDCAMS DEFINE DATA** statement used to create the control data sets as key range. Use **IDCAMS REPRO** to create a new non-key-range multicluster control data set.

When DFSMSHsm is started, DFSMSHsm dynamically calculates the key boundaries for each cluster. Use the **QUERY CONTROLDATASETS** command to display both the low and high keys that DFSMSHsm calculates for each cluster.

For more information on multi-cluster control data sets, refer to the *DFSMSHsm Implementation and Customization Guide*, SC35-0418. Additional information can be found in the IBM Redbook *DFSMSHsm Primer*, SG24-5272.

## 2.3 CDS allocation parameters

A typical (non-multicluster) CDS can be allocated using IDCAMS and a parameter set as shown in Figure 2-1.

```

//STEP1 EXEC PGM=IDCAMS,REGION=512K
//SYSPRINT DD SYSOUT=A
//SYSUDUMP DD SYSOUT=A
//SYSIN DD *
DEFINE CLUSTER -
    NAME(HSM.MCDS) -
    VOLUME(volser) -
    STORCLAS(SCHSM) -
    RECSZ(200 2040) -
    SPEED -
    NOIMBED -
    NOREPLICATE -
    BUFSPC(530432) -
    FSPC(0 0) -
    INDEXED KEYS(44 0) -
    SHAREOPTIONS(3 3) -
    NOWRITECHECK) -
    DATA(NAME(HSM.MCDS.DATA) -
    CISIZE(4096) -
    CYLINDERS(50)) -
    INDEX(NAME(HSM.MCDS.INDEX) -
    CISIZE(4096) -
    CYLINDERS(1))

```

Figure 2-1 Typical CDS allocation DEFINE CLUSTER command

To allocate a multi-cluster CDS, simply repeat the DEFINE CLUSTER statement, allocating another VSAM non-key-range data set with a unique name. As an example, the first MCDS may be named HSM.MCDS1 and the second MCDS may be named HSM.MCDS2. Modify your DFSMSHsm startup procedure in SYS1.PROCLIB and in other JCL that references the CDS to reflect the multi-cluster CDS names. There must be a separate DD card for each cluster of a multi-cluster CDS. The ddnames should be MIGCAT, MIGCAT2, MIGCAT3, and MIGCAT4 or BAKCAT, BAKCAT2, BAKCAT3, and BAKCAT4.

## 2.4 Primer record

When a new CDS is created it must be primed with a dummy record that has the highest possible key in order for DFSMSHsm to open the CDS for the first time. Figure 2-2 shows the JCL you can use to create the primer record. More information about the primer record is found in the *DFSMSHsm Implementation and Customization Guide*, SC35-0418.

[illegible]

*Figure 2-2 DFMSHsm CDS primer record*

You can copy the primer record into the newly defined control data sets using **IDCAMS REPRO**. Figure 2-3 shows the IDCAMS JCL used to prime the control data set.

```
//REPRO EXEC PGM=IDCAMS
//SYSPRINT DD SYSOUT=*
//SYSIN DD *
REPRO FROMDATASET(CDS.PRIMER)
TODATASET(HSM.MCDS)
```

Figure 2-3 JCL to REPRO the primer record into the CDS

## 2.5 CDS backup

The DFSMSHsm control data sets should be backed up regularly, at least once per day. DFSMSHsm automatically backs up the control data sets when automatic backup is run. The SETSYS CDSVERSIONBACKUP parameter defines the number of backup versions that DFSMSHsm maintains for each CDS and journal file.

There are a variety of ways to backup these data sets, depending on your hardware and system level. During backup all DFSMSHsm activity and all JES2 and JES3 setup and initialization activity is suspended. Because of this, the performance of the system is degraded whenever a CDS backup is performed, which is why it is important to choose a fast backup method such as concurrent copy or SnapShot.

## 2.5.1 Concurrent copy

Concurrent copy is a hardware, licensed internal code (LIC), and software systems management tool that creates data copies in just a few seconds. This allows processing to continue with very little interruption.

DFSMSHsm can automatically invoke concurrent copy to back up the control data sets. The system is suspended for as long as it takes to initialize a concurrent copy session for each of the control data sets and to backup and nullify the journal data set. The journal file cannot be backed up using concurrent copy, it is backed up by DFSMSHsm.

Once the concurrent copy session is established for each control data set, DFSMSHsm releases the control data sets for other processing. The backup continues in the background.

To enable concurrent copy for CDS backup, you'll need to implement the following:

- ▶ Allocate the control data sets on DFSMS managed volumes.
- ▶ The volumes must be concurrent copy enabled.
- ▶ Assign a management class with the BACKUP COPY TECHNIQUE set to CONCURRENT PREFERRED.
- ▶ Issue the SETSYS command to specify that DFSMSHsm use DFSMSdss as the data mover.
- ▶ Use the PARALLEL option if you are writing the backup copies to tape.

Concurrent copy may also be invoked in DFSMSdss JCL. You may use DFSMSdss to back up the control data sets outside of DFSMSHsm. These additional backups may be required for disaster recovery purposes.

## 2.5.2 SnapShot

SnapShot provides a copy operation very similar to concurrent copy. If your DASD devices are RVA, SnapShot is automatically invoked when **CONCURRENT PREFERRED** or **REQUIRED** is specified in the DFSMS Management Class. The same is true when using DFSMSdss JCL to backup the control data set.

## 2.6 Reorganizing control data sets

Like any other VSAM structure, control data sets may need to be reorganized. The ARC0909I message can be captured and read to see if the CDS needs to be

reorganized. This message is issued when the threshold specified in the **SETSYS MONITOR** command has been exceeded. The **QUERY CONTROLDATASETS** command displays the current status. The status is shown in the ARC0148I message.

- ▶ You do not need to reorganize control data sets to consolidate splits or for performance reasons. In fact, the performance of the CDS is degraded immediately following reorganization until splits are taken and normalized the CDS. This may take up to three weeks depending on the activity in the environment.
- ▶ Only reorganize the CDS when you need to increase the size or when moving a CDS to another device.
- ▶ DFSMSHsm must be shut down on all hosts that share the CDS. Always allocate the CDS with DISP=OLD.
- ▶ Never code freespace (use FREESPACE (0 0) ) and let the splits occur naturally. You cannot predict where the splits will take place and therefore, coding freespace in every CI or in every CA is a waste of space.

## 2.7 CDS forward recovery

A VSAM structural error, a hardware error, or a logical software error may cause one or more control data sets to become corrupted, requiring the restoration of a backup copy and a forward recovery using the journal data set. How much time elapses between backup copies of the control data sets is the variable in how long it takes to recover the CDS. This may be a consideration when you are deciding how often to back up the control data sets.

There are two ways to perform CDS recovery, one using the **UPDATEC** command and the other is the enhanced CDS recovery function provided by DFSMSHsm. For performance reasons, the preferred method of CDS recovery is the enhanced CDS recovery function. A process that begins with a program that copies the current active journal file and any older journal backups to a file for subsequent processing, sorts the records by key and merges the file into an exported backup version of the control data set.

Information on using the **UPDATEC** command to recover your control data sets and more information on the enhanced CDS recovery function can be found documented in the *DFSMSHsm Storage Administration Guide*, SC35-0421.

### 2.7.1 Enhanced CDS recovery function

DFSMSHsm must be stopped on all systems while the CDSs are being recovered. This procedure assumes that you have specified the **SETSYS JOURNAL**

command because an up-to-date journal is necessary as input to the enhanced CDS recovery function.

The process begins with the program ARCBJRNL. This program copies the current active journal file, and any older journal file backups to an output file for subsequent processing. If older journal file backups are not used then the BKUPJRNL DD is removed as well as the **PARM='BOTH'**. Figure 2-4 shows the ARCBJRNL program to copy the current active journal and any older journal file backups to an output file specified in DD JRNLBK.

Older journal files are normally only used when you are recovering using an older version of the control data set backup. This may be because the more current CDS backup is also corrupted and cannot be used.

```
//STEP01 EXEC PGM=ARCBJRNL,PARM='BOTH'  
//STEPLIB DD DSN=location where program ARCBJRNL resides,  
// DISP=SHR  
//SYSPRINT DD SYSOUT=*  
//JRNL0 DD DSN=current journal data set name,  
// DISP=SHR  
//BKUPJRNL DD DSN=backup copy name and version,DISP=OLD  
// DD DSN=backup copy name and version,DISP=OLD  
// DD DSN=backup copy name and version,DISP=OLD  
//JRNLBK DD DSN=new output file name,  
// UNIT=valid unit name,  
// VOL=SER=volume where backup journal will reside,  
// SPACE=adequate space for journal copy,  
// DISP=(,CATLG,DELETE)
```

Figure 2-4 JCL to run ARCBJRNL

After the output file is created in the first step, the second step is to sort the output file by key using the EQUALS parameter as shown in Figure 2-5. The JCL assumes you have a SORTD cataloged procedure. The SORTD procedure is created by running the DFSORT™ SORTPROC job that is distributed with DFSORT in SYS1.SAMPLIB. If you do not have the SORTD procedure, modify the EXEC card in the JCL to identify the sort program you want to use.

```

//STEP02 EXEC SORTD
//SORTLIB DD SYS1.SORTLIB,DISP=SHR
//SYSOUT DD SYSOUT=*
//SORTWK01 DD appropriate unit and space
//SORTWK02 DD appropriate unit and space
//SORTWK03 DD appropriate unit and space
//SORTWKnn DD etc.
//SORTIN DD DSN=backup journal name from Job1,
// DISP=SHR
//SORTOUT DD DSN=sorted backup journal name,
// UNIT=valid unit name,
// VOL=SER=volume for the sorted backed up journal,
// SPACE=adequate space for sorted backed up journal,
// DCB=(RECFM=VBS,BLKSIZE=0,LRECL=6560),
// DISP=(,CATLG,DELETE)
//SYSIN DD *
SORT FIELDS=(17,44,CH,A)
OPTION EQUALS
/*

```

*Figure 2-5 JCL to sort the journal output file by key*

The next step, shown in Figure 2-6, restores the backup version (or the last good backup) that was made by DFSMSDss. If the backup copy was made by DFSMSHsm, that backup copy is already in the form of a portable exported data set.

```

//STEP1 EXEC PGM=ADRDSSU
//SYSPRINT DD SYSOUT=*
//CDSBKUP DD DSN=uid.CDS.BACKUP.Dnnnnnnn,DISP=(OLD,KEEP)
/* Specify the name of the best available CDS backup version */
//CDSCOPY DD UNIT=a valid DASD unit name,
// VOL=SER=an appropriate volume,
// DISP=OLD
//SYSIN DD *
RESTORE DATASET(-
INCLUDE(uid.dsname))-
INDDNAME(CDSBKUP)-
OUTDDNAME(CDSCOPY)-
REPLACE
/*

```

*Figure 2-6 JCL to restore the backed up version of the CDS*

The next step, shown in Figure 2-7, exports the restored CDS to an output file that is used as input to the ARCIMPRT program which merges the journal file records with the exported CDS and creates a new CDS.

```
//JOB4 JOB accounting information,MSGCLASS=A,REGION=0M
//STEP1 EXEC PGM=IDCAMS
//SYSPRINT DD SYSOUT=*
//OUTDD1 DD DISP=(,CATLG,DELETE),BUFNO=26,BLKSIZE=28332,
//        SPACE=(CYL,(200,20),RLSE),
//        UNIT=a valid unit name ,
//        VOL=SER=an appropriate volume ,
//        DSN=temp_dsname
//SYSIN DD *
        EXPORT dsname -
        OUTFILE(OUTDD1) -
        TEMPORARY
```

*Figure 2-7 JCL to export the restored CDS to an output file*

The last step is to import and merge the files and create a new, updated, control data set. The input is the journal file we created and sorted in steps one and two and the restored control data set that we exported to an output file. You can increase the size of the CDS, if desired, by issuing the **DEFINE** command before the **IMPORT** command. Any valid IDCAMS command may be specified in the **SYSIN** stream. The **IMPORT** command must use the **INFILE** parameter; using the **INDATASET** parameters results in an error return code.

The JCL shown in Figure 2-8 shows how to re-create an MCDS. To re-create a backup control data set or an offline control data set, make changes to the **PARM=** in the **EXEC** statement and to the **EXPORT DD** statement **DSNAME** that reflects the control data set name. Change the **OUTDATASET** name in the IDCAMS control statement to reflect the control data set name to be re-created.



```

//STEP04 EXEC PGM=ARCIMPRT,PARM='MCDS'
//STEPLIB DD DSNAME=where ARCIMPRT resides,
// DISP=SHR
//SYSPRINT DD SYSOUT=*
//AMSDUMP DD SYSOUT=*
//JOURNAL DD DSNAME=sorted backup journal from Job 2,
// DISP=SHR
//SYSIN DD *
DELETE (MCDS cluster name) CLUSTER PURGE
IMPORT INDATASET(name of exported copy from Job 4) -
OUTDATASET(MCDS cluster name)
/*

```

*Figure 2-8 JCL to merge journal and restored CDS*

ARCIMPRT can recover non-key-range data sets. It uses the key boundaries to determine which journal records need to be applied to the CDS being recovered. If you reorganize your multicluster control data sets, you should immediately back them up. If you do not, and you attempt to perform CDS recovery using a CDS backup version created before the CDS was reorganized, the recovery fails with message ARC0466I.

After these steps have been successfully run, you may restart DFSMSHsm. You may want to verify that the CDSs are correct by running an audit. See Chapter 5, “Using FastAudit/390” on page 39 or Chapter 9, “Auditing DFSMSHsm CDSs using command audit” on page 237.





# Control data set records

This chapter describes the various control data set (CDS) record types.

### 3.1 CDS record types, functions and relationships

There are as many as 26 different types of records in the DFSMSHsm CDSs. A few record types are related to optional functions and may not be present in some environments. Two examples are the type Q (ABR) record that is absent if ABARS is not being used and the type W records that only exist if at least one dumpclass has been defined.

Table 3-1 provides a table of the record type, the common record name and a short description. The short description gives you some idea of where the record is stored.

*Table 3-1 CDS record table*

| Record Type | Record Name                       | Record Description                                                                                                                                        |
|-------------|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| A           | MCA                               | MCDS alias entry record                                                                                                                                   |
| B           | MCB                               | BCDS data set record (true name)                                                                                                                          |
| C           | MCC                               | BCDS version record                                                                                                                                       |
| D           | MCD                               | MCDS data set record (true name)                                                                                                                          |
| G           | DGN                               | BCDS dump generation record                                                                                                                               |
| L           | MCL                               | BCDS backup of a migrated data set                                                                                                                        |
| M           | MCM                               | BCDS move backup version record                                                                                                                           |
| N           | VAC                               | MCDS JES3 volume activity count                                                                                                                           |
| O           | MCO                               | MCDS VSAM associations record                                                                                                                             |
| P           | MCP                               | BCDS eligible volume record                                                                                                                               |
| Q           | ABR                               | BCDS ABARS Record                                                                                                                                         |
| R           | BCR<br>BVR<br>DCR                 | BCDS backup control record<br>BCDS backup cycle volume record<br>BCDS Dump control record                                                                 |
| S           | DSR<br>L2CR<br>MCR<br>MHCR<br>VSR | MCDS daily statistics record<br>MCDS ML2 record<br>MCDS management control record<br>MCDS multiple host processor record<br>MCDS volume statistics record |
| T           | TTC (or TTOC)                     | OCDS Tape Table of Contents                                                                                                                               |
| U           | MCU                               | MCDS user record                                                                                                                                          |

| Record Type | Record Name | Record Description        |
|-------------|-------------|---------------------------|
| V           | MCV         | MCDS volume record        |
| W           | DCL         | BCDS dump class record    |
| X           | MCT         | BCDS backup volume record |
| Y           | DVL         | BCDS dump volume record   |
| 1           | MC1         | MCDS ML1 FREESPACE record |

### 3.1.1 DFSMShsm record type extended description

Of the 26 record types, only ten are part of any DFSMShsm audit and are discussed further. Those include:

|               |                                    |
|---------------|------------------------------------|
| MCA           | MCDS Alias Entry Record            |
| MCB           | BCDS Data Set Record (true name)   |
| MCC           | BCDS Backup Version Record         |
| MCD           | MCDS Data Set Record (true name)   |
| MCO           | MCDS VSAM Association Record       |
| MCP           | BCDS Eligible Volume Record        |
| MCV           | MCDS Volume Record                 |
| MCT           | BCDS Backup Volume Record          |
| TTC (or TTOC) | OCDS Tape Table of Contents Record |
| ABR           | ABARS Aggregate Record             |

All 26 record types are described in detail, including record layouts, in the manual *DFSMShsm Diagnosis Reference*, LY35-0111, a licensed manual not part of the general manual set.

## 3.2 Migration records

There are several types of DFSMShsm migration records which we describe here.

### 3.2.1 MCD record

The key of the MCD record is the true name of a data set that has been migrated. There are nine fields in that record that are especially important to be able to recall a data set, and thus important to audit. Those fields are:

- ▶ The record key, which is the true name of the data set
- ▶ The volume on which the data set resides (as far as DFSMSHsm knows)
- ▶ The migration status, that is Primary, ML1 or ML2
- ▶ Whether VSAM or non-VSAM
- ▶ Whether migrated to an Small Data Set Packing data set (SDSP)
- ▶ The DFSMSHsm generated name for the migration copy (see MCA record below).
- ▶ The number of blocks that the migrated data set occupies on the migration volume
- ▶ The relative block ID if the data set is migrated to tape
- ▶ The TTOC relative record number that contains the reference to the migrated data set if migrated to tape.

The MCA or Migration Control Alias Record has a key matching the name that DFSMSHsm generated when it migrated the data set. That generated name is of the format:

```
prefix.HMIG.Tssmmhh.userqual1.userqual2.iydd
```

Where:

- ▶ `prefix` is the migration prefix defined for DFSMSHsm migrated data sets.
- ▶ `HMIG` is a fixed value.
- ▶ `Tssmmhh` is the time of migration in reversed form. In order to assure unique keys, the T may be replaced by another character if another dataset was migrated within the same second of the same day and had identical first two qualifiers.
- ▶ `userqual1` and `userqual2` are the first two qualifiers of the true name of the data set that has been migrated.
- ▶ `iydd` is the date of migration where `i` represents the decade, `y` represents the year and `ddd` is the Julian date. Thus a data set migrated on January 30, 1989 would have a date field of H9030 (H is the 8th letter of the alphabet). The Julian date 2002135 would have a DFSMSHsm date field of J2135.

The MCA record has only four fields that are of any significance to audit. Those are:

- ▶ The record key, which is the generated name.
- ▶ Recall intercept type which indicates whether the record is for a VSAM object or a non-VSAM data set. If this field indicates it is a VSAM data set, then one or more MCO records should exist.

- ▶ If VSAM, whether the record is for a cluster or an AIX®.
- ▶ The true name of the data set that was migrated.

In effect, the true name record (MCD) and the alias record (MCA) have pointers to each other. Where the pointer back from the MCA to the MCD is especially important is when an audit of tape records (TTOC) has to identify the true name of a data set migrated to ML2, since the TTOC contains only references to DFSMSHsm generated names.

There is an MCA record for each component that exists for a VSAM cluster, that is cluster, data, index, AIX, and path.

### 3.2.2 MCO record

The MCO record is called the VSAM associations record. It records details of the components of a VSAM cluster. The key of the MCO Record is the DFSMSHsm generated name for the VSAM cluster. Remember that DFSMSHsm migrates and recalls a VSAM data set at the cluster level.

The MCO record indicates the name of each component of the cluster that in effect is a pointer to each related MCA record. These fields plus the record key are the only fields that are examined by audit.

Figure 3-1 shows the relationship between MCD, MCA, and MCO records as well as the volume and TTOC records described later. All of these records are in the MCDS except the TTOC record that is in the OCDS.

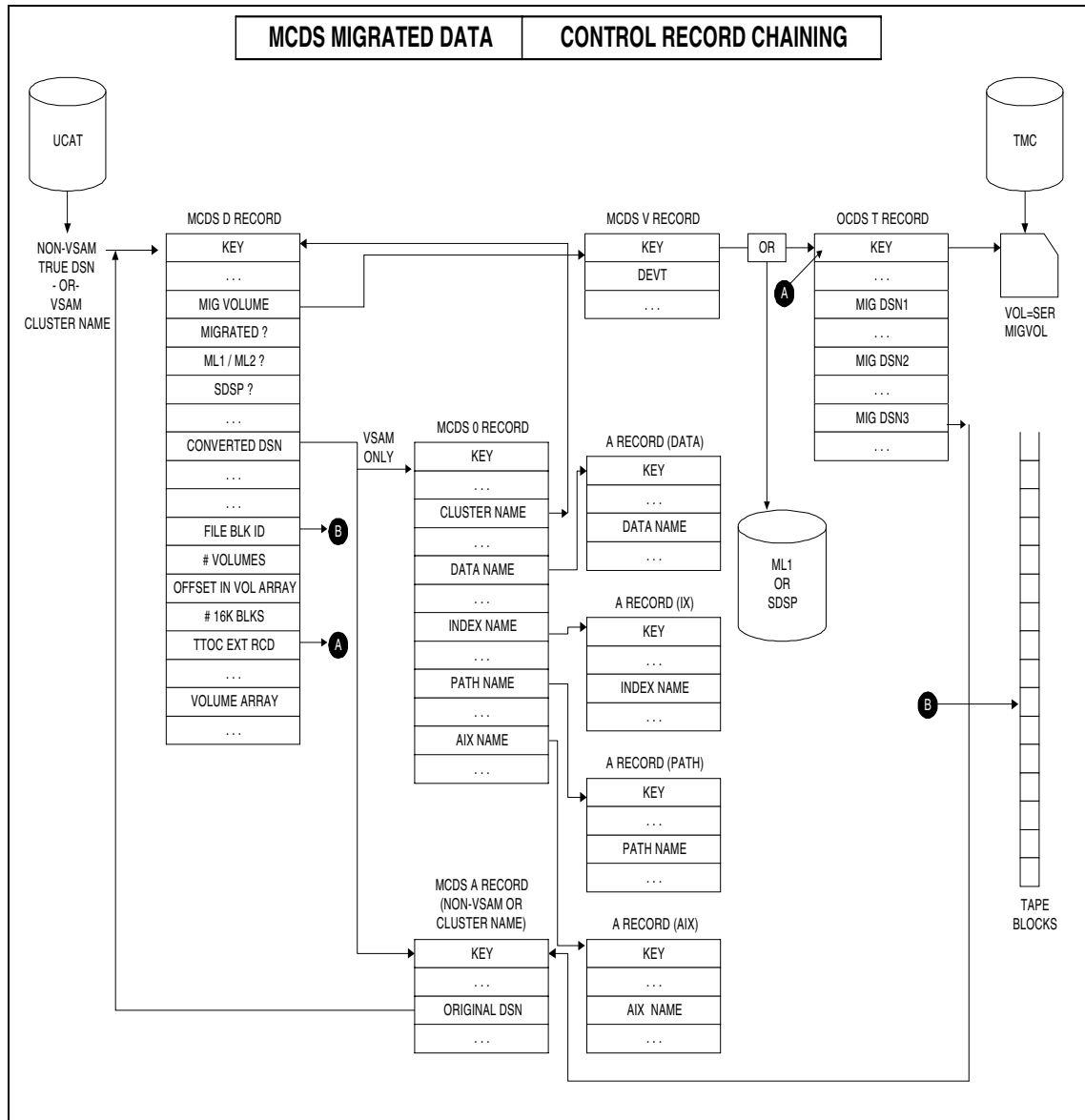


Figure 3-1 Relationships between various migration related records



## 3.3 Backup records

To maintain its inventory of backup versions and backup volumes, DFSMSHsm uses information from at least three types of BCDS records, and four types if the backup is on tape. Figure 3-2 shows the relationship of those records.

### 3.3.1 MCB record

The key of the MCB record is the true name of a data set for which there is at least one backup copy.

Only three fields of the MCB are examined during an audit. They are:

- ▶ The record key, which is the true name of a data set that has been backed up by DFSMSHsm
- ▶ The DFSMSHsm generated name for each of the one to 100 backup versions that currently exist
- ▶ Total number of backup versions that exist for the dataset

Those DFSMSHsm generated names for each of the backup versions are similar to the generated name for the migration copy described above. The generated name is of the format:

`prefix.BACK.Tssmmhh.userqual1.userqual2.iydd`

Where:

- ▶ `prefix` is the backup prefix defined for data sets backed up by DFSMSHsm.
- ▶ `BACK` is a fixed value for backup version names.
- ▶ `Tssmmhh` is the time of backup in reversed form. To maintain unique keys, DFSMSHsm will change the T to some other value if another dataset has been migrated within the same second of the same day and has the same first two qualifiers.
- ▶ `userqual1` and `userqual2` are the first two qualifiers of the true name of the data set that has been migrated.
- ▶ `iydd` is the date of backup where `i` represents the decade, `y` represents the year and `ddd` is the Julian date. Thus a backup version created on January 10, 1999 would have a date field of I9010 (I is the 9th letter of the alphabet). The Julian date 2001100 would have a DFSMSHsm date field of J1100.

The generated names point to the MCC record(s) for each backup version, whether created by DFSMSHsm's automatic backup process or by a user command backup.

There is but one MCB record for any true name, but there are potentially multiple MCC records reflecting multiple stored backup versions. The MCC record length is fixed, but the MCB record length varies with the number of backup versions that currently exist for the true name dataset. The current limit for backup versions is 100, so MCB records could be quite large and point to many MCC records. All those relationships must be verified during a BCDS or complete AUDIT.

### 3.3.2 MCC record

The key of an MCC record is the DFSMSHsm generated name for a particular backup version of some data set. There are 5 fields in the MCC record that are essential for recovery of a data set from a backup version and thus of interest to audit. They are:

- ▶ The record key that is the DFSMSHsm generated name for a particular backup version which in effect points back to the MCB record
- ▶ The volume on which the backup version resides
- ▶ If backed up to tape rather than DASD, the relative block ID where the version is written on the backup tape
- ▶ The number of blocks occupied by this backup version
- ▶ The TTOC relative record number that contains the reference to the backup version

Figure 3-2 depicts the several CDS records involved with DFSMSHsm backup copies. They include the MCB record and the several MCC records that represent the backup versions plus the volume and TTOC records. The TTOC records are in the OCDS and the other records are in the BCDS.



### 3.4.1 TTOC record

The Tape Table of Contents (TTC or TTOC) records, which are stored in the OCDS, are arguably the most complex of the several CDS records that are processed by audit. They are complicated for several reasons, including the fact that DFSMSHsm allows data sets to span tapes, and because there may be, and almost always will be, several TTOC records required to describe the contents of a single tape. There are several count fields and chains between records that must exist to adequately maintain the inventory of tape data, be it migrated data sets or backup copies. As an aside, there is no logical difference between TTOC records for migrated data and backup tapes.

Let us look at the spanned tape issue. Current migration and backup tapes may contain several hundred to several thousand data sets. Let us assume that the current tape that is the target for migration has 100 blocks remaining free at the end. Further, the DFSMSHsm parameters allow a file that requires 100 blocks or more to span from one tape to another. The next data set to be migrated has 150 blocks, so DFSMSHsm begins writing to the nearly full tape, and when necessary mounts a new tape to write the remainder.

The TTOC for the first tape must include an indication that the second tape is a continuation of the first. The TTOC for the second tape must indicate that the first tape is a predecessor. Both TTOC records must record that the data set in question exists on their volumes, one at the end of the first volume, and the other at the beginning of the second volume. If one of those tapes becomes a candidate for RECYCLE, both tapes will be recycled. It is not unusual to have four or five tapes chained together because of spanned data sets.

Audit also checks the block counts indicated in the several relative records and the number of "valid" data sets shown in the several relative records and verifies that related fields are correct.

During an AUDIT MEDIACONTROLS, DFSMSHsm reads the entire contents of the backup or migration tape and verifies that most fields in the TTOC records are accurate.

Figure 3-3 shows the relationship between TTOC records for the same volume as well as records for related volumes with a data set that spans two volumes.

## TTOC RECORD CHAINING

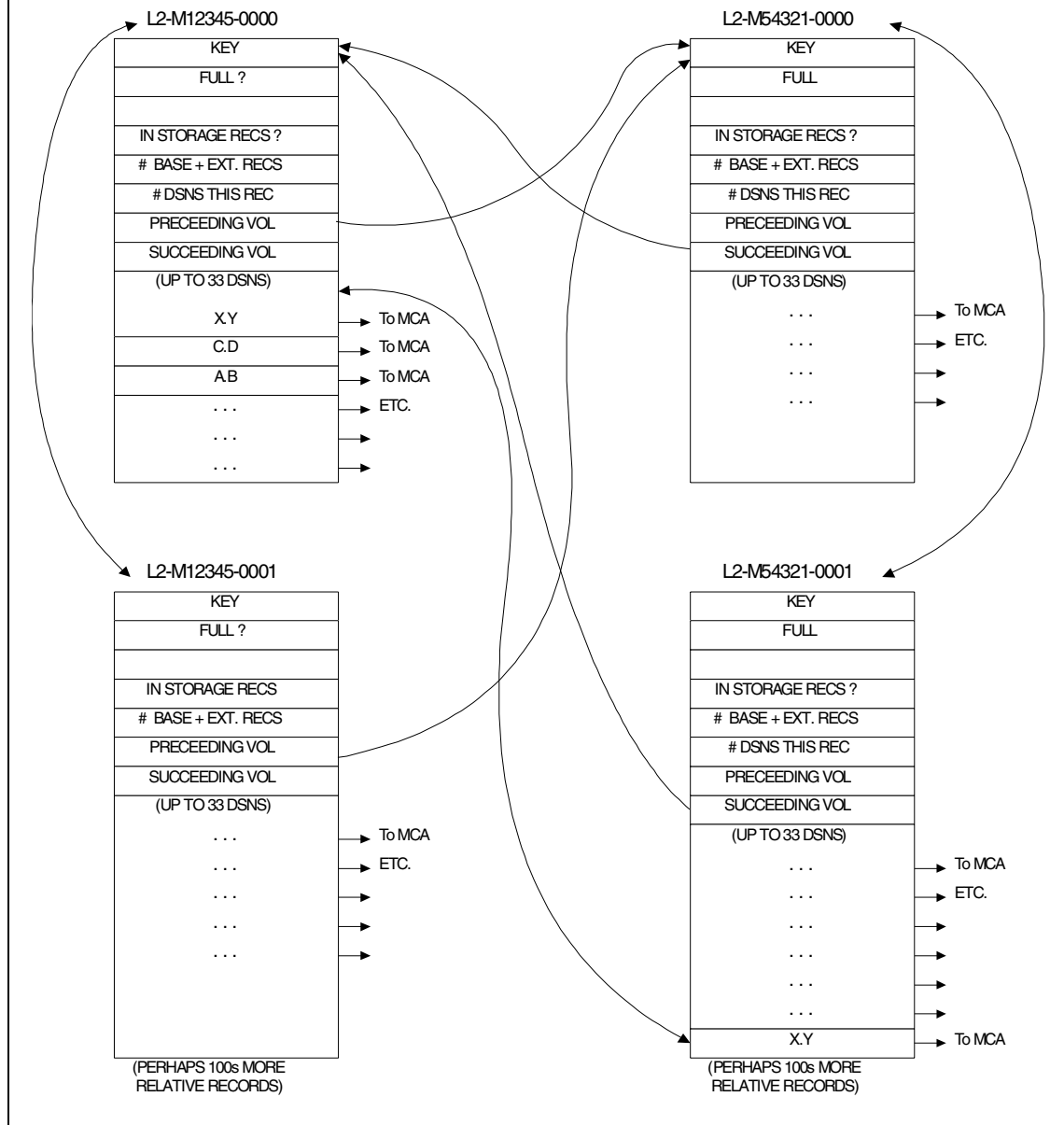


Figure 3-3 Relationship between TTOC records

## 3.5 Tapes in failed create status

At any given time the TTOC record(s) for a tape that is currently in use may be, in fact likely is, incorrect. This is because of the way that DFSMSShsm attempts to reduce I/O to the OCDS. If DFSMSShsm is migrating to a tape, it may write several hundred small data sets within a very, very short time. To reduce the I/O to the OCDS, DFSMSShsm accumulates in storage up to 32 changes to any TTOC record or one hour's worth of changes whichever occurs first, before doing the I/O to update a TTOC record for that volume.

As it begins the accumulation of changes, it sets a flag indicating that in-storage records exist and it turns that flag off after it has updated the TTOC with those record changes. If the system fails in the mean time, or if DFSMSShsm abends with certain types of errors, several TTOC records in the OCDS may not include all data sets that have been written successfully to their respective tapes.

A tape that is in Failed Create status cannot be recycled. Recycle sees the "Failed Create" bit and ends without moving any data.

DFSMSShsm alerts you to the existence of those incompletely updated TTOC records with an ARC0378I message at the time it next initializes. This is one of the few cases where a message recommends you execute an audit, in this case AUDIT MEDIACONTROLS.

### 3.5.1 ABR record

The Type Q or ABR record describes a specific backup version created by Aggregate Backup And Recovery Support (ABARS). The primary value of the ABARS record is to simplify the execution of an ABARS recovery at a disaster site. If an ABR record exists, the recovery command does not have to include the ABARS control file name and volume serial, it can be simply "ARECOVER aggrname".

One DFSMSShsm audit option checks that any output files created in a particular ABARS backup are cataloged. Another audit option flags the record in error if the DFSMSDss data file name field is null. Mainstar's *FastAudit/390* includes a comparison of tape volumes listed in ABR records versus tape records in the tape management catalogs of DFSMSrmm, CA-1, or ZARA.

ABARS backup tapes do not have TTOC records like other DFSMSShsm backup tapes do.

### 3.5.2 MCP, MCT and MCV records

An MCP, or BCDS eligible backup volume record, exists for any DASD volume that has been backed up or dumped by DFSMSHsm. During some audit operations DFSMSHsm verifies whether the volume still exists, actually whether it is online. For example while auditing the contents of a DFSMSHsm ML1 volume, when a VTOCCOPY from a primary volume is found, DFSMSHsm verifies that there is still an MCP record for the primary volume. AUDIT identifies at least 10 different inconsistencies between MCP records and other records or volume contents.

An MCT, or BCDS backup volume record, exists for every backup (tape) volume in DFSMSHsm's inventory that contains backup versions. Here are two examples of where this record is checked:

- ▶ During a BCDS or MCDS AUDIT, AUDIT verifies that volumes in DFSMSHsm's RACF® TAPEVOL list all have an MCT record.
- ▶ During an audit of tape volumes, AUDIT verifies that an MCT record exists for every tape for which there is a TTOC record.

There are at least 10 different errors that AUDIT may flag involving MCT records.

An MCV, or MCDS volume record, exists for every primary or migration volume, both Migration Level 1 and Migration Level 2. When an AUDIT is examining a migration tape volume it verifies that there is an MCV record for each volume. When AUDIT is examining migrated dataset records (MCD) it verifies that the DEVT in the MCD matches the DEVT in the MCV record. There are in all about 12 error situations that AUDIT might detect involving MCV records.







## Why audit the DFSMSHsm control data sets?

What are the conditions under which a DFSMSHsm control data set (CDS) AUDIT is required or recommended? It may seem obvious that one would submit an audit when DFSMSHsm alerts you to a CDS error. But DFSMSHsm does not often just pop up an error message that says "Control Data Set Error". In fact there is only one such message, ARC0184I, that mentions CDS error and it is an "I" type message. A better answer is whenever you know or suspect that the control data sets or related metadata sources like user catalogs, VTOCs, or tape catalogs have been damaged and may no longer be in sync.

## 4.1 Reasons for auditing the CDSs

Some possible conditions that suggest those data sources might be in error or should be audited include:

- ▶ Master or User Catalog error messages
- ▶ Recovery of a Master or User Catalog
- ▶ VSAM error messages for a CDS
- ▶ Recovery of a DFSMSHsm CDS
- ▶ Recovery of a Primary DASD volume
- ▶ Loss or recovery of an ML1 volume
- ▶ Loss or recovery of a DFSMSHsm SDSP
- ▶ Loss of a DFSMSHsm owned ML2, Backup, or ABARS tape
- ▶ Tape library errors possibly affecting DFSMSHsm tapes
- ▶ Receipt of one of several DFSMSHsm messages
- ▶ During a disaster recovery test or an actual recovery
- ▶ Following a DFSMSHsm or system crash, especially when DFSMSHsm was very busy
- ▶ After accidental deletion of CDS records
- ▶ After eliminating a DFSMSHsm aggregate from SMS
- ▶ After consolidating two or more DFSMSHsm environments or splitting a DFSMSHsm environment into two or more parts

Let us look at a few examples of those error conditions to see why an audit of the CDSs and the other meta data sources should be done.

## 4.2 Recovery of a master or user catalog

DFSMSHsm relies on the catalog information to detect that a data set is migrated and needs to be recalled. If the catalog entry for a data set has a volume serial of MIGRAT, DFSMSHsm is invoked to recall that file from migration level 1 or migration level 2 based on information in the DFSMSHsm MCDS. If the data set's catalog entry is missing, or does not say MIGRAT, DFSMSHsm cannot be invoked to do the recall.

Catalog entries can be incorrect for a variety of reasons including inappropriate human actions or program errors. They can also be due to incomplete recovery of a damaged catalog. As one of the final actions to the recovery of any catalog,

especially one that might contain data sets under DFSMSHsm control, an AUDIT should be done. That audit should compare catalog with CDS information, and for non-migrated data the volume contents as well. Further, it should analyze in both directions, for example, from catalog entry to MCDS and from MCDS entry back to catalog.

Mainstar's *FastAudit/390* provides those capabilities as well as the ability to compare master and user catalogs with the Tape Management Catalogs of DFSMSrmm, CA-1, and ZARA.

### 4.3 VSAM errors in a DFSMSHsm CDS

DFSMSHsm CDSs are VSAM key sequenced data sets (KSDS). One of the features of the KSDS structure is the index which points to sequence set indices (SSI), which in turn indicate the range of record keys in the VSAM control areas (CAs), which contain the actual data. When a VSAM KSDS is new, the logical and physical structure of the file looks like Figure 4-1, with just four CAs for simplicity in our discussion.

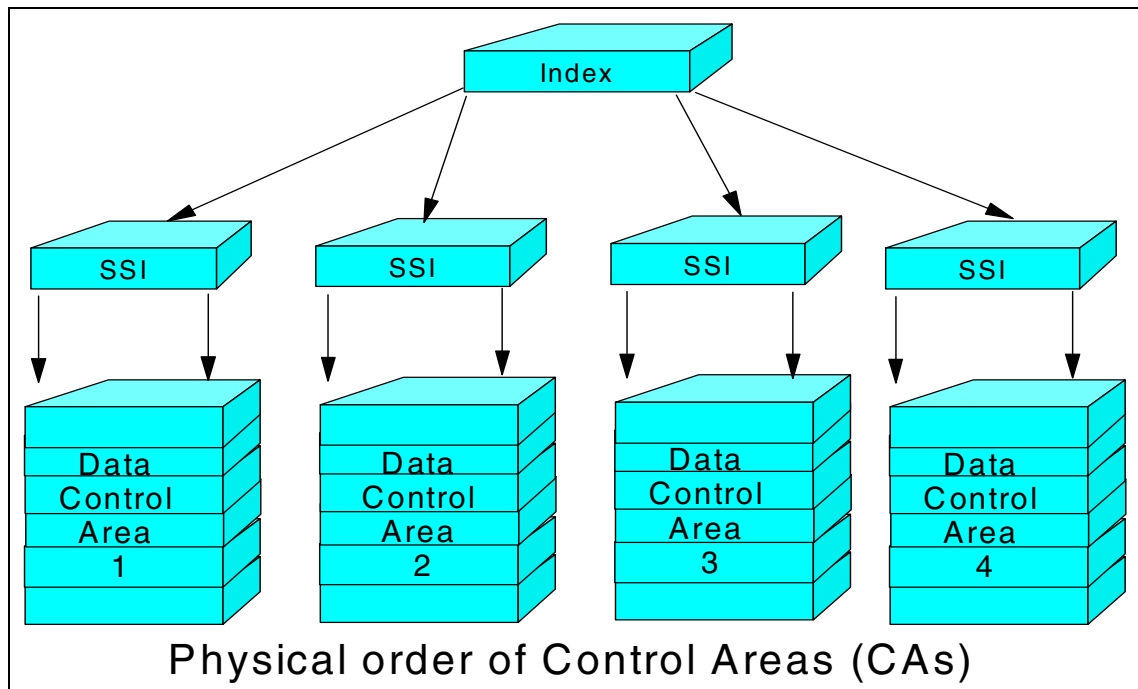


Figure 4-1 VSAM KSDS physical structure before CA splits have occurred

Once records are inserted and a CA split has occurred, in this example in the first CA, the VSAM structure looks logically as in Figure 4-2. Horizontal Chain Pointers (HCPs), shown by the arrows, connect the sequence set indices (SSIs) to help maintain the logical sequence of records. The CA on the right is now logical CA number 2, but is still physically the fourth CA in the data portion of the file.

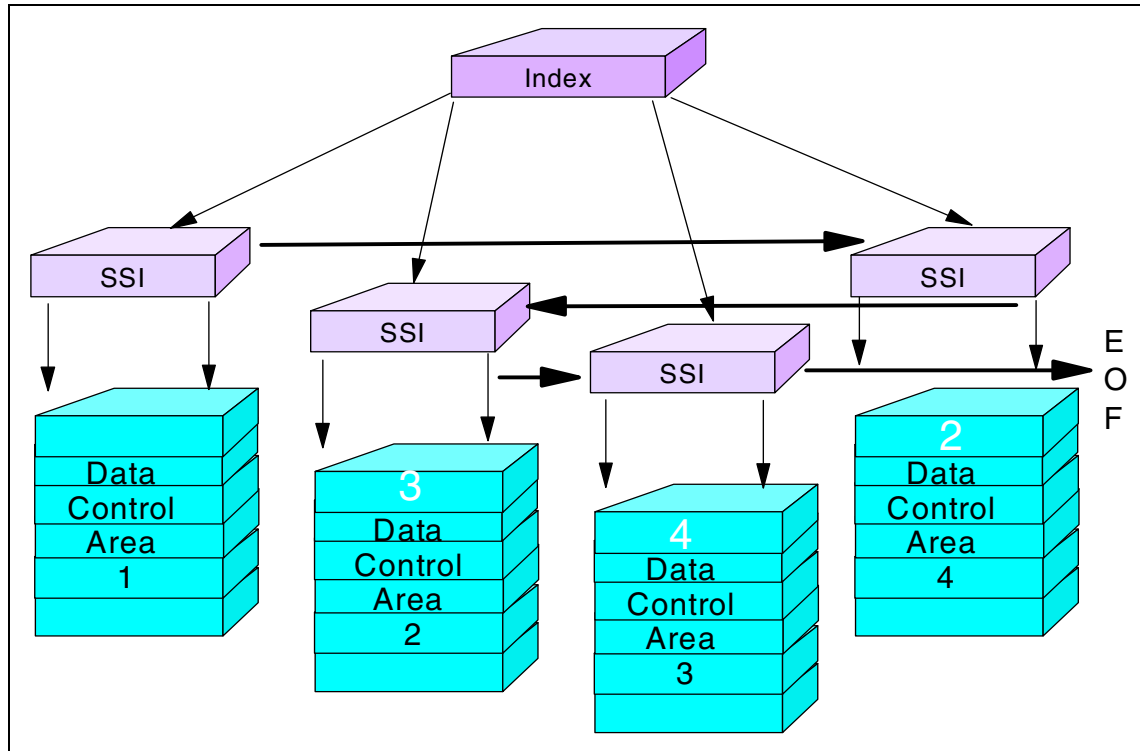


Figure 4-2 VSAM logical structure after a CA split in the first data CA

If the file were being processed in usual VSAM fashion, a request to read or write a record would go to the cluster's index, then to the appropriate SSI, then to the control interval (CI) that contains the record. If a record is being inserted that causes another CA split, the horizontal chain pointers are rewritten to reflect the new logical structure. The HCPs (arrows) allow keyed VSAM processing and allow utilities like REPRO to maintain the data records in the correct order when accessed through the index of the KSDS.

If the horizontal chain pointers are damaged as in Figure 4-3, one or more CAs may be inaccessible, making the records in those CAs impossible to reach by normal keyed VSAM processing. In the figure, the SSI for the third data CA is no longer part of the SSI horizontal chain.

A backup of the cluster with REPRO, which follows those chains, will miss records because of the inaccessible CA. A backup using DFSMSdss or other utilities that backup at the track level will restore the file with the chain pointers broken.

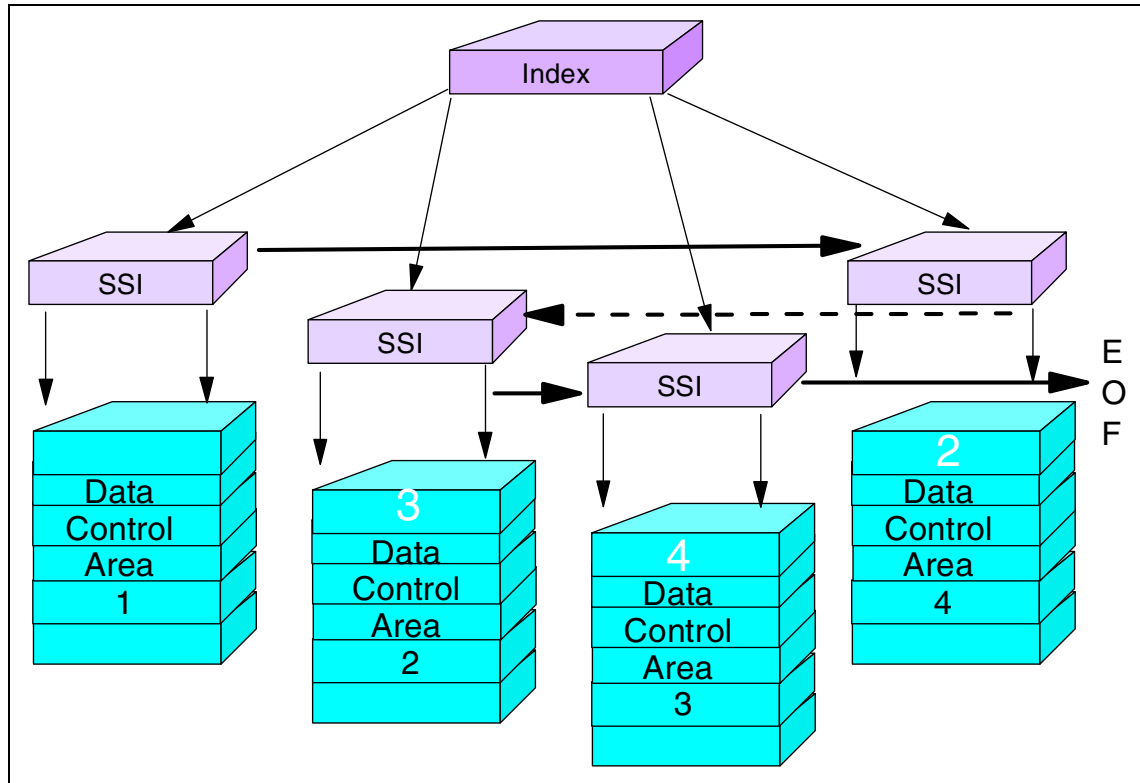


Figure 4-3 VSAM KSDS with horizontal chain pointer damage

If one is aware of the HCP damage, a REPRO of the data portion only will capture all the records, since it reads the data portion sequentially rather than through the index. But the key is, one must be aware of the condition since REPRO of the data portion only is not the usual way that REPRO is used.

As an example, one organization was using the DFSMSHsm CDS version backup to retain seven backup versions. One weekend during a reorganization of the CDS they inadvertently had DFSMSHsm still active on one image in the JES complex. This caused an HCP error in the reorganized CDS which they did not recognize until the ninth day after it occurred. They were using REPRO of the VSAM cluster to make the backup rather than DFSMSdss. By the time they recognized the error, they no longer had a backup copy that contained all the

data. Based on record counts from previous backups, they estimated that well over 85,000 records were lost.

*FastAudit/390* identifies horizontal chain pointer damage so that the storage administrator can take action, like REPRO of the data portion, to restore the CDS in a way that recovers all the data records.

## 4.4 Audit during a disaster recovery

Backup of some or all of an organization's user catalogs is an integral part of a disaster recovery plan. Ideally those catalog backups are synchronized with the backup of the data sets that those catalogs point to. But real world experience shows that it is very difficult to synchronize all data and catalog backups, especially when relying on full volume dumps for recovery of some data sets. Just one unsynchronized condition could significantly delay the recovery of several critical applications or even make them unrecoverable.

For those organizations using ABARS, the intent of that process is that the user catalogs will reflect only those data sets that were actually recovered. If those organizations cannot begin with empty catalogs, there will invariably be many entries in the catalogs that do not exist. JCL errors caused by DISP=OLD data sets that do not exist, or new data sets that are already cataloged will confound troubleshooting during recovery.

No matter what processes are used for recovery, using an audit to identify discrepancies between catalogs and volumes and DFSMSHsm CDSs so that orderly cleanup can be done will help speed the resumption of production processing.



## Using *FastAudit/390*

*FastAudit/390* is a complete metadata interrogation tool that helps keep your z/OS environment trouble-free with fast, accurate, and flexible audits. Audits of some, or all, of the z/OS metadata environment can be run frequently, even daily, to identify and fix problems early, avoiding loss of access to data or overlaid data.

The selectable unit design allows you to choose only the groups of functionality that are meaningful to you. Comprehensive audit support is provided in the following key areas:

- ▶ HSM *FastAudit*
- ▶ Catalog & DASD Audit
- ▶ Tape Audit

## 5.1 Authorization codes

Mainstar Software Corporation maintains authorization codes that enable the product to perform its functions. The product may be purchased or run on a trial basis. Implementation of those functions requires an update to the FAUTH member of a PARMLIB library shipped with the product. Figure 5-1 shows the authorization syntax.

```
EDIT          FAUDITX.QABETA.PARMLIB(FAUTH) - 01.03          Columns 00001 00072
000001 *****
000002 *   PRODUCT SECURITY                                   *
000003 *****
000004 :PRODUCT_SECURITY
000005   AUTHCODE           = AAAAAAAAAA
000006   EXPIRATION_DATE    = 00NNNN          /* 00/00/2000          */
***** ***** Bottom of Data *****
```

Figure 5-1 Authorization syntax

## 5.2 Master catalogs

The accuracy of master catalogs, user catalogs and alias names related to the user catalogs is critical for many audits. If your environment has an individual master catalog for each MVS™ image in a global complex, with shared DASD, DFSMSHsm, and tape library catalog, then the differences between the master catalogs should be reconciled. This process should precede any other audits involving the catalogs. *FastAudit/390* provides a cataloged procedure, *FAUDITMC*, to do this. If you are not the person who is responsible for maintaining the master and user catalogs, you should work closely with whomever is before correcting any of the conditions found. You must determine a base master catalog and audit all other master catalogs against the base.

## 5.3 User catalogs

User catalogs contain basic information such as data set name, device type, volume serial, volume sequence and file sequence numbers. In other words it simply points to where data should be. It is the “*should be*” part that concerns *FastAudit/390*. For example, if a user catalog indicates a data set is on VOLSER MIGRAT, is it really migrated? Conversely if a data set is migrated and valid, is the VOLSER cataloged to VOLSER MIGRAT?



The forward and backward (two-way) audit concept is very important to understand with *FastAudit/390*. In addition to the user catalog and MIGRAT relationship, the user catalog to DASD, to DFSMSrmm and to CA-1 TMS can be audited. Again, both the catalog contents and the subject media contents are audited against each other. This results in the most comprehensive auditing facility available.

Use cataloged procedure FAUDITCO to audit the contents of user catalogs.

Before launching audits of DFSMSHsm and others against the user catalogs it is wise to audit the catalogs themselves. User catalogs can contain old residual data that is no longer accurate or not needed. Here are some examples:

- ▶ A user catalog contains information about a data set but there is no alias defined to the catalog from the master catalog (an orphaned entry).
- ▶ Other problems exist, such as data set is cataloged to a DASD volume that is not online to the system.
- ▶ Generation data set (GDS) sphere record is defined but has no entries.
- ▶ Catalog exists but has no aliases defined to it (orphaned).

## 5.4 DFSMSHsm CDSs

The DFSMSHsm system provides functions for archiving (migrating), backing-up and dumping data on primary volumes along with recall, recover, and restore, respectively. In addition the aggregate backup and recovery system (ABARS) records are stored in the Backup Control data set (BCDS). The amount of data under the control of DFSMSHsm can be as much as 80% of a corporation's entire inventory. It is critically important that DFSMSHsm is functioning properly and its inventory of data is recorded accurately. Unfortunately, problems are frequently discovered at the most inopportune times. Furthermore, the problems may be the "tip of the iceberg" where similar errors exist for other data.

IBM provides audit and enhanced audit programs to audit records in DFSMSHsm and in many cases to automatically fix them. So why use a separate auditing product? How can it save money?

The answer to the first question is speed. *FastAudit/390* audits DFSMSHsm CDS records faster than the IBM provided audits and hundreds of times faster than the user catalog to DFSMSHsm audits. Because of the speed, the user may be pro-active in discovering and repairing problems before they are encountered. In addition, when problems are discovered at critical times, they can be repaired faster, thus minimizing the outage.

The answer to the second question is two fold. Because the audits run fast, orphaned data can be eliminated which frees up CDS space. A great deal of data is stored on tape. These tapes, under DFSMSHsm control, usually have an interface to other systems such as DFSMSrmm or CA-1 that control the return of the tapes to the scratch pool. *FastAudit/390* provides audits to analyze the relationship of DFSMSHsm to DFSMSrmm and DFSMSHsm to CA-1 TMS. Cataloged procedures FAUDITRM and FAUDITTP, respectively, are the audits to run. These audits have recovered millions of dollars in tape inventory, because the tapes were never returned to the scratch pool.

## 5.5 Direct access devices

At the heart of any system is the storage containing the data and programs that make the system run. It is not uncommon for the amount of DASD installed to exceed 40–50 terabytes for a single image. Some installations exceed 10,000 DASD volumes in a single environment. The quote, “if storage were free and unlimited, it would still need to be managed”, is coming true. Yet there are too few tools than can operate effectively in an environment this size. An IDCAMS **LISTCATALOG ALL** command can and does run for hours. If meaningful information is to be gleaned, additional programs using SAS or REXX are required to isolate or capture DASD entries, which in turn, adds to the run time. Also retrieving information from the VTOC of the volumes is time consuming. Comparing the results of a **LISTVTOC** or **DCOLLECT** extraction to a catalog extraction adds still more time until the results are no longer current and dangerous to use. *FastAudit/390* provides an audit that can be run in about 1.5 hours in a system with thousands of volumes representing millions of data sets while still verifying the accuracy of the errors. Cataloged procedure FAUDITDA and its little brother FAUDITDR provide a convenient method of identifying and correcting catalog-DASD conflicts.

## 5.6 Tape management catalogs for RMM and CA-1 TMS

The term tape management catalog (TMC) is used generically. It represents the database containing information about tapes. *FastAudit/390* has catalog to tape audits for DFSMSrmm (IBM’s Removeable Media Manager) and CA-1 TMS (Computer Associate’s Tape Management System). These audits must not be confused with the DFSMSHsm to TMC audits which were described earlier. TMCs were developed to insure that valuable data was not overwritten if a tape were inadvertently mounted as scratch. If a real scratch tape is mounted and has an un-expired date on its internal label, the TMC can over-ride it automatically. These two functions imply elaborate management criteria, such as when a tape becomes scratch, when a tape goes “off site” (vaulted), and many others. A

criteria common to both systems is “scratch when un-cataloged”. Phrased differently, it is called catalog retention. Typically this is used in generation data set groups (GDGs) processing where the oldest data set in the group becomes un-cataloged and the tape it is on can be returned to the scratch pool. Tapes can contain many files, and files on the tapes can span to other volumes. If one data set on a tape remains cataloged for a long period of time and all other data sets become un-cataloged, none of the tapes become eligible for scratch until they are all un-cataloged. It becomes prudent for the user community to design processes where multi-file volumes all become un-cataloged at similar times or have the same expiration date. Otherwise valuable resources can be wasted.

The relationship between the catalogs and the TMC is very important and it is for this relationship that the tape audits were designed. The procedures to use are FAUDITCR for the DFSMSrmm to catalog audit, and FAUDITTC for the CA-1 TMS to catalog audit.

## 5.7 Diagnostic reporting methods

To analyze and correct the diagnostics that *FastAudit/390* provides, a series of CLISTs and panels are provided. To display the primary fix panel, execute the CLIST FIXAUDIT using the command:

```
EX 'PFX.FAUDIT.CLIST(FIXAUDIT)'
```

Figure 5-2 shows the primary fix panel.

```
<V6 R1.01> FAST AUDIT.  CHOOSE WHICH AUDIT TO FIX
COMMAND==> 1
1 MCDS    CREATED BY PROCS FAUDITMD, FAUDITCT OR FAUDITAL
2 BCDS    CREATED BY PROC FAUDITBD
3 OCDS    CREATED BY PROC FAUDITOD
4 TAPE    CREATED BY PROCS FAUDITTP, FAUDITZR OR FAUDITRM
5 CAT     CREATED BY PROC FAUDITCO
6 DASD    CREATED BY PROC FAUDITDA OR FAUDITDE
7 MASTCAT CREATED BY PROC FAUDITMC
8 CATRMM  CREATED BY PROC FAUDITCR
9 CATTMS  CREATED BY PROC FAUDITTC

ENTER  END  TO TERMINATE ONLINE FUNCTION
```

Figure 5-2 Primary Fix panel

The audits, numbered 1 through 9, create an error summary panel of each type of audit. The audit must have been run previously for the summary to be displayed. Refer to *FastAudit/390* User Guide for complete instructions on

running the audit. To view the summary of each of the audits, type the corresponding number of the audit at the command prompt.

In this example, the MCDS audit from the cataloged procedure FAUDITAL was selected. Each diagnostic has a corresponding diagnostic number, a count of the number of occurrences and a brief description of each problem. The diagnostic numbers end with a C, W, V or I.

- ▶ **C** means critical. The data set in question cannot be recalled or otherwise processed.
- ▶ **W** means warning. The data set probably can be processed but something is not quite right.
- ▶ **V** is a volume summary of a diagnostic. In Figure 5-3, error 1107W which occurred 597 times has a corresponding 1107V message that occurred once. In other words all the errors were for one volume.
- ▶ **I** means information. While processing records for audit purposes, there are numerous times when information is interesting to know as opposed to actually being in error.

|                                                |       |                                                  |               |
|------------------------------------------------|-------|--------------------------------------------------|---------------|
| <V6 R1.01> FASTAUDIT MCDS SUMMARY              |       |                                                  | Row 1 of 31   |
| OPTION ==>                                     |       |                                                  | SCROLL==> CSR |
| ENTER F TO DISPLAY FIXES OR B TO BROWSE ERRORS |       |                                                  |               |
| S NUMBER                                       | COUNT | MESSAGE                                          |               |
| B 11D1C                                        | 1     | MCD ENTRY IS NOT CATALOGED                       |               |
| 11V1C                                          | 0     | MCO VSAM COMPONENT IS NOT CATALOGED              |               |
| 11D2C                                          | 1     | MCD ENTRY IS CATALOGED ON DIFFERENT VOLUME       |               |
| -----                                          |       |                                                  |               |
| 1103W                                          | 0     | MCD ENTRY IS MISSING THE MCA ENTRY               |               |
| 1106C                                          | 0     | MCD IS ON VOLUME WHICH HAS NO MCV ENTRY          |               |
| 1106V                                          | 0     | SUMMARY OF VOLUMES HAVING NO MCV ENTRY           |               |
| 1107W                                          | 597   | MCD LEVEL 2 IS ON VOLUME WHICH HAS NO TTOC ENTRY |               |
| 1107V                                          | 1     | SUMMARY OF VOLUMES MISSING IN TTOC               |               |
| -----                                          |       |                                                  |               |
| 1125W                                          | 0     | TTOC ENTRY MARKED INVALID, MCD/MCC EXISTS        |               |
| 1126I                                          | 0     | SUMMARY OF TTOC VOLUMES WITH INVALID DSNS        |               |

Figure 5-3 FastAudit/390 MCDS error summary panel

## 5.7.1 Diagnosing error 11D1C

Looking up the error in the *FastAudit/390* user guide, we get a more detailed explanation of the error 11D1C MCD Entry is Not Cataloged.

*FastAudit/390* has compared the MCDS MCD extraction against the catalog extraction and has found an MCD entry with no catalog entry. When this

condition occurs, *FastAudit/390* verifies the catalog status by doing a direct catalog look-up.

Using the *FastAudit/390* ISPF interface, we can browse the details of the error. Type a **B** to the left of the error message and press Enter to display the browse detail (Figure 5-4).

|       |                    |                                   |        |    |    |
|-------|--------------------|-----------------------------------|--------|----|----|
| F1    | F2                 | F3                                | F4     | F5 | F6 |
| 11D1C | P390B.ISPF.ISPPROF | HSM.HMIG.T364718.P390B.ISPF.J1034 | MIGRAT | N  | 1  |

Figure 5-4 11D1C MCD error detail

Use the Field Description Box, provided in the product documentation, shown in Table 5-1 for an explanation of the fields shown in the browse detail display.

Table 5-1 Diagnostic 11D1C field descriptions

| Field # / Description      | Example                           |
|----------------------------|-----------------------------------|
| Field 1 ERROR NUMBER       | 11D1C                             |
| Field 2 DSNAME             | P390B.ISPF.ISPPROF                |
| Field 3 HSM GENERATED NAME | HSM.HMIG.T364718.P390B.ISPF.J1034 |
| Field 4 HSM VOLSER         | MIGRAT                            |
| Field 5 VSAM/NVSAM         | N                                 |
| Field 6 HSM LEVEL          | 1                                 |

All error detail is presented in this manner and the product documentation provides a Field Description Box for each error. While diagnosing errors, it is important to have the user guide open to the appropriate page describing the error. The fields are positional and are described as field 1, 2, and so on. In the example above, field 5 indicates the data set is non-VSAM and field 6 indicates it is migration level 1.

Repairing errors may require a variety of actions. Also troubleshooting information (TSI) may be needed to determine the correct repair action. We have spent hundreds of hours type boarding commands and many more hours writing SAS code to implement commands that correspond to printed instructions. We found that effort aggravating and wanted to use data that was being reported to format commands for action. To do this we devised data dictionaries describing the fields in the error text. These dictionaries contain the field names, offsets, and lengths. They correspond to the layout of the detail shown above.

Knowing the names (tags) of the fields makes it possible to format the associated values into a command format. For example to create a command to list the contents of the catalog for a data set name, the parameter “TDSN” (true data set name), may be substituted into the command **LISTC ENT(‘/TDSN’) ALL**. Whenever the fix process is run for an error that is mapped by the dictionary for the MCDS audit (DMCDS), the value at offset 6 for 44 bytes is substituted whenever it finds /TDSN in the command template. The forward slash (/) indicates the beginning of a symbolic parameter. When a chosen fix is processed, the command **LISTC ENT(‘/TDSN’) ALL** is substituted with the data set name in the error text, resulting in the **LISTC ENT(‘P390B.ISPF.ISPPROF’) ALL** command.

There is a dictionary for every type of audit. The dictionaries reside in the provided *FastAudit/390* CNTL data set as individual members. Figure 5-5 shows member DMCDS, the dictionary for the MCDS audit.

|          |                                         |
|----------|-----------------------------------------|
| * TAG    | TY LNG                                  |
| *        | PE TH OFFSET                            |
| *01      | @31@34 @39                              |
| ERRNUM   | 01 0005 00000 <==ERROR NUMBER           |
| TDSN     | 01 0044 00006 <==TRUE DSNNAME           |
| CLUSTER  | 01 0044 00006 <==VSAM CLUSTER NAME      |
| COMPNAME | 01 0044 00051 <==VSAM COMPONENT NAME    |
| ODSN     | 01 0044 00096 <==MIGRATION ALIAS NAME   |
| MCOTYPE  | 01 0001 00141 <==MCO TYPE C,D,I,G,R,P   |
| ADSN     | 01 0044 00051 <==ALIAS DSNNAME          |
| XDSN     | 01 0044 00114 <==MISMATCH ALIAS         |
| CVOLSER  | 01 0006 00051 <==CATALOGGED TO VOLSER   |
| UCAT     | 01 0044 00058 <==USER CATALOG           |
| SDSPDSN  | 01 0044 00051 <==NAME OF SDSP CLUSTER   |
| VOLSER   | 01 0006 00096 <==HSM VOLSER OR MIGRAT   |
| DSORG    | 01 0001 00103 <==DSORG VSAM OR NONVSAM  |
| LEVEL    | 02 0001 00105 <==MIGRATION LEVEL 1 OR 2 |
| MFBID    | 02 0006 00107 <==MCDS FBID              |
| TFBID    | 02 0006 00114 <==TTOC FBID              |
| MEXTN    | 02 0004 00121 <==TTOC XTN FROM MCD      |
| TEXTN    | 02 0004 00126 <==TTOC XTN FROM TTOC     |
| VTOCERR  | 01 0001 00107 <==ERR IND FROM VTOC      |
| TVOL     | 01 0006 00006 <==TAPE VOLSER            |
| TLEVEL   | 01 0002 00013 <==TAPE LEVEL             |
| TTCVOL   | 01 0006 00006 <==VOL FOR 1124 ERR       |
| TTCTYP   | 01 0002 00013 <==TYPE OF VOL            |
| TTCCNT   | 01 0007 00016 <==COUNTED BLOCKS W/SIGN  |
| TTCRES   | 01 0007 00025 <==RESIDENT BLOCKS W/SIGN |
| TTCXCNT  | 01 0008 00034 <==HEX COUNTED BLKS       |
| TTCXRES  | 01 0008 00043 <==HEX RESIDENT BLOCKS    |
| ITTCXTN  | 01 0004 00016 <==TTOC XTN FOR 1125W     |
| ITVFLAG  | 01 0002 00021 <==TTOC DSN FLAG BYTE     |
| ITPFLAG  | 01 0002 00024 <==TTOC DSN PATCH BYTE    |
| ITDOFF   | 01 0004 00027 <==OFFSET INTO DSN RECORD |
| ITFOFF   | 01 0004 00032 <==OFFSET INTO FLAG BYTE  |
| IANAME   | 01 0044 00037 <==ALIAS NAME FOR 1125W   |
| ITNAME   | 01 0044 00081 <==TRUE NAME FOR 1125W    |

Figure 5-5 Diagnostic data dictionary information for the MCDS audit

The member names and descriptions of all the dictionaries follow:

|              |                                                            |
|--------------|------------------------------------------------------------|
| <b>DMCDS</b> | Dictionary for the MCDS audit (FAUDITMD FAUDITCT FAUDITAL) |
| <b>DBCDS</b> | Dictionary for the BCDS audit (FAUDITBD)                   |
| <b>DOCDS</b> | Dictionary for the OCDS audit (FAUDITOD)                   |

|                 |                                                                 |
|-----------------|-----------------------------------------------------------------|
| <b>DMASTCAT</b> | Dictionary for the Master Catalog audit (FAUDITMC)              |
| <b>DCAT</b>     | Dictionary for the User Catalog audit (FAUDITCO)                |
| <b>DCATRMM</b>  | Dictionary for the Catalog to RMM audit (FAUDITCR)              |
| <b>DCATTMS</b>  | Dictionary for the Catalog to CA1 TMS (FAUDITTC)                |
| <b>DDASD</b>    | Dictionary for the Catalog to DASD audit (FAUDITDA)             |
| <b>DTAPE</b>    | Dictionary for the DFSMSHsm to TMC audit (FAUDITRM<br>FAUDITTP) |

Troubleshooting and fixing problems requires some action by the user. We have seen earlier how to browse the contents of the errors. To generate a fix, type an **F** to the left of the error in question and press Enter, as shown in Figure 5-6.

|                                                |       |                                                  |               |
|------------------------------------------------|-------|--------------------------------------------------|---------------|
| <V6 R1.01>                                     |       | FASTAUDIT MCDS SUMMARY                           | Row 1f 31     |
| OPTION ==>                                     |       |                                                  | SCROLL==> CSR |
| ENTER F TO DISPLAY FIXES OR B TO BROWSE ERRORS |       |                                                  |               |
| S NUMBER                                       | COUNT | MESSAGE                                          |               |
| <b>F</b> 11D1C                                 | 1     | MCD ENTRY IS NOT CATALOGED                       |               |
| 11V1C                                          | 0     | MCO VSAM COMPONENT IS NOT CATALOGED              |               |
| 11D2C                                          | 1     | MCD ENTRY IS CATALOGED ON DIFFERENT VOLUME       |               |
| -----                                          |       |                                                  |               |
| 1103W                                          | 0     | MCD ENTRY IS MISSING THE MCA ENTRY               |               |
| 1106C                                          | 0     | MCD IS ON VOLUME WHICH HAS NO MCV ENTRY          |               |
| 1106V                                          | 0     | SUMMARY OF VOLUMES HAVING NO MCV ENTRY           |               |
| 1107W                                          | 597   | MCD LEVEL 2 IS ON VOLUME WHICH HAS NO TTOC ENTRY |               |
| 1107V                                          | 1     | SUMMARY OF VOLUMES MISSING IN TTOC               |               |
| -----                                          |       |                                                  |               |
| 1125W                                          | 0     | TTOC ENTRY MARKED INVALID, MCD/MCC EXISTS        |               |
| 1126I                                          | 0     | SUMMARY OF TTOC VOLUMES WITH INVALID DSNS        |               |

Figure 5-6 MCDS error summary panel

A list of fixes related to the error is shown. Each one has a number directly associated with the error diagnostic (for example, F11D1CA and F11D1CB). A terse description describing what the fix does is displayed. To review what was discussed in the previous pages, edit one of the fixes by typing an **E** to the left of the fix number and pressing Enter, as shown in Figure 5-7.



```

<V6 R1.01>                                     Row 1 of 5
      FAST AUDIT FIX PLAN MEMBER LIST
OPTION ===>                                     SCROLL==> CSR
YOU MAY BROWSE (B), EDIT (E), FIX (F) OR VIEW (V) RESULTS
COMMAND  MEMBER  ACTION  TITLE
E        F11D1CA  TSO CATALOG A MIGRATED data set TO VOLSER MIGRAT
          F11D1CB  TSO HSEND DELETE MIGRATED DSN MISSING CATALOG
          F11D1CC  TSO LISTC FOR DATA SET NOT CATALOGED
          F11D1CD  HSA DISPLAY DUMP OF MCD RECORD
          F11D1CR1 HSM DELETE MCD RECORD WHEN UNABLE TO CATALOG

```

Figure 5-7 MCDS Fix member list

An ISPF EDIT panel shown in Figure 5-8 is displayed. It is important to understand the FIX format and its syntax requirements. The user may create additional fixes to fit any situation they desire.

```

EDIT      FAUDITX.QABETA.CNTL(F11D1CA) - 01.00      Columns 00001 00072
Command ===>                                     Scroll ==> CSR
***** ***** Top of Data *****
000001 TSO CATALOG A MIGRATED data set TO VOLSER MIGRAT
000002 FORMAT FOLLOWS
000003 CATALOG A MIGRATED data set TO MIGRAT
000004 DEF NVSAM +
000005 (NAME('/TDSN') +
000006 VOL(MIGRAT) DEVT(SYSALLDA) +
000007 RECATALOG)

```

Figure 5-8 Fix F11D1CA syntax

The member name must follow the conventions of the diagnostic. In Figure 5-8, member F11D1C is associated with diagnostic 11D1C. This is done so that when the process to display available fixes is run, only those members that match the fix number are displayed.

- Line one has a special requirement. The first three characters indicate special utility processing.
  - HSA** Process Special *FastAudit* CDS record display
  - HSM** Process DFSMSHsm commands
  - IDC** Process native IDCAMS commands
  - RMM** Process RMM commands
  - TMC** Process CA1 TMS commands
  - TSO** Process TSO commands

**VTC** Process VTOC utility commands

**ZRA** Process ZARA utility commands

The remainder of line one is free form comments up to 60 characters.

- ▶ Line two must be coded as shown. **FORMAT FOLLOWS** is a trigger mechanism that causes the format process to start looking for the format symbolic parameter.
- ▶ Line three is another comment line. It is free form and is used primarily for tracking fixes and supporting comments to line 1.
- ▶ Line four is the start of the command to format. The limit is eight lines (8 through 12).

## 5.8 Create a new fix

There may be additional fixes that you want. New fixes are easy to create. For example, in the list of fixes for 11D1C, no fix is available to **FIXCDS DELETE** the alias name record from the MCDS. In the fix list below, **F11D1CR1** has a comment about deleting an MCD record. Edit that member to use as an example by typing an **E** to the left of the fix (Figure 5-9).

|                                                           |          |                                                  |               |
|-----------------------------------------------------------|----------|--------------------------------------------------|---------------|
| <V6 R1.01>                                                |          |                                                  | Row 1 of 5    |
| FAST AUDIT FIX PLAN MEMBER LIST                           |          |                                                  |               |
| OPTION ==>                                                |          |                                                  | SCROLL==> CSR |
| YOU MAY BROWSE (B), EDIT (E), FIX (F) OR VIEW (V) RESULTS |          |                                                  |               |
| COMMAND                                                   | MEMBER   | ACTION                                           | TITLE         |
|                                                           | F11D1CA  | TSO CATALOG A MIGRATED data set TO VOLSER MIGRAT |               |
|                                                           | F11D1CB  | TSO HSEND DELETE MIGRATED DSN MISSING CATALOG    |               |
|                                                           | F11D1CC  | TSO LISTC FOR DATA SET NOT CATALOGED             |               |
|                                                           | F11D1CD  | HSA DISPLAY DUMP OF MCD RECORD                   |               |
| E                                                         | F11D1CR1 | HSM DELETE MCD RECORD WHEN UNABLE TO CATALOG     |               |

*Figure 5-9 MCDS Fix member list*

In ISPF EDIT, type **CREATE** for member name **F11D1CR2** and copy all the statements in the displayed fix. Use **PF3** to exit the fix list and type **F** beside 11D1C again (Figure 5-10).

```

EDIT          FAUDITX.QABETA.CNTL(F11D1CR1) - 01.00      Columns 00001 00072
Command ==> CREATE F11D1CR2                               Scroll ==> CSR
***** ***** Top of Data *****
CC01 HSM DELETE MCD RECORD WHEN UNABLE TO CATALOG MIGRAT
000002 FORMAT FOLLOWS
000003  DELETE MCD
000004  HSEND FIXCDS D '/TDSN' DELETE +
CC05  ODS('/PFX.FIXAUDIT.F11D1CR1') LOGONLY
***** ***** Bottom of Data *****

```

Figure 5-10 Fix F11D1CR2 syntax

F11D1CR2 has been added but its contents are the same as F11D1CR1. To change the contents, edit F11D1CR2 by typing an **E** to the left of the fix and pressing Enter, as shown in Figure 5-11.

```

<V6 R1.01>                                                    Row 1 of 5
          FAST AUDIT FIX PLAN MEMBER LIST
OPTION ==>                                                    SCROLL==> CSR
YOU MAY BROWSE (B), EDIT (E), FIX (F) OR VIEW (V) RESULTS
COMMAND  MEMBER  ACTION          TITLE
          F11D1CA      TSO CATALOG A MIGRATED data set TO VOLSER MIGRAT
          F11D1CB      TSO HSEND DELETE MIGRATED DSN MISSING CATALOG
          F11D1CC      TSO LISTC FOR DATA SET NOT CATALOGED
          F11D1CD      HSA DISPLAY DUMP OF MCD RECORD
          F11D1CR1     HSM DELETE MCD RECORD WHEN UNABLE TO CATALOG
E        F11D1CR2     HSM DELETE MCD RECORD WHEN UNABLE TO CATALOG

```

Figure 5-11 MCDS Fix member list - 2

The stated purpose of this exercise was to create a FIXCDS DELETE of the alias name record (A). Change the comments to reflect that. Statement 5 needs to be changed to reflect the proper fix number (F11D1CR2). Also, statement 4 needs to be changed to **HSEND FIXCDS A**. The alias name symbolic parameter is not TDSN so you will need to look at the data dictionary (DMCDS) for the correct alias name symbolic parameter. A sample of the dictionary provided in *FastAudit/390* is shown in Figure 5-12.

|          |                                       |
|----------|---------------------------------------|
| * TAG    | TY LNG                                |
| *        | PE TH OFFSET                          |
| *01      | @31@34 @39                            |
| ERRNUM   | 01 0005 00000 <==ERROR NUMBER         |
| TDSN     | 01 0044 00006 <==TRUE DSNAME          |
| CLUSTER  | 01 0044 00006 <==VSAM CLUSTER NAME    |
| COMPNAME | 01 0044 00051 <==VSAM COMPONENT NAME  |
| ODSN     | 01 0044 00096 <==MIGRATION ALIAS NAME |
| MCOTYPE  | 01 0001 00141 <==MCO TYPE C,D,I,G,R,P |
| ADSN     | 01 0044 00051 <==ALIAS DSNAME         |
| XDSN     | 01 0044 00114 <==MISMATCH ALIAS       |
| CVOLSER  | 01 0006 00051 <==CATALOGGED TO VOLSER |

Figure 5-12 MCDS data dictionary

The symbolic parameter for the alias name is ADSN. Type the symbolic parameter /ADSN in the command as shown on line 4 and exit edit, saving your changes. This action returns you to the fix list as shown in Figure 5-13.

```

EDIT      FAUDITX.QABETA.CNTL(F11D1CR2) - 01.00      Columns 00001 00072
Command ==>                                         Scroll ==> CSR
***** ***** Top of Data *****
000001 HSM DELETE MCD RECORD WHEN UNABLE TO CATALOG MIGRAT ←
OLD
000001 HSM DELETE MCA RECORD FOR UNCATALOGED data set ← NEW
000002 FORMAT FOLLOWS
000003 DELETE MCD ← OLD
000003 DELETE MCA ← NEW
000004 HSEND FIXCDS D '/TDSN' DELETE + ← OLD
000004 HSEND FIXCDS A /ADSN DELETE + ← NEW
000005 ODS('/PFX.FIXAUDIT.F11D1CR1) LOGONLY ← old
000005 ODS('/PFX.FIXAUDIT.F11D1CR2) LOGONLY ← new
***** ***** Bottom of Data *****

```

Figure 5-13 Fix syntax

Now the fix, F11D1CR2 has the correct comments and symbolic parameters to perform the desired function. To generate the FIX JCL, type an **F** to the left of the fix name and press Enter. The JCL shown in Figure 5-14 is displayed.

```

<V6 R1.01>
Row 1 of 6
      FAST AUDIT FIX PLAN MEMBER LIST
OPTION ==>                                SCROLL==> CSR
YOU MAY BROWSE (B), EDIT (E), FIX (F) OR VIEW (V) RESULTS
COMMAND MEMBER ACTION TITLE
      F11D1CA      TSO CATALOG A MIGRATED data set TO VOLSER MIGRAT
      F11D1CB      TSO HSEND DELETE MIGRATED DSN MISSING CATALOG
      F11D1CC      TSO LISTC FOR DATA SET NOT CATALOGED
      F11D1CD      HSA DISPLAY DUMP OF MCD RECORD
      F11D1CR1     HSM DELETE MCD RECORD WHEN UNABLE TO CATALOG
      F11D1CR2     HSM DELETE MCA RECORD FOR UNCATALOGED data set
F
***** Bottom of data *****

```

Figure 5-14 MCDS Fix member list - 3

The generated results create HSEND FIXCDS for all data sets in error. This could be one or thousands. Notice in Figure 5-15, that all the JCL and control statements are generated and the alias name substituted. To execute the fix, simply type **SUBMIT** on the command line and press Enter.

```

EDIT      DCD.H134157                      Columns 00001 00072
Command ==>                                Scroll ==> CSR
***** Top of Data *****
000001 //DCDMC JOB      , 'DCD.SIS - - OBC.07S', CLASS=A, MSGLEVEL=(1,1),
000002 //      MSGCLASS=X, NOTIFY=DCD
000003 //*
000004 //STEP1 EXEC PGM=IKJEFT01,
000005 // DYNAMNBR=25, REGION=4096K
000006 //* CHANGE THE FOLLOWING STATEMENT TO
000007 //* DSN=PPFX.CLIST WHERE PPFX IS THE PREFIX ASSIGNED TO
000008 //* THE FASTAUDIT PRODUCT
000009 //*
000010 //SYSPROC DD DSN=FAUDITX.QABETA.CLIST, DISP=SHR
000011 //*      ****
000012 //SYSOUT DD SYSOUT=*
000013 //SYSTSPT DD SYSOUT=*
000014 //SYSTSIN DD *
000015 HSEND FIXCDS A HSM.HMIG.T364718.P390B.ISPF.J1034 DELETE +
000016 ODS('DCD.FIXAUDIT.F11D1CR2') LOGONLY
***** Bottom of Data *****

```

Figure 5-15 MCDS Fix member list





## Auditing DFSMShsm using HSM *FastAudit*

The DFSMShsm Control Data Set audits are provided in the *FastAudit/390* selectable unit, HSM *FastAudit*. In this chapter we discuss how to use the HSM *FastAudit* selectable component of *FastAudit/390*.

The HSM *FastAudit* product audits the internal structure of the DFSMShsm control data sets (CDSs). These include the Migration Control data set (MCDS), Backup Control data set (BCDS) and the Offline Control data set (OCDS).

## 6.1 Migration Control Data Set audits

The MCDS contains information about migrated data organized into sundry records. The records for which HSM *FastAudit* performs audits functions are listed here.

- ▶ Migration Control Data Set data set record (MCD) *True name record* contains:
  - DFSMSHsm generated alias name
  - Migration level (1 or 2)
  - Small data set packing (SDSP) indicator
  - VSAM or non-VSAM indicator
  - Migration volume number
  - Number of valid blocks
  - TTOC extent (if level 2)
- ▶ Migration Control Data Set Alias Entry record (MCA) *Alias name record* contains:
  - True name
  - Component name (if VSAM)
  - VSAM type (if VSAM) - D=Data, I=Index, R=Path
  - VSAM association - C=Cluster, G=AIX
- ▶ Migration Control Data Set VSAM Associations Record (MCO) *VSAM component names* contains:
  - Index name
  - Data name
  - Migration alias name with hexadecimal 12 as first character
- ▶ Migration Control Data Set Volume Record (MCV) contains:
  - VOLSER

In addition to the records in the MCDS, data from the OCDS (TTOC including the VOLSER and DFSMSHsm generated alias name), VTOC, small data set packing (SDSP) and catalogs are extracted for auditing purposes.

The audits are both forward and backward. For example, the MCD record contains the DFSMSHsm generated alias name and the MCA record contains the true data set name. Their pointers should match up to one another (forwards and backwards). Error conditions occur when they do not match or one is missing.

Figure 6-1 graphically represents the pointer relationships.



|     |   |   |                              |
|-----|---|---|------------------------------|
| MCD | ← | → | MCA                          |
| MCD | ← | → | MCO (if VSAM)                |
| MCD | ← | → | MCV                          |
| MCD | ← | → | TTOC VOLUME (from OCDS)      |
| MCD | ← | → | TTOC ALIAS (from OCDS)       |
| MCD | ← | → | VTOC (if ML1)                |
| MCD | ← | → | CATALOG if MIGRAT            |
| MCO | ← | → | CATALOG (if VSAM and MIGRAT) |
| MCO | ← | → | MCA                          |
| MCD | ← | → | SDSP (if SDSP)               |

*Figure 6-1 DFSMSHsm record audit relationships*

The cataloged procedures available for the MCDS audit are:

- FAUDITMD** Procedure is used to audit the MCDS structures. It audits everything except the catalog relationships.
- FAUDITCT** Procedure audits only the elements in the MCDS that should have cataloged references.
- FAUDITAL** Procedure audits all the catalog and MCDS internal structures.

Before executing any of the HSM Audits, it is recommend that you execute the Master Catalog and User Catalog audits and resolve those errors.

Prior to executing the audit JCL, FAUDITMD, the following symbolic parameters need to be replaced with your customized values.

- PPFX** Product Prefix as installed. For example if the product has been installed with MSC.FAUDIT as a prefix to the libraries, then PPFX=MSC.FAUDIT must be coded in the EXEC PROC statement.
- PFX** Prefix represents the Runtime Prefix the user has chosen for this cataloged procedure to allocate work files that contain the audit results. Depending on the SMS treatment of new allocations, the work data sets may be deleted within a day or so. Choose a prefix that is retained for an appropriate amount of time so the results of the audit may be reviewed and repaired.
- PUNIT** Permanent unit for allocating data sets containing the audit results. The default is SYSDA. If this unit allocates on work volumes or other volumes that scratch data in a short period of time, you may want to change it to a different unit name to retain the data longer.

|                |                                                                                                                                                                                              |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>WRKUNIT</b> | Work Unit for allocating temporary data sets deleted at the end of the job. The default is SYSDA. If your installation has a different standard, code the different name.                    |
| <b>ATP</b>     | Allocation Type. Allocation is given in units such as PRI and SEC. For the larger installations the unit TRKS sometimes is not enough. If your installation is large, change the ATP to CYL. |
| <b>PRI</b>     | The number of primary allocation units. The default is 100. This value must be adjusted to accommodate the size of your DFSMSHsm MCDS and the number of records.                             |
| <b>SEC</b>     | The number of secondary allocation units. The default is 200. This parameter is important to adjust for the size of your MCDS.                                                               |
| <b>OPT</b>     | MCD is the default. Don't change this for FAUDITMD.                                                                                                                                          |
| <b>TOPT</b>    | YES is the default. This option controls the audit functions for OCDS extraction comparisons. If TOPT = NO, is coded, diagnostics 1124W and 1125W are suppressed.                            |
| <b>MCDS</b>    | MCDS is required. It is the first split of the DFSMSHsm MCDS data set or the MCDS data set name if the MCDS is not split.                                                                    |
| <b>MCDS1</b>   | MCDS1 is the second split of the MCDS. The default is NULLFILE.                                                                                                                              |
| <b>MCDS2</b>   | MCDS2 is the third split of the MCDS. The default is NULLFILE.                                                                                                                               |
| <b>MCDS3</b>   | MCDS3 is the fourth split of the MCDS. The default is NULLFILE.                                                                                                                              |
| <b>OCDS</b>    | OCDS is the name of the Offline Control Data Set.                                                                                                                                            |
| <b>LOAD</b>    | LOAD is the name of the load library as installed with <i>FastAudit/390</i> . It must be APF authorized.                                                                                     |

Figure 6-2 shows the symbolic parameters in the JCL used to execute procedure FAUDITMD for the MCDS. The figure shows a multicluseter MCDS with two splits. The OCDS is named SIS.HSM.OCDS. The product prefix is FAUDITX.QABETA and the runtime prefix is DES2.

```

000001 //FAUDITMD PROC PFX=PFX,          <==RUN TIME PREFIX
000002 /* FASTAUDIT <V6 R1.01> SEPTEMBER, 2001
000003 //      PPFX=PPFX,          <==INSTALLED PRODUCT PREFIX
000004 //      PUNIT=SYSDA,        <==UNIT FOR PERMANENT ALLOCATION
000005 //      SOUT=*,             <==PREFERRED SYSOUT CLASS
000006 //      WRKUNIT=SYSDA,      <==UNIT FOR TEMPORARY ALLOCATION
000007 //      ATP=TRK,           <==ALLOCATION TYPE TRK/CYL
000008 //      PRI=100,           <==PRIMARY ALLOCATION FOR FAUDIT
000009 //      SEC=200,            <==SECONDARY SPACE ALLOCATION
000010 //      OPT=MCD,            <==DO MCD AUDIT ONLY
000011 //      TOPT=YES,          <==AUDIT THE TTOC ALIAS BLKS
000016 //*                      <==OTHER WISE NO
000017 //      MCDS=HSM.MCDS,      <==INSTALLATION MCDS DSNAME
000018 //      MCDS1=NULLFILE,     <==1ST MCDS SPLIT
000019 //      MCDS2=NULLFILE,     <==2ND MCDS SPLIT
000020 //      MCDS3=NULLFILE,     <==3RD MCDS SPLIT
000021 //      OCDS=HSM.OCDS,      <==INSTALLATION OCDS DSNAME
000022 //      OCDS1=NULLFILE,     <==1ST OCDS SPLIT
000023 //      OCDS2=NULLFILE,     <==2ND OCDS SPLIT
000024 //      OCDS3=NULLFILE,     <==3RD OCDS SPLIT
000025 //      LOAD=PFX.LOAD       <==LOAD LIBRARY AS INSTALLED
- - - - - 338 Line(s) not Displayed
000350 //SORTWK03 DD  UNIT=&WRKUNIT,SPACE=(CYL,(&PRI))
000351 //      PEND
000352 //PROC1      EXEC FAUDITMD,
000353 //      MCDS=SIS.HSM.SPLIT.MCDS1,
000354 //      MCDS1=SIS.HSM.SPLIT.MCDS2,
000355 //      OCDS=SIS.HSM.OCDS,
000356 //      LOAD=FAUDITX.QABETA.LOAD,
000357 //      PPFX=FAUDITX.QABETA,
000358 //      PFX=DES2

```

Figure 6-2 Cataloged procedure FAUDITMD with symbolic substitutions

### 6.1.1 Setup for FAUDITAL

The cataloged procedure setup for FAUDITAL is identical for FAUDITMD with the exception of the OPT parameter.

**OPT** ALL is the default. Do not change this for FAUDITAL.

Figure 6-3 shows the symbolic parameters in the JCL used to execute procedure FAUDITAL for an MCDS and the catalogs. The sample shows a multicluster MCDS with two splits and an OCDS of SIS.HSM.OCDS. The product prefix is FAUDITX.QABETA, the load library is FAUDITX.QABETA.LOAD and the run time prefix is DES2.

```

000001 //FAUDITAL PROC PFX=PFX,          <==RUN TIME PREFIX
000002 //* FASTAUDIT <V6 R1.01> SEPTEMBER, 2001
000003 //      PPFX=PPFX,                  <==PRODUCT PREFIX
000004 //      PUNIT=SYSDA,                  <==UNIT FOR PERMANENT ALLOCATION
000005 //      SOUT=*,                        <==PREFERED SYSOUT CLASS
000006 //      WRKUNIT=SYSDA,                <==UNIT FOR TEMPORARY ALLOCATION
000007 //      ATP=TRK,                       <==TRK OR CYL ALLOCATION
000008 //      PRI=100,                       <==PRIMARY ALLOCATION FOR FAUDIT
000015 //      SEC=200,                     <==SECONDARY SPACE ALLOCATION
000016 //      OPT=ALL,                      <==AUDIT ALL COMPONENTS
000017 //      TOPT=YES,                      <==AUDIT THE TTOC ALIAS BLKS
000018 //*                                  <==OTHER WISE NO
000019 //      MCDS=HSM.MCDS,                  <==INSTALLATION MCDS DSNAME
000020 //      MCDS1=NULLFILE,                 <==1ST MCDS SPLIT
000021 //      MCDS2=NULLFILE,                 <==2ND MCDS SPLIT
000022 //      MCDS3=NULLFILE,                 <==3RD MCDS SPLIT
000023 //      OCDS=HSM.OCDS,                  <==INSTALLATION OCDS DSNAME
000024 //      OCDS1=NULLFILE,                 <==1ST OCDS SPLIT
000025 //      OCDS2=NULLFILE,                 <==2ND OCDS SPLIT
000026 //      OCDS3=NULLFILE,                 <==3RD OCDS SPLIT
000027 //      LOAD=PPFX.LOAD                  <==LOAD LIBRARY AS INSTALLED
- - - - -
000487 //PROC1      EXEC FAUDITAL,
000488 //              MCDS=SIS.HSM.SPLIT.MCDS1,
000489 //              MCDS1=SIS.HSM.SPLIT.MCDS2,
000490 //              OCDS=SIS.HSM.OCDS,
000491 //              LOAD=FAUDITX.QABETA.LOAD,
000492 //              PPFX=FAUDITX.QABETA,
000493 //              PFX=DES2

```

Figure 6-3 Cataloged procedure FAUDITAL showing symbolic substitutions

## Setup for FAUDITCT

Prior to executing the audit JCL, FAUDITCT, the following symbolic parameters need to be replaced with your customized values.

**PPFX** Product Prefix as installed. For example if the product has been installed with MSC.FAUDIT as a prefix to the libraries, then PPFX=MSC.FAUDIT must be coded in the EXEC PROC statement.

**PFX** Prefix represents the Runtime Prefix the user has chosen for this cataloged procedure to allocate work files that contain the audit results. Depending on the SMS treatment of new allocations, the work data sets may be deleted within a day or so. Choose a prefix that is retained for an appropriate amount of time so the results of the audit may be reviewed and repaired.

|                |                                                                                                                                                                                                                                                                                |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>PUNIT</b>   | Permanent unit for allocating data sets containing the audit results. The default is SYSDA. If this unit allocates on work volumes or other volumes that scratch data in a short period of time, you may want to change it to a different unit name to retain the data longer. |
| <b>WRKUNIT</b> | Work Unit for allocating temporary data sets that will be deleted at the end of the job. The default is SYSDA. If your installation has a different standard, code the different name.                                                                                         |
| <b>ATP</b>     | Allocation Type. Allocation is given in units such as PRI and SEC. For the larger installations the unit TRKS sometimes is not enough. If your installation is large, change the ATP to CYL.                                                                                   |
| <b>PRI</b>     | The number of primary allocation units. The default is 100. This value must be adjusted to accommodate the size of your DFSMSHsm MCDS and the number of records.                                                                                                               |
| <b>SEC</b>     | The number of secondary allocation units. The default is 200. This parameter is important to adjust for the size of your MCDS.                                                                                                                                                 |
| <b>OPT</b>     | CAT is the default. Do not change this for FAUDITCT.                                                                                                                                                                                                                           |
| <b>MCDS</b>    | MCDS is required. It is the first split of the DFSMSHsm MCDS data set or the only split if the MCDS is not split.                                                                                                                                                              |
| <b>MCDS1</b>   | MCDS1 is the second split of the MCDS. The default is NULLFILE.                                                                                                                                                                                                                |
| <b>MCDS2</b>   | MCDS2 is the third split of the MCDS. The default is NULLFILE.                                                                                                                                                                                                                 |
| <b>MCDS3</b>   | MCDS3 is the fourth split of the MCDS. The default is NULLFILE.                                                                                                                                                                                                                |
| <b>LOAD</b>    | LOAD is the name of the load library as installed with <i>FastAudit/390</i> . It must be authorized.                                                                                                                                                                           |

Figure 6-4 shows the symbolic parameters in the JCL used to execute procedure FAUDITCT for an MCDS and the catalogs. This procedure audits only those elements of the MCDS that have corresponding catalog entries. The sample shows a multicluster MCDS with two splits. The product prefix is FAUDITX.QABETA, the load library is FAUDITX.QABETA.LOAD and the run time prefix is DES2.

```

000007 //FAUDITCT PROC PFX=PFX,          <==RUN TIME PREFIX
000008 //* FASTAUDIT <V6 R1.01> SEPTEMBER,2001
000009 //      PPFX=PPFX,                  <==PRODUCT PREFIX
000010 //      PUNIT=SYSDA,                 <==UNIT FOR PERMANENT ALLOCATION
000011 //      SOUT=*,                      <==PREFERED SYSOUT CLASS
000012 //      WRKUNIT=SYSDA,              <==UNIT FOR TEMPORARY ALLOCATION
000013 //      ATP=TRK,                     <==TYPE OF ALLOCATION TRK/CYL
000014 //      PRI=100,                    <==PRIMARY ALLOCATION FOR FAUDIT
000015 //      SEC=200,                     <==SECONDARY SPACE ALLOCATION
000016 //      OPT=CAT,                     <==AUDIT CATALOG COMPONENTS
000017 //      MCDS=HSM.MCDS,                 <==INSTALLATION MCDS DSNAME
000018 //      MCDS1=NULLFILE,               <==1ST MCDS SPLIT
000019 //      MCDS2=NULLFILE,               <==2ND MCDS SPLIT
000020 //      MCDS3=NULLFILE,               <==3RD MCDS SPLIT
000021 //      LOAD=PPFX.LOAD                 <==LOAD LIBRARY AS INSTALLED
000022 //*
- - - - -
000230 //PROC1 EXEC FAUDITCT,
000231 //      MCDS=SIS.HSM.SPLIT.MCDS1,
000232 //      MCDS1=SIS.HSM.SPLIT.MCDS2,
000233 //      LOAD=FAUDITX.QABETA.LOAD,
000234 //      PPFX=FAUDITX.QABETA,
000235 //      PFX=DES2

```

Figure 6-4 Cataloged procedure FAUDITCT showing symbolic substitutions

After the audit job completes, execute the CLIST FIXAUDIT. This CLIST displays the main panel, Fast Audit Choose Which Audit To Fix, shown in Figure 6-5, and the various audit results options available.

```

<V6 R1.01> FAST AUDIT.  CHOOSE WHICH AUDIT TO FIX
COMMAND==> 1
1  MCDS   CREATED BY PROCS FAUDITMD, FAUDITCT OR FAUDITAL
2  BCDS   CREATED BY PROC FAUDITBD
3  OCDS   CREATED BY PROC FAUDITOD
4  TAPE   CREATED BY PROCS FAUDITTP, FAUDITZR OR FAUDITRM
5  CAT    CREATED BY PROC FAUDITCO
6  DASD   CREATED BY PROC FAUDITDA OR FAUDITDE
7  MASTCAT CREATED BY PROC FAUDITMC
8  CATRMM CREATED BY PROC FAUDITCR
9  CATTMS CREATED BY PROC FAUDITTC

ENTER  END  TO TERMINATE ONLINE FUNCTION

```

Figure 6-5 Primary Fix panel

To view the results of FAUDITMD, FAUDITCT or FAUDITAL, select option 1 MCDS by typing the number one on the command line and pressing Enter. The FastAudit MCDS Summary panel is displayed in Figure 6-6.

|                                                |        |               |                                                  |
|------------------------------------------------|--------|---------------|--------------------------------------------------|
| <V6 R1.01>                                     |        |               | Row 1 of 31                                      |
| FASTAUDIT MCDS SUMMARY                         |        |               |                                                  |
| OPTION ==>                                     |        | SCROLL==> CSR |                                                  |
| ENTER F TO DISPLAY FIXES OR B TO BROWSE ERRORS |        |               |                                                  |
| S                                              | NUMBER | COUNT         | MESSAGE                                          |
|                                                | 11D1C  | 2             | MCD ENTRY IS NOT CATALOGED                       |
|                                                | 11V1C  | 0             | MCO VSAM COMPONENT IS NOT CATALOGED              |
|                                                | 11D2C  | 1             | MCD ENTRY IS CATALOGED ON DIFFERENT VOLUME       |
|                                                | 11V2C  | 0             | MCO VSAM COMPONENT ON DIFFERENT VOLUME           |
|                                                | 1103W  | 0             | MCD ENTRY IS MISSING THE MCA ENTRY               |
|                                                | 1104C  | 0             | MCD LEVEL 1 ENTRY HAS NO VTOC ENTRY ON L1 VOLUME |
|                                                | 1105W  | 0             | MCD VSAM BASE NAME IS MISSING IN MCO             |
|                                                | 1106C  | 0             | MCD IS ON VOLUME WHICH HAS NO MCV ENTRY          |
|                                                | 1106V  | 0             | SUMMARY OF VOLUMES HAVING NO MCV ENTRY           |
|                                                | 1107W  | 595           | MCD LEVEL 2 IS ON VOLUME WHICH HAS NO TTOC ENTRY |
|                                                | 1107V  | 1             | SUMMARY OF VOLUMES MISSING IN TTOC               |
|                                                | 1108W  | 0             | MCA ENTRY IS MISSING MCO VSAM COMPONENT ENTRY    |
|                                                | 1109W  | 0             | MCD ALIAS IS NOT ON TTOC                         |
|                                                | 1109V  | 0             | SUMMARY OF VOLUMES WHERE ALIAS IS NOT IN TTOC    |
|                                                | 1110W  | 0             | MCD HAS DUPLICATE MCA ENTRY                      |
|                                                | 1111W  | 27            | MCA ENTRY IS MISSING MCD ENTRY                   |
|                                                | 1112W  | 0             | MCO VSAM BASE NAME IS MISSING IN MCD             |
|                                                | 1113W  | 0             | MCO VSAM COMPONENT NAME IS MISSING IN MCA        |
|                                                | 1114W  | 0             | ALIAS NAME EXISTS ON VTOC, MCA IS MISSING        |
|                                                | 1115W  | 0             | VTOC ALIAS NAME HAS VTOC ERROR                   |
|                                                | 1116C  | 0             | CATALOG ENTRY HAS NO MCD/MCO ENTRY               |
|                                                | 1117C  | 0             | CATALOG ENTRY IS NOT CONNECTED TO MASTER CAT     |
|                                                | 1118C  | 0             | MIGRATED DATA SET IS MISSING IN SDSP CLUSTER     |
|                                                | 1119W  | 29            | ENTRY IN SDSP DATA SET IS MISSING IN MCDS        |
|                                                | 1120I  | 0             | TTOC VOLUME IS EMPTY                             |
|                                                | 1121W  | 0             | MIGRATION LEVEL 2 VOLUME HAS NO TTOC ENTRY       |
|                                                | 1122W  | 0             | MCD ALIAS IS OWNED BY DIFFERENT MCD              |
|                                                | 1123W  | 216           | MCA TRUENAME RECORD IS MISMATCHED                |
|                                                | 1124W  | 0             | VALID BLOCK COUNT IN TTOC DOES NOT AGREE         |
|                                                | 1125W  | 0             | TTOC ENTRY MARKED INVALID, MCD/MCC EXISTS        |
|                                                | 1126I  | 0             | SUMMARY OF TTOC VOLUMES WITH INVALID DSNS        |

Figure 6-6 MCDS error summary panel

There are subsequent panels behind each error number listed. These panels provide the list of FIXES to either diagnose or resolve the various errors. In all cases, you can view the list of available fixes by typing the letter F next to the error number and pressing Enter. The Fast Audit Fix Plan Member List for that error number is displayed and lists the available fixes for the error as shown in Figure 6-7.

```

FAST AUDIT FIX PLAN MEMBER LIST
OPTION ==>
YOU MAY BROWSE (B), EDIT (E), FIX (F) OR VIEW (V) RESULTS
COMMAND MEMBER ACTION TITLE
F11D1CA TSO CATALOG A MIGRATED data set TO VOLSER MIGRAT
F11D1CB TSO HSEND DELETE MIGRATED DSN MISSING CATALOG
F11D1CC TSO LISTC FOR DATA SET NOT CATALOGED
F11D1CD HSA DISPLAY DUMP OF MCD RECORD
F11D1CR1 HSM DELETE MCD RECORD WHEN UNABLE TO CATALOG MIGRAT
F11D1CR2 HSM DELETE MCA RECORD FOR UNCATALOGED data set
F11D1CR3 HSM FIXCDS PATCH ASSIGNED AND NEEDS SCRATCH
***** Bottom of data *****

```

Figure 6-7 Fix Plan Member List for error 11D1C

*FastAudit/390* automatically generates the required JCL and FIX statements for each FIX listed when you type the letter F next to the FIX member name and press Enter. Figure 6-8 is an example of the JCL and FIX statements generated for fix F11D1CA.

```

EDIT CG.H163212 Columns 00001 00072
Command ==> Scroll ==> CSR
***** ***** Top of Data *****
000001 //CGA JOB (0),COLGOR,
000002 // CLASS=A,MSGCLASS=X,NOTIFY=COLGOR
000003 //*
000004 //STEP1 EXEC PGM=IKJEFT01,
000005 // DYNAMNBR=25,REGION=4096K
000010 //SYSPROC DD DSN=FAUDITX.QABETA.CLIST,DISP=SHR
000011 //* ****
000012 //SYSOUT DD SYSOUT=*
000013 //SYSTSPRT DD DSN=CG.FIXAUDIT.F11D1CA,DISP=SHR
000014 //SYSTSIN DD *
000015 DEF NVSAM +
000016 (NAME('CRP1.CAT111.MIGRAT.NMSGDG.G0001V00') +
000017 VOL(MIGRAT) DEVT(SYSALLDA) +
000018 RECATALOG)

```

Figure 6-8 Fix F11D1CA define non-VSAM statements and JCL

You can use ISPF edit commands to edit the member and remove any records that you do not want to FIX or diagnose. Once your editing is complete or if no editing is desired, execute the FIX JCL by typing the word SUBMIT or the short version SUB on the command line in the edit entry panel and pressing Enter. A batch job is submitted to execute the FIX commands.



When the FIX batch job has completed you can view the results by typing the letter V next to the FIX member name and pressing Enter. This action results in an ISPF view of the FIX results.

6.1.2 FastAudit/390 audit error 11D1C

We will look at FastAudit/390 error 11D1C MCD ENTRY IS NOT CATALOGED.

Diagnosing the error

Looking up the error in the FastAudit/390 User Guide, you find the following explanation:

FastAudit/390 has compared the MCDS MCD extraction against the catalog extraction and has found an MCD entry with no catalog entry. When this condition occurs, FastAudit/390 verifies the catalog status by doing a direct catalog look-up.

Because there is no catalog entry for the data sets with migration copies, the copies are most likely old or obsolete and for some reason the MCD record remains. However, this is not always the case. Recent catalog work or human error could cause catalog entries to be inadvertently deleted. The data could be valid data and the catalog entries may need to be restored. Once verified, you can proceed with applying the appropriate FIX to resolve the error.

To view the error detail, type B on the FastAudit MCDS Summary panel next to the error and press Enter. The error detail is displayed in Figure 6-9.

|       |                 |                                   |        |    |    |
|-------|-----------------|-----------------------------------|--------|----|----|
| f1    | f2              | f3                                | f4     | f5 | f6 |
| 11D1C | BMOHN.ISPF.ISPP | HSM.HMIG.T350200.BMOHN.ISPF.I4088 | HM0069 | N  | 2  |

Figure 6-9 Error detail for error 11D1C

The Field Description Box provided in the documentation and shown in Table 6-1 provides an explanation of each field.

Table 6-1 Diagnostic 11D1C field descriptions

| Dictionary symbol | Example                           |
|-------------------|-----------------------------------|
| Field 1 /ERRNUM   | 11D1C                             |
| Field 2 /TDSN     | BMOHN.ISPF.ISPPROFB               |
| Field 3 /ADSN     | HSM.HMIG.T350200.BMOHN.ISPF.I4088 |
| Field 4 /VOLSER   | HM0069                            |

| Dictionary symbol | Example |
|-------------------|---------|
| Field 5 /DSORG    | N       |
| Field 6 /LEVEL    | 2       |

### Resolving the error

If you find that the errant MCDS records need to be deleted, there are two possible methods you can use to resolve this error.

1. Execute fixes F11D1CB and F11D1CR2 to delete the MCD and MCA records in the MCDS immediately.
2. Execute fix F11D1CR3 to patch the migration copy and its D record for cleanup the next time DFSMSHsm automatic migration is executed. The FIXCDS command, FIXCDS D dsname PATCH command resets the assigned flag (MCDFASN) bit to a value of 0 (off) and the scratch flag (MCDFNSCR) bit to 1(on).

If the data sets are valid and the catalog records need to be recreated, execute fix F11D1CA to rebuild the catalog entry as a non-VSAM data set with a volume serial of MIGRAT.

To view the list of available fixes, type F next to the error message and press Enter. The Fast Audit Fix Plan Member List is displayed in Figure 6-10.

```
FAST AUDIT FIX PLAN MEMBER LIST
OPTION ==>                                SCROLL==> PAGE
YOU MAY BROWSE (B), EDIT (E), FIX (F) OR VIEW (V) RESULTS
COMMAND MEMBER ACTION TITLE
F11D1CA      TSO CATALOG A MIGRATED data set TO VOLSER MIGRAT
F11D1CB      TSO HSEND DELETE MIGRATED DSN MISSING CATALOG
F11D1CC      TSO LISTC FOR DATA SET NOT CATALOGED
F11D1CD      HSA DISPLAY DUMP OF MCD RECORD
F11D1CR1     HSM DELETE MCD RECORD WHEN UNABLE TO CATALOG MIGRAT
F11D1CR2     HSM DELETE MCA RECORD FOR UNCATALOGED data set
F11D1CR3     HSM FIXCDS PATCH ASSIGNED AND NEEDS SCRATCH
***** Bottom of data *****
```

Figure 6-10 FAST Audit fix plan member list for F11D1C

### 6.1.3 FastAudit/390 audit error 11V1C

The next error we will look at is *FastAudit/390 11V1C MCO VSAM COMPONENT IS NOT CATALOGED*.

## Diagnosing the error

Looking up the error in the *FastAudit/390* User Guide, you find the following explanation:

*FastAudit/390* has compared the MCDS MCO extraction against the catalog extraction and has found an MCO VSAM component with no catalog entry. When this condition occurs, *FastAudit/390* verifies the catalog status by doing a direct catalog look-up. When DFSMSHsm migrates a VSAM cluster it creates an MCO record and an MCA record of the cluster's components. Each component 'looks like' a single migrated data set but it really is not. Reference to the MCO entry is required if any single one of the components is referenced.

The VSAM data set is uncataloged but a migration copy still exists. This error is similar to 11D1C, but refers to uncataloged VSAM data sets rather than non-VSAM. This could be another case where a corruption occurred due to DFSMSHsm being abnormally terminated and the VSAM data set was recalled before DFSMSHsm cleanup, uncataloging of a migrated data set, or any other cause for DFSMSHsm not performing its cleanup of ghost records.

To view the error detail, type B on the FastAudit MCDS Summary panel next to the error and press Enter. The error detail is displayed in Figure 6-11.

|       |                                    |                               |
|-------|------------------------------------|-------------------------------|
| f1    | f2                                 | f3                            |
| 11V1C | CRP1.CAT111.MIGRAT.NSMSKSDS        | CRP1.CAT111.MIGRAT.NSMSKSDS.D |
| f4    | HSM.HMIG.T050514.CRP1.CAT111.J2143 |                               |

Figure 6-11 Error detail for error 11V1C

The Field Description Box provided in the documentation and shown in Figure 6-2 provides an explanation of each field.

Table 6-2 Diagnostic 11V1C field descriptions

| Field # / Description       | Example                            |
|-----------------------------|------------------------------------|
| Field 1 Error number        | 11V1C                              |
| Field 2 Cluster Name        | CRP1.CAT111.MIGRAT.NSMSKSDS        |
| Field 3 Component Name      | CRP1.CAT111.MIGRAT.NSMSKSDS.D      |
| Field 4 HSM Generated Alias | HSM.HMIG.T050514.CRP1.CAT111.J2143 |

## Resolving the error

To view the list of available fixes, type **F** next to the error message and press Enter. The Fast Audit Fix Plan Member List is displayed in Figure 6-12.

```
<V6 R1.01>                                     Row 1 of 11
      FAST AUDIT FIX PLAN MEMBER LIST
OPTION ===>                                     SCROLL==> PAGE
YOU MAY BROWSE (B), EDIT (E), FIX (F) OR VIEW (V) RESULTS
COMMAND  MEMBER  ACTION  TITLE
F11V1CA      TSO CATALOG A MIGRATED data set TO VOLSER MIGRAT
F11V1CB      TSO HSEND DELETE MIGRATED DSN MISSING CATALOG
F11V1CC      TSO LISTC FOR DATA SET NOT CATALOGED
F11V1CD      HSA DISPLAY DUMP OF MCO RECORD
F11V1CE      HSM FIXCDS DELETE OF UNCATALOGED COMPONENT
F11V1CF      HSA DISPLAY DUMP OF ALIAS RECORD
F11V1CG      HSM FIXCDS DELETE OF MCO RECORD
F11V1CH      TSO RECALL JUST THE CLUSTER NAME
F11V1CR1     HSM FIXCDS DELETE OF MCD FOR UNCAT COMP
F11V1CR2     HSM DELETE ORPHANED MCO RECORDS
F11V1CR3     HSM DELETE ORPHANED MCA RECORDS
***** Bottom of data *****
```

Figure 6-12 Fast Audit Fix Plan Member List for error 11V1C

If it is determined that the uncataloged VSAM data sets in migration can be deleted, use fixes F11V1CR1, F11V1CR2 and F11V1CR3 to delete the MCD record for the uncataloged data set and delete the orphaned MCO and MCA records. If the uncataloged VSAM data sets are still valid use fix F11V1CA to catalog the data set to VOLSER MIGRAT.

### 6.1.4 FastAudit/390 audit error 11V1C

The next error message we will look at is *FastAudit/390* Audit Error 11V1C MCO VSAM COMPONENT IS NOT CATALOGED

#### Diagnosing the error

Looking up the error in the *FastAudit/390* User Guide, you find the following explanation:

*FastAudit/390* has compared the MCDS MCO extraction against the Catalog extraction and has found an MCO VSAM component with no catalog entry. When this condition occurs, *FastAudit/390* verifies the catalog status by doing a direct catalog look-up. When DFSMSHsm migrates a VSAM cluster it creates an MCO record and an MCA record of the cluster's components. Each component 'looks like' a single migrated data set but it really is not.

Reference to the MCO entry is required if any single one of the components is referenced.

To view the error detail, type B on the FastAudit MCDS Summary panel next to the error and press Enter. The error detail is displayed in Figure 6-13.

|       |                     |                  |                                    |
|-------|---------------------|------------------|------------------------------------|
| f1    | f2                  | f3               | f4                                 |
| 11V1C | USERID.VSAM.CLUSTER | USERID.VSAM.DATA | HSM.HMIG.T352004.USERID.VSAM.I1261 |

Figure 6-13 Error detail for 11D2C

The Field Description Box provided in the documentation and shown in Table 6-3 provides an explanation of each field.

Table 6-3 Diagnostic 11V1C field descriptions

| Dictionary symbol | Example                            |
|-------------------|------------------------------------|
| Field 1 /ERRNUM   | 11V1C                              |
| Field 2 /Cluster  | USERID.VSAM.CLUSTER                |
| Field 3 /CompName | USERID1.VSAM.DATA                  |
| Field 4 /ADSN     | HSM.HMIG.T352004.USERID.VSAM.I1261 |

Resolving the error

To view the list of available fixes, type F next to the error message and press Enter. The Fast Audit Fix Plan Member List is displayed in Figure 6-14.

```
<V6 R1.01>                                     Row 1 of 11
      FAST AUDIT FIX PLAN MEMBER LIST
OPTION ==>                                     SCROLL==> PAGE
YOU MAY BROWSE (B), EDIT (E), FIX (F) OR VIEW (V) RESULTS
COMMAND  MEMBER  ACTION  TITLE
      F11V1CA      TSO CATALOG A MIGRATED data set TO VOLSER MIGRAT
      F11V1CB      TSO HSEND DELETE MIGRATED DSN MISSING CATALOG
      F11V1CC      TSO LISTC FOR DATA SET NOT CATALOGED
      F11V1CD      HSA DISPLAY DUMP OF MCO RECORD
      F11V1CE      HSM FIXCDS DELETE OF UNCATALOGED COMPONENT
      F11V1CF      HSA DISPLAY DUMP OF ALIAS RECORD
      F11V1CG      HSM FIXCDS DELETE OF MCO RECORD
      F11V1CH      TSO RECALL JUST THE CLUSTER NAME
      F11V1CR1     HSM FIXCDS DELETE OF MCD FOR UNCAT COMP
      F11V1CR2     HSM DELETE ORPHANED MCO RECORDS
      F11V1CR3     HSM DELETE ORPHANED MCA RECORDS
***** Bottom of data *****
```

Figure 6-14 Fast Audit Fix Plan Member List for error 11V1C

If your organization is in the habit of allocating and using VSAM components as opposed to VSAM clusters then this diagnostic could be very important and the user should explore the current status of the cluster by executing fix F11V1CC. If the cluster resides on primary volumes then this orphaned VSAM component can be safely deleted with fixes F11V1CDR2 and F11V1CR3 to delete the orphaned MCO and MCA records. If the cluster name is not listed (missing) then the MCO component may safely be deleted. If the cluster name is cataloged to VOLSER MIGRAT, the data set may be recalled by the cluster name by executing fix F11V1CH. DFSMShsm's subsequent cleanup processes clear the error.

If the data sets are valid and should not be deleted, the chosen fix is to re-catalog the data set to a VOLSER of MIGRAT using fix F11V1CA. Once the data sets are cataloged to MIGRATE, you can test the results by performing recall on a few selected data sets. If the recall is successful then all attributes of the files in error were re-cataloged. If the test is unable to re-catalog the files successfully then check for a backup and the data sets, delete the data set using fix F11V1R2 and F11V1CR2 and recover the data sets with the DFSMShsm HRECOVER command.

### 6.1.5 FastAudit/390 audit error 11D2C

The next error we will look at is *FastAudit/390* Audit Error 11D2C MCD ENTRY IS CATALOGED ON DIFFERENT VOLUME.

### Diagnosing the error

Looking up the error in the *FastAudit/390* User Guide, you find the following explanation:

*FastAudit/390* has compared the MCDS MCD extraction against the catalog extraction and has found an MCD entry but the catalog indicates that the data set is on a different volume than MIGRAT.

This situation can occur when an older version of the data set exists in migration, but is uncataloged and a newer version of the data set is cataloged on a primary DASD volume. In most cases, the data set on the primary DASD volume is the correct or newer version of the data set and the migrated copy needs to be deleted.

Other possibilities are that the catalog may have been offline to DFSMSHsm when the catalog entry for the data set in question was replaced with one for a non-migrated (or recalled) data set. The condition could have also been caused by a DELETE NOSCRATCH operation. All MCDS records for the data sets have serial numbers of the volume from which the data set was migrated.

To view the error detail, type B on the FastAudit MCDS Summary panel next to the error and press Enter. The error detail is displayed in Figure 6-15.

|        |               |                                      |        |    |    |
|--------|---------------|--------------------------------------|--------|----|----|
| f1     | f2            | f3                                   | f4     | f5 | f6 |
| 11D2C  | IP01.PPROCLIB | HSM.HMIG.T080300.IP01.PPROCLIB.I2075 | HM0069 | N  | 2  |
| f7     |               |                                      |        |    |    |
| MVSCAT |               |                                      |        |    |    |

Figure 6-15 Error detail for 11D2C

The Field Description Box provided in the documentation and shown in Table 6-4 provides an explanation of each field.

Table 6-4 Diagnostic 11D2C field descriptions

| Dictionary symbol | Example                              |
|-------------------|--------------------------------------|
| Field 1 /ERRNUM   | 11D2C                                |
| Field 2 /TDSN     | IPO1.PPROCLIB                        |
| Field 3 /ADSN     | HSM.HMIG.T080300.IPO1.PPROCLIB.I2075 |
| Field 4 /VOLSER   | HM0069                               |
| Field 5 /DSORG    | N                                    |
| Field 6 /LEVEL    | 2                                    |

| Dictionary symbol | Example |
|-------------------|---------|
| Field 7 /CVOLSER  | MVSCAT  |

### Resolving the error

To view the list of available fixes, type F next to the error message and press Enter. The Fast Audit Fix Plan Member List is displayed in Figure 6-16.

```
FAST AUDIT FIX PLAN MEMBER LIST
OPTION ==>                                SCROLL==> PAGE
YOU MAY BROWSE (B), EDIT (E), FIX (F) OR VIEW (V) RESULTS
COMMAND MEMBER ACTION TITLE
      F11D2CA      TSO DELETE AND CATALOG DSN TO VOLSER MIGRAT
      F11D2CB      TSO DELETE AN UNCATALOGED MIGRATED data set
      F11D2CC      HSM FIXCDS DELETE OF A AND D RECORD
      F11D2CR1     HSM DELETE MCD RECORD WHEN data set IS CATALOGED ONLINE
      F11D2CR3     HSM FIXCDS PATCH ASSIGNED AND NEEDS SCRATCH
***** Bottom of data *****
```

Figure 6-16 Fast Audit Fix Plan Member List for error F11D2C

If you can determine that the current catalog entry has the correct status for the data set, you have several choices for resolving the error:

- Option 1. Execute fix F11D2CR3, to issue an **HSM FIXCDS PATCH ASSIGNED AND NEEDS SCRATCH** command to reset the assigned flag (MCDFASN) and to set the scratch flag (MCDFNSCR) in the MCDS record. This causes the next migration cleanup execution to remove the migration copy.
- Option 2. Execute fix F11D2CB to delete the uncataloged migrated data set immediately.

### 6.1.6 FastAudit/390 audit error 11V2C

The next error we will look at is *FastAudit/390* Audit Error 11V2C MCO VSAM COMPONENT ON DIFFERENT VOLUME.

### Diagnosing the error

Looking up the error in the *FastAudit/390* User Guide, you find the following explanation:

*FastAudit/390* has compared the MCDS MCO extraction against the Catalog extraction and has found an MCO entry but the catalog indicates that the VSAM component is on a different volume than MIGRAT.



To view the error detail, type B on the FastAudit MCDS Summary panel next to the error and press Enter. The error detail is displayed in Figure 6-17.

|       |                     |                  |        |
|-------|---------------------|------------------|--------|
| f1    | f2                  | f3               | f4     |
| 11V2C | USERID.VSAM.CLUSTER | USERID.VSAM.DATA | TS0002 |

Figure 6-17 Error detail for 11V2C

The Field Description Box provided in the documentation and shown in Table 6-5 provides an explanation of each field.

Table 6-5 Diagnostic 11V2C field descriptions

| Dictionary symbol | Example             |
|-------------------|---------------------|
| Field 1 /ERRNUM   | 11V2C               |
| Field 2 /Cluster  | USERID.VSAM.CLUSTER |
| Field 3 /CompName | USERID1.VSAM.DATA   |
| Field 4 /CVOLSER  | TSO002              |

Resolving the error

To view the list of available fixes, type F next to the error message and press Enter. The Fast Audit Fix Plan Member List is displayed in Figure 6-18.

|                                                           |         |                                                     |
|-----------------------------------------------------------|---------|-----------------------------------------------------|
| <V6 R1.01>                                                |         | Row 1 of 4                                          |
| FAST AUDIT FIX PLAN MEMBER LIST                           |         |                                                     |
| OPTION ==>                                                |         | SCROLL==> PAGE                                      |
| YOU MAY BROWSE (B), EDIT (E), FIX (F) OR VIEW (V) RESULTS |         |                                                     |
| COMMAND                                                   | MEMBER  | ACTION TITLE                                        |
|                                                           | F11V2CA | TSO DELETE AND CATALOG DSN TO VOLSER MIGRAT         |
|                                                           | F11V2CB | TSO DELETE AN UNCATALOGED MIGRATED data set         |
|                                                           | F11V2CE | HSA DISPLAY DUMP OF MCO RECORD                      |
|                                                           | F11V2CF | HSM GENERATE FIXCDS DELETE FOR CONFLICT MCO AND MCD |
| ***** Bottom of data *****                                |         |                                                     |

Figure 6-18 Fast Audit fix plan member list for F11V2C

If your organization is in the habit of allocating and using VSAM components as opposed to VSAM clusters then this diagnostic could be very important and the user should explore the current status of the cluster (LIST CATALOG). If the cluster resides on primary volumes then this orphaned VSAM component can be safely deleted with a FIXCDS DELETE command executed in fix F11V2CF. If the cluster name is not listed (missing) then the MCO component may safely be deleted.

First, you may want to verify that the MCO record exists by issuing fix F11V2CE. If both components of the record are present then fix F11V2CA can be executed to resolve the problem. If the MCA or MCO record is not present then a DELETE of the record is the only solution. If desired, fix F11V2CB can be tried and if unsuccessful, then fix F11V2CF performs a FIXCDS MCA and MCO record DELETE.

6.1.7 FastAudit/390 audit error 1103W

The next error we will look at is *FastAudit/390* Audit Error 1103W MCD ENTRY IS MISSING THE MCA ENTRY.

Diagnosing the error

Looking up the error in the *FastAudit/390* User Guide, you find the following explanation:

*FastAudit/390* has compared the MCDS MCD extraction against the MCDS MCA extraction and has found an MCD entry but the MCA entry is missing. Although information in the MCA is not used during recall, it may be an indicator of the health of the MCDS inventory. A missing MCA can cause recycle to run very slowly because recycle must verify missing MCA entries. Furthermore for ML1 (DASD) migration alias names identify VTOC names and the VTOC copy could be orphaned or missing.

To view the error detail, type B on the FastAudit MCDS Summary panel next to the error and press Enter. The error detail is displayed in Figure 6-19.

|       |                    |                                 |        |    |    |
|-------|--------------------|---------------------------------|--------|----|----|
| f1    | f2                 | f3                              | f4     | f5 | f6 |
| 1103W | DRS.ICS91102.CLIST | HSM.HMIG.T472104.ICS91102.I1257 | HM0017 | N  | 2  |

Figure 6-19 Diagnostic 1103W field descriptions

The Field Description Box provided in the documentation and shown in Table 6-6 provides an explanation of each field.

Table 6-6 Diagnostic 1103W field descriptions

| Dictionary symbol | Example                             |
|-------------------|-------------------------------------|
| Field 1 /ERRNUM   | 1103W                               |
| Field 2 /TDSN     | DRS.ICS91102.CLIST                  |
| Field 3 /ADSN     | HSM.HMIG.T472104.DRS.ICS91102.I1257 |
| Field 4 /VOLSER   | HM0017                              |

| Dictionary symbol | Example |
|-------------------|---------|
| Field 5 /DSORG    | N       |
| Field 6 /LEVEL    | 2       |

### Resolving the error

To view the list of available fixes, type **F** next to the error message and press Enter. The Fast Audit Fix Plan Member List is displayed in Figure 6-20.

```
<V6 R1.01>                                     Row 1 of 5
      FAST AUDIT FIX PLAN MEMBER LIST
OPTION ==>                                     SCROLL==> PAGE
YOU MAY BROWSE (B), EDIT (E), FIX (F) OR VIEW (V) RESULTS
COMMAND MEMBER ACTION TITLE
      F1103WA      HSM GENERATE FIXCDS DISPLAY FOR MISSING MCA RECORD
      F1103WB      HSA VERIFY MCD AND MCA RECORDS
      F1103WR1     HSM GENERATE RECALL FOR data setS
      F1103WR2     HSM GENERATE HMIG FOR data setS
      F1103WR3     HSM GENERATE HSEND DELETES FOR data setS
***** Bottom of data *****
```

Figure 6-20 Fix Plan Member List for error 1103W

The first option is to execute fix F1103WR1 to recall the data set. This allows DFSMSHsm to maintain all of its internal records. If a recall for this data set fails and no other diagnostic exists related to the data set, a DFSMSHsm recovery of the data set is most likely the best and easiest way of clean up.

Another option is to verify that the MCA record is missing, by executing fix F1103WA or F1103WB. If the MCA record is missing then only a delete of the record can be performed using fix F1103WR3. If both the MCA and MCD record are present, you may recall the data set using fix F1103WR1, then fix F1103WR2 can be executed to migrate the recalled data set to tape. If either of the records is missing then only a delete of the data set can be performed using fix F1103WR3.

### 6.1.8 FastAudit/390 audit error 1104C

The next error message we will look at is HSM *FastAudit* Error 1104C MCD LEVEL 1 ENTRY HAS NO VTOC ENTRY ON L1 VOLUME.

### Diagnosing the error

Looking up the error in the *FastAudit/390* User Guide, you find the following explanation:

*FastAudit/390* has compared the MCDS MCD Level 1 extraction against the HSM LEVEL 1 volume DCOLLECT extraction and has found an MCD entry but no VTOC entry. *FastAudit/390* verifies this by doing an OBTAIN against the volume in question.

To view the error detail, type B on the FastAudit MCDS Summary panel next to the error and press Enter. The error detail is displayed in Figure 6-21.

|       |                    |                                       |        |    |    |
|-------|--------------------|---------------------------------------|--------|----|----|
| f1    | f2                 | f3                                    | f4     | f5 | f6 |
| 1104C | XXX.XXXXXXXXXX.DSN | HSM.HMIG.T443206.XXX.XXXXXXXXXX.I1257 | HM0017 | N  | 1  |

Figure 6-21 Error detail for 1104C

The Field Description Box provided in the documentation and shown in Table 6-7 provides an explanation of each field.

Table 6-7 Diagnostic 1104C field descriptions

| Dictionary symbol | Example                               |
|-------------------|---------------------------------------|
| Field 1 /ERRNUM   | 1104C                                 |
| Field 2 /TDSN     | XXX.XXXXXXXXXX.DSN                    |
| Field 3 /ADSN     | HSM.HMIG.T443206.XXX.XXXXXXXXXX.I1257 |
| Field 4 /VOLSER   | HM0017                                |
| Field 5 /DSORG    | N                                     |
| Field 6 /LEVEL    | 1                                     |

**Resolving the error**

To view the list of available fixes, type F next to the error message and press Enter. The Fast Audit Fix Plan Member List is displayed in Figure 6-22.

```

<V6 R1.01>
FAST AUDIT FIX PLAN MEMBER LIST
OPTION ==>
YOU MAY BROWSE (B), EDIT (E), FIX (F) OR VIEW (V) RESULTS
COMMAND MEMBER ACTION TITLE
F1104CA TSO DELETE AND RECOVER ML1 DSN
F1104CB HSA DISPLAY DUMP OF MCD RECORD MISSING ML1
F1104CC HSM FIXCDS COMMAND TO DELTE DSN MISSING ML1 VOL
F1104CR1 HSM RR-DELETE MCD RECORD WHEN ML1 MCV MISSING
F1104CR2 TSO LISTC ENTRIES OF data setS
F1104CR3 HSM HLIST data set NAME
***** Bottom of data *****

```

Figure 6-22 Fix Plan Member List for error 1104C

It is always recommended to perform additional verification first to know exactly the extent of what record is missing. To verify the error you can execute fixes F1104CB, F1104CR2 and F1104CR4. If a DFSMSHsm backup copy of the data set is present, then the recommended fix is to delete the migration copy and recover the data set by executing fix F1104CA. If the MCD or MCA record is missing and there is no DFSMSHsm backup available, then a delete of the record must be done by executing fix F1104CC and F1104CR1.

## 6.1.9 FastAudit/390 audit error 1105W

The next error message we will look at is *FastAudit/390* Audit Error 1005W MCD VSAM BASE IS MISSING IN MCO.

### Diagnosing the error

Looking up the error in the *FastAudit/390* Guide, you find the following explanation:

*FastAudit/390* has compared the MCDS MCD extraction against the HSM MCO and was unable to find an MCO entry that contained the Base Cluster Name associated with a VSAM data set. The data set can probably be recalled but this might be a health indicator of the VSAM inventory. Typically VSAM components are not recalled independent of the cluster name, however, if they are the recall might fail.

To view the error detail, type B on the FastAudit MCDS Summary panel next to the error and press Enter. The error detail is displayed in Figure 6-23.

|       |                  |                                 |        |    |    |
|-------|------------------|---------------------------------|--------|----|----|
| f1    | f2               | f3                              | f4     | f5 | f6 |
| 1105W | DSN.VSAM.CLUSTER | HSM.HMIG.T233206.DSN.VSAM.I1257 | HM0017 | V  | 1  |

Figure 6-23 Error detail for 1105W

The Field Description Box provided in the documentation and shown in Table 6-8 provides an explanation of each field.

Table 6-8 Diagnostic 1105W field descriptions.

| Dictionary symbol | Example                         |
|-------------------|---------------------------------|
| Field 1 /ERRNUM   | 1105W                           |
| Field 2 /TDSN     | DSN.VSAM.CLUSTER                |
| Field 3 /ADSN     | HSM.HMIG.T233206.DSN.VSAM.I1257 |
| Field 4 /VOLSER   | MIGR01                          |
| Field 5 /DSORG    | V                               |
| Field 6 /LEVEL    | 1                               |

### Resolving the error

To view the list of available fixes, type **F** next to the error message and press Enter. The Fast Audit Fix Plan Member List is displayed in Figure 6-24.

|                                                           |         |                                          |
|-----------------------------------------------------------|---------|------------------------------------------|
| <V6 R1.01>                                                |         | Row 1 of 3                               |
| FAST AUDIT FIX PLAN MEMBER LIST                           |         |                                          |
| OPTION ==>                                                |         | SCROLL==> PAGE                           |
| YOU MAY BROWSE (B), EDIT (E), FIX (F) OR VIEW (V) RESULTS |         |                                          |
| COMMAND                                                   | MEMBER  | ACTION TITLE                             |
|                                                           | F1105WA | TSO DELETE AND RECOVER MISSING VSAM MCO  |
|                                                           | F1105WB | HSA DISPLAY MCO RECORD MISSING BASE NAME |
|                                                           | F1105WC | HSM GENERATE RECALL FOR data setS        |
| ***** Bottom of data *****                                |         |                                          |

Figure 6-24 Fix Plan Member List for error 1105W

One option is to recall the data set by executing fix F1105WC. This allows DFSMSHsm to maintain all of its internal records. A subsequent migration of the data set creates the new MCO record that is presently missing. If the recall is unsuccessful then check for a DFSMSHsm backup copy of the data set in error. If a backup copy exists, execute fix F1105WA to perform the TSO delete and DFSMSHsm recover of the data set.

6.1.10 FastAudit/390 audit error 1106C and 1106V

The next error message we will look at is *FastAudit/390* Audit Error 1106C and 1106V MCD IS ON VOLUME WHICH HAS NO MCV ENTRY.

*FastAudit/390* reports data set errors on tape volumes with an error message code that ends with the letter V for volume. Data set errors on DASD volumes have an error message code that ends with either C for critical, W for warning and I for informational only. In this case, error code 1106C are for MCD errors for data sets that resided on DASD volumes, and 1106V are for MCD errors for data sets that resided on tape volumes. In your diagnostic efforts, you will need to review both the 1106C errors and the 1106V errors to know all the data sets and volumes reported in this error category. Fixes for both of these errors are available in fix 1106V.

Diagnosing the error

Looking up the error in the *FastAudit/390* User Guide, you find the following explanation:

1106C: *FastAudit/390* has compared the MCDS MCD extraction against the HSM MCV extraction and has found an entry on a migration volume that is no longer managed by HSM. This is a critical error. The data set cannot be recalled. See 1106V for a summary of the volumes with this condition.

1106V: This error is a summary by volumes of error 1106C. Research of this problem is easier to do at a volume level. Supporting fixes and interrogations at a volume level is supplied for this error.

To view the error detail, type B on the *FastAudit* MCDS Summary panel next to the error 1106C and 1106V and press Enter. The error detail is displayed in Figure 6-25 and Figure 6-26.

|       |               |                                  |        |    |    |
|-------|---------------|----------------------------------|--------|----|----|
| f1    | f2            | f3                               | f4     | f5 | f6 |
| 1106C | DSN.NOMCV.DSN | HSM.HMIG.T433206.DSN.NOMCV.I1257 | HM0017 | V  | 1  |

Figure 6-25 Error detail for 1106C

|       |        |    |        |
|-------|--------|----|--------|
| f1    | f2     | f3 | f4     |
| 1106V | HM0061 | L2 | 000009 |

Figure 6-26 Error detail for 1106V

The Field Description Boxes provided in the documentation and shown in Table 6-9 and Table 6-10 provides an explanation of each field.

Table 6-9 Diagnostic 1106C field descriptions

| Dictionary symbol | Example                          |
|-------------------|----------------------------------|
| Field 1 /ERRNUM   | 1106C                            |
| Field 2 /TDSN     | DSN.NOMCV.DSN                    |
| Field 3 /ADSN     | HSM.HMIG.T433206.DSN.NOMCV.I1257 |
| Field 4 /VOLSER   | HM0017                           |
| Field 5 /DSORG    | V                                |
| Field 6 /LEVEL    | 1                                |

Table 6-10 Diagnostic 1106V field descriptions

| Dictionary symbol  | Example                                  |
|--------------------|------------------------------------------|
| Field 1 /ERRNUM    | 1106V                                    |
| Field 2 /VOLSER    | HM0061                                   |
| Field 3 /TTCTYP    | L2 (indicates level 2 tape)              |
| Field 4 DSN Number | 000009 (number of dsns referencing tape) |

Resolving the error

To view the list of available fixes, type F next to the error message and press Enter. The Fast Audit Fix Plan Member List is displayed in Figure 6-27.

```
<V6 R1.01>                                     Row 1 of 4
      FAST AUDIT FIX PLAN MEMBER LIST
OPTION ====                                     SCROLL==> PAGE
  YOU MAY BROWSE (B), EDIT (E), FIX (F) OR VIEW (V) RESULTS
COMMAND MEMBER ACTION TITLE
      F1106VA      HSM FIXCDS COMMAND TO VERIFY MISSING MCV RECORD
      F1106VB      TMS INTERROGATE TMS FOR HSM TAPES MISSING MCV
      F1106VC      HSM FIXCDS TO ADD NEW MCV LEVEL 2 RECORD
      F1106VD      HSM FIXCDS TO DELETE MCV LEVEL 2 RECORD
***** Bottom of data *****
```

Figure 6-27 Fix Plan Member List for error 1106V

The *FastAudit/390* 1106V error is the same as 1106C. *FastAudit* has been modified so that the 1106C errors are reported at the data set level, and the 1106V errors are reported at the volume level. Review both errors to know all the data sets and volumes reported in error. All fixes can be executed in the 1106V



fix panel where the code has been modified to remove duplicate volume VOLSERS.

It is recommended that you execute fix F1106VA and F1106VB to verify the MCV record is in error. This is a redundant step as FastAudit automatically verifies it by a direct catalog lookup. You can choose to skip this step if desired. You can then perform a FIXCDS CREATE/PATCH by executing fix F1106VC then execute fix F1106VD to remove any extraneous records added.

6.1.11 **FastAudit/390 audit error 1107V**

The next error message we will look at is *FastAudit/390 Audit Error 1107V* SUMMARY OF VOLUMES MISSING IN TTOC.

**Diagnosing the error**

Looking up the error in the *FastAudit/390 User Guide*, you find the following explanation:

*FastAudit/390* has compared the MCDS MCD extraction (Level 2) against the HSM OCDS (TTOC) extraction and has found an entry on a volume that is not in the TTOC inventory. In other words, the root entry for the volume is not in the OCDS. The TTOC information is not used for Recall, however these errors may reflect the general health of the HSM environment.

Errors reported can indicate possible data lost if the tape volume has been returned to scratch status.

This error is a summary by volumes of error 1107W. This error is easier to research and resolve at the volume level. Supporting fixes and interrogations at a volume level is supplied for this error.

To view the error detail, type B on the FastAudit MCDS Summary panel next to the error and press Enter. The error detail is displayed in Figure 6-28.

| f1    | f2     | f3 | f4     |
|-------|--------|----|--------|
| 1107V | HM0060 | L2 | 000009 |

Figure 6-28 Error detail for 1106V

The Field Description Box provided in the documentation and shown in Table 6-11 provides an explanation of each field.

Table 6-11 Diagnostic 1107V field descriptions.

| Dictionary symbol  | Example                                  |
|--------------------|------------------------------------------|
| Field 1 /ERRNUM    | 1107V                                    |
| Field 2 /VOLSER    | HM0060                                   |
| Field 3 /TTCTYP    | L2 (indicates level 2 tape)              |
| Field 4 DSN Number | 000009 (number of dsns referencing tape) |

### Resolving the error

To view the list of available fixes, type **F** next to the error message and press Enter. The Fast Audit Fix Plan Member List is displayed in Figure 6-29.

```
<V6 R1.01>                                     Row 1 of 11
      FAST AUDIT FIX PLAN MEMBER LIST
OPTION ==>                                     SCROLL==> PAGE
YOU MAY BROWSE (B), EDIT (E), FIX (F) OR VIEW (V) RESULTS
COMMAND MEMBER ACTION TITLE
      F1107VA      HSM FIXCDS COMMAND TO VERIFY MISSING TTOC RECORD
      F1107VB      HSM LIST TTOC FOR MISSING VOLUME
      F1107VC      RMM INTERROGATE RMM FOR HSM TAPES
      F1107VD      HSM FIXCDS DISPLAY TTOC RECORD EXTENTS 3 - 5
      F1107VE      HSM FIXCDS DISPLAY TTOC RECORD EXTENTS 6 - 8
      F1107VF      HSM FIXCDS DISPLAY TTOC RECORD EXTENTS 9 - 11
      F1107VG      HSM FIXCDS DISPLAY TTOC RECORD EXTENTS 12 - 14
      F1107VH      HSM FIXCDS DISPLAY TTOC RECORD EXTENTS 15- 17
      F1107VI      TMS INTERROGATE TMS FOR HSM TAPES MISSING IN OCDS
      F1107VJ      HSM AUDIT MEDAICONTROLS FOR MISSING TTOC
      F1107VK      HSM FIXCDS TO ADD NEW MCV LEVEL 2 RECORD
***** Bottom of data *****
```

Figure 6-29 Fix Plan Member List for error 1107V

*FastAudit/390* is the only tool that detects and reports these errors. Standard DFSMSHsm command audit doesn't find or report these errors. As in all cases you should determine if additional verification is needed. Some of the additional verifications that can be performed can interrogate your tape management system further. To do this, choose fixes F1107VA, F1107VB, F1107VC, F1107VD, F1107VE, F1107VF, F1107VG, F1107VH or F1107VI. These fixes display additional information in different detailed reports. The only resolution to correcting the tape in error is fix F1107VJ, to execute AUDITMEDIAICONTROLS for the tape in question.

6.1.12 FastAudit/390 audit error 1107W

The next error we will look at is *FastAudit/390* Audit Error 1107W MCD LEVEL 2 IS ON VOLUME WHICH HAS NO TTOC ENTRY.

Diagnosing the error

Looking up the error in the *FastAudit/390* User Guide, you find the following explanation:

*FastAudit/390* has compared the MCDS MCD extraction (Level 2) against the HSM OCDS (TTOC) extraction and has found an entry on a volume that is not in the TTOC inventory. In other words, the root entry for the volume is not in the OCDS. The TTOC information is not used for Recall, however these errors may reflect the general health of the HSM environment.

This error could indicate possible loss of data if the tape volume had been returned to scratch status. See error message 1107V for a summary of volumes with this condition.

To view the error detail, type B on the FastAudit MCDS Summary panel next to the error and press Enter. The error detail is displayed in Figure 6-30.

|       |                  |                                    |        |    |    |
|-------|------------------|------------------------------------|--------|----|----|
| f1    | f2               | f3                                 | f4     | f5 | f6 |
| 1107W | XAS.IXPV3R2.LOAD | HSM.HMIG.T040202.XAS.IXPV3R2.I4142 | HM0348 | N  | 2  |

Figure 6-30 Error detail for 1107W

The Field Description Box provided in the documentation and shown in Table 6-12 provides an explanation of each field.

Table 6-12 Diagnostic 1107W field descriptions

| Dictionary symbol | Example                            |
|-------------------|------------------------------------|
| Field 1 /ERRNUM   | 1107W                              |
| Field 2 /TDSN     | XAS.IXPV3R2.LOAD                   |
| Field 3 /ADSN     | HSM.HMIG.T040202.XAS.IXPV3R2.I4142 |
| Field 4 /VOLSER   | HM0348                             |
| Field 5 /DSORG    | N                                  |
| Field 6 /LEVEL    | 2                                  |

### Resolving the error

To view the list of available fixes, type **F** next to the error message and press Enter. The Fast Audit Fix Plan Member List is displayed in Figure 6-31.

```
<V6 R1.01>                                     Row 1 of 5
      FAST AUDIT FIX PLAN MEMBER LIST
OPTION ==>                                     SCROLL==> PAGE
YOU MAY BROWSE (B), EDIT (E), FIX (F) OR VIEW (V) RESULTS
COMMAND  MEMBER  ACTION  TITLE
F1107WA      HSM LIST TTOC COMMAND TO VERIFY MISSING TTOC
F1107WB      HSM FIXCDS DISPLAY TTOC EXTN FOR MISSING MCA
F1107WC      HSM FIXCDS DELETE MCD RECORD MISSING TTOC
F1107WD      TSO DELETE MIGRATION ENTRY MISSING THE TTOC
F1107WE      TSO RECALL THE DATA SET ANYWAY
***** Bottom of data *****
```

Figure 6-31 Fix plan member list for error 1107W

See error message 1107V for a summary of volumes with this condition.

To correct tape error, execute fix F1107VJ to execute AUDITMEDIACONTROLS for the tape in question.

### 6.1.13 FastAudit/390 audit error 1108W

The next error we will look at is *FastAudit/390* Audit Error 1108W MCA ENTRY IS MISSING MCO VSAM COMPONENT ENTRY.

#### Diagnosing the error

Looking up the error in the *FastAudit/390* User Guide, you find the following explanation:

When auditing the MCA extraction against the MCD extraction, a VSAM component name was found but the cluster name was missing. This is not a critical error for Recall but if there are many of these it indicates that the MCDS inventory may be corrupted.

To view the error detail, type **B** on the FastAudit MCDS Summary panel next to the error and press Enter. The error detail is displayed in Figure 6-32.

| f1    | f2                   | f3                | f4  | f5 | f6 |
|-------|----------------------|-------------------|-----|----|----|
| 1108W | DRS.ICS91102.CLISTER | DRS.ICS91102.DATA | UNK | V  | U  |

Figure 6-32 Error detail for 1108W

The Field Description Box provided in the documentation and shown in Table 6-13 provides an explanation of each field.

Table 6-13 Diagnostic 1108W field descriptions

| Dictionary symbol | Example              |
|-------------------|----------------------|
| Field 1 /ERRNUM   | 1108W                |
| Field 2 /TDSN     | DRS.ICS91102.CLUSTER |
| Field 3 /ADSN     | DRS.ICS91102.DATA    |
| Field 4 /VOLSER   | UNK                  |
| Field 5 /DSORG    | V                    |
| Field 6 /LEVEL    | U                    |

### Resolving the error

To view the list of available fixes, type **F** next to the error message and press Enter. The Fast Audit Fix Plan Member List is displayed in Figure 6-33.

```
<V6 R1.01>                                     Row 1 of 2
      FAST AUDIT FIX PLAN MEMBER LIST
OPTION ==>                                     SCROLL==> PAGE
YOU MAY BROWSE (B), EDIT (E), FIX (F) OR VIEW (V) RESULTS
COMMAND MEMBER ACTION TITLE
      F1108WA      HSM FIXCDS DISPLAY MCA RECORD MISSING MCO COMPONENT
      F1108WB      HSM FIXCDS DELETE MCA MISSING MCO COMPONENT
***** Bottom of data *****
```

Figure 6-33 Fix Plan Member List for error 1108W

If it is determined that the VSAM cluster is missing you will want to check for a possible DFSMSHsm backup or any other backup of the data. The only corrective action is to execute fix F1108WB to delete the component record and subsequent recovery of the data set. If the data set is needed and a backup is available, follow your standard backup recovery procedures.

### 6.1.14 FastAudit/390 audit error 1109V

The next error we will look at is *FastAudit/390* 1109V SUMMARY OF VOLUMES WHERE ALIAS IS NOT IN TTOC.

## Diagnosing the error

Looking up the error in the *FastAudit/390* User Guide, you find the following explanation:

This error is a summary by volumes of error 1109W. Research of this problem is easier to do at a volume level. Supporting fixes and interrogations at a volume level is supplied for this error.

To view the error detail, type B on the FastAudit MCDS Summary panel next to the error and press Enter. The error detail is displayed in Figure 6-34.

|       |        |    |        |
|-------|--------|----|--------|
| f1    | f2     | f3 | f4     |
| 1109V | HM0061 | L2 | 000009 |

Figure 6-34 Error detail for 1109V

The Field Description Box provided in the documentation and shown in Table 6-14 provides an explanation of each field.

Table 6-14 Diagnostic 1109V field descriptions

| Dictionary symbol  | Example                                  |
|--------------------|------------------------------------------|
| Field 1 /ERRNUM    | 1109V                                    |
| Field 2 /VOLSER    | HM0061                                   |
| Field 3 /TTCTYP    | L2 (indicates level 2 tape)              |
| Field 4 DSN Number | 000009 (number of DSNs referencing tape) |

## Resolving the error

To view the list of available fixes, type F next to the error message and press Enter. The Fast Audit Fix Plan Member List is displayed in Figure 6-35.

|                                                           |         |                                                   |
|-----------------------------------------------------------|---------|---------------------------------------------------|
| <V6 R1.01>                                                |         | Row 1 of 4                                        |
| FAST AUDIT FIX PLAN MEMBER LIST                           |         |                                                   |
| OPTION ==>                                                |         | SCROLL==> PAGE                                    |
| YOU MAY BROWSE (B), EDIT (E), FIX (F) OR VIEW (V) RESULTS |         |                                                   |
| COMMAND                                                   | MEMBER  | ACTION TITLE                                      |
|                                                           | F1109VA | HSM LIST TTOC FOR VOLUME HAVING MISSING MCA ENTRY |
|                                                           | F1109VB | HSM FIXCDS DISPLAY T FOR VOL MISSING MCA ENTRY Y  |
|                                                           | F1109VC | HSM LIST TTOC VOLUMES FOR FAILEDCREATE            |
|                                                           | F1109VD | HSM AUDIT MEDIA CONTROL FOR MCA MISSING IN TTOC   |
| ***** Bottom of data *****                                |         |                                                   |

Figure 6-35 Fix Plan Member List for error 1109V

You can research all the information concerning the tapes and the missing records by executing fixes F1109VA, F1109VB, and F1109VC. After viewing the executed fixes you will know the full extent of the errors encountered. Fix F1109VD executes the **AUDIT MEDIACONTROLS VOLUMES (VOLSER)** DFSMSHsm fix command against the tapes in error.

6.1.15 *FastAudit/390* audit error 1109W

The next error we will look at is *FastAudit/390* Audit Error 1109W MCD ALIAS IS NOT ON TTOC.

Diagnosing the error

Looking up the error in the *FastAudit/390* User Guide, you find the following explanation:

*FastAudit/390* has compared the MCDS MCD extraction (Level 2) against the HSM OCDS TTOC extraction and has been unable to find a matching HSM generated alias name in the TTOC. DFSMSHsm does not use the TTOC information during recall but these errors may indicate a general health problem in the HSM environment. This error can also be the result of an incomplete TTOC update resulting from an DFSMSHsm ABEND or system crash or an IPL over a working DFSMSHsm. Errors in this category can cause recycle to fail.

**Note:** *FastAudit/390* does *not* verify these errors in real-time. If recycle was running while the audit was extracting information, these errors may be erroneous. For best results make sure recycle is not running at the same time as *FastAudit/390*.

To view the error detail, type B on the *FastAudit* MCDS Summary panel next to the error and press Enter. The error detail is displayed in Figure 6-36.

|       |                    |                                     |        |    |    |
|-------|--------------------|-------------------------------------|--------|----|----|
| f1    | f2                 | f3                                  | f4     | f5 | f6 |
| 1109W | DRS.ICS91102.CLIST | HSM.HMIG.T472104.DRS.ICS91102.I1257 | HM0017 | V  | 2  |

Figure 6-36 Diagnostic 1109W field descriptions

The Field Description Box provided in the documentation and shown in Table 6-15 provides an explanation of each field.

Table 6-15 .Diagnostic 1109W Field descriptions

| Dictionary symbol | Example |
|-------------------|---------|
| Field 1 /ERRNUM   | 1109W   |

| Dictionary symbol | Example                             |
|-------------------|-------------------------------------|
| Field 2 /TDSN     | DRS.ICS91102.CLIST                  |
| Field 3 /ADSN     | HSM.HMIG.T472104.DRS.ICS91102.I1257 |
| Field 4 /VOLSER   | HM0017                              |
| Field 5 /DSORG    | V                                   |
| Field 6 /LEVEL    | 2                                   |

### Resolving the error

To view the list of available fixes, type **F** next to the error message and press Enter. The Fast Audit Fix Plan Member List is displayed in Figure 6-37.

```
<V6 R1.01> Row 1 of 4
      FAST AUDIT FIX PLAN MEMBER LIST
OPTION ==> SCROLL==> PAGE
YOU MAY BROWSE (B), EDIT (E), FIX (F) OR VIEW (V) RESULTS
COMMAND MEMBER ACTION TITLE
      F1109VA      HSM LIST TTOC FOR VOLUME HAVING MISSING MCA ENTRY
      F1109VB      HSM FIXCDS DISPLAY T FOR VOL MISSING MCA ENTRY Y
      F1109VC      HSM LIST TTOC VOLUMES FOR FAILEDCREATE
      F1109VD      HSM AUDIT MEDIA CONTROL FOR MCA MISSING IN TTOC
***** Bottom of data *****
```

Figure 6-37 Fix plan member list for error 1106V

Refer to the fixes provided in error F1109V for all F1109W errors. Fixes provided in F1109W are optional verifications of the errors.

### 6.1.16 FastAudit/390 audit error 1110W

The next error we will look at is *FastAudit/390* Audit Error 1110W MCD HAS DUPLICATE ENTRY.

### Diagnosing the error

Looking up the error in the *FastAudit/390* User Guide, you find the following explanation:

*FastAudit/390* has scanned the MCDS MCD alias names for duplicate names. This condition can happen on very fast CPUs when the alias name is generated. The seconds portion of the time stamp does not turn over before the alias name is generated. It is possible that the data set cannot be recalled. This entry is presented in pairs of two but is counted only once.



To view the error detail, type B on the FastAudit MCDS Summary panel next to the error and press Enter. The error detail is displayed in Figure 6-38.

| f1    | f2               | f3                                 | f4     | f5 | f6 |
|-------|------------------|------------------------------------|--------|----|----|
| 1110W | DCD.HFA2201.LOAD | HSM.HMIG.T001300.DCD.HFA2201.J1251 | L1000D | N  | 1  |
| 1110W | DCD.HFA2201.LAST | HSM.HMIG.T001300.DCD.HFA2201.J1251 | ML1001 | N  | 1  |

Figure 6-38 Error detail for error 1110W

The Field Description Box provided in the documentation and shown in Table 6-16 provides an explanation of each field.

Table 6-16 Diagnostic 1110W field descriptions

| Dictionary symbol | Example                        |
|-------------------|--------------------------------|
| Field 1 /ERRNUM   | 1110W                          |
| Field 2 /TDSN     | DCD.HFA2201.LOAD               |
| Field 3 /ADSN     | HSM.HMIG.T001300.HFA2201.J1251 |
| Field 4 /VOLSER   | L100D                          |
| Field 5 /DSORG    | N                              |
| Field 6 /LEVEL    | 1                              |

### Resolving the error

To view the list of available fixes, type F next to the error message and press Enter. The Fast Audit Fix Plan Member List is displayed in Figure 6-39.

|                                                           |         |                                                        |       |
|-----------------------------------------------------------|---------|--------------------------------------------------------|-------|
| <V6 R1.01>                                                |         | Row 1 of 2                                             |       |
| FAST AUDIT FIX PLAN MEMBER LIST                           |         |                                                        |       |
| OPTION ==>                                                |         | SCROLL==> PAGE                                         |       |
| YOU MAY BROWSE (B), EDIT (E), FIX (F) OR VIEW (V) RESULTS |         |                                                        |       |
| COMMAND                                                   | MEMBER  | ACTION                                                 | TITLE |
|                                                           | F1110WA | TSO LISTC FIXCDS D AND FIXCDS A TO ANALYZE MCA DUPLICA |       |
|                                                           | F1110WB | HSM RECALL FOR CONFLICTING ALIAS NAMES                 |       |
| ***** Bottom of data *****                                |         |                                                        |       |

Figure 6-39 Fix plan member list for error 111)W

The fix F1110WA performs additional diagnostics LISTC FICXCDS A and D records. The final fix is to recall the data sets in error. Should this fail, execute the DFSMSHsm command **HLIST DSN('data setname') BCDS** to list available backup copies. If backup copies are present, HDELETE and HRECOVER the data set. If

no backup is present, a HDELETE of the record is the only way to resolve the error.

6.1.17 **FastAudit/390 audit error 1111W**

The next error we will look at is *FastAudit/390* Audit Error 1111W MCA ENTRY IS MISSING MCD ENTRY.

**Diagnosing the error**

Looking up the error in the *FastAudit* User Guide, you find the following explanation:

*FastAudit/390* has compared the MCDS MCA extraction against the MCDS MCD extraction and has been unable to find a matching MCD true name. These errors cause recycle of tapes to fail. In the field explanations below, the MCA has no VOLSER information and no HSM level information. Therefore, field 4 and field 6 values are unknown.

To view the error detail, type B on the FastAudit MCDS Summary panel next to the error and press Enter. The error detail is displayed in Figure 6-40.

|       |                        |                             |     |    |    |
|-------|------------------------|-----------------------------|-----|----|----|
| f1    | f2                     | f3                          | f4  | f5 | f6 |
| 1111W | EM.Z.PRESTAGE.INPUTJCL | HSM.HMIG.T210408.EM.Z.J0230 | UNK | N  | U  |

Figure 6-40 Error detail for error 1111W

The Field Description Box provided in the documentation and shown in Figure 6-17 provides an explanation of each field.

Table 6-17 Diagnostic 1111W field descriptions

| Dictionary symbol | Example                     |
|-------------------|-----------------------------|
| Field 1 /ERRNUM   | 1111W                       |
| Field 2 /TDSN     | EM.Z.PRESTAGE.INPUTJCL      |
| Field 3 /ADSN     | HSM.HMIG.T210408.EM.Z.J0230 |
| Field 4 /VOLSER   | UNK                         |
| Field 5 /DSORG    | N                           |
| Field 6 /LEVEL    | U                           |

## Resolving the error

To view the list of available fixes, type **F** next to the error message and press Enter. The Fast Audit Fix Plan Member List is displayed in Figure 6-41.

```
<V6 R1.01> Row 1 of 8
      FAST AUDIT FIX PLAN MEMBER LIST
OPTION ===>
YOU MAY BROWSE (B), EDIT (E), FIX (F) OR VIEW (V) RESULTS
COMMAND MEMBER ACTION TITLE
F1111WA HSM FIXCDS A, D AND O TO EXAMINE MISS MCD/MCO
F1111WB HSM LIST TRUE NAME IN MCDS
F1111WC HSM FIXCDS DELETE ORPHANED MCA RECORD
F1111WD HSA VERIFY A AND D RECORD FOR EXAMINATION
F1111WR1 HSM FIXCDS D TO EXAMINE MISSING MCD/MCO
F1111WR2 HSM FIX A/D/O DISP MISSING MCD/MCO ** ADD LOGONLY
F1111WR3 HSM FIX A/O DELETE AFTER VERIFY D MISSING
F1111WR4 FAUDITX.QABETA.CNTL(F1111WR4) - 01.00
***** Bottom of data *****
```

Figure 6-41 Fix Plan Member List for error 1110W

You can execute the additional fixes F1111WA, F1111WB, and F1111WR1 to verify and understand better the total affect of the error and data sets in question. If you do not want to delete these orphaned data sets, the IBM Command Audit commands AUDIT DSCTL(mig) or DIRCTL can be executed. You can choose to execute fix F111WR1 and F111WR3 to delete the records that are actually missing the D record.

### 6.1.18 FastAudit/390 audit error 1112W

The next error we will look at is *FastAudit/390* Audit Error 1112W MCO VSAM BASE NAME IS MISSING IN MCD.

#### Diagnosing the error

Looking up the error in the *FastAudit* User Guide, you find the following explanation:

*FastAudit/390* has compared the MCDS MCO extraction against the MCDS MCD extraction and has been unable to find a matching VSAM Base name. This means that the MCO entry is orphaned and should be deleted.

To view the error detail, type **B** on the FastAudit MCDS Summary panel next to the error and press Enter. The error detail is displayed in Figure 6-42.

|       |                  |               |                                |
|-------|------------------|---------------|--------------------------------|
| f1    | f2               | f3            | f4                             |
| 1112W | MCO.BASE.CLUSTER | MCO.BASE.DATA | HSM.HMIG.T47311.MCO.BASE.I1132 |

Figure 6-42 Error detail for error 1112W

The Field Description Box provided in the documentation and shown in Table 6-18 provides an explanation of each field.

Table 6-18 Diagnostic 1112W field descriptions

| Dictionary symbol | Example                         |
|-------------------|---------------------------------|
| Field 1 /ERRNUM   | 1112W                           |
| Field 2 /CLUSTER  | MCO.BASE.CLUSTER                |
| Field 3 /COMPNAME | MCO.BASE.DATA                   |
| Field 4 /ADSN     | HSM.HMIG.T473100.MCO.BASE.I1132 |

Resolving the error

To view the list of available fixes, type F next to the error message and press Enter. The Fast Audit Fix Plan Member List is displayed in Figure 6-43.

|                                                           |                                             |
|-----------------------------------------------------------|---------------------------------------------|
| V6 R1.01>                                                 | Row 1 of 3                                  |
| FAST AUDIT FIX PLAN MEMBER LIST                           |                                             |
| OPTION ==>                                                | SCROLL==> PAGE                              |
| YOU MAY BROWSE (B), EDIT (E), FIX (F) OR VIEW (V) RESULTS |                                             |
| COMMAND MEMBER ACTION TITLE                               |                                             |
| F1112WA                                                   | HSM FIXCDS 0 DELETE FOR ORPHANED MCO RECORD |
| F1112WB                                                   | HSM LIST TRUE NAME IN MCDS                  |
| F1112WC                                                   | HSA DISPLAY DUMP OF MCO RECORD              |
| ***** Bottom of data *****                                |                                             |

Figure 6-43 Fix Plan Member List for error 1112W

Fixes F1112WB and F1112WC perform additional diagnostics detailing if the “O” record is present. If the results of this diagnostics is that no actual “O” record exists, a delete of the erroneous “O” record entry resolves this error.

6.1.19 FastAudit/390 audit error 1113W

The next error message we will look at is FastAudit/390 Audit Error 1113W MCO VSAM COMPONENT NAME IS MISSING IN MCA.

## Diagnosing the error

Looking up the error in the *FastAudit/390* User Guide, you find the following explanation:

*FastAudit/390* has compared the MCDS MCO extraction against the MCDS MCA extraction and has been unable to find a matching VSAM component. This means that the MCO entry is orphaned and should be deleted.

To view the error detail, type B on the FastAudit MCDS Summary panel next to the error and press Enter. The error detail is displayed in Figure 6-44.

| f1    | f2               | f3            | f4                             |
|-------|------------------|---------------|--------------------------------|
| 1113W | MCO.BASE.CLUSTER | MCO.BASE.DATA | HSM.HMIG.T47300.MCO.BASE.I1132 |

Figure 6-44 Error detail for error 1113W

The Field Description Box provided in the documentation and shown in Table 6-19 provides an explanation of each field.

Table 6-19 Diagnostic 1113W field descriptions

| Dictionary symbol | Example                         |
|-------------------|---------------------------------|
| Field 1 /ERRNUM   | 1113W                           |
| Field 2 /CLUSTER  | MCO.BASE.CLUSTER                |
| Field 3 /COMPNAME | MCO.BASE.DATA                   |
| Field 4 /ADSN     | HSM.HMIG.T473100.MCO.BASE.I1132 |

## Resolving the error

To view the list of available fixes, type F next to the error message and press Enter. The Fast Audit Fix Plan Member List is displayed in Figure 6-45.

|                                                           |         |                                                 |
|-----------------------------------------------------------|---------|-------------------------------------------------|
| <V6 R1.01>                                                |         | Row 1 of 2                                      |
| FAST AUDIT FIX PLAN MEMBER LIST                           |         |                                                 |
| OPTION ==>                                                |         | SCROLL==> PAGE                                  |
| YOU MAY BROWSE (B), EDIT (E), FIX (F) OR VIEW (V) RESULTS |         |                                                 |
| COMMAND                                                   | MEMBER  | TITLE                                           |
|                                                           | F1113WA | TSO DELETE AND RECOVER VSAM CLUSTER MISSING MCA |
|                                                           | F1113WB | HSM FIXCDS DELETE OF MCO RECORD                 |
| ***** Bottom of data *****                                |         |                                                 |

Figure 6-45 Fix Plan Member List for error 1113W

If a DFSMSHsm backup is available for the data set in error, execute fix F1113WA to delete and recover the VSAM cluster missing the MCA record. If there is no DFSMSHsm backup, fix F1113WB must be executed to delete the extraneous MCO record.

6.1.20 *FastAudit/390* audit error 1114W

The next error we will look at is *FastAudit/390* Audit Error 1114W ALIAS NAME EXISTS IN VTOC, MCA IS MISSING.

Diagnosing the error

Looking up the error in the *FastAudit/390* User Guide, you find the following explanation:

*FastAudit/390* has compared the VTOC extraction against the alias name found in the MCDS record (as opposed to the native MCA record). When *FastAudit/390* verifies the problem, an MCA record really might be found. When this occurs, the true name field is replaced (usually it is unknown). However, this is still an orphaned record because the MCD (true name in DFSMSHsm MCDS) doesn't match. This means that the entry in the VTOC is orphaned and should be deleted.

To view the error detail, type B on the FastAudit MCDS Summary panel next to the error and press Enter. The error detail is displayed in Figure 6-46.

|       |         |                                       |        |    |    |
|-------|---------|---------------------------------------|--------|----|----|
| f1    | f2      | f3                                    | f4     | f5 | f6 |
| 1114W | UNKNOWN | HSM.HMIG.T07109.SOURCE.ASSEMBLR.I3115 | HSM003 | 1  | 0  |

Figure 6-46 Error detail for error 1114W

The Field Description Box provided in the documentation and shown in Table 6-20 provides an explanation of each field.

Table 6-20 Diagnostic 1114W field descriptions.

| Dictionary symbol | Example                                |
|-------------------|----------------------------------------|
| Field 1 /ERRNUM   | 1114W                                  |
| Field 2 /TDSN     | UNKNOWN                                |
| Field 3 /ADSN     | HSM.HMIG.T070109.SOURCE.ASSEMBLR.I3115 |
| Field 4 /VOLSER   | HSM003                                 |
| Field 5 /DSORG    | 1                                      |

| Dictionary symbol | Example |
|-------------------|---------|
| Field 6 /LEVEL    | 0       |

### Resolving the error

To view the list of available fixes, type F next to the error message and press Enter. The Fast Audit Fix Plan Member List is displayed in Figure 6-47.

```
<V6 R1.01>                                     Row 1 of 3
      FAST AUDIT FIX PLAN MEMBER LIST
OPTION ==>                                     SCROLL==> PAGE
YOU MAY BROWSE (B), EDIT (E), FIX (F) OR VIEW (V) RESULTS
COMMAND  MEMBER  ACTION          TITLE
      F1114WA      HSM VERIFY MISSING MCA RECORD
      F1114WB      TSO CATALOG AND DELETE ORPHANED L1 HSM ALIAS
      F1114WR1     TSO CATALOG AND DELETE ORPHANED L1 HSM ALIAS W/O RECAT
***** Bottom of data *****
```

Figure 6-47 Fix Plan Member List for error 1114W

Errors like this are usually old data sets that were unable to be cleaned up by DFSMSHsm's normal cleanup work. You should notice that either the data set is not needed or that a newer version has been created. To correct this error simply execute fix F1114WB or F1114WR1 to delete the data sets in error.

### 6.1.21 FastAudit/390 audit error 1115W

The next error we will look at is *FastAudit/390* Audit Error 1115W VTOC ALIAS NAME HAS VTOC ERROR.

### Diagnosing the error

Looking up the error in the *FastAudit/390* User Guide, you find the following explanation:

*FastAudit/390* has compared the VTOC extraction against the MCDS MCA extraction and has detected a Format 1 DSCB error on an ML1 volume. This means that even though the data set exists, the data set probably cannot be recalled.

To view the error detail, type B on the FastAudit MCDS Summary panel next to the error and press Enter. The error detail is displayed in Figure 6-48.

| f1    | f2                      | f3                                     | f4     | f5 | f6 |
|-------|-------------------------|----------------------------------------|--------|----|----|
| 1115W | SOURCE.PLSTOB2.PROGRAMS | HSM.HMIG.T473104.SOURCE.PLSTCOB2.I3115 | HSM003 | 1  | 1  |

Figure 6-48 Error detail for error 1115W

The Field Description Box provided in the documentation and shown in Table 6-21 provides an explanation of each field.

Table 6-21 Diagnostic 1115W field descriptions

| Dictionary symbol | Example                                |
|-------------------|----------------------------------------|
| Field 1 /ERRNUM   | 1115W                                  |
| Field 2 /TDSN     | SOURCE.PLSTCOB2.PROGRAMS               |
| Field 3 /ADSN     | HSM.HMIG.T473104.SOURCE.PLSTCOB2.I3115 |
| Field 4 /VOLSER   | HSM003                                 |
| Field 5 /LEVEL    | 1                                      |
| Field 6 /VTOCERR  | 1                                      |

Resolving the error

To view the list of available fixes, type F next to the error message and press Enter. The Fast Audit Fix Plan Member List is displayed in Figure 6-49.

|                                                           |                |                                    |
|-----------------------------------------------------------|----------------|------------------------------------|
| <V6 R1.01>                                                |                | Row 1 of 2                         |
| FAST AUDIT FIX PLAN MEMBER LIST                           |                |                                    |
| OPTION ==>                                                | SCROLL==> PAGE |                                    |
| YOU MAY BROWSE (B), EDIT (E), FIX (F) OR VIEW (V) RESULTS |                |                                    |
| COMMAND                                                   | MEMBER         | ACTION TITLE                       |
|                                                           | F1115WA        | TSO DEL NVR FOR FORMAT1 DSCB ERROR |
|                                                           | F1115WB        | VTC SCRATCH ENTRY FROM VTOC        |
| ***** Bottom of data *****                                |                |                                    |

Figure 6-49 Fast Audit Fix Plan Member List for error 1115W

Here is another example where *FastAudit/390* is finding and reporting errors not found with DFSMSHsm Command Audit commands. Errors like this are usually due to data sets that are no longer needed. Fixes F1115WA and F1115WB delete or scratch the record from the VTOC or VVDS.



### 6.1.22 FastAudit/390 audit error 1116C

The next error we will look at is *FastAudit/390* Audit Error 1116C CATALOG ENTRY HAS NO MCD/MCO ENTRY.

#### Diagnosing the error

Looking up the error in the *FastAudit/390* User Guide, you find the following explanation:

*FastAudit/390* has compared the catalog extraction against the MCDS MCD extraction and has determined that even though the data set is cataloged, the MCD does not exist and the data set cannot be recalled. These entries should be deleted from the catalog. The data set could be recovered if a backup copy exists.

To view the error detail, type B on the FastAudit MCDS Summary panel next to the error and press Enter. The error detail is displayed in Figure 6-50.

|       |                    |        |                         |         |
|-------|--------------------|--------|-------------------------|---------|
| F1    | f2                 | f3     | f4                      | f5      |
| 1116C | ACFNET.CNM.AAUVSPL | MIGRAT | CATALOG.MVSICF1.VPROD06 | 2000342 |

Figure 6-50 Error detail for error 1116C

The Field Description Box provided in the documentation and shown in Table 6-22 164 provides an explanation of each field.

Table 6-22 Diagnostic 1116C field descriptions

| Dictionary symbol    | Example                 |
|----------------------|-------------------------|
| Field 1 /ERRNUM      | 1116C                   |
| Field 2 /TDSN        | ACFNET.CNM.AAUVSPL      |
| Field 3/VOLSER       | MIGRAT                  |
| Field 4/UCAT         | CATALOG.MVSICF1.VPROD06 |
| Filed 5CREATION DATE | 2000342                 |

#### Resolving the error

To view the list of available fixes, type F next to the error message and press Enter. The Fast Audit Fix Plan Member List is displayed in Figure 6-51.

```
<V6 R1.01> Row 1 of 3
FAST AUDIT FIX PLAN MEMBER LIST
OPTION ==> SCROLL==> PAGE
YOU MAY BROWSE (B), EDIT (E), FIX (F) OR VIEW (V) RESULTS
COMMAND MEMBER ACTION TITLE
F1116CA TSO DELETE NOSCRATCH FOR MIGRAT DSNS MISSING MCD
F1116CB TSO LISTC TO VERIFY CATALOG STATUS OF DATA SET
F1116CC HSM LIST COMMANDS TO VERIFY MISSING MCD
***** Bottom of data *****.
```

Figure 6-51 Fix Plan Member List for error 1116C

This is a *critical* error meaning that the data sets listed cannot be accessed. In your analysis, you will find that these data sets are usually old and obsolete. In most cases fix F1116CA corrects the error by deleting the data set. If the data set is needed you can check for available backup copies in the BCDS and issue HRECOVER to recover the data.

6.1.23 FastAudit/390 audit error 1117C

The next error we will look at is *FastAudit/390* Audit Error 1117C CATALOG ENTRY IS NOT CONNECTED TO MASTER CAT.

Diagnosing the error

Looking up the error in the *FastAudit/390* User Guide, you find the following explanation:

*FastAudit/390* has compared the catalog extraction against the MCDS MCD extraction and has determined that even though the data set is cataloged, and the MCD record exists, the high level node(s) of the data set is not connected to the master catalog and therefore the data set cannot be recalled.

To view the error detail, type B on the FastAudit MCDS Summary panel next to the error and press Enter. The error detail is displayed in Figure 6-52.

|       |              |        |           |         |
|-------|--------------|--------|-----------|---------|
| F1    | f2           | f3     | f4        | f5      |
| 1117C | XAS.VSAMMLIB | MIGRAT | UCAT.TECH | 2000198 |

Figure 6-52 Error detail for error 1117C

The Field Description Box provided in the documentation and shown in Figure 6-52 provides an explanation of each field.

Table 6-23 Diagnostic 1117C field descriptions

| Dictionary symbol     | Example      |
|-----------------------|--------------|
| Field 1 /ERRNUM       | 1117C        |
| Field 2 /TDSN         | XAS.VSAMMLIB |
| Field 3 /VOLSER       | MIGRAT       |
| Field 4 /UCAT         | UCAT.TECH    |
| Field 5 cREATION DATE | 2000198      |

### Resolving the error

To view the list of available fixes, type **F** next to the error message and press Enter. The Fast Audit Fix Plan Member List is displayed in Figure 6-53.

```
<V6 R1.01>                                     Row 1 of 2
      FAST AUDIT FIX PLAN MEMBER LIST
OPTION ==>                                     SCROLL==> PAGE
YOU MAY BROWSE (B), EDIT (E), FIX (F) OR VIEW (V) RESULTS
COMMAND MEMBER ACTION TITLE
      F1117CA      TSO DEL WITH CAT STATEMENT
      F1117CB      TSO LISTC FOR DSN MISSING CONNECTOR
***** Bottom of data *****.
```

Figure 6-53 Fast Audit Fix Plan Member List for error 1117C

This is a *critical* error meaning that the data sets listed cannot be accessed. In your analysis, you will find that these data sets are usually old or obsolete. In most cases executing fix F1117CA corrects the error by deleting the data set. If the data set is needed you can check for available DFSMSHsm backup copies in the BCDS using the **HLIST DSN('data setname') BCDS** command and issue an **HRECOVER** command to recover the data set.

### FastAudit/390 audit error 1118C

The next error we will look at is *FastAudit/390* Audit Error 1118C Migrated Data Set Is Missing In SDSP Cluster.

### Diagnosing the error

Looking up the error in the *FastAudit/390* User Guide, you find the following explanation:

*FastAudit/390* has compared the MCDS extraction against the Small data set Packing (SDSP) extraction and has determined that the data set is missing in

the SDSP VSAM cluster on the Migration Level 1 volume. The data set cannot be recalled.

To view the error detail, type **B** on the FastAudit MCDS Summary panel next to the error and press Enter. The error detail is displayed in Figure 6-54.

|       |                              |        |    |    |
|-------|------------------------------|--------|----|----|
| f1    | f2                           | f3     | f4 | f5 |
| 1118C | PO0.TOT.DMSACT.FILE.G4864V00 | ML1001 | 1  | N  |

Figure 6-54 Error detail for error 1118C

The Field Description Box provided in the documentation and shown in Table 6-24 provides an explanation of each field.

Table 6-24 Diagnostic 1118C field descriptions

| Dictionary symbol | Example                      |
|-------------------|------------------------------|
| Field 1 /ERRNUM   | 1118C                        |
| Field 2 /TDSN     | PO0.TOT.DMSACT.FILE.G4864V00 |
| Field 3 /VOLSER   | ML1001                       |
| Field 4 /LEVEL    | 1                            |
| FIELD 5 /DSORG    | N                            |

Resolving the error

To view the list of available fixes, type **F** next to the error message and press Enter. The Fast Audit Fix Plan Member List is displayed in Figure 6-55.

|                                                           |         |                                         |
|-----------------------------------------------------------|---------|-----------------------------------------|
| <V6 R1.01>                                                |         | Row 1 of 3                              |
| FAST AUDIT FIX PLAN MEMBER LIST                           |         |                                         |
| OPTION ==>                                                |         | SCROLL==> PAGE                          |
| YOU MAY BROWSE (B), EDIT (E), FIX (F) OR VIEW (V) RESULTS |         |                                         |
| COMMAND                                                   | MEMBER  | ACTION TITLE                            |
|                                                           | F1118CA | TSO DELETE AND RECOVER FOR SDSP MISSING |
|                                                           | F1118CB | TSO UNCATALOG DSN FOR SDSP MISSING      |
|                                                           | F1118CC | HSM FIXCDS D DELETE FOR SDSP MISSING    |
| ***** Bottom of data *****                                |         |                                         |

Figure 6-55 Fast Audit Fix Plan Member List for error 1118C

This is a *critical* error meaning that the data sets cannot be accessed. These data sets are usually old or obsolete, but you will need to investigate to be sure. In most cases fix F1118CB or F1118CC corrects the error by uncataloging the

data set and deleting the MCD record. If the data set is needed and it is in a SDSP (field 4) fix F1118CA deletes and recovers the data set.

6.1.24 **FastAudit/390 audit error 1119W**

The next error message we will look at is *FastAudit/390 Audit Error 1119W Entry In SDSP DATA SET IS MISSING IN MCDS*.

**Diagnosing the error**

Looking up the error in the *FastAudit User Guide*, you find the following explanation:

*FastAudit/390* has compared the small data set packing (SDSP) extraction against the MCDS extraction and has determined that a data set is in the SDSP but has no MCD entry in the MCDS. The data set cannot be recalled.

To view the error detail, type **B** on the FastAudit MCDS Summary panel next to the error and press Enter. The error detail is displayed in Figure 6-56.

|       |                       |                     |        |    |
|-------|-----------------------|---------------------|--------|----|
| f1    | f2                    | f3                  | f4     | f5 |
| 1119W | PO0.CSB.QMFCERT.INUSE | HSM.SMALLDS.VML1003 | ML1001 | 1  |

Figure 6-56 Error detail for error 1119W

The Field Description Box provided in the documentation and shown in Table 6-25 provides an explanation of each field.

Table 6-25 Diagnostic 1119W field descriptions

| Dictionary symbol | Example               |
|-------------------|-----------------------|
| Field 1 /ERRNUM   | 1119W                 |
| Field 2 /TDSN     | PO0.CSB.QMFCERT.INUSE |
| Field 3 /SDSPDSN  | HSM.SMALLDS.VML1003   |
| Field 4 /VOLSER   | ML1001                |
| Field 5 /LEVEL    | 1                     |

**Resolving the error**

To view the list of available fixes, type **F** next to the error message and press Enter. The Fast Audit Fix Plan Member List is displayed in Figure 6-57.

```
FAST AUDIT FIX PLAN MEMBER LIST
OPTION ==>
YOU MAY BROWSE (B), EDIT (E), FIX (F) OR VIEW (V) RESULTS
COMMAND MEMBER ACTION TITLE
F1119WA HSM FIXCDS D FOR TO VERIFY MISSING MCD IS REALLY SDSP
F1119WB HSM LIST TRUE NAME IN MCDS
F1119WC TSO LISTC FOR SDSP data set
F1119WD HSM AUDIT MEDCTL FOR SDSP MISSING MCD
F1119WE HSA VERIFY MCD RECORD MISSING FOR SDSP CLUSTER
F1119WF HSA DUMP MCD RECORD MISSING FOR SDSP CLUSTER
```

Figure 6-57 Fast Audit fix plan member list for error 1119W

This is another example where *FastAudit/390* provides additional diagnostic aids to help you research the extent of the error. Once your diagnostics is complete, the error can be corrected by reading the TTOC extension record and updating the MCDS to reflect what is on the tape. Fix F1119WD should correct this error by executing AUDIT MEDIACONTROLS(sdsp) VOLUMES(volser) FIX command for the SDSP on the ML1 volume.

6.1.25 FastAudit/390 audit error 1120I

The next error we will look at it *FastAudit/390* Audit Error 1120I TTOC Volume Is Empty.

Diagnosing the error

Looking up the error in the *FastAudit/390* User Guide, you find the following explanation:

*FastAudit/390* has found a tape in the TTOC inventory that has no entries. This is probably not an error but if there are a large number of these it is possible that the HSM ARCTVEXT exit for CA-1 is not working properly. This tape information is useful to interrogate CA-1 to determine the DSN assigned to it. If CA1 indicates that the data set is other than *HLQ.HMIGTAPE...*, then there may be a conflict if DFSMSshm chooses one of these tapes to use in migration.

To view the error detail, type **B** on the FastAudit MCDS Summary panel next to the error and press Enter. The error detail is displayed in Figure 6-58.

```
f1 f2 f3
1120I 010018 L2
```

Figure 6-58 Error detail for error 1120I

The Field Description Box provided in the documentation and shown in Table 6-26 provides an explanation of each field.

Table 6-26 Diagnostic 1120I field descriptions

| Dictionary symbol | Example |
|-------------------|---------|
| Field 1 /ERRNUM   | 1120I   |
| Field 2 /VOLSER   | 010018  |
| Field 3 /TTCTYP   | L2      |

### Resolving the error

To view the list of available fixes, type **F** next to the error message and press Enter. The Fast Audit Fix Plan Member List is displayed in Figure 6-59.

|                                                           |         |                                   |
|-----------------------------------------------------------|---------|-----------------------------------|
| <V6 R1.01>                                                |         | Row 1 of 4                        |
| FAST AUDIT FIX PLAN MEMBER LIST                           |         |                                   |
| OPTION ==>                                                |         | SCROLL==> PAGE                    |
| YOU MAY BROWSE (B), EDIT (E), FIX (F) OR VIEW (V) RESULTS |         |                                   |
| COMMAND                                                   | MEMBER  | ACTION TITLE                      |
|                                                           | F1120IA | HSM LIST TTOC FOR EMPTY VOLUME    |
|                                                           | F1120IB | TMS LIST EMPTY VOLSER IN TMS      |
|                                                           | F1120IC | RMM INTERROGATE RMM FOR HSM TAPES |
|                                                           | F1120ID | HSM RECYCLE EMPTY VOLUMES         |

Figure 6-59 Fix plan member list for error 1120I

*FastAudit/390* provides several different diagnostic fixes depending on the tape management system used. The diagnostic fixes further check the status of the tape and its status. Once verified, the tapes can be returned to scratch by executing FIX F1120ID which issues RECYCLE commands on the tapes.

### 6.1.26 FastAudit/390 audit error 1121W

The next error message we will look at is *FastAudit/390* Audit Error 1121W Migration Level 2 Volume Has No TTOC Entry.

### Diagnosing the error

Looking up the error in the *FastAudit/390* User Guide, you find the following explanation:

This error is similar to 1107V except that these volumes contain no migrated data. A tape is in the MCV inventory but no root record exists in the TTOC. The information in this error can be used to interrogate CA1 and the TTOC information on the tape.

To view the error detail, type **B** on the FastAudit MCDS Summary panel next to the error and press Enter. The error detail is displayed in Figure 6-60.

|       |        |    |
|-------|--------|----|
| f1    | f2     | f3 |
| 1121W | 010018 | L2 |

Figure 6-60 Error detail for error 1121W

The Field Description Box provided in the documentation and shown in Table 6-27 provides an explanation of each field.

Table 6-27 Diagnostic 1121W field descriptions

| Dictionary symbol | Example |
|-------------------|---------|
| Field 1 /ERRNUM   | 1121W   |
| Field 2 /VOLSER   | 010018  |
| Field 3 /TTCTYP   | L2      |

### Resolving the error

To view the list of available fixes, type **F** next to the error message and press Enter. The Fast Audit Fix Plan Member List is displayed in Figure 6-61.

|                                                           |         |                                                  |
|-----------------------------------------------------------|---------|--------------------------------------------------|
| <V6 R1.01>                                                |         | Row 1 of 4                                       |
| FAST AUDIT FIX PLAN MEMBER LIST                           |         |                                                  |
| OPTION ==>                                                |         | SCROLL==> PAGE                                   |
| YOU MAY BROWSE (B), EDIT (E), FIX (F) OR VIEW (V) RESULTS |         |                                                  |
| COMMAND                                                   | MEMBER  | ACTION TITLE                                     |
|                                                           | F1121WA | TMS INTERROGATE LEVEL2 TAPE MISSING TTOC         |
|                                                           | F1121WB | HSM FIXCDS COMMAND TO VERIFY MISSING TTOC RECORD |
|                                                           | F1121WC | HSM DELVOL PURGE FOR ML2 VOLUMES                 |

Figure 6-61 Fix Plan Member List for error 1121W

This type of error is indicating that DFSMSHsm thinks it owns a tape, but there is no record of the tape in the tape management database. If there is a physical tape present, fix F1121WB should be executed to verify if the TTOC record is missing. If the TTOC record is missing, an ADDVOL of the tape must be performed to create the missing TTOC. Then a RECYCLE FORCE command for the tape can be executed. If the TTOC is present just a RECYCLE FORCE command will need to be executed. If the physical tape does not exist, then fix F1121WC issues the DFSMSHsm command DELVOL PURGE to remove the tape from DFSMSHsm control.



An alternative solution is to issue an `AUDIT DATASETCONTROLS(MIGRATION)` to ensure no MCD records point to the tape. If no MCD entries are present the tape can be removed from DFSMSHsm control using the `DELVOL PURGE` command. If MCD entries are found, and this is a migration tape, check for the last good backup copy. Then delete the migrated data set and recover the data set from the backup copy. This must be done for all the MCD entries found. If this is a backup MCD record, recall the data set and issue the `DFSMSHsm HBACKDS` command to create a new backup copy. Again, this must be done for all the MCD entries found.

6.1.27 **FastAudit/390 audit error 1122W**

The next error message we will look at is *FastAudit/390 Audit Error 1122W* MCD ALIAS IS OWNED BY DIFFERENT MCD.

**Diagnosing the error**

Looking up the error in the *FastAudit/390 User Guide*, you find the following explanation:

*FastAudit/390* has found an MCD record whose alias name (the MCA) record is owned by a different MCD.

To view the error detail, type **B** on the FastAudit MCDS Summary panel next to the error and press Enter. The error detail is displayed in Figure 6-62.

|       |                          |                                        |        |    |    |
|-------|--------------------------|----------------------------------------|--------|----|----|
| f1    | f2                       | f3                                     | f4     | f5 | f6 |
| 1122W | SOURCE.PLSTCOB2.PROGRAMS | HSM.HMIG.T473104.SOURCE.PLSTCOB2.I3115 | HSM003 | 1  | 1  |
| f7    | SOURCE.PLSCOB2.MACROS    |                                        |        |    |    |

Figure 6-62 Error detail for error 1122W

The Field Description Box provided in the documentation and shown in Table 6-28 provides an explanation of each field.

Table 6-28 Diagnostic 1122W field descriptions

| Dictionary symbol | Example                                |
|-------------------|----------------------------------------|
| Field 1 /ERRNUM   | 1122W                                  |
| Field 2 /TDSN     | SOURCE.PLSTCOB2.PROGRAMS               |
| Field 3 /ADSN     | HSM.HMIG.T473104.SOURCE.PLSTCOB2.I3115 |
| Field 4 /VOLSER   | HSM003                                 |

| Dictionary symbol | Example                |
|-------------------|------------------------|
| Field 5 /Level    | 1                      |
| Field 6 /VTOCERR  | 1                      |
| Field 7 /XDSN     | SOURCE.PLSTCOB2.MACROS |

### Resolving the error

To view the list of available fixes, type **F** next to the error message and press Enter. The Fast Audit Fix Plan Member List is displayed in Figure 6-63.

|                                                           |         |                                           |
|-----------------------------------------------------------|---------|-------------------------------------------|
| <V6 R1.01>                                                |         | Row 1 of 4                                |
| FAST AUDIT FIX PLAN MEMBER LIST                           |         |                                           |
| OPTION ==>                                                |         | SCROLL==> PAGE                            |
| YOU MAY BROWSE (B), EDIT (E), FIX (F) OR VIEW (V) RESULTS |         |                                           |
| COMMAND                                                   | MEMBER  | ACTION TITLE                              |
|                                                           | F1122WA | HSM AUDIT MEDIA CONTROL STATEMENT FOR HSM |
|                                                           | F1122WB | HSM GENERATE RECALL FOR data setS         |

Figure 6-63 Fix Plan Member List for error 1122W

To diagnose this error you may first want to try and recall the data sets listed in error. If there were errors returned by recall an AUDIT MEDIACONTROLS command, fix F1122WA can fix the error.

### 6.1.28 FastAudit/390 audit error 1123W

The next error message we will look at is *FastAudit/390 Audit Error 1123W MCA Truename Record Is Mismatched*.

#### Diagnosing the error

Looking up the error in the *FastAudit/390 User Guide*, you find the following explanation:

*FastAudit/390* has found an MCA record whose true name (the MCD) exists. The alias name in the MCD record contains a different alias name. That alias name in turn exists and points to the MCD. In other words, these MCA records are orphaned and should be deleted.

To view the error detail, type **B** on the FastAudit MCDS Summary panel next to the error and press Enter. The error detail is displayed in Figure 6-64.

|       |                                  |                                  |     |    |    |
|-------|----------------------------------|----------------------------------|-----|----|----|
| f1    | f2                               | f3                               | f4  | f5 | f6 |
| 1123W | PLB.PFLOG.SRCELIB.TEST           | HSM.HMIG.T533301.PLB.PFLOG.I7324 | UNK | N  | U  |
| f7    | HSM.HMIG.T004701.PLB.PFLOG.I8070 |                                  |     |    |    |

Figure 6-64 Error detail for error 1123W

The Field Description Box provided in the documentation and shown in Table 6-29 provides an explanation of each field.

Table 6-29 Diagnostic 1123W field descriptions

| Dictionary symbol | Example                          |
|-------------------|----------------------------------|
| Field 1 /ERRNUM   | 1123W                            |
| Field 2 /TDSN     | PLB.PFLOG.SRCELIB.TEST           |
| Field 3 /ADSN     | HSM.HMIG.T533301.PLB.PFLOG.I7324 |
| Field 4 /VOLSER   | UNK                              |
| Field 5 /DSORG    | N                                |
| Field 6 /LEVEL    | U                                |
| Field 7 /XDSN     | HSM.HMIG.T004701.PLB.PFLOG.I8070 |

Resolving the error

To view the list of available fixes, type F next to the error message and press Enter. The Fast Audit Fix Plan Member List is displayed in Figure 6-65.

|                                                           |         |                                        |                |
|-----------------------------------------------------------|---------|----------------------------------------|----------------|
| <V6 R1.01>                                                |         |                                        | Row 1 of 4     |
| FAST AUDIT FIX PLAN MEMBER LIST                           |         |                                        |                |
| OPTION ==>                                                |         |                                        | SCROLL==> PAGE |
| YOU MAY BROWSE (B), EDIT (E), FIX (F) OR VIEW (V) RESULTS |         |                                        |                |
| COMMAND                                                   | MEMBER  | ACTION                                 | TITLE          |
|                                                           | F1123WA | HSA DUMP MCD AND MCA RECORDS           |                |
|                                                           | F1123WB | HSM FIXCDS DELETE CONFLICTING A RECORD |                |

Figure 6-65 Fix Plan Member List for error 1123W

Using fix F1123WA to dump the MCA and the MCD records provides details if either or both records are missing. In most cases one of these records is missing so fix F1123WB can execute the FIXCDS DELETE command to remove the extraneous record and, in turn, correct the problem.

6.1.29 FastAudit/390 audit error 1124W

The next error we will look at is *FastAudit/390* Audit Error 1124W Valid Block Count In TTOC Does Not Agree.

Diagnosing the error

Looking up the error in the *FastAudit/390* User Guide, you find the following explanation:

When the customer has requested TTOC contents verification in FAUDITAL, FAUDITMD or FAUDITBD (TOPT=YES), FastAudit/390 counts all the valid records in the TTOC extraction and compares it to the original block count from the TTOC record. It reports on tapes that are ten days old or older. It then verifies that the record count is still off and reports this error.

To view the error detail, type **B** on the FastAudit MCDS Summary panel next to the error and press Enter. The error detail is displayed in Figure 6-66.

|       |        |    |         |         |          |          |         |         |
|-------|--------|----|---------|---------|----------|----------|---------|---------|
| f1    | f2     | f3 | f4      | f5      | f6       | f7       | f8      | f9      |
| 1124W | A01135 | L2 | 1513240 | 1513228 | 00171718 | 001716FA | 2002072 | 2002044 |

Figure 6-66 Error detail for error 1124W

The Field Description Box provided in the documentation and shown in Table 6-30 provides an explanation of each field.

Table 6-30 Diagnostic 1124W field descriptions

| Dictionary symbol           | Example  |
|-----------------------------|----------|
| Field 1 /ERRNUM             | 1124W    |
| Field 2 /TVOL               | A01135   |
| Field 3 /TTCTYP             | L2       |
| Field 4 /TTCCNT             | 1513240  |
| Field 5 /TTCRES             | 1513228  |
| Field 6 /TTCXCNT            | 00171718 |
| Field 7 /TTCXRES            | 001716FA |
| Field 8 Date of TTOC update | 2002072  |
| Field 9 Date of TTOC create | 2002044  |

### Resolving the error

To view the list of available fixes, type **F** next to the error message and press Enter. The Fast Audit Fix Plan Member List is displayed in Figure 6-67.

|                                                           |         |                |                                                       |
|-----------------------------------------------------------|---------|----------------|-------------------------------------------------------|
| <V6 R1.01>                                                |         |                | Row 1 of 4                                            |
| FAST AUDIT FIX PLAN MEMBER LIST                           |         |                |                                                       |
| OPTION ==>                                                |         | SCROLL==> PAGE |                                                       |
| YOU MAY BROWSE (B), EDIT (E), FIX (F) OR VIEW (V) RESULTS |         |                |                                                       |
| COMMAND                                                   | MEMBER  | ACTION         | TITLE                                                 |
|                                                           | F1124WA |                | HSA DISPLAY DUMP OF TTOC WITH MISS MATCH VALID BLOCKS |
|                                                           | F1124WB |                | HSM FIXCDS PATCH T FOR VOL WITH MIS MATCH BLOCK COUNT |
|                                                           | F1124WC |                | HSM AUDIT DIRCTL FOR MISS MATCH BLOCK COUNT           |

Figure 6-67 Fix Plan Member List for error 1124W

You can perform additional diagnostics on the errors using fixes F1124WA. Tape errors like these can be corrected by executing fix F1124WC, executing the DFSMSHsm command AUDIT DIRECTORYCONTROLS to update the CDS.

### 6.1.30 FastAudit/390 audit error 1125W

The next error we will look at is *FastAudit/390* Audit Error 1125W TTOC Entry Marked Invalid, MCD/MCC Exists.

### Diagnosing the error

Looking up the error in the *FastAudit/390* User Guide, you find the following explanation:

*FastAudit/390* has compared the MCDS MCD extraction or BCDS MCC extraction against the extracted TTOC and has found DSNAMES in the TTOC that are marked invalid but the MCD/MCC exists. This can cause HSM to drop valid data if the tape is recycled. Special fields are captured in this error to allow the user to patch the TTOC to be valid. Field 5 is the hex representation of the DSNAME flags as they exist in the TTOC. Field 6 is the hex representation of the DSNAME flags as they would be if patched valid. Field 7 is the offset in the TTOC extension record to the DSNAME in question and Field 8 is the offset in the TTOC to the Flags byte for the DSNAME.

To view the error detail, type **B** on the FastAudit MCDS Summary panel next to the error and press Enter. The error detail is displayed in Figure 6-68.

| f1               | f2     | f3 | f4   | f5 | f6 | f7   | f8   | f9                                 |
|------------------|--------|----|------|----|----|------|------|------------------------------------|
| 1125W            | HM0222 | L2 | 0001 | 00 | 04 | 0724 | 0758 | HSM.HMIG.T060400.DDAVENP.ZIP.I9133 |
| f10              |        |    |      |    |    |      |      |                                    |
| DDAVENP.ZIP.EXEC |        |    |      |    |    |      |      |                                    |

Figure 6-68 Diagnostic 1107W field descriptions

The Field Description Box provided in the documentation and shown in Table 6-31 provides an explanation of each field.

Table 6-31 Diagnostic 1125W Field descriptions

| Dictionary symbol | Example                            |
|-------------------|------------------------------------|
| Field 1 /ERRNUM   | 1125W                              |
| Field 2 /VOLSER   | HM0222                             |
| Field 3 /TTCTYP   | L2                                 |
| Field 4 /ITTCXTN  | 0001                               |
| Field 5 /ITVFLAG  | 00                                 |
| FIELD 6 /ITPFLAG  | 04                                 |
| Field 7 /ITDOFF   | 0724                               |
| Field 8 /ITFOFF   | 0758                               |
| Field 9 /IANAME   | HSM.HMIG.T060400.DDAVENP.ZIP.I9133 |
| Field 10 /ITNAME  | DDAVENP.ZIP.EXEC                   |

Resolving the error

To view the list of available fixes, type F next to the error message and press Enter. The Fast Audit Fix Plan Member List is displayed in Figure 6-69.

```
<V6 R1.01> Row 1 of 4
FAST AUDIT FIX PLAN MEMBER LIST
OPTION ==> SCROLL==> PAGE
YOU MAY BROWSE (B), EDIT (E), FIX (F) OR VIEW (V) RESULTS
COMMAND MEMBER ACTION TITLE
F1125WA HSM FIXCDS DISPLAY TT0C RECORD EXTENTION OF TAPE
F1125WB HSM FIXCDS DISPLAY MCD RECORD
F1125WC HSM FIXCDS DISPLAY MCA RECORD
F1125WD HSM FIXCDS PATCH T MARK DSN VALID
F1125WR1 HSM HSEND LIST data set
```

Figure 6-69 Fix Plan Member List for error 1125W

You can perform further diagnostics by executing fixes F1125WA, F1125WB, and F1125WC. These diagnostic fixes provide additional information reporting the exact records in error and if so verify the existence of the errors. Fix F1125WD can be executed to re-patch the data set name as valid to the tape, correcting the error.

6.1.31 FastAudit/390 audit error 1126I

The next error we will look at is *FastAudit/390 Audit Error 1126I Summary Of TT0C Volumes With Invalid DSNS.*

Diagnosing the error

Looking up the error in the *FastAudit/390 User Guide*, you find the following explanation:

*FastAudit/390* has examined the TT0C extraction and has found volumes containing DSNames marked invalid. This is an unexpected condition since DFSMSHsm no longer maintains the valid bit in the TT0C (since DFSMSHsm 2.4). This is an informational message and allows the user to research this condition.

To view the error detail, type B on the FastAudit MCDS Summary panel next to the error and press Enter. The error detail is displayed in Figure 6-70.

```
f1 f2 f3
1126I HM0002 L2
```

Figure 6-70 Error detail for error 1126I

The Field Description Box provided in the documentation and shown in Table 6-32 provides an explanation of each field.

Table 6-32 Diagnostic 1125W field descriptions

| Dictionary symbol | Example |
|-------------------|---------|
| Field 1 /ERRNUM   | 1126I   |
| Field 2 /Volser   | HM0002  |
| Field 3 /TTCTYP   | L2      |

### Resolving the error

This is an informational message only. No action is required.

## 6.2 Backup Control Data Set audits

Auditing the DFSMSHsm BCDS is just as critical as auditing the DFSMSHsm MCDS (migration control data set) for insuring recoverability of corrupted, lost and migrated data. The BCDS contains a record of backup data set activity and aggregate backup and recovery activity. Records being audited by *FastAudit/390* are the MCB (BCDS data set record), MCC (backup version record), MCD (MCDS data set record), MCT (backup volume record), TTOC (tape table of contents), and VTOC (DCOLLECT of migration level one).

*FastAudit/390* matches the MCB records to the MCC, TTOC, and VTOC records, the MCD records to the TTOC records extracted, and verifies that all data and structures are complete and constant.

**Note:** EXPIREBV and FREEVOL BACKUP should be suspended during the actual audit process.

What you will notice is when working with *FastAudit/390*, the tool will find, report, and correct more errors than the standard DFSMSHsm command audits.

### 6.2.1 Setup for FAUDITBD

The *FastAudit/390* job that audits the BCDS is FAUDITBD found in the *hlq.userdefined.JCL* library. Some JCL user customization is required as follows.

**PPFX** Product Prefix as installed. For example, if the product has been installed with MSC.FAUDIT as a prefix to the libraries, then



|                |                                                                                                                                                                                                                                                                                                                                                                        |
|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                | PPFX=MSC.FAUDIT must be coded in the EXEC PROC statement.                                                                                                                                                                                                                                                                                                              |
| <b>PFX</b>     | Prefix represents the Runtime Prefix the user has chosen for this cataloged procedure to allocate work files that contain the audit results. Depending on the SMS treatment of new allocations, the work data sets may be deleted within a day or so. Choose a prefix that will not be scratched for a while so the results of the audit may be reviewed and repaired. |
| <b>PUNIT</b>   | Permanent unit for allocating data sets containing the audit results. The default is SYSDA. This unit should allocate data sets to a device that retains data for several days or weeks.                                                                                                                                                                               |
| <b>WRKUNIT</b> | Work Unit for allocating temporary data sets deleted at the end of the job. The default is SYSDA. If your installation has a different standard, code the different name.                                                                                                                                                                                              |
| <b>ATP</b>     | Allocation Type. Allocation is given in units such as PRI and SEC. For the larger shops the unit TRKS sometimes is not enough. If your installation is large, change the ATP to CYL.                                                                                                                                                                                   |
| <b>PRI</b>     | The number of primary allocation units. The default is 100. This value must be adjusted to accommodate the size of your DFSMSHsm MCDS and the number of records.                                                                                                                                                                                                       |
| <b>SEC</b>     | The number of secondary allocation units. The default is 200. This parameter is important to adjust for the size of your MCDS.                                                                                                                                                                                                                                         |
| <b>SOUT</b>    | SOUT is the parameter representing the SYSOUT class the user wants the system information written to.                                                                                                                                                                                                                                                                  |
| <b>LOAD</b>    | LOAD is the name of the load library as installed with <i>FastAudit/390</i> . It must be APF authorized.                                                                                                                                                                                                                                                               |

Figure 6-71 shows the JCL with the symbolic parameters used to execute procedure FAUDITBD for the BCDS audit. The product prefix is FAUDITX.QABETA and the runtime prefix is DES2.

```

000011 //FAUDITBD PROC PFX=PFX,          <==RUN TIME PREFIX
000012 //* FASTAUDIT <V6 R1.01> SEPTEMBER, 2001
000013 //          PPFX=PPFX,              <==PREFIX FOR INSTALLED PRODUCT
000014 //          PUNIT=SYSDA,             <==UNIT FOR PERMANENT ALLOCATION
000015 //          SOUT=*,                   <==PREFERRED SYSOUT CLASS
000016 //          WRKUNIT=SYSDA,           <==UNIT FOR TEMPORARY ALLOCATION
000017 //          ATP=TRK,                  <==ALLOCATION TYPE TRK/CYL
000018 //          PRI=100,                 <==PRIMARY ALLOCATION FOR FAUDIT
000019 //          SEC=200,                 <==SECONDARY SPACE ALLOCATION
000020 //          LOAD=PPFX.LOAD            <==LOAD LIBRARY AS INSTALLED
000021 //*
000022 //* THIS INSTREAM PROCESS AUDITS THE BCDS -
-
- - - - -
- -
000219 //TRACE   DD DSN=&PFX..FAUDIT.BCDS.ERRORS,DISP=(,CATLG),
000220 //          DCB=(RECFM=FB,LRECL=200,BLKSIZE=23000),
000221 //          UNIT=&PUNIT,SPACE=(&ATP,(&PRI,&SEC),RLSE)
000222 //*
000223 //          PEND
000224 //PROC1     EXEC FAUDITBD,
000225 //          LOAD=FAUDITX.QABETA.LOAD,
000226 //          PPFX=FAUDITX.QABETA,
000227 //          PFX=SC

```

Figure 6-71 FAUDITBD symbolic parameters

After the audit job completes, execute the CLIST FIXAUDIT. This CLIST displays the main panel, Fast Audit Choose Which Audit To Fix, shown in Figure 6-72, and the various audit results options available.

```

<V6 R1.01> FAST AUDIT. CHOOSE WHICH AUDIT TO FIX
COMMAND==> 2
1 MCDS   CREATED BY PROCS FAUDITMD, FAUDITCT OR FAUDITAL
2 BCDS   CREATED BY PROC FAUDITBD
3 OCDS   CREATED BY PROC FAUDITOD
4 TAPE   CREATED BY PROCS FAUDITTP, FAUDITZR OR FAUDITRM
5 CAT     CREATED BY PROC FAUDITCO
6 DASD   CREATED BY PROC FAUDITDA OR FAUDITDE
7 MASTCAT CREATED BY PROC FAUDITMC
8 CATRMM  CREATED BY PROC FAUDITCR
9 CATTMS  CREATED BY PROC FAUDITTC

ENTER  END  TO TERMINATE ONLINE FUNCTION

```

Figure 6-72 Fast Audit primary fix panel

Select option 2, BCDS by typing the number 2 on the command line and pressing Enter. The BCDS Error Summary Panel is displayed in Figure 6-73.

```

<V6 R1.01>                                     Row 1 of
2
                                FASTAUDIT BCDS SUMMARY
OPTION ==>                                     SCROLL==> PAGE
ENTER F TO DISPLAY FIXES OR B TO BROWSE ERRORS
S NUMBER   COUNT   MESSAGE
2101W      13      MCB IS CONTAMINATED
2102W      156     MCB HAS DUPLICATE MCC ENTRY
2103C        2      MCB ENTRY IS MISSING THE MCC ENTRY
2104C     1657     MCC BKUP TO LEVEL 1 HAS NO VTOC ENTRY
2105W        0      DUPLICATE BACK-UP VOLUMES FOUND
2106C      400     MCC IS ON VOLUME WHICH HAS NO MCT ENTRY
2106V        0      SUMMARY OF VOLUMES MISSING MCT ENTRY
2107W      34      MCC TAPE VOLUME HAS NO TTOC ENTRY
2107V        0      SUMMARY OF VOLUMES MISSING TTOC ENTRY
2109W        0      MCC ALIAS IS NOT ON TTOC
2109V        2      SUMMARY OF VOLUMES MCC ALIAS MISSING TTOC ENTRY
2111W        0      MCC ENTRY IS MISSING MCB ENTRY
2113I        0      BACK-UP TTOC VOLUME IS EMPTY
2114W        0      BACKUP NAME EXISTS ON VTOC, MCC IS MISSING
2115W        0      VTOC BACKUP NAME HAS VTOC ERROR
2116W        1      MCB TRUENAME IS MISMATCHED WITH MCC ENTRY
2117W        0      MCC ENTRY IS OWNED BY DIFFERENT MCB
1124W        0      VALID BLOCK COUNT IN TTOC DOES NOT AGREE
1125W        0      TTOC ENTRY MARKED INVALID, MCD/MCC EXISTS
1126I        0      SUMMARY OF TTOC VOLUMES WITH INVALID DSNS

```

Figure 6-73 BCDS error summary panel

There are subsequent panels behind each error number listed. These panels provide the list of FIXES to either diagnose or resolve the various errors. In all cases, you can view the list of available fixes by typing the letter F next to the error number and pressing Enter. The Fast Audit Fix Plan Member List for that error number is displayed and lists the available fixes for the error as shown in Figure 6-74.

```

<V6 R1.01>                                     Row 1 of 2
          FAST AUDIT FIX PLAN MEMBER LIST
OPTION ==>                                     SCROLL==> PAGE
YOU MAY BROWSE (B), EDIT (E), FIX (F) OR VIEW (V) RESULTS
COMMAND MEMBER ACTION TITLE
      F2104CA      TSO HBDELETE BACKUP VERSION MISSING VTOC
      F2104CB      HSA DUMP MCB AND MCC RECORD
***** Bottom of data *****

```

Figure 6-74 Fast Audit fix plan member list

*FastAudit/390* automatically generates the required JCL and FIX statements for each FIX listed when you type the letter F next to the FIX member name and press Enter. Figure 6-75 is an example of the JCL and FIX statements generated for fix F2104CA.

```

EDIT      CG.H093446                               Columns 00001 00072
Command ==>                                     Scroll ==> CSR
***** ***** Top of Data *****
000001 //CGA JOB (0),COLGOR,
000002 //          CLASS=A,MSGCLASS=X,NOTIFY=COLGOR
000003 //*
000004 //STEP1 EXEC PGM=IKJEFT01,
000005 // DYNAMNBR=25,REGION=4096K
000006 //* CHANGE THE FOLLOWING STATEMENT TO
000007 //* DSN=PPFX.CLIST WHERE PPFX IS THE PREFIX ASSIGNED TO
000008 //* THE FASTAUDIT PRODUCT
000009 //*
000010 //SYSPROC DD DSN=FAUDITX.QABETA.CLIST,DISP=SHR
000011 //*          ****
000012 //SYSOUT DD SYSOUT=*
000013 //SYSTSPRT DD DSN=CG.FIXAUDIT.F2104CA,DISP=SHR
000014 //SYSTSIN DD *
000015 HSEND BDELETE 'AMP2.PTF02217.XFR' +
000016 VER(001)
000017 HSEND BDELETE 'EDP4.LAB.RENAME.Q0008.G000A.Q000B.Q000C' +
000018 VER(003)
000019 HSEND BDELETE 'AMP2.QABETA.AMPASAP.BTCHSMF.ENTRIES' +

```

Figure 6-75 HSEND BDELETE statements automatically generated for fix F2104CA

You can use ISPF edit commands to edit the member and remove any records that you do not want to FIX or diagnose. Once your editing is complete or if no editing is desired, execute the FIX JCL by typing the word SUBMIT or the short version SUB on the command line in the edit entry panel and pressing Enter. A batch job is submitted to execute the FIX commands.

When the FIX batch job has completed you can view the results by typing the letter V next to the FIX member name and pressing Enter. This action results in an ISPF view of the FIX results.

The error messages are identified as follows:

- ▶ **(C)** - Critical means that a data set cannot be recovered using IBM provided recover facilities.
- ▶ **(V)** - Volume is summaries of error detail volumes.
- ▶ **(I)** - Information only.
- ▶ **(W)** - Warning means that the data set might not be recoverable and further investigation is required to determine the status of the data set.

The BCDS error messages we will cover are listed in Table 6-33.

*Table 6-33 BCDS error messages summary*

|       |                                                 |
|-------|-------------------------------------------------|
| 1124W | VALID BLOCK COUNT IN TTOC DOES NOT AGREE        |
| 2101W | MCB IS CONTAMINATED                             |
| 2102W | MCB HAS DUPLICATE MCC ENTRY                     |
| 2103C | MCB ENTRY IS MISSING THE MCC ENTRY              |
| 2104C | MCC BKUP TO LEVEL 1 HAS NO VTOC ENTRY           |
| 2106C | MCC IS ON VOLUME WHICH HAS NO MCT ENTRY         |
| 2106V | SUMMARY OF VOLUMES MISSING MCT ENTRY            |
| 2107V | SUMMARY OF VOLUMES MISSING TTOC ENTRY           |
| 2107W | MCC TAPE VOLUME HAS NO TTOC ENTRY               |
| 2109V | SUMMARY OF VOLUMES MCC ALIAS MISSING TTOC ENTRY |
| 2109W | MCC ALIAS IS NOT ON TTOC                        |
| 2111W | MCC ENTRY IS MISSING MCB ENTRY                  |
| 2113I | BACKUP TTOC VOLUME IS EMPTY                     |
| 2114W | BACKUP NAME EXISTS ON VTOC, MCC IS MISSING      |
| 2115W | VTOC BACKUP NAME HAS VTOC ERROR                 |
| 2116W | MCB TRUENAME IS MISMATCHED WITH MCC ENTRY       |
| 2117W | MCC ENTRY IS OWNED BY DIFFERENT MCB             |

## 6.2.2 *FastAudit/390* audit error 1124W

The first BCDS audit error we will look at is *FastAudit/390* audit error 1124W  
VALID BLOCK COUNT IN TTOC DOES NOT AGREE.

### Diagnosing the error

Looking up the error in the *FastAudit/390* User Guide, you find the following explanation:

When the customer has requested the block count verification in FAUDITAL, FAUDITMD or FAUDITBD (TOPT=YES), *FastAudit/390* counts all the valid records in the TTOC extraction and compares it to the original block count from the TTOC record. It reports on tapes that are ten days old or older. It then verifies that the record count is still off and reports this error.

To view the error detail, type **B** on the FastAudit BCDS Summary panel next to the error and press Enter. The Field Description Box provided in the documentation and shown in Table 6-34 provides an explanation of each field.

Table 6-34 Diagnostic 1124W field descriptions

| Field # / Description         | Example   |
|-------------------------------|-----------|
| Field 1 Error Number          | 1124W     |
| Field 2 TAPE VOLUME           | 010018    |
| Field 3 TAPE TYPE             | L2        |
| Field 4 COUNTED VALID BLOCKS  | 0000005   |
| Field 5 BLOCK COUNT (TTOC)    | 0000004-  |
| Field 6 COUNTED BLKS (IN HEX) | 00000005  |
| Field 7 BLK CNT (TTOC IN HEX) | FFFFFFFF3 |
| Field 8 Date of TTOC update   | 1997266   |
| Field 9 Date of TTOC create   | 1997252   |

### Resolving the error

To view the list of available fixes, type **F** next to the error message and press Enter. The Fast Audit Fix Plan Member List is displayed in Figure 6-76.

|                                                           |         |                |                                                   |
|-----------------------------------------------------------|---------|----------------|---------------------------------------------------|
| <V6 R1.01>                                                |         |                | Row 1 of 3                                        |
| FAST AUDIT FIX PLAN MEMBER LIST                           |         |                |                                                   |
| OPTION ==>                                                |         | SCROLL==> PAGE |                                                   |
| YOU MAY BROWSE (B), EDIT (E), FIX (F) OR VIEW (V) RESULTS |         |                |                                                   |
| COMMAND                                                   | MEMBER  | ACTION         | TITLE                                             |
|                                                           | F1124WA | HSA            | DISPLAY DUMP OF TTOC WITH MISS MATCH VALID BLOCKS |
|                                                           | F1124WB | HSM            | FIXCDS PATCH T FOR VOL WITH MISS MATCH BLK COUNT  |
|                                                           | F1124WC | HSM            | AUDIT DIRCTL FOR MISS MATCH BLOCK COUNT           |
| ***** Bottom of data *****                                |         |                |                                                   |

Figure 6-76 Fix plan member list for error 1124W

The first action you can perform is verification against the data sets reported in error. Execute fix F1124WA (HSA DISPLAY DUMP OF TTOC WITH MISS MATCH VALID BLOCKS) to display and verify that there is a mismatch between the TTOC and CDS. Once this verification is completed and you are satisfied with the report, you can execute fix F1124WC (HSM AUDIT DIRCTL FOR MISS MATCH BLOCK COUNT) against the data sets reported in error.

After the completion of this fix rerun the FAUDITBD audit and review this warning message for any data sets still in error and any new data sets found in error. If there are still data sets reported in error after the execution of fix F1124WC, you can execute fix F1124WB (HSM FIXCDS PATCH T FOR VOL WITH MISS MATCH BLOCK COUNT). If there are still errors being reported then a RECYCLE VOLUME(volser) FORCE of the volume can be performed.

### 6.2.3 FastAudit/390 audit error 2101W

The next error we will look at is *FastAudit/390* audit error 2101W MCB Is Contaminated.

#### Diagnosing the error

Looking up the error in the *FastAudit/390* User Guide, you find the following explanation:

When *FastAudit/390* was extracting the MCB records from the BCDS it found an entry that had the current number of backup copies field with a negative number. The alias of the first backup entry is presented but the remaining entries, if any, are not audited.

To view the error detail, type **B** on the FastAudit BCDS Summary panel next to the error and press Enter. The Field Description Box provided in the documentation and shown in Table 6-35 provides an explanation of each field.

Table 6-35 Diagnostic 2101W field descriptions

| Field # / Description      | Example                                |
|----------------------------|----------------------------------------|
| Field 1 Error number       | 2101W                                  |
| Field 2 dsname             | BMOHN.ISPF.ISPPROFB                    |
| Field 3 HSM Generated name | HSM.BACK.T350200.BMOHN.ISPF.I4088      |
| Field 4 DATE OF BACKUP     | 1994273                                |
| Field 5 VSAM/NONVSAM       | N                                      |
| Field 6 BACKUP VERSION     | 001 (version of this backup)           |
| Field 7 FROMVOL            | V39999 (volume of the original dsname) |

### Resolving the error

To view the list of available fixes, type **F** next to the error message and press Enter. The Fast Audit Fix Plan Member List is displayed in Figure 6-77.

```
R1.01> Row 1 of 1
      FAST AUDIT FIX PLAN MEMBER LIST
OPTION ==> SCROLL==> PAGE
YOU MAY BROWSE (B), EDIT (E), FIX (F) OR VIEW (V) RESULTS
COMMAND MEMBER ACTION TITLE
      F2101WA      HSM FIXCDS DISPLAY FOR CONTAMINATED MCB RECORD
***** Bottom of data *****
```

Figure 6-77 Fix plan member list for error 2101W

You can verify the data sets reported in the error by executing fix F2101WA (HSM FIXCDS DISPLAY FOR CONTAMINATED MCB RECORD) to display and verify that there is a mismatch in the BCDS.

There is no documented fix for this error. You can execute a **HSEND LIST DATASETNAME(h1q.data.set) BCDS** command to view the backup generations of the data set and if the version in error is not the most current backup copy, you can delete it. If the backup generation is the most current, you can recall the data set and execute the HBACKS command to create a new backup copy.

### 6.2.4 FastAudit/390 audit error 2102W

The next error message we will look at is *FastAudit/390* audit error 2102W MCB HAS DUPLICATE MCC ENTRY.



## Diagnosing the error

Looking up the error in the *FastAudit/390* User Guide, you find the following explanation:

*FastAudit/390* for the BCDS has found a DFSMSHsm generated alias in the MCB that is a duplicate for a different MCB. The DFSMSHsm recover command for these data sets might not be possible. In the error detail, this error is presented in two lines. Line 1 shows the true name record and the generated alias name. Line 2 shows the true name and the duplicate generated alias name.

To view the error detail, type **B** on the FastAudit BCDS Summary panel next to the error and press Enter. The Field Description Box provided in the documentation and shown in Table 6-36 provides an explanation of each field.

Table 6-36 Diagnostic 2102W field descriptions

| Field # / Description      | Example                           |
|----------------------------|-----------------------------------|
| Line 1                     |                                   |
| Field 1 Error number       | 2102W                             |
| Field 2 dsname             | BMOHN.ISPF.CNTL.FILE              |
| Field 3 HSM Generated name | HSM.BACK.T350200.BMOHN.ISPF.I4088 |
| Line 2                     |                                   |
| Field 1 Error number       | 2102W                             |
| Field 2 dsname             | BMOHN.ISPF.CNTL.FILE              |
| Field 3 HSM Generated name | HSM.BACK.T350200.BMOHN.ISPF.I4088 |
| Field 4 DATE OF BACKUP     | 1994273                           |
| Field 5 VSAM/NONVSAM       | N                                 |

## Resolving the error

To view the list of available fixes, type **F** next to the error message and press Enter. The Fast Audit Fix Plan Member List is displayed in Figure 6-78.

```
<V6 R1.01>                                     Row 1 of 1
      FAST AUDIT FIX PLAN MEMBER LIST
OPTION ==>                                     SCROLL==> PAGE
YOU MAY BROWSE (B), EDIT (E), FIX (F) OR VIEW (V) RESULTS
COMMAND MEMBER ACTION TITLE
      F2102WA      HSM FIXCDS DISPLAY FOR DUPLICATE MCC ENTRIES
***** Bottom of data *****
```

Figure 6-78 Fix plan member list for error 2102W

You can verify the data sets reported in error by executing fix F2102WA (HSM FIXCDS DISPLAY FOR DUPLICATE MCC ENTRIES). This fix displays and verifies that there is a duplicate record in the BCDS.

There is no documented fix for this error. You can execute a **HSEND LIST DATASETNAME(h1q.data.set) BCDS** command to view the backup generations of the data set and if the version in error is not the most current backup copy, you can delete it. If the backup generation is the most current, you can recall the data set and execute the HBACKS command to create a new backup copy.

6.2.5 FastAudit/390 audit error 2103C

The next error we will look at is *FastAudit/390* audit error 2103C MCB Is Missing The MCC Entry.

Diagnosing the error

Looking up the error in the *FastAudit/390* User Guide, you find the following explanation:

*FastAudit/390* failed to find the corresponding MCC (HSM generated alias name) in the BCDS. Information to recover the data set is contained in the MCC. The DSN cannot be recovered.

To view the error detail, type **B** on the FastAudit BCDS Summary panel next to the error and press Enter. The Field Description Box provided in the documentation and shown in Table 6-37 provides an explanation of each field.

Table 6-37 Diagnostic 2103W field descriptions

| Field # / Description      | Example                               |
|----------------------------|---------------------------------------|
| Field 1 Error number       | 2103C                                 |
| Field 2 dsname             | APPL1.CRITICAL.FILE                   |
| Field 3 HSM Generated name | HSM.BACK.T350200.APPL1.CRITICAL.I4088 |

| Field # / Description  | Example |
|------------------------|---------|
| Field 4 DATE OF BACKUP | 1994273 |
| Field 5 VSAM/NONVSAM   | N       |
| Field 6 BACKUP VERSION | 001     |

### Resolving the error

To view the list of available fixes, type **F** next to the error message and press Enter. The Fast Audit Fix Plan Member List is displayed in Figure 6-79.

```
FAST AUDIT FIX PLAN MEMBER LIST
OPTION ==>                                SCROLL==> PAGE
YOU MAY BROWSE (B), EDIT (E), FIX (F) OR VIEW (V) RESULTS
COMMAND MEMBER ACTION TITLE
      F2103CA      HSM FIXCDS DISPLAY FOR MCB MISSING MCC ENTRY
      F2103CB      TSO BDELETE BACKUP
      F2103CC      HSM LIST COMMANDS TO VERIFY MISSING MCC
      F2103CR1     HSM HBDELETES FOR MCB MISSING MCC ENTRY W/VER #
      F2103CR2     HSM FIXCDS DISPLAY FOR MCB MISSING MCC ENTRY
      F2103CR3     TSO LIST CAT FOR MCB MISSING MCC ENTRY
      F2103CR4     HSM RECALL FOR MCB MISSING MCC ENTRY
      F2103CR5     HSM BACKUP FOR MCB MISSING MCC ENTRY
      F2103CR6     HSA VERIFY MISSING MCC RECORDS
      F2103CR7     HSA DUMP MCB MISSING MCC RECORDS
***** Bottom of data *****
```

Figure 6-79 Fix plan member list for error 1106V

You can verify the data sets reported to be in error by executing fix F2103CA (HSM FIXCDS DISPLAY FOR MCB MISSING MCC ENTRY) to display and verify that there is a missing MCC record in the BCDS.

Other verification processes are provided in these fixes:

- ▶ F2103CC (HSM LIST COMMANDS TO VERIFY MISSING MCC)
- ▶ F2103CR2 (HSM FIXCDS DISPLAY FOR MCB MISSING MCC ENTRY)
- ▶ F2103CR3 (TSO LIST CAT FOR MCB MISSING MCC ENTRY)
- ▶ F2103CR6 (HSA VERIFY MISSING MCC RECORDS)
- ▶ F2103CR7 (HSA DUMP MCB MISSING MCC RECORDS)

Once this verification is completed and you are satisfied with the results you can try any of the following fixes:

- ▶ F2103CB (TSO BDELETE BACKUP)

- ▶ F2103CR1 (HSM HBDELETES FOR MCB MISSING MCC ENTRY W/VER #)
- ▶ F2103CR4 (HSM RECALL FOR MCB MISSING MCC ENTRY)
- ▶ F2103CR5 (HSM BACKUP FOR MCB MISSING MCC ENTRY)

Rerun the audit, FAUDITBD before continuing with the additional fixes.

If, after rerunning the audit, data sets are still showing up as an error in this category, then two additional corrective actions in the form of DFSMSHsm commands can be used. Use one of the following:

- ▶ **AUDIT DATASETCONTROLS(MIGRATION)**
- ▶ **AUDIT DATASETCONTROLS(BACKUP)**

Rerun the audit, FAUDITBD, to ensure that the errors have been resolved.

## 6.2.6 *FastAudit/390* audit error 2104C

The next error we will look at is *FastAudit/390* audit error 2104C MCC Backup To Level 1 Has No VTOC Entry.

### Diagnosing the error

Looking up the error in the *FastAudit/390* User Guide, you find the following explanation:

A data set that has been backed up to an DFSMSHsm Migration Level 1 volume has no VTOC entry for the DFSMSHsm generated alias name. This data set cannot be recovered.

To view the error detail, type **B** on the FastAudit BCDS Summary panel next to the error and press Enter. The Field Description Box provided in the documentation and shown in Table 6-38 provides an explanation of each field.

*Table 6-38 Diagnostic 2104C field descriptions*

| Field # / Description      | Example                               |
|----------------------------|---------------------------------------|
| Field 1 Error number       | 2104C                                 |
| Field 2 dsname             | APPL1.CRITICAL.FILE                   |
| Field 3 HSM Generated name | HSM.BACK.T350200.APPL1.CRITICAL.I4088 |
| Field 4 Volume Serial      | MIGR01                                |
| Field 5 DATE OF BACKUP     | 1994273                               |
| Field 6 VSAM/NONVSAM       | N                                     |

| Field # / Description       | Example |
|-----------------------------|---------|
| Field 7 TAPE or DASD        | D       |
| Field 8 BACKUP VERSION      | 001     |
| Field 9 BACKUP GENERATION   | 03      |
| Field 10 Type of Bkup Tape  | SP™     |
| Field 11 TTOC EXTENSION NUM | 0002    |
| Field 12 Primary volume     | STR002  |

### Resolving the error

To view the list of available fixes, type **F** next to the error message and press Enter. The Fast Audit Fix Plan Member List is displayed in Figure 6-80.

|                                                           |         |                                          |
|-----------------------------------------------------------|---------|------------------------------------------|
| <V6 R1.01>                                                |         | Row 1 of 2                               |
| FAST AUDIT FIX PLAN MEMBER LIST                           |         |                                          |
| OPTION ==>                                                |         | SCROLL==> PAGE                           |
| YOU MAY BROWSE (B), EDIT (E), FIX (F) OR VIEW (V) RESULTS |         |                                          |
| COMMAND                                                   | MEMBER  | ACTION TITLE                             |
|                                                           | F2104CA | TSO HBDELETE BACKUP VERSION MISSING VTOC |
|                                                           | F2104CB | HSA DUMP MCB AND MCC RECORD              |
| ***** Bottom of data *****                                |         |                                          |

Figure 6-80 Fix plan member list for error 2104C

You can verify the errors by executing fix F2104CB (HSA DUMP MCB AND MCC RECORD) to dump and display the MCB and MCC records.

Once the results are verified you can execute fix F2104CA (TSO HBDELETE BACKUP VERSION MISSING VTOC) to delete the backup version being reported in error. Rerun the audit, FAUDITBD, to verify that the error has been resolved.

If the error has not been resolved, you can issue the DFSMSHsm command **HSEND LIST DATASETNAME(hlq.data.set) BCDS**. Check the version of the data set and decide if the backup version can be deleted. If the backup version is current, you can delete the backup version, recall, and issue the DFSMSHsm **HBACKDS** command to backup the data set.

### 6.2.7 FastAudit/390 audit error 2106C

The next error we will look at is *FastAudit/390* audit error 2106C MCC Is On Volume Which Has No MCT Entry.

## Diagnosing the error

Looking up the error in the *FastAudit/390* User Guide, you find the following explanation:

A data set that has been backed up to an HSM backup volume, which has no BCDS MCT record. The volume must exist in the MCT before the volume can be mounted for recover.

To view the error detail, type **B** on the FastAudit BCDS Summary panel next to the error and press Enter. The Field Description Box provided in the documentation and shown in Table 6-39 provides an explanation of each field.

Table 6-39 Fix plan member list for error 1106V

| Field # / Description       | Example                               |
|-----------------------------|---------------------------------------|
| Field 1 Error number        | 2106C                                 |
| Field 2 dsname              | APPL1.CRITICAL.FILE                   |
| Field 3 HSM Generated name  | HSM.BACK.T350200.APPL1.CRITICAL.I4088 |
| Field 4 Volume Serial       | 025346                                |
| Field 5 DATE OF BACKUP      | 1994273                               |
| Field 6 VSAM/NONVSAM        | N                                     |
| Field 7 TAPE or DASD        | T                                     |
| Field 8 BACKUP VERSION      | 001                                   |
| Field 9 BACKUP GENERATION   | 03                                    |
| Field 10 Type of Bkup Tape  | SP                                    |
| Field 11 TTOC EXTENSION NUM | 0002                                  |
| Field 12 Primary volume     | STR002                                |

## Resolving the error

To view the list of available fixes, type **F** next to the error message and press Enter. The Fast Audit Fix Plan Member List is displayed in Figure 6-81.

```

<V6 R1.01>                                     Row 1 of 6
      FAST AUDIT FIX PLAN MEMBER LIST
OPTION ==>                                     SCROLL==> PAGE
YOU MAY BROWSE (B), EDIT (E), FIX (F) OR VIEW (V) RESULTS
COMMAND  MEMBER  ACTION  TITLE
      F2106CA      TSO BDELETE BACK-UP VERSION MISSING MCT
      F2106CB      HSM FIXCDS DISPLAY X FOR MISSING MCT
      F2106CC      HSA VERIFY MCT RECORD MISSING
      F2106CD      TSO BDELETE BACKUP VER USING VER AND FROMVOL
      F2106CR1     HSM RECALL MIGRATED DATASET
      F2106CR2     HSM BACKUP DATASET
***** Bottom of data *****

```

Figure 6-81 Fix plan member list for error 2106C

You can verify the missing MCT records by executing fix F2106CB (HSM FIXCDS DISPLAY X FOR MISSING MCT) or F2106CC (HSA VERIFY MCT RECORD MISSING).

Once satisfied with the verification you can delete the backup version in error. Execution of fix F2106CA (TSO BDELETE BACK-UP VERSION MISSING MCT) or F2106CD (TSO BDELETE BACKUP VER USING VER AND FROMVOL) deletes the backup version with the bad MCT record.

Once deleted you can create a new backup version of the data set by executing fixes F2106CR1 (HSM RECALL MIGRATED DATASET) to recall the data set and F2106CR2 (HSM BACKUP DATASET) to create a new backup copy.

## 6.2.8 FastAudit/390 audit error 2106V

The next error we will look at is *FastAudit/390* audit error 2106V Summary of Volumes Missing MCT Entry.

### Diagnosing the error

Looking up the error in the *FastAudit/390* User Guide, you find the following explanation:

This error is a summary by volumes of error 2106C. Research of this problem is easier to do at a volume level. Supporting fixes and interrogations at a volume level is supplied for this error.

To view the error detail, type **B** on the FastAudit BCDS Summary panel next to the error and press Enter. The Field Description Box provided in the documentation and shown in Table 6-40 provides an explanation of each field.

Table 6-40 Diagnostic 2106V field descriptions

| Field # / Description | Example                                  |
|-----------------------|------------------------------------------|
| Field 1 Error number  | 2106V                                    |
| Field 2 VOLSER        | HB0021                                   |
| Field 3 Tape Type     | SP (SP-spill, 01-31 cycle tapes)         |
| Field 4 DSN Number    | 000009 (number of dsns referencing tape) |

### Resolving the error

To view the list of available fixes, type **F** next to the error message and press Enter. The Fast Audit Fix Plan Member List is displayed in Figure 6-82.

```
<V6 R1.01>                                     Row 1 of 4
      FAST AUDIT FIX PLAN MEMBER LIST
OPTION ==>                                     SCROLL==> PAGE
YOU MAY BROWSE (B), EDIT (E), FIX (F) OR VIEW (V) RESULTS
COMMAND MEMBER ACTION TITLE
      F2106VA      TMS INTERROGATE TMS FOR HSM TAPES MISSING MCT
      F2106VB      HSM FIXCDS DISPLAY X FOR MISSING MCT
      F2106VC      HSM FIXCDS TO ADD NEW MCT RECORD
      F2106VD      RMM INTERROGATE RMM FOR BACK-UP MISSING MCT
***** Bottom of data *****
```

Figure 6-82 Fix plan member list for error 2106V

You can verify the missing MCT records by executing fix F2106VB (HSM FIXCDS DISPLAY X FOR MISSING MCT), then either fix F2106VA (TMS INTERROGATE TMS FOR HSM TAPES MISSING MCT) for a TMS environment or F2106VD (RMM INTERROGATE RMM FOR BACK-UP MISSING MCT) for a RMM environment.

The fix F2106VC (HSM FIXCDS TO ADD NEW MCT RECORD) can be used to add a MCD record by issuing a patch command associated to a tape.

### 6.2.9 FastAudit/390 audit errors 2107W and 2107V

The next errors we will look at are *FastAudit/390* audit errors 2107W and 2107V Summary Of Volumes Missing TT0C Entry.

### Diagnosing the error

Looking up the error in the *FastAudit/3900* User Guide, you find the following explanation:



2107W: A data set has been backed up to an HSM tape backup volume, which is not currently in the OCDSTTOC volume inventory. The data set on this volume might be able to be recovered, however, it is possible that this volume has been returned to the system scratch pool and has been written over.

2107V: This error is a summary by volumes of error 2107W. Research of this problem is easier to do at a volume level. Supporting fixes and interrogations at a volume level is supplied for this error.

To view the error detail, type **B** on the FastAudit BCDS Summary panel next to the error and press Enter. The Field Description Box provided in the documentation and shown in Table 6-41 and Table 6-41 provides an explanation of each field.

Table 6-41 Diagnostic Diagnostic 2107W field descriptions

| Field # / Description       | Example                               |
|-----------------------------|---------------------------------------|
| Field 1 Error number        | 2107W                                 |
| Field 2 dsname              | APPL1.CRITICAL.FILE                   |
| Field 3 HSM Generated name  | HSM.BACK.T350200.APPL1.CRITICAL.I4088 |
| Field 4 Volume Serial       | 048751                                |
| Field 5 DATE OF BACKUP      | 1994273                               |
| Field 6 VSAM/NONVSAM        | N                                     |
| Field 7 TAPE or DASD        | T                                     |
| Field 8 BACKUP VERSION      | 001                                   |
| Field 9 BACKUP GENERATION   | 03                                    |
| Field 10 Type of Bkup Tape  | SP                                    |
| Field 11 TTOC EXTENSION NUM | 0002                                  |
| Field 12 Primary volume     | STR002                                |

Table 6-42 Diagnostic 2107V field descriptions

| Field # / Description | Example |
|-----------------------|---------|
| Field 1 Error number  | 2107V   |
| Field 2 VOLSER        | HB0022  |

| Field # / Description | Example                                  |
|-----------------------|------------------------------------------|
| Field 3 Tape Type     | SP (SP-spill, 01-31 cycle tapes)         |
| Field 4 DSN Number    | 000003 (number of dsns referencing tape) |

### Resolving the error

To view the list of available fixes, type **F** next to the error message and press Enter. The Fast Audit Fix Plan Member List is displayed in Figure 6-83.

```
<V6 R1.01>                                     Row 1 of 3
      FAST AUDIT FIX PLAN MEMBER LIST
OPTION ==>                                     SCROLL==> PAGE
YOU MAY BROWSE (B), EDIT (E), FIX (F) OR VIEW (V) RESULTS
COMMAND  MEMBER  ACTION  TITLE
F2107VA   HSM LIST TTOC FOR BACK-UP DSN MISSING TTOC
F2107VB   TMS DISPLAY TMS STATUS FOR MISSING TTOC VOL
F2107VC   RMM INTERROGATE RMM FOR HSM TAPES
***** Bottom of data *****
```

Figure 6-83 Fix plan member list for error 2107V

Errors 2107W and 2107V are the same error. Error 2107W reports at a data set level and error 2107V reports data sets on a volume level. It is recommended that you use the diagnostics and fixes from 2107V as the duplicate VOLSERS are removed.

To verify the error, execute fix F2107VC (RMM INTERROGATE RMM FOR HSM TAPES) if this was a RMM environment to perform additional interrogations on the reported records reported in error.

CA-1 users can execute fix F2107VB (CA1 DISPLAY TMS STATUS FOR MISSING TTOC VOL) should be used to perform additional interrogation against the tape volumes.

Fix F2107VA (HSM LIST TTOC FOR BACK-UP DSN MISSING TTOC) performs a LIST TTOC command and spools to an output dataset. From your diagnostics, you can decide to force a recycle of the tape or delete the records in error.

### 6.2.10 FastAudit/390 audit errors 2109W and 2109V

The next errors we will look at are the *FastAudit/390* audit errors 2109W and 2109V Summary Of Volumes MCC Alias Missing TTOC Entry.

# Diagnosing the error

Looking up the error in the *FastAudit/390* User Guide, you find the following explanation:

2109W: A data set has been backed up to an HSM tape backup volume. The HSM generated alias name is not in the alias name inventory in the TTOC for that volume. This can happen if HSM has abnormally terminated or the system has been IPLed before the TTOC information has been updated. The data set can probably be recovered but when the tape is recycled an error may occur. Also, if tape recycle was running, this condition might not be true. For best results, stop recycle while running the audit.

2109V: This error is a summary by volumes of error 2109W. Research of this problem is easier to do at a volume level. Supporting fixes and interrogations at a volume level is supplied for this error.

**Note:** *FastAudit/390* does *not* verify these errors in real-time. If recycle was running while the audit was extracting information these errors may be erroneous. For best results make sure recycle is not running at the same time as *FastAudit/390*.

To view the error detail, type **B** on the FastAudit BCDS Summary panel next to the error and press Enter. The Field Description Box provided in the documentation and shown in Table 6-43 and Table 6-44 provides an explanation of each field.

Table 6-43 Diagnostic 2109W field descriptions

| Field # / Description      | Example                               |
|----------------------------|---------------------------------------|
| Field 1 Error number       | 2109W                                 |
| Field 2 dsname             | APPL1.CRITICAL.FILE                   |
| Field 3 HSM Generated name | HSM.BACK.T350200.APPL1.CRITICAL.I4088 |
| Field 4 Volume Serial      | 048751                                |
| Field 5 DATE OF BACKUP     | 1994273                               |
| Field 6 VSAM/NONVSAM       | N                                     |
| Field 7 TAPE or DASD       | T                                     |
| Field 8 BACKUP VERSION     | 001                                   |
| Field 9 BACKUP GENERATION  | 03                                    |
| Field 10 Type of Bkup Tape | SP                                    |

| Field # / Description       | Example |
|-----------------------------|---------|
| Field 11 TTOC EXTENSION NUM | 0002    |
| Field 12 Primary volume     | STR002  |

Table 6-44 Diagnostic 2109V field descriptions

| Field # / Description | Example                                  |
|-----------------------|------------------------------------------|
| Field 1 Error number  | 2109V                                    |
| Field 2 VOLSER        | HB0023                                   |
| Field 3 Tape Type     | SP (SP-spill, 01-31 cycle tapes)         |
| Field 4 DSN Number    | 000003 (number of dsns referencing tape) |

### Resolving the error

To view the list of available fixes, type **F** next to the error message and press Enter. The Fast Audit Fix Plan Member List is displayed in Figure 6-84.

```

<V6 R1.01>                                     Row 1 of 3
      FAST AUDIT FIX PLAN MEMBER LIST
OPTION ==>                                     SCROLL==> PAGE
YOU MAY BROWSE (B), EDIT (E), FIX (F) OR VIEW (V) RESULTS
COMMAND MEMBER ACTION TITLE
      F2109VA      HSM LIST TTOC VOLUME FOR MCC MISSING TTOC ENTRY
      F2109VB      HSM LIST TTOC FAILEDCREATE
      F2109VC      HSM AUDIT MEDIA CONTROL FOR MCC MISSING IN TTOC
***** Bottom of data *****

```

Figure 6-84 Fix plan member list for error 2109V

Errors 2109W and 2109V are the same error. Error 2109W reports at a dataset level and error 2109V reports data sets on a volume level. It is recommended that you use the diagnostics and fixes from 2109V as the duplicate VOLSERs are removed.

To verify the missing MCC record, execute fix F2109VA (HSM LIST TTOC VOLUME FOR MCC MISSING TTOC ENTRY) or fix F2109VB (HSM LIST TTOC FAILEDCREATE).

Once this is done and the records are determined in error you can execute fix F2109VC (HSM AUDIT MEDIA CONTROL FOR MCC MISSING IN TTOC) to

execute an AUDIT MEDIACONTROLS VOLUME against the selected tape volumes.

### 6.2.11 *FastAudit/390* audit error 2111W

The next error we will look at is *FastAudit/390* audit error 2111W MCC Entry Is Missing MCB Entry.

Looking up the error in the *FastAudit/390* User Guide, you find the following explanation:

*FastAudit/390* has found an orphaned MCC entry. Either the MCB is missing altogether or the referenced MCC is not contained in the array of MCCs within the MCB record. The MCC is a record in the BCDS whose key is the HSM generated alias name of the backup. The MCC is required for a successful RECOVER of a data set. It contains all the information about what tape it is on, its size, etc. In the MCC is a reference back to the true data set name or the MCB. The MCB is a record in the BCDS whose key is the true name of the data set that was backed up. It contains an array of DFSMSHsm generated alias names which are the keys of the MCC record. The recycle of the tape volumes containing this data set might fail. If it is successful, it will recycle data that is not recoverable. Since the true name does not exist, a RECOVER for this backup version will fail because DFSMSHsm cannot find the true name.

To view the error detail, type **B** on the *FastAudit* BCDS Summary panel next to the error and press Enter. The Field Description Box provided in the documentation and shown in Table 6-45 provides an explanation of each field.

Table 6-45 *Diagnostic 2111W field descriptions*

| Field # / Description      | Example                               |
|----------------------------|---------------------------------------|
| Field 1 Error number       | 2111W                                 |
| Field 2 dsname             | APPL1.CRITICAL.FILE                   |
| Field 3 HSM Generated name | HSM.BACK.T350200.APPL1.CRITICAL.I4088 |
| Field 4 Volume Serial      | 048751                                |
| Field 5 DATE OF BACKUP     | 1994273                               |
| Field 6 VSAM/NONVSAM       | N                                     |
| Field 7 TAPE or DASD       | T                                     |
| Field 8 BACKUP VERSION     | 007                                   |
| Field 9 BACKUP GENERATION  | 03                                    |

| Field # / Description       | Example |
|-----------------------------|---------|
| Field 10 Type of Bkup Tape  | SP      |
| Field 11 TTOC EXTENSION NUM | 0002    |
| Field 12 Primary volume     | STR002  |

### Resolving the error

To view the list of available fixes, type **F** next to the error message and press Enter. The Fast Audit Fix Plan Member List is displayed in Figure 6-85.

```
<V6 R1.01> Row 1 of 2
FAST AUDIT FIX PLAN MEMBER LIST
OPTION ==> SCROLL==> PAGE
YOU MAY BROWSE (B), EDIT (E), FIX (F) OR VIEW (V) RESULTS
COMMAND MEMBER ACTION TITLE
F2111WA HSM FIXCDS DISPLAY B FOR MCC MISSING MCB
F2111WB HSM FIXCDS DELETE C FOR MCC MISSING MCB
***** Bottom of data *****
```

Figure 6-85 Fix plan member list for error 2111W

To verify the error, execute fix F2111WA (HSM FIXCDS DISPLAY B FOR MCC MISSING MCB) to analyze the “B” record (of the MCB) to the MCC record.

Once satisfied with the diagnostics, a FIXCDS DELETE of the “C” record in the MCB can be done by executing fix F2111WB (HSM FIXCDS DELETE C FOR MCC MISSING MCB).

### 6.2.12 FastAudit/390 audit error 2113I

The next error message we will look at is *FastAudit/390* audit error 2113I Backup TTOC Volume is Empty.

Looking up the error in the *FastAudit/390* User Guide, you find the following explanation:

*FastAudit/390* has found a volume entry in the TTOC that has no backup data sets. This is for information only. If there are a large number of these, they may need to be released for scratch.

To view the error detail, type **B** on the FastAudit BCDS Summary panel next to the error and press Enter. The Field Description Box provided in the documentation and shown in Table 6-46 provides an explanation of each field.

Table 6-46 Diagnostic 2113I field descriptions

| Field # / Description | Example |
|-----------------------|---------|
| Field 1 Error number  | 2113I   |
| Field 2 VOLSER        | HB0003  |
| Field 3 TAPE TYPE     | SP      |

### Resolving the error

To view the list of available fixes, type **F** next to the error message and press Enter. The Fast Audit Fix Plan Member List is displayed in Figure 6-86.

```
<V6 R1.01> Row 1 of 2
      FAST AUDIT FIX PLAN MEMBER LIST
OPTION ==> SCROLL==> PAGE
YOU MAY BROWSE (B), EDIT (E), FIX (F) OR VIEW (V) RESULTS
COMMAND MEMBER ACTION TITLE
      F2111WA      HSM FIXCDS DISPLAY B FOR MCC MISSING MCB
      F2111WB      HSM FIXCDS DELETE C FOR MCC MISSING MCB
***** Bottom of data *****
```

Figure 6-86 Fix plan member list for error 2111W

To verify the error, execute fix F2113IA (HSM LIST TTOC CONTENTS FOR EMPTY VOLUME). These same volumes are listed in the 2107V and the 2109V errors and are corrected with those corrective actions. Rerun the FAUDITBD after executing the corrective actions to verify that these errors have been resolved.

### 6.2.13 FastAudit/390 audit error 2114W

The next error we will look at is *FastAudit/390* audit error 2114W Backup Name Exists On VTOC, MCC Is Missing.

#### Diagnosing the error

Looking up the error in the *FastAudit/390* User Guide, you find the following explanation:

*FastAudit/390* has found an DFSMSHsm generated alias name on an ML1 volume VTOC for a backup version that has no corresponding entry in the BCDS MCC inventory. The alias name needs to be deleted from the VTOC.

To view the error detail, type **B** on the FastAudit BCDS Summary panel next to the error and press Enter. The Field Description Box provided in the documentation and shown in Table 6-47 provides an explanation of each field.

Table 6-47 Diagnostic 2114W field descriptions

| Field # / Description      | Example                               |
|----------------------------|---------------------------------------|
| Field 1 Error number       | 2114W                                 |
| Field 2 dsname             | UNKNOWN (can not resolve true name)   |
| Field 3 HSM Generated name | HSM.BACK.T350200.APPL1.CRITICAL.I4088 |
| Field 4 Volume Serial      | MIGRL1                                |
| Field 5 DATE OF BACKUP     | 1994273                               |
| Field 6 VSAM/NONVSAM       | N                                     |
| Field 7 TAPE or DASD       | D                                     |
| Field 8 VTOC ERR           | 0 (0 INDICATES NO ERROR)              |

**Resolving the error**

To view the list of available fixes, type **F** next to the error message and press Enter. The Fast Audit Fix Plan Member List is displayed in Figure 6-87.

```
<V6 R1.01>                                     Row 1 of 1
      FAST AUDIT FIX PLAN MEMBER LIST
OPTION ===>                                     SCROLL==> PAGE
  YOU MAY BROWSE (B), EDIT (E), FIX (F) OR VIEW (V) RESULTS
COMMAND MEMBER ACTION          TITLE
      F2114WA      TSO CATALOG AND DELETE ORPHANED L1 MCC ALIAS
***** Bottom of data *****
```

Figure 6-87 Fix plan member list for error 2114W

To resolve the error, execute fix F2114WA (TSO CATALOG AND DELETE ORPHANED L1 MCC ALIAS). This fix defines the data set as non-VSAM and data set deletes the alias entry from the VTOC.

**6.2.14 FastAudit/390 audit error 2115W**

The next error we will look at is *FastAudit/390* audit error 2115W VTOC Backup Name Has VTOC Error.



### Diagnosing the error

Looking up the error in the *FastAudit/390* User Guide, you find the following explanation:

*FastAudit/390* has found an HSM generated alias name for a backup version. There is an error in the VTOC. The alias name needs to be deleted from the VTOC.

To view the error detail, type **B** on the FastAudit BCDS Summary panel next to the error and press Enter. The Field Description Box provided in the documentation and shown in Table 6-48 provides an explanation of each field.

Table 6-48 Diagnostic 2115W field descriptions

| Field # / Description      | Example                               |
|----------------------------|---------------------------------------|
| Field 1 Error number       | 2115W                                 |
| Field 2 dsname             | APPL1.CRITICAL.FILE                   |
| Field 3 HSM Generated name | HSM.BACK.T350200.APPL1.CRITICAL.I4088 |
| Field 4 Volume Serial      | MIGRL1                                |
| Field 5 DATE OF BACKUP     | 1994273                               |
| Field 6 VSAM/NONVSAM       | N                                     |
| Field 7 TAPE or DASD       | D                                     |
| Field 8 VTOC ERR           | 5 (NON ZERO INDICATES VTOC ERROR)     |

### Resolving the error

To view the list of available fixes, type **F** next to the error message and press Enter. The Fast Audit Fix Plan Member List is displayed in Figure 6-88.

|                                                           |         |                                               |
|-----------------------------------------------------------|---------|-----------------------------------------------|
| <V6 R1.01>                                                |         | Row 1 of 2                                    |
| FAST AUDIT FIX PLAN MEMBER LIST                           |         |                                               |
| OPTION ==>                                                |         | SCROLL==> PAGE                                |
| YOU MAY BROWSE (B), EDIT (E), FIX (F) OR VIEW (V) RESULTS |         |                                               |
| COMMAND                                                   | MEMBER  | ACTION TITLE                                  |
|                                                           | F2115WA | TSO DELETE BACK-UP DSN ON ML1 WITH VTOC ERROR |
|                                                           | F2115WB | TSO DELETE BACK-UP VER AND FROM VOL           |
| ***** Bottom of data *****                                |         |                                               |

Figure 6-88 Fix plan member list for error 2115W

To resolve the error, execute fix F2114WA (TSO CATALOG AND DELETE ORPHANED L1 MCC ALIAS). This fix defines the data sets in error as non-VSAM and deletes the alias from the VTOC.

6.2.15 *FastAudit/390* audit error 2116W

The next error message we will look at is *FastAudit/390* audit error 2116W MCB TRUENAME IS MISMATCHED WITH MCC.

**Diagnosing the error**

Looking up the error in the *FastAudit/390* User Guide, you find the following explanation:

*FastAudit/390* has found a backup true name (MCB) whose generated alias name is mismatched. The MCC record indicates that a different MCB true name owns it.

To view the error detail, type **B** on the FastAudit BCDS Summary panel next to the error and press Enter. The Field Description Box provided in the documentation and shown in Table 6-49 provides an explanation of each field.

Table 6-49 *Diagnostic 2116W field descriptions*

| Field # / Description      | Example                                 |
|----------------------------|-----------------------------------------|
| Field 1 Error number       | 2116W                                   |
| Field 2 dsname             | PO0.ISI.PDB2PARM.CDBTRM.QBKOF.G0025V00  |
| Field 3 HSM Generated name | HSM.BACK.T000105.PO0.ISI.I8108          |
| Field 4 DATE OF BACKUP     | 1998108                                 |
| Field 5 Tape or DASD       | D                                       |
| Field 6 VSAM/NONVSAM       | N                                       |
| Field 7 BACKUP VERSION NUM | 001                                     |
| Field 8 REAL OWNER OF MCC  | PO0.ISI.PDB2PARM.PAYTHF.DBKUP. G2128V00 |

**Resolving the error**

To view the list of available fixes, type **F** next to the error message and press Enter. The Fast Audit Fix Plan Member List is displayed in Figure 6-89.

```

<V6 R1.01>
FAST AUDIT FIX PLAN MEMBER LIST
OPTION ==>
YOU MAY BROWSE (B), EDIT (E), FIX (F) OR VIEW (V) RESULTS
COMMAND MEMBER ACTION TITLE
F2116WA HSM FIXCDS DISPLAY B FOR MISMATCH MCC
F2116WB HSM FIXCDS DISPLAY C FOR MCB MISMATCH
***** Bottom of data *****
Row 1 of 2
SCROLL==> PAGE

```

Figure 6-89 Fix plan member list for error 2116W

To verify the errors, execute fixes F2116WA (HSM FIXCDS DISPLAY B FOR MISMATCH MCC) and fix F2116WB (HSM FIXCDS DISPLAY C FOR MCB MISMATCH). These fixes interrogate the MCB and MCC records and report where the record is matched.

Once verified you can decide to delete the backup copy and to recall the data set and create a new backup copy using the DFSMSHsm HBACKDS command.

## 6.2.16 FastAudit/390 audit error 2117W

The next error we will look at is *FastAudit/390* audit error 2117W MCC ENTRY IS OWNED BY DIFFERENT MCB.

Looking up the error in the *FastAudit/390* User Guide, you find the following explanation:

*FastAudit/390* has found a backup generated alias name (an MCC) whose true name record exists, but the MCC is owned by a different MCB than that indicated in the MCC.

To view the error detail, type **B** on the FastAudit BCDS Summary panel next to the error and press Enter. The Field Description Box provided in the documentation and shown in Table 6-50 provides an explanation of each field.

Table 6-50 Diagnostic 2117W field descriptions

| Field # / Description      | Example                           |
|----------------------------|-----------------------------------|
| Field 1 Error number       | 2117W                             |
| Field 2 DSNAME             | CXTT.CLK.DB2ADM.CREV05.VSYSB.CMDS |
| Field 3 HSM Generated name | HSM.BACK.T000105.CXTT.CLK.I8247   |
| Field 4 VOLSER             | 035228                            |
| Field 5 DATE OF BACKUP     | 1994273                           |

| Field # / Description      | Example                           |
|----------------------------|-----------------------------------|
| Field 6 VSAM/NONVSAM       | N                                 |
| Field 7 Tape or DASD       | T                                 |
| Field 8 BACKUP VERSION NUM | 001                               |
| Field 9 BACKUP GEN NUM     | UN                                |
| Field 10 REAL OWNER OF MCC | CXTT.CLK.DB2ADM.CREV05.VSYSB.CMDA |

### Resolving the error

To view the list of available fixes, type **F** next to the error message and press Enter. The Fast Audit Fix Plan Member List is displayed in Figure 6-90.

```

<V6 R1.01>                                     Row 1 of 3
      FAST AUDIT FIX PLAN MEMBER LIST
OPTION ==>                                     SCROLL==> PAGE
YOU MAY BROWSE (B), EDIT (E), FIX (F) OR VIEW (V) RESULTS
COMMAND MEMBER ACTION TITLE
      F2117WA      HSM FIXCDS DISPLAY B FOR MCC CONFLICT
      F2117WB      HSM FIXCDS DISPLAY C FOR MCB MISMATCH
      F2117WC      HSM FIXCDS PATCH C WITH CORRECT DSN
***** Bottom of data *****

```

Figure 6-90 Fix plan member list for error 2117W

To verify the errors, execute fix F2117WA (HSM FIXCDS DISPLAY B FOR MCC CONFLICT) and fix F2117WB (HSM FIXCDS DISPLAY C FOR MCB MISMATCH). These fixes interrogate and verify the MCB and MCC records.

Once you have reviewed and verified that the MCC record is owned by a different MCB record, you can execute fix F2117WC (HSM FIXCDS PATCH C WITH CORRECT DSN) to patch the “C” record to the correct “B” record using a FIXCDS.

If the FIXCDS fails to correct the error you can delete the MCB record (or backup record) and re-issue a new backup.

### 6.3 Output Control Data Set audits

The OCDS contains information about data sets on tape for both backup and migrated data. The Tape Table of Contents Record (TTOC) and its extensions are audited to ensure all pointers, forward and backward, are correct and all

extension records are accounted for. All valid data sets in each extension and their 16K blocks are counted. This count is compared to the value in extension zero (0000) for accuracy. In addition *FastAudit/390* looks for tapes that are in the TTOC, are older than 11 days and have never been marked full. DFSMSHsm assigns tapes for migration and for backup tasks. Each assigned tape could take several days to fill up considering the capacity of the higher density tape cartridges. Eleven days is a reasonable but arbitrary value.

### 6.3.1 FAUDITOD audit procedure

FAUDITOD audits the DFSMSHsm OCDS volume pointer chains forward and backward. This audit is necessary to ensure that the tapes in the OCDS inventory can recycle properly.

With DFSMSHsm Release 2.6, IBM tuned the DFSMSHsm system by eliminating many of the OCDS updates that occurred when data sets were recalled or deleted. Consequently, many of the audit checks that are provided with the DFSMSHsm command audit, are no longer valid.

Recycle processes ignore much of the information about valid bits. However, there are many examples where the forward and backward volume pointers, shown in Figure 6-91, are incorrect and recycle processes fail without diagnostics. HSM *FastAudit* not only examines the correct pointers, but it verifies each dataset in the pointer conflict, to determine the correct pointer, and provides appropriate fixes.

|          |   |                            |
|----------|---|----------------------------|
| TTCVBLK  | ↔ | Actual Count               |
| TTCENTRY | ↔ | First and Last Alias names |
| TTCPREVL | ↔ | TTCSUCVL                   |
| TTCNUM   | ↔ | Actual Extension numbers   |

Figure 6-91 Forward and backward record relationships

The HSM *FastAudit* cataloged procedure, FAUDITOD, creates two data sets that are used in error analysis and correction. The following are changes that need to be made to the FAUDITOD procedure based on your installation standards.

- PPFX** Product Prefix as installed. For example, if the product has been installed with MSC.FAUDIT as a prefix to the libraries, then PPFX=MSC.FAUDIT must be coded in the EXEC PROC statement.
- PFX** Prefix represents the Runtime Prefix the user has chosen for this cataloged procedure to allocate work files that contain the audit results. Depending on the SMS treatment of new allocations, the

work data sets may be deleted within a day or so. Choose a prefix that will not be scratched for a while so the results of the audit may be reviewed and repaired.

|                |                                                                                                                                                                                          |
|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>PUNIT</b>   | Permanent unit for allocating data sets containing the audit results. The default is SYSDA. This unit should allocate data sets to a device that retains data for several days or weeks. |
| <b>WRKUNIT</b> | Work Unit for allocating temporary data sets deleted at the end of the job. The default is SYSDA. If your installation has a different standard, code the different name.                |
| <b>SOUT</b>    | SYSOOT CLASS. The default is an asterisk (*) indicating same as MSGCLASS on the JOBCARD. Change this class if your installation requires different classes.                              |
| <b>ATP</b>     | Allocation Type. Allocation is given in units such as PRI and SEC. For larger installations the unit TRKS sometimes is not enough. If your installation is large, change the ATP to CYL. |
| <b>PRI</b>     | The number of primary allocation units. The default is 100. This value must be adjusted to accommodate the size of your DFSMSHsm MCDS and the number of records.                         |
| <b>SEC</b>     | The number of secondary allocation units. The default is 200. This parameter is important to adjust for the size of your MCDS.                                                           |
| <b>OCDS</b>    | OCDS is required. It is a single VSAM cluster and is the only CDS that cannot be split.                                                                                                  |
| <b>MCDS</b>    | MCDS is required. It is the first split of the DFSMSHsm MCDS data set or the only split if the MCDS is not split.                                                                        |
| <b>MCDS1</b>   | MCDS1 is the second split of the MCDS. The default is NULLFILE                                                                                                                           |
| <b>MCDS2</b>   | MCDS2 is the third split of the MCDS. The default is NULLFILE.                                                                                                                           |
| <b>MCDS3</b>   | MCDS3 is the fourth split of the MCDS. The default is NULLFILE.                                                                                                                          |
| <b>BCDS</b>    | BCDS is required. It is the first split of the DFSMSHsm BCDS data set or the only split if the BCDS is not split.                                                                        |
| <b>BCDS1</b>   | BCDS1 is the second split of the BCDS. The default is NULLFILE                                                                                                                           |
| <b>BCDS2</b>   | BCDS2 is the third split of the BCDS. The default is NULLFILE.                                                                                                                           |
| <b>BCDS3</b>   | BCDS3 is the fourth split of the BCDS. The default is NULLFILE.                                                                                                                          |
| <b>LOAD</b>    | LOAD is the name of the load library as installed with <i>FastAudit/390</i> . It must be authorized.                                                                                     |

A sample FAUDITOD procedure is shown in Figure 6-92.

```

//FAUDITOD PROC PFX=PFX,          <==RUN TIME PREFIX
/* FASTAUDIT <V6 R1.01> SEPTEMBER, 2001
//      PPFX=PPFX,                <==PRODUCT PREFIX
//      PUNIT=SYSDA,              <==UNIT FOR PERMANENT ALLOCATION
//      SOUT=*,                   <==PREFERED SYSOUT CLASS
//      WRKUNIT=SYSDA,            <==UNIT FOR TEMPORARY ALLOCATION
//      ATP=TRK,                  <==ALLOCATION TYPE TRK/CYL
//      PRI=100,                  <==PRIMARY ALLOCATION FOR FAUDIT
//      SEC=200,                  <==SECONDARY SPACE ALLOCATION
//      MCDS=HSM.MCDS,            <==INSTALLATION MCDS DSNAME
//      MCDS1=NULLFILE,           <==USED WITH SPLIT MCDS
//      MCDS2=NULLFILE,           <==USED WITH SPLIT MCDS
//      MCDS3=NULLFILE,           <==USED WITH SPLIT MCDS
//      BCDS=HSM.BCDS,            <==INSTALLATION BCDS DSNAME
//      BCDS1=NULLFILE,           <==USED WITH SPLIT BCDS
//      BCDS2=NULLFILE,           <==USED WITH SPLIT BCDS
//      BCDS3=NULLFILE,           <==USED WITH SPLIT BCDS
//      OCDS=HSM.OCDS,            <==INSTALLATION OCDS DSNAME
//      OCDS1=NULLFILE,           <==USED WITH SPLIT BCDS
//      OCDS2=NULLFILE,           <==USED WITH SPLIT BCDS
//      OCDS3=NULLFILE,           <==USED WITH SPLIT BCDS
//      LOAD=USERID.LOAD          <==LOAD LIBRARY AS INSTALLED
- - - - -
//TTCAUDIT DD   DSN=&PFX..FAUDIT.OCDS.TTCAUDIT,DISP=SHR
//TTCVAUDT DD   DSN=&PFX..FAUDIT.OCDS.TTCVAUDT,DISP=SHR,
//   DCB=BUFNO=10
//*
//      PEND
//PROC1      EXEC FAUDITOD,
//      BCDS=SIS.HSM.SPLIT.BCDS1,
//      BCDS1=SIS.HSM.SPLIT.BCDS2,
//      MCDS=SIS.HSM.SPLIT.MCDS1,
//      MCDS1=SIS.HSM.SPLIT.MCDS2,
//      OCDS=SIS.HSM.OCDS,
//      LOAD=FAUDITX.QABETA.LOAD,
//      PPFX=FAUDITX.QABETA,
//      PFX=DES2

```

Figure 6-92 Cataloged procedure FAUDITOD JCL

### 6.3.2 OCDS TTOC audit errors detail

All errors are contained in the *PFX.FAUDIT.OCDS.ERRORS* data set after the cataloged procedure FAUDITOD has been run. The following are the field descriptions and an explanation of how the error was detected.

After the audit job completes, execute the CLIST FIXAUDIT. This CLIST displays the main panel, Fast Audit Choose Which Audit To Fix, shown in Figure 6-93, and the various audit results options available.

```
<V6 R1.01> FAST AUDIT. CHOOSE WHICH AUDIT TO FIX
COMMAND==> 3
1 MCDS    CREATED BY PROCS FAUDITMD, FAUDITCT OR FAUDITAL
2 BCDS    CREATED BY PROC FAUDITBD
3 OCDS    CREATED BY PROC FAUDITOD
4 TAPE    CREATED BY PROCS FAUDITTP, FAUDITZR OR FAUDITRM
5 CAT     CREATED BY PROC FAUDITCO
6 DASD    CREATED BY PROC FAUDITDA OR FAUDITDE
7 MASTCAT CREATED BY PROC FAUDITMC
8 CATRMM  CREATED BY PROC FAUDITCR
9 CATTMS  CREATED BY PROC FAUDITTC

ENTER  END  TO TERMINATE ONLINE FUNCTION
```

Figure 6-93 Primary Fix Panel

Select option 3, OCDS by typing the number 3 on the command line and pressing Enter. The OCDS Error Summary Panel is displayed in Figure 6-94.

```
<V6 R1.01> Row 1 of 12
FASTAUDIT OCDS SUMMARY
OPTION ==>
ENTER F TO DISPLAY FIXES OR B TO BROWSE ERRORS
SCROLL==> CSR
S NUMBER  COUNT  MESSAGE
4001W      0  PREVIOUS VOLUME NOT FOUND IN OCDS
4002W      0  SUCCESSOR VOLUME NOT FOUND IN OCDS
4003W      0  PREVIOUS VOLUME POINTER IS INCORRECT
4004W      0  SUCCESSOR VOLUME POINTER IS INCORRECT
4007C      1  TTOC RECORD HAS BROKEN SEQUENCE SET
4007V      1  VOLUME CONTAINS BROKEN SEQUENCE SET/SETS
4008C      1  TTOC EXTENTION RECORDS ARE LESS THAN EXPECTED
4009C      0  TTOC EXTENTION RECORDS ARE MORE THAN EXPECTED
4010W      1  16K BLOCK COUNT DOES NOT MATCH COMPUTED BLKCNT
4011W      1  TAPE HAS FAILED RECYCLE
4012C     31  TAPE IS NOT FULL AND WILL NOT BE SELECTED FOR RECYCLE
4013I      7  TAPE IS ELIGIBLE FOR RECYCLE
4014I     38  TAPES CREATED TODAY. NOT AN ERROR.
```

Figure 6-94 OCDS Error Summary Panel

There are subsequent panels behind each error number listed. These panels provide the list of FIXES to either diagnose or resolve the various errors. In all



cases, you can view the list of available fixes by typing the letter F next to the error number and pressing Enter. The Fast Audit Fix Plan Member List for that error number is displayed and lists the available fixes for the error as shown in Figure 6-95.

```

<V6 R1.01>                                     Row 1 of 5
          FAST AUDIT FIX PLAN MEMBER LIST
OPTION ==>                                     SCROLL==> PAGE
YOU MAY BROWSE (B), EDIT (E), FIX (F) OR VIEW (V) RESULTS
COMMAND MEMBER ACTION TITLE
          F4012CA      TSO RECYCLE TAPE NOT FULL
          F4012CB      HSM LIST CONTENTS OF TAPE
          F4012CR1 FIX  TSO MARKFULL AND RECYCLE ML2 TAPES
          F4012CR2      TSO MARKFULL AND RECYCLE BACKUP TAPES
          F4012CR3      TSO MARKFULL AND RECYCLE DUMP TAPES
***** Bottom of data *****

```

Figure 6-95 Fix plan member list for error 4012C

FastAudit/390 automatically generates the required JCL and FIX statements for each FIX listed when you type the letter F next to the FIX member name and press Enter. Figure 6-96 is an example of the JCL and FIX statements generated for fix F4012CR2.

```

File Edit Edit_Settings Menu Utilities Compilers Test Help

EDIT          CG.H154615                      Columns 00001 00072
Command ==>                                     Scroll ==> CSR
***** ***** Top of Data *****
==MSG> -Warning- The UNDO command is not available until you change
==MSG>          your edit profile using the command RECOVERY ON.
000001 //CGA   JOB (0),COLGOR,
000002 //          CLASS=A,MSGCLASS=X,NOTIFY=COLGOR
000003 //*
000004 //STEP1 EXEC PGM=IKJEFT01,
000005 // DYNAMNBR=25,REGION=4096K
000006 //* CHANGE THE FOLLOWING STATEMENT TO
000007 //* DSN=PPFX.CLIST WHERE PPFX IS THE PREFIX ASSIGNED TO
000008 //* THE FASTAUDIT PRODUCT
000009 //*
000010 //SYSPROC DD DSN=FAUDITX.QABETA.CLIST,DISP=SHR
000011 //*          ****
000012 //SYSOUT DD SYSOUT=*
000013 //SYSTSPRT DD DSN=CG.FIXAUDIT.F4012CR1,DISP=SHR
000014 //SYSTSIN DD *
000015 HSEND DELVOL CART05 MIGRATION(MARKFULL)
000016 HSEND RECYCLE VOL(CART05) EXEC
000017 HSEND DELVOL 003625 MIGRATION(MARKFULL)

```

Figure 6-96 JCL and fix statements automatically generated for fix F4012CR2

You can use ISPF edit commands to edit the member and remove any records that you do not want to FIX or diagnose. Once your editing is complete or if no editing is desired, execute the FIX JCL by typing the word SUBMIT or the short version SUB on the command line in the edit entry panel and pressing Enter. A batch job is submitted to execute the FIX commands.

When the FIX batch job has completed you can view the results by typing the letter V next to the FIX member name and pressing Enter. This action results in an ISPF view of the FIX results.

The error messages are identified as:

- ▶ **(C)** - Critical means that a data set cannot be recovered using IBM provided recover facilities.
- ▶ **(V)** - Volume is summaries of error detail volumes.
- ▶ **(I)** - Information only.
- ▶ **(W)** - Warning means that the data set might not be recoverable and further investigation is required to determine the status of the data set.

Table 6-51 shows the *FastAudit/390* OCDS error messages we will look at.

Table 6-51 *FastAudit/390* OCDS error messages

|       |                                                       |
|-------|-------------------------------------------------------|
| 4001W | PREVIOUS VOLUME NOT FOUND IN OCDS                     |
| 4002W | SUCCESSOR VOLUME NOT FOUND IN OCDS                    |
| 4003W | PREVIOUS VOLUME POINTER IS INCORRECT                  |
| 4004W | SUCCESSOR VOLUME POINTER IS INCORRECT                 |
| 4005C | TAPE HAS FAILED CREATE STATUS                         |
| 4007C | TTOC RECORD HAS BROKEN SEQUENCE SET                   |
| 4007V | VOLUME CONTAINS BROKEN SEQUENCE SET/SETS              |
| 4008C | TTOC EXTENTION RECORDS ARE LESS THAN EXPECTED         |
| 4009C | TTOC EXTENTION RECORDS ARE MORE THAN EXPECTED         |
| 4010C | 16K BLOCK COUNT DOES NOT MATCH COMPUTED BLKCNT        |
| 4011W | TAPE HAS FAILED RECYCLE                               |
| 4012C | TAPE IS NOT FULL AND WILL NOT BE SELECTED FOR RECYCLE |
| 4013I | TAPE IS ELIGIBLE FOR RECYCLE                          |

### 6.3.3 *FastAudit/390* audit error 4001W

The first OCDS error message we will look at is *FastAudit/390* audit error 4001W PREVIOUS VOLUME NOT FOUND IN OCDS.

Looking up the error in the *FastAudit/390* User Guide, you find the following explanation:

In the OCDS each tape has a root record called extension 0000. In this extension is a previous volume pointer, indicating that the first data set on the tape is continued from a previous volume. *FastAudit/390* has looked up the detail of that first dataset and has determined that it is valid and is continued from a previous volume. In this case the previous volume is different than the PVOL field in the OCDS.

To view the error detail, type **B** on the *FastAudit* BCDS Summary panel next to the error and press Enter. The Field Description Box provided in the documentation and shown in Table 6-52 provides an explanation of each field.

Table 6-52 Diagnostic 4001W field descriptions

| Field # / Description       | Example                              |
|-----------------------------|--------------------------------------|
| Field 1 /errnum             | 4001W                                |
| Field 2 /VOLSER             | 014673                               |
| Field 3 /TYPE               | L2 (L2, SP OR CYCLE DAY)             |
| Field 4 /cred               | 1997214                              |
| Field 5 /PVOL               | 014926                               |
| Field 6 /cvol (correct vol) | F0F1F4F5F4F5                         |
| Field 7 /adsn (FIRST DSN)   | HSM.HMIG.T033102.TDB2.ARCHLOG1.I4068 |

Resolving the error

To view the list of available fixes, type **F** next to the error message and press Enter. The Fast Audit Fix Plan Member List is displayed in Figure 6-97.

```
<V6 R1.01>                                     Row 1 of 1
      FAST AUDIT FIX PLAN MEMBER LIST
OPTION ==>                                     SCROLL==> PAGE
YOU MAY BROWSE (B), EDIT (E), FIX (F) OR VIEW (V) RESULTS
COMMAND MEMBER ACTION TITLE
      F4001WA      HSM FIXCDS COMMAND TO CORRECT PREVIOUS VOL MISSING
***** Bottom of data *****
```

Figure 6-97 Fix plan member list for error 4001WV

To resolve this error, execute fix F4001WA. The fix patches the PVOL field to the correct VOLSER using the FIXCDS PATCH command.

6.3.4 FastAudit/390 audit error 4002W

The next OCDS error message we will look at is *FastAudit/390* audit error 4002W SUCCESSOR VOLUME NOT FOUND IN OCDS.

Diagnosing the error

Looking up the error in the *FastAudit/390* User Guide, you find the following explanation:

In the OCDS each tape has a root record called extension 0000. In this extension is a successor volume pointer, indicating that the last data set on the tape is continued to a successor volume. *FastAudit/390* has looked up the detail of that last data set and has determined that it is valid and is continued

to a successor volume. In this case the successor volume is different than the SVOL field in the OCDS.

To view the error detail, type **B** on the FastAudit BCDS Summary panel next to the error and press Enter. The Field Description Box provided in the documentation and shown in Table 6-53 provides an explanation of each field.

Table 6-53 Diagnostic 4002W field descriptions

| Field # / Description       | Example                              |
|-----------------------------|--------------------------------------|
| Field 1 /errnum             | 4002W                                |
| Field 2 /VOLSER             | 014869                               |
| Field 3 /TYPE               | L2 (L2, SP OR CYCLE DAY)             |
| Field 4 /cred               | 1997214                              |
| Field 5 /PVOL               | 026898                               |
| Field 6 /cvol (correct vol) | F0F1F7F3F4F5                         |
| Field 7 /adsn (LAST DSN)    | HSM.HMIG.T304202.TDB2.ARCHLOG1.I5289 |

Resolving the error

To view the list of available fixes, type **F** next to the error message and press Enter. The Fast Audit Fix Plan Member List is displayed in Figure 6-98.

```
<V6 R1.01>                                     Row 1 of 1
      FAST AUDIT FIX PLAN MEMBER LIST
OPTION ==>                                     SCROLL==> PAGE
YOU MAY BROWSE (B), EDIT (E), FIX (F) OR VIEW (V) RESULTS
COMMAND MEMBER ACTION TITLE
      F4002WA      HSM FIXCDS COMMAND TO CORRECT SUCCESSOR VOL MISSING
***** Bottom of data *****
```

Figure 6-98 Fix plan member list for error 1106V

To resolve this error, execute fix F4002WA. The fix patches the correct VOLSER into the successor volume field.

6.3.5 FastAudit/390 audit error 4003W

The next OCDS error message we will look at is FastAudit/390 audit error 4003W PREVIOUS VOLUME POINTER IS INCORRECT.

## Diagnosing the error

Looking up the error in the *FastAudit/390* User Guide, you find the following explanation:

This error messages differs from 4001W because the previous volume pointer was found but was incorrect. In error 4001W, the previous volume was not in the OCDS inventory. *FastAudit/390* has compared the first DSN on the extension 0000 record with the last DSN on the previous volume and has found a conflict. *FastAudit/390* has looked up the detail about the dataset and has determined that the PVOL field in the OCDS is incorrect.

To view the error detail, type **B** on the FastAudit BCDS Summary panel next to the error and press Enter. The Field Description Box provided in the documentation and shown in Table 6-54 provides an explanation of each field.

Table 6-54 Diagnostic 4003W field descriptions

| Field # / Description       | Example                  |
|-----------------------------|--------------------------|
| Field 1 /errnum             | 4003W                    |
| Field 2 /VOLSER             | 011487                   |
| Field 3 /TYPE               | L2 (L2, SP OR CYCLE DAY) |
| Field 4 /credt              | 1997214                  |
| Field 5 /PVOL               | 012247                   |
| Field 6 /cvol (correct vol) | F0F2F5F8F5F0             |

## Resolving the error

To view the list of available fixes, type **F** next to the error message and press Enter. The Fast Audit Fix Plan Member List is displayed in Figure 6-99.

```
<V6 R1.01> Row 1 of 1
      FAST AUDIT FIX PLAN MEMBER LIST
OPTION ==>          SCROLL==> PAGE
  YOU MAY BROWSE (B), EDIT (E), FIX (F) OR VIEW (V) RESULTS
COMMAND MEMBER ACTION TITLE
      F4003WA      HSM FIXCDS COMMAND TO CORRECT PREVIOUS VOL INCORRECT
***** Bottom of data *****
```

Figure 6-99 Fix plan member list for error 4003W

To resolve this error, execute fix F4003WA. The fix patches the correct previous volume.

### 6.3.6 *FastAudit/390* audit error 4004W

The next OCDS error message we will look at is *FastAudit/390* audit error 4004W SUCCESSOR VOLUME POINTER IS INCORRECT.

#### Diagnosing the error

Looking up the error in the *FastAudit/390* User Guide, you find the following explanation:

This error messages differs from 4002W because the successor volume pointer was found but was incorrect. In error 4002W, the successor volume was not in the OCDS inventory. *FastAudit/390* has compared the last DSN on the extension 0000 record with the first DSN on the successor volume and has found a conflict. *FastAudit/390* has looked up the detail about the dataset and has determined that the SVOL field in the OCDS is incorrect.

To view the error detail, type **B** on the FastAudit BCDS Summary panel next to the error and press Enter. The Field Description Box provided in the documentation and shown in Table 6-55 provides an explanation of each field.

Table 6-55 Diagnostic 4004W field descriptions

| Field # / Description       | Example                  |
|-----------------------------|--------------------------|
| Field 1 /errnum             | 4004W                    |
| Field 2 /VOLSER             | 011487                   |
| Field 3 /TYPE               | L2 (L2, SP OR CYCLE DAY) |
| Field 4 /cred               | 1997214                  |
| Field 5 /PVOL               | 012247                   |
| Field 6 /cvol (correct vol) | F0F2F5F8F5F0             |

#### Resolving the error

To view the list of available fixes, type **F** next to the error message and press Enter. The Fast Audit Fix Plan Member List is displayed in Figure 6-100.

```
<V6 R1.01> Row 1 of 1
      FAST AUDIT FIX PLAN MEMBER LIST
OPTION ==> SCROLL==> PAGE
YOU MAY BROWSE (B), EDIT (E), FIX (F) OR VIEW (V) RESULTS
COMMAND MEMBER ACTION TITLE
      F4004WA HSM FIXCDS COMMAND TO CORRECT SUCCESSOR VOL INCORRECT
***** Bottom of data *****
```

Figure 6-100 Fix plan member list for error 4004W

To resolve this error, execute fix F4004WA. This fix patches the successor volume field with the correct volser.

6.3.7 FastAudit/390 audit error 4005C

The next OCDS error message we will look at is *FastAudit/390* audit error 4005C TAPE HAS FAILED CREATE STATUS.

Diagnosing the error

Looking up the error in the *FastAudit/390* User Guide, you find the following explanation:

The TTOC extraction process has found a tape marked as “active in storage” and the last activity was more than 24 hours prior to the current time. TTOC activity is updated every 20 minutes or less. This is the criteria that **LIST TTOC SELECT (FAILEDCREATE)** uses to determine failed create status.

To view the error detail, type **B** on the FastAudit BCDS Summary panel next to the error and press Enter. The Field Description Box provided in the documentation and shown in Table 6-56 provides an explanation of each field.

Table 6-56 Diagnostic 4005C field descriptions

| Field # / Description | Example                  |
|-----------------------|--------------------------|
| Field 1 /errnum       | 4005C                    |
| Field 2 /VOLSER       | 011529                   |
| Field 3 /TYPE         | SP (L2, SP OR CYCLE DAY) |
| Field 4 /cred         | 1997214                  |

Resolving the error

To view the list of available fixes, type **F** next to the error message and press Enter. The Fast Audit Fix Plan Member List is displayed in Figure 6-101.



|                                                           |         |                                                  |
|-----------------------------------------------------------|---------|--------------------------------------------------|
| <V6 R1.01>                                                |         | Row 1 of 1                                       |
| FAST AUDIT FIX PLAN MEMBER LIST                           |         |                                                  |
| OPTION ==>                                                |         | SCROLL==> PAGE                                   |
| YOU MAY BROWSE (B), EDIT (E), FIX (F) OR VIEW (V) RESULTS |         |                                                  |
| COMMAND                                                   | MEMBER  | ACTION TITLE                                     |
|                                                           | F4005CA | HSM AUDIT MEDIA CONTROL FOR SELECT FAILED CREATE |

Figure 6-101 Fix plan member list for error 4005C

Tapes in this status can only be reclaimed by performing an AUDIT MEDCTL. This diagnostic can be verified by submitting a:

LIST TTOC SELECT(FAILEDCREATE)

To resolve this error, execute fix F4005CA AUDIT MEDCTL.

6.3.8 FastAudit/390 audit error 4007C

The next OCDS error message we will look at is *FastAudit/390* audit error 4007C  
TTOC RECORD HAS BROKEN SEQUENCE SET.

Diagnosing the error

Looking up the error in the *FastAudit/390* User Guide, you find the following explanation:

Each tape in the OCDS inventory can have many extension records starting from extension 0000 to 9999. When *FastAudit/390* processes the extensions for a volume, it tracks the numerical sequence numbers. If the count is different than the number in the extension record then, diagnostic 4007C is declared. This error is critical because recycle will fail for this volume.

To view the error detail, type **B** on the FastAudit BCDS Summary panel next to the error and press Enter. The Field Description Box provided in the documentation and shown in Table 6-57 provides an explanation of each field.

Table 6-57 Diagnostic 4007C filed descriptions

| Field # / Description        | Example                  |
|------------------------------|--------------------------|
| Field 1 /errnum              | 4007C                    |
| Field 2 /VOLSER              | 011487                   |
| Field 3 /TYPE                | 01 (L2, SP OR CYCLE DAY) |
| Field 4 /CREDIT              | 1997214                  |
| Field 5 /EXTN (EXPECTED SEQ) | 00013                    |

| Field # / Description      | Example |
|----------------------------|---------|
| Field 6 /AXTN (ACTUAL SEQ) | 00015   |

### Resolving the error

To view the list of available fixes, type **F** next to the error message and press Enter. The Fast Audit Fix Plan Member List is displayed in Figure 6-102.

|                                                           |         |                                            |
|-----------------------------------------------------------|---------|--------------------------------------------|
| <V6 R1.01>                                                |         | Row 1 of 2                                 |
| FAST AUDIT FIX PLAN MEMBER LIST                           |         |                                            |
| OPTION ==>                                                |         | SCROLL==> PAGE                             |
| YOU MAY BROWSE (B), EDIT (E), FIX (F) OR VIEW (V) RESULTS |         |                                            |
| COMMAND                                                   | MEMBER  | ACTION TITLE                               |
|                                                           | F4007CB | HSM FIXCDS DELETE OF TTOC EXTENTION RECORD |
|                                                           | F4007CC | HSA DISPLAY DUMP OF TTOC                   |
| ***** Bottom of data *****                                |         |                                            |

Figure 6-102 Fix plan member list for error 4007C

This error may be corrected by running **AUDIT MEDIACONTROLS** for the volume in error. An alternative is to delete the TTOC extension record by executing fix F4007CB.

### 6.3.9 FastAudit/390 audit error 4008C

The next OCDS error message we will look at is *FastAudit/390* audit error 4008C TTOC EXTENSION RECORDS ARE LESS THAN EXPECTED.

### Diagnosing the error

Looking up the error in the *FastAudit/390* User Guide, you find the following explanation:

Each tape in the OCDS inventory has an expected extension number in extension 0000. While *FastAudit/390* was reading the extension records it encountered the end of extensions before the expected count was finished. This volume will probably fail recycle.

To view the error detail, type **B** on the FastAudit BCDS Summary panel next to the error and press Enter. The Field Description Box provided in the documentation and shown in Table 6-58 provides an explanation of each field.

Table 6-58 Diagnostic 4008C field descriptions

| Field # / Description | Example |
|-----------------------|---------|
| Field 1 /errnum       | 4008C   |

| Field # / Description        | Example                  |
|------------------------------|--------------------------|
| Field 2 /VOLSER              | 067487                   |
| Field 3 /TYPE                | 02 (L2, SP OR CYCLE DAY) |
| Field 4 /CREDIT              | 1997214                  |
| Field 5 /EXTN (EXPECTED SEQ) | 00013                    |
| Field 6 /AXTN (LAST READ)    | 00009                    |

### Resolving the error

To view the list of available fixes, type **F** next to the error message and press Enter. The Fast Audit Fix Plan Member List is displayed in Figure 6-103.

```
<V6 R1.01> Row 1 of 1
FAST AUDIT FIX PLAN MEMBER LIST
OPTION ==> SCROLL==> PAGE
YOU MAY BROWSE (B), EDIT (E), FIX (F) OR VIEW (V) RESULTS
COMMAND MEMBER ACTION TITLE
F4008CA HSM AUDIT MEDIA CONTROL FOR EXTENTION ERROR
***** Bottom of data *****
```

Figure 6-103 Fix plan member list for error 4008C

This error may be corrected by running AUDIT MEDIACONTROLS for the volume in error. To resolve this error, execute fix F4008CA to perform (AUDIT MEDIACONTROLS).

### 6.3.10 FastAudit/390 audit error 4009C

The next OCDS error message we will look at is *FastAudit/390* audit error 4009C  
TTOC EXTENSION RECORDS ARE MORE THAN EXPECTED.

### Diagnosing the error

Looking up the error in the *FastAudit/390* User Guide, you find the following explanation:

Each tape in the OCDS inventory has an expected extension number in extension 0000. While *FastAudit/390* was reading the extension records it found more extensions than was expected. This volume will probably fail recycle.

To view the error detail, type **B** on the FastAudit BCDS Summary panel next to the error and press Enter. The Field Description Box provided in the documentation and shown in Table 6-58 provides an explanation of each field.

Table 6-59 Diagnostic 4009C field descriptions

| Field # / Description        | Example                  |
|------------------------------|--------------------------|
| Field 1 /errnum              | 4009C                    |
| Field 2 /VOLSER              | 067487                   |
| Field 3 /TYPE                | 02 (L2, SP OR CYCLE DAY) |
| Field 4 /CREDIT              | 1997214                  |
| Field 5 /EXTN (EXPECTED SEQ) | 00013                    |
| Field 6 /AXTN (LAST READ)    | 00016                    |

**Resolving the error**

To view the list of available fixes, type **F** next to the error message and press Enter. The Fast Audit Fix Plan Member List is displayed in Figure 6-104.

|                                                           |                |
|-----------------------------------------------------------|----------------|
| <V6 R1.01>                                                | Row 1 of 1     |
| FAST AUDIT FIX PLAN MEMBER LIST                           |                |
| OPTION ===>                                               | SCROLL==> PAGE |
| YOU MAY BROWSE (B), EDIT (E), FIX (F) OR VIEW (V) RESULTS |                |
| COMMAND MEMBER ACTION TITLE                               |                |
| F4009CA HSM AUDIT MEDIA CONTROL FOR EXTENSION ERROR       |                |
| ***** Bottom of data *****                                |                |

Figure 6-104 Fix plan member list for error 4009C

This error may be corrected by executing fix F4009CA to run **AUDIT MEDIACONTROLS** for the volume in error.

**6.3.11 FastAudit/390 audit error 4010C**

The next OCDS error message we will look at is *FastAudit/390* audit error 4010C 16K BLOCK COUNT DOES NOT MATCH COMPUTED BLKCNT.

**Diagnosing the error**

Looking up the error in the *FastAudit/390* User Guide, you find the following explanation:

Extension 0000 for each volume in the OCDS contains a field indicating how many 16K blocks of data is written on that tape. *FastAudit/390* has read the

detail about the number of 16K blocks for each data set and has accumulated the totals. At the end of the extensions, these fields are compared and are found to have a mismatch.

To view the error detail, type **B** on the FastAudit BCDS Summary panel next to the error and press Enter. The Field Description Box provided in the documentation and shown in Table 6-60 provides an explanation of each field.

Table 6-60 Diagnostic 4010C field descriptions

| Field # / Description | Example                  |
|-----------------------|--------------------------|
| Field 1 /errnum       | 4010C                    |
| Field 2 /VOLSER       | 069260                   |
| Field 3 /TYPE         | 30 (L2, SP OR CYCLE DAY) |
| Field 4 /CREDIT       | 1997214                  |
| Field 5 /EBLKCNT      | 019013                   |
| Field 6 /ABLKCNT      | 018393                   |

Resolving the error

To view the list of available fixes, type **F** next to the error message and press Enter. The Fast Audit Fix Plan Member List is displayed in Figure 6-105.

```
<V6 R1.01>                                     Row 1 of 1
      FAST AUDIT FIX PLAN MEMBER LIST
OPTION ===>                                     SCROLL==> PAGE
  YOU MAY BROWSE (B), EDIT (E), FIX (F) OR VIEW (V) RESULTS
COMMAND MEMBER ACTION TITLE
      F4010WA      HSM AUDIT DIRECTORY CONTROL FOR MISMATCH BLKCNT
***** Bottom of data *****
```

Figure 6-105 Fix plan member list for error 4010C

This tape either will not recycle or some of the data may not be recalled/recovered. To resolve this error, execute fix F4010CA to run AUDIT DIRECTORYCONTROLS to examine and correct this tape volume.

6.3.12 FastAudit/390 audit error 4011W

The next OCDS error message we will look at is *FastAudit/390* audit error 4011W TAPE HAS FAILED RECYCLE.

## Diagnosing the error

Looking up the error in the *FastAudit/390* User Guide, you find the following explanation:

When *FastAudit/390* reads the OCDS inventory, in each TTOC record 0000 for each tape, it examines the field that indicates whether the tape has failed recycle. This is an indicator that something is in error on the tape and some corrective action is required before the tape can be successfully recycled.

To view the error detail, type **B** on the FastAudit BCDS Summary panel next to the error and press Enter. The Field Description Box provided in the documentation and shown in Table 6-61 provides an explanation of each field.

Table 6-61 Diagnostic 4011W field descriptions

| Field # / Description | Example                  |
|-----------------------|--------------------------|
| Field 1 /errnum       | 4011W                    |
| Field 2 /VOLSER       | 069260                   |
| Field 3 /TYPE         | SP (L2, SP OR CYCLE DAY) |
| Field 4 /CREDIT       | 1997214                  |

## Resolving the error

To view the list of available fixes, type **F** next to the error message and press Enter. The Fast Audit Fix Plan Member List is displayed in Figure 6-106.

|                                                           |          |                                            |
|-----------------------------------------------------------|----------|--------------------------------------------|
| <V6 R1.01>                                                |          | Row 1 of 2                                 |
| FAST AUDIT FIX PLAN MEMBER LIST                           |          |                                            |
| OPTION ==>                                                |          | SCROLL==> PAGE                             |
| YOU MAY BROWSE (B), EDIT (E), FIX (F) OR VIEW (V) RESULTS |          |                                            |
| COMMAND                                                   | MEMBER   | ACTION TITLE                               |
|                                                           | F4011WA  | HSM AUDIT MEDIA CONTROL FOR RECYCLE FAILED |
|                                                           | F4011WR1 | HSM RECYCLE VOLUME BEFORE AUDIT            |
| ***** Bottom of data *****                                |          |                                            |

Figure 6-106 Fix plan member list for error 4011W

To resolve this error, execute fix F4011WA to run AUDIT MEDIACONTROLS to examine and correct this tape volume.

### 6.3.13 FastAudit/390 audit error 4012C

The next OCDS error message we will look at is *FastAudit/390* audit error 4012C TAPE IS NOT FULL AND WILL NOT BE SELECTED FOR RECYCLE.

### Diagnosing the error

Looking up the error in the *FastAudit/390* User Guide, you find the following explanation:

When *FastAudit/390* reads the OCDS inventory, it looks for tapes that are eleven days old or older that have not been marked full. When the recycle process looks for tapes to recycle, it looks only for full tapes. The volumes with these errors will never be marked eligible for recycle because they are not full.

To view the error detail, type **B** on the FastAudit BCDS Summary panel next to the error and press Enter. The Field Description Box provided in the documentation and shown in Table 6-62 provides an explanation of each field.

Table 6-62 Diagnostic 4012C field descriptions

| Field # / Description | Example                  |
|-----------------------|--------------------------|
| Field 1 /errnum       | 4012C                    |
| Field 2 /VOLSER       | 045092                   |
| Field 3 /TYPE         | 1997214                  |
| Field 4 /CREDIT       | SP (L2, SP OR CYCLE DAY) |

### Resolving the error

To view the list of available fixes, type **F** next to the error message and press Enter. The Fast Audit Fix Plan Member List is displayed in Figure 6-107.

```

<V6 R1.01>
FAST AUDIT FIX PLAN MEMBER LIST
OPTION ==>
YOU MAY BROWSE (B), EDIT (E), FIX (F) OR VIEW (V) RESULTS
COMMAND MEMBER ACTION TITLE
F4012CA TSO RECYCLE TAPE NOT FULL
F4012CB HSM LIST CONTENTS OF TAPE
F4012CR1 TSO MARKFULL AND RECYCLE ML2 TAPES
F4012CR2 TSO MARKFULL AND RECYCLE BACKUP TAPES
F4012CR3 TSO MARKFULL AND RECYCLE DUMP TAPES
***** Bottom of data *****

```

Figure 6-107 Fix plan member list for error 4012CV

To resolve this error, execute fix F4012CA to recycle the tape.

6.3.14 FastAudit/390 audit error 4013I

The next OCDS error message we will look at is *FastAudit/390* audit error 4013I  
TAPE IS ELIGIBLE FOR RECYCLE.

Diagnosing the error

Looking up the error in the *FastAudit/390* User Guide, you find the following explanation:

When *FastAudit/390* reads the OCDS inventory, it looks for tapes that are marked eligible for recycle. This is an information only message and is intended to be used to indicate the health of the recycle process. This does not mean that the prescribed percent valid has been reached. It means that the tape has been marked full and HSM has been told that this is a healthy tape that can be selected for recycle

To view the error detail, type **B** on the FastAudit BCDS Summary panel next to the error and press Enter. The Field Description Box provided in the documentation and shown in Table 6-63 provides an explanation of each field.

Table 6-63 Diagnostic 4013I field descriptions

| Field # / Description | Example                  |
|-----------------------|--------------------------|
| Field 1 /errnum       | 4013I                    |
| Field 2 /VOLSER       | 069260                   |
| Field 3 /TYPE         | 1997214                  |
| Field 4 /CREDIT       | SP (L2, SP OR CYCLE DAY) |

Resolving the error

To view the list of available fixes, type **F** next to the error message and press Enter. The Fast Audit Fix Plan Member List is displayed in Figure 6-108.

```
<V6 R1.01> Row 1 of 1
      FAST AUDIT FIX PLAN MEMBER LIST
OPTION ==> SCROLL==> PAGE
YOU MAY BROWSE (B), EDIT (E), FIX (F) OR VIEW (V) RESULTS
COMMAND MEMBER ACTION TITLE
      F4013IA TSO RECYCLE ELIGIBLE TAPE
***** Bottom of data *****
```

Figure 6-108 Fix plan member list for error 4013I

Execute fix F4013IA to recycle the eligible tapes.





## Auditing your catalogs using Catalog & DASD Audit

The master catalog audits are provided in the *FastAudit/390* selectable unit, Catalog & DASD Audit. The purpose of the master catalog audit is to reconcile differences between master catalogs in a shared DASD environment. In this chapter we discuss how to use the Catalog & DASD Audit component of *FastAudit/390*.

## 7.1 Master catalog audits

The master catalog audits are provided in the *FastAudit/390* selectable unit, Catalog & DASD Audit. The purpose of the master catalog audit is to reconcile differences between master catalogs in a shared DASD environment. None of the audits involving catalog extractions can be accurate if one or more of the master catalogs contain disagreeing or missing elements. These elements might be that a user catalog is missing or is IMPORT CONNECTed to the wrong volume or the ALIAS DEFINITIONS disagree or are missing to the user catalogs.

For the **FAUDITMC** procedure we recommend that errors involving user catalogs definitions (1011W through 1016W) be corrected before processing alias errors (1017W- 1020W). It is recommended that the audits execute repeatedly until all the differences are resolved.

Prior to executing the audit JCL, the following symbolic parameters need to be replaced in the JCL shown in Figure 7-1 with your customized values. Instructions on where to find the audit JCL and how to customize the product are documented in the *FastAudit/390* Installation and Maintenance Guide. The following symbolic values are replaced by your values in the audit JCL.

|                |                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>PPFX</b>    | Product Prefix as installed. For example, if the product has been installed with MSC.FAUDIT as a prefix to the libraries, then PPFX=MSC.FAUDIT must be coded in the EXEC PROC statement.                                                                                                                                                                                                                                                         |
| <b>PFX</b>     | Prefix represents the Runtime Prefix you have chosen for this cataloged procedure to allocate work files that contain the audit results. Depending on the DFSMS treatment of new allocations, it is possible that the work data sets may be deleted fairly quickly, within a day or so. We recommend that you choose a prefix value that is retained for an appropriate amount of time so the results of the audit may be reviewed and repaired. |
| <b>PUNIT</b>   | Permanent unit for allocating data sets containing the audit results. The default is SYSDA. If this unit allocation allocates on a work volume or another volume that scratches data in a short period of time, you may want to change it to a different unit name to allocate on volumes with a longer retention.                                                                                                                               |
| <b>SOUT</b>    | SYSOUT class is desirable for your installation.                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>WRKUNIT</b> | Work Unit for allocating temporary data sets that are deleted at the end of the job. The default is SYSDA. If your installation has a different standard, code the different name.                                                                                                                                                                                                                                                               |
| <b>LOAD</b>    | Load library as installed for <i>FastAudit/390</i> . This library must be APF authorized.                                                                                                                                                                                                                                                                                                                                                        |

Figure 7-1 shows the symbolic parameters replaced in the JCL used to execute procedure FAUDITMC. Our sample shows load library FAUDITX.QABETA.LOAD, product prefix FAUDITX.QABETA, and run time prefix DES2.

```

000005 //FAUDITMC PROC PFX=PFX,          <==RUN TIME PREFIX
000006 //* FASTAUDIT <V6 R1.01> SEPTEMBER, 2001
000007 //          PPFX=PPFX,          <==PRODUCT PREFIX
000008 //          SOUT=*,              <==PREFERED SYSOUT CLASS
000009 //          PUNIT=SYSDA,         <==UNIT FOR PERMANENT ALLOCATION
000010 //          WRKUNIT=SYSDA,       <==UNIT FOR TEMPORARY ALLOCATION
000011 //          PRI=5,              <==PRIMARY ALLOCATION FOR FAUDIT
000012 //          SEC=5,              <==SECONDARY SPACE ALLOCATION
000013 //          LOAD=PPFX.LOAD      <==LOAD LIBRARY AS INSTALLED
- - - - -
000014 //*
000140 //PROC1      EXEC FAUDITMC,
000141 //          LOAD=FAUDITX.QABETA.LOAD,
000142 //          PPFX=FAUDITX.QABETA,
000143 //          PFX=DES2

```

*Figure 7-1 Example of FAUDITMC cataloged procedure*

In the product CNTL library, member LISTMC needs to be modified to indicate the name of your master catalog. Example 7-1 shows a sample LISTMC member.

*Example 7-1 Sample LISTMC member*

---

```

LISTC UCAT ALL OUTFILE(MCAT1) +
      CAT(SYST80.MASTER.CATALOG)
LISTC UCAT ALL OUTFILE(MCAT2) +
      CAT(CATALOG.OS390.MASTER)

```

---

For the first execution of the audit, change the CAT statements for MCAT1 and MCAT2 to your installation names. MCAT1 is the base master catalog in this example. Change the MCAT1 and MCAT2 catalog names before running the audit for the first time. After correcting any errors found as a result of the first audit, change the MCAT2 statement for all other master catalogs and re-execute the audit job.

After the audit has run, execute the CLIST named FIXAUDIT. The main panel, Choose Which Audit to Fix, displays the names of all of the various audits provided in *FastAudit/390*. To view the results of the FAUDITMC audit, select option 7 on this panel, as shown in Figure 7-2, and press Enter.

```
<V6 R1.01> FAST AUDIT. CHOOSE WHICH AUDIT TO FIX
COMMAND==> 7

1 MCDS    CREATED BY PROCS FAUDITMD, FAUDITCT OR FAUDITAL
2 BCDS    CREATED BY PROC FAUDITBD
3 OCDS    CREATED BY PROC FAUDITOD
4 TAPE    CREATED BY PROCS FAUDITTP, FAUDITZR OR FAUDITRM
5 CAT     CREATED BY PROC FAUDITCO
6 DASD    CREATED BY PROC FAUDITDA OR FAUDITDE
7 MASTCAT CREATED BY PROC FAUDITMC
8 CATRMM  CREATED BY PROC FAUDITCR
9 CATTMS  CREATED BY PROC FAUDITTC

ENTER END TO TERMINATE ONLINE FUNCTION
```

Figure 7-2 Primary Fix panel

The error summary panel is shown in Figure 7-3. This panel provides an opportunity to show the count of errors by error number and a brief explanation of each. In addition you may format fixes and troubleshooting commands to aid in repairing the problems.

```
<V6 R1.01> Row 1 of 9
FASTAUDIT MASTCAT SUMMARY
OPTION ==> SCROLL==> CSR
ENTER F TO DISPLAY FIXES OR B TO BROWSE ERRORS
S NUMBER COUNT MESSAGE
1011W 72 UCAT IN PRIMARY MASTER NOT CONNECTED TO SECONDARY
1012W 7 UCAT IN SECOND MASTER NOT CONNECTED TO PRIMARY
1013W 0 UCAT VOLSER IN PRIMARY NOT SAME IN SECONDARY
1015W 60 UNABLE TO ACCESS USER CAT DEFINED IN PRIMARY
1016W 1 UNABLE TO ACCESS USER CAT DEFINED IN SECONDARY
1017W 7366 HLQ IN PRIMARY MASTER NOT DEFINED TO SECONDARY
1018W 17 HLQ IN SECONDARY MASTER NOT DEFINED TO PRIMARY
1019W 45 HLQ IN PRIMARY MASTER CONFLICTS WITH SECONDARY
1020W 45 HLQ IN SECONDARY MASTER CONFLICTS WITH PRIMARY
***** Bottom of data *****
```

Figure 7-3 Master catalog error summary panel

## 7.2 Master catalog error messages

Now we will look at the error messages produced by the FAUDITMC shown in Figure 7-3, in detail. The error messages are listed, followed by ways to diagnose the error.

7.2.1 Master catalog audit error 1101W

The first error is 1101W, UCAT IN PRIMARY MASTER NOT CONNECTED TO SECONDARY. There are 72 occurrences of this error.

Diagnosing the error

Our first source of diagnostic information is the *FastAudit/390* User Guide. Looking up this error, you find the following explanation:

*FastAudit/390* has compared the user catalog connections in the primary master catalog against secondary master catalog and has found a user catalog not connected to the secondary master catalog.

In our example, 72 user catalogs have been defined to the primary master catalog through the IMPORT CONNECT command, but the secondary master catalog has no connection to it.

One obvious solution to this problem may be to IMPORT CONNECT the user catalogs to the secondary master catalog, however, there may be a reason the catalogs are not connected and you need to eliminate this possibility first.

Check with your system programmer to determine if the catalogs should be connected to the secondary master catalog. If there is not a reason why they should not be connected, the next step would be to ensure they are viable catalogs and that they are in use by the system. Other possibilities may be that the primary master catalog is connected to them by mistake, one or more catalogs are spelled incorrectly, or they were never IMPORT CONNECTed when they were created.

To view the error detail for this error, type B next to the error in the *FastAudit* MASTCAT Summary panel. The error detail is displayed in Figure 7-4. Our example shows the first catalog in the list, we have suppressed the other 71 entries.

|       |        |      |               |                      |                      |
|-------|--------|------|---------------|----------------------|----------------------|
| f1    | f2     | f3   | f4            | f5                   | f6                   |
| 1011W | SMS002 | 3390 | CAT.ICF.USER2 | CATALOG.OS390.MASTER | SYT80.MASTER.CATALOG |

Figure 7-4 Error message 1011W details

In the *FastAudit/390* User Guide, the Field Description Box provided in the documentation and shown in Table 7-1 provides the explanations of each field.

Table 7-1 Diagnostic 1011W field descriptions

| Dictionary Symbol | Example              |
|-------------------|----------------------|
| Field 1 / ERRNUM  | 1011W                |
| Field 2 / VOLSER  | SMS002               |
| Field 3 / DEVTYPE | 3390                 |
| Field 4 / UCAT    | CAT.ICF.USER2        |
| Field 5 / ALTMAS  | CATALOG.OS390.MASTER |
| Field 6 /MASTCAT  | SYT80.MASTER.CATALOG |

## Resolving the error

One possible resolution may be to IMPORT CONNECT the catalogs to the secondary master catalog if you find that there are no reasons why they should not be connected.

If your decision is to IMPORT CONNECT the catalogs to the secondary master catalog, *FastAudit/390* can do this with fix F1011WB. You can view all of the available fixes by typing an **F** next to the error message 1011W in the FastAudit MASTCAT Summary panel and pressing Enter. The FastAudit Fix Plan Member List is shown in Figure 7-5.

|                                                           |         |                                                   |       |
|-----------------------------------------------------------|---------|---------------------------------------------------|-------|
| FAST AUDIT FIX PLAN MEMBER LIST                           |         |                                                   |       |
| OPTION ==>                                                |         | SCROLL==> PAGE                                    |       |
| YOU MAY BROWSE (B), EDIT (E), FIX (F) OR VIEW (V) RESULTS |         |                                                   |       |
| COMMAND                                                   | MEMBER  | ACTION                                            | TITLE |
|                                                           | F1011WA | TSO LISTC FOR USER CAT NOT CONNECTED              |       |
| F                                                         | F1011WB | TSO IMPORT CONNECT USER CAT TO ALTERNATE MASTER   |       |
|                                                           | F1011WC | TSO EXPORT DISCONNECT USERCAT FROM PRIMARY MASTER |       |

Figure 7-5 Fast Audt Fix Plan Member List for error 1101W

To generate the JCL and fix statements, type **F** next to F1011WB in this panel and press Enter. The JCL and the fix statements generated for all of the catalogs in error is shown in Figure 7-6.

```
//DCDMC JOB      ,DCD.SIS - - OBC.07S',CLASS=A,MSGLEVEL=(1,1),
//      MSGCLASS=X,NOTIFY=DCD
//*
//STEP1 EXEC PGM=IKJEFT01,
//  DYNAMNBR=25,REGION=4096K
//SYSTSPRT DD  DSN=DCD.FIXAUDIT.F1011WB,DISP=SHR
//SYSTSIN DD  *
IMPORT OBJECTS(('CAT.ICF.USER2' VOLUME(SMS002) +
  DEVT(3390))) CONNECT +
  CATALOG('CATALOG.OS390.MASTER')
-----
IMPORT OBJECTS(('EMTST3.FAC.UCAT0027' VOLUME(SC0003) +
  DEVT(3390))) CONNECT +
  CATALOG('CATALOG.OS390.MASTER')
IMPORT OBJECTS(('USERCAT.EDPTST3' VOLUME(SMS001) +
  DEVT(3390))) CONNECT +
  CATALOG('CATALOG.OS390.MASTER')
IMPORT OBJECTS(('USERCAT.VIPLT80' VOLUME(IPLT80) +
  DEVT(3390))) CONNECT +
  CATALOG('CATALOG.OS390.MASTER')
```

*Figure 7-6 Generated fix error correction syntax*

The fix generated is an IMPORT CONNECT command for each catalog. This fix cannot automatically define the aliases to the user catalogs, and you will need to do this in a subsequent step. In addition, you should note that the fix commands are generated for all of the catalogs in the list. If you do not want to include all of the catalogs in the fix command, you can remove the unwanted IMPORT statements from the job by deleting the appropriate statements.

Execute the fix by typing **SUBMIT** on the command line and pressing Enter.

The results of the IMPORT CONNECT commands can be viewed in the product output data set pfx.FA6101.FAUDIT.F1011WB and in the sysout of the job.

After the IMPORT CONNECT command and all resulting connections are complete, you need to re-run the audit to determine the 1017W HLQ IN PRIMARYMASTER NOT DEFINED TO SECONDARY errors and define the alias records as required. An example of this fix is described later in this chapter.

## 7.2.2 Master catalog audit error 1012W

The next error message we will look at is Master Catalog Audit error 1012W UCAT IN SECONDARY MASTER NOT CONNECTED TO PRIMARY.

### Diagnosing the error

Looking up the error in the *FastAudit/390* User Guide, you find the following explanation:

*FastAudit/390* has compared the user catalog connections in the primary master catalog against secondary master catalog and has found a user catalog not connected to the primary master catalog.

This error results when a user catalog is connected to the alternate master catalog but not the primary. This may be the result of a test system catalog or another scenario and is intended. Check with your system programmer before proceeding with the fix.

When you type a B on the FastAudit MASTCAT Summary panel and press Enter, the error detail is displayed in Figure 7-7. Our example shows the first catalog in the list, and we have suppressed the other entries.

| f1    | f2     | f3   | f4            | f5                   | f6                    |
|-------|--------|------|---------------|----------------------|-----------------------|
| 1012W | NONSMS | 3390 | USERCAT.VCR01 | CATALOG.OS390.MASTER | SYST80.MASTER.CATALOG |

Figure 7-7 Error message 1012W detail

The Field Description Box provided in the documentation and shown in Table 7-2, provides the explanations of each field.

Table 7-2 Diagnostic 1012W field descriptions

| Dictionary symbol | Example               |
|-------------------|-----------------------|
| Field 1 /ERRNUM   | 1012W                 |
| Field 2 /VOLSER   | NONSMS                |
| Field 3 /DEVType  | 3390                  |
| Field 4 /UCAT     | USERCAT.VCR01         |
| Field 5 /ALTMAS   | CATALOG.OS390.MASTER  |
| Field 6 /MASTCAT  | SYST80.MASTER.CATALOG |

### Resolving the error

If you find that the catalogs should be connected to the primary master catalog, we can use *FastAudit/390* to generate the fix for the error. Type F next to the error message and press Enter to display the Fast Audit Fix Plan Member List.



Select FIX F1012WB TSO IMPORT CONNECT USER CAT TO PRIMARY MASTER by typing an F next to the FIX, as shown in Figure 7-8, and pressing Enter.

| FAST AUDIT FIX PLAN MEMBER LIST                           |         |                |                                               |
|-----------------------------------------------------------|---------|----------------|-----------------------------------------------|
| OPTION ==>                                                |         | SCROLL==> PAGE |                                               |
| YOU MAY BROWSE (B), EDIT (E), FIX (F) OR VIEW (V) RESULTS |         |                |                                               |
| COMMAND                                                   | MEMBER  | ACTION         | TITLE                                         |
|                                                           | F1012WA |                | TSO LISTC FOR USER CAT NOT CONNECTED          |
| F                                                         | F1012WB |                | TSO IMPORT CONNECT USER CAT TO PRIMARY MASTER |
|                                                           | F1012WC |                | TSO EXPORT DISCONNECT USERCAT FROM ALTMAS     |

Figure 7-8 Fast Audit Fix Plan Member List for F1012WB

To generate the JCL and fix statements, type F next to F1012WB in this panel and press Enter. The JCL and the fix statements generated for all of the catalogs in error is shown in Figure 7-9.

```

EDIT          CG.H091440                      Columns 00001 00072
Command ==>                                     Scroll ==> CSR
***** ***** Top of Data *****
000001 //CGA   JOB (0),COLGOR,
000002 //              CLASS=A,MSGCLASS=X,NOTIFY=COLGOR
000003 //*
000004 //STEP1 EXEC PGM=IKJEFT01,
000005 // DYNAMNBR=25,REGION=4096K
000009 //*
000010 //SYSPROC DD DSN=FAUDITX.QABETA.CLIST,DISP=SHR
000011 //*          ****
000012 //SYSOUT DD SYSOUT=*
000013 //SYSTSPRT DD DSN=CG.FIXAUDIT.F1012WB,DISP=SHR
000014 //SYSTSIN DD *
000015 IMPORT OBJECTS(('CAT.ICF.USER2' VOLUME(SMS002) +
000016 DEVT(3390))) CONNECT +
000017 CATALOG('SYSD80.MASTER.CATALOG')

```

Figure 7-9 JCL and FIX statements generated for error 1012W

The fix generated is an IMPORT CONNECT command for each catalog to the primary master catalog. Like error 1101W previously, this fix cannot automatically define the aliases to the user catalogs. You will need to do this in a subsequent step.

To execute this fix, type **SUBMIT** on the command line and press Enter. The results of the IMPORT CONNECT commands can be viewed in the product output data set pfx.FA6101.FAUDIT.F1012WB and in the sysout of the job.

A note of caution about the provided fix, F1012WC; TSO EXPORT DISCONNECT USERCAT FROM ALTMAS. This fix will EXPORT DISCONNECT the catalogs from an alternate master catalog. When this fix is executed, all of the aliases for those catalogs defined to the alternate master catalog are removed.

### 7.2.3 Master catalog audit error 1013W

The next error message is Master Catalog Audit error 1013W UCAT VOLSER IN PRIMARY NOT SAME IN SECONDARY.

#### Diagnosing the error

Looking up this error in the *FastAudit/390* User Guide, you find the following explanation.

*FastAudit/390* has compared the user catalog connections in the primary master catalog against secondary master catalog and has found a user catalog connected to both the primary master catalog and the secondary master catalog but the VOLSER of the user catalog is not the same.

This is one of those strange but true situations. The condition may happen because the IMPORT CONNECT of a user catalog does not verify the catalog exists on the referenced volume. The user must decide which master catalog has the correct connection. To view the error detail for this error, type B next to the error in the *FastAudit* MASTCAT Summary panel and press Enter.

The error detail is displayed in Figure 7-10.

|       |        |      |       |      |             |                |                       |
|-------|--------|------|-------|------|-------------|----------------|-----------------------|
| f1    | f2     | f3   | f4    | f5   | f6          | f7             | f8                    |
| 1013W | SCPMV5 | 3390 | EH001 | 3380 | CAT.DEXAMPL | RESCUE.MASTCAT | SYST80.MASTER.CATALOG |

Figure 7-10 Error detail for 1013W

In the *FastAudit/390* User Guide, the Field Description Box provided in the documentation and shown in Table 7-3 provides the explanations of each field.

Table 7-3 Diagnostic 1013W field descriptions

| Dictionary symbol | Example |
|-------------------|---------|
| Field 1 /ERRNUM   | 1013W   |
| Field 2 /VOLSER   | SCPMV5  |
| Field 3 /DEVTYPE  | 3390    |
| Field 4 /CVOLSER  | EH001   |

| Dictionary symbol | Example               |
|-------------------|-----------------------|
| Field 5 /CUNIT    | 3380                  |
| Field 6 /UCAT     | CAT.DEXAMPL           |
| Field 7 /ALTMAT   | RESCUE.MASTCAT        |
| Field 8 /MASTCAT  | SYST80.MASTER.CATALOG |

### Resolving the error

To resolve this error, you need to determine what the correct VOLSER is for the catalog and which catalog has it defined incorrectly.

You will need to disconnect the user catalog with the incorrect VOLSER and re-connect it again. Remember that when we disconnect the catalog, the alias definitions are removed. You will need to run this FIX to disconnect the catalog and re-connect it, re-run the audit and view the 1017W HLQ IN PRIMARY MASTER NOT DEFINED TO SECONDARY errors, and resolve the alias definitions with a fix for that error message. We cover error message 1017W and the fixes for that error later in this chapter.

To view the available fixes, type F next to the 1013W error and press Enter. The Fast Audit Fix Plan Member List is shown in Figure 7-11.

|                                                           |         |                |                                                     |
|-----------------------------------------------------------|---------|----------------|-----------------------------------------------------|
| <V6 R1.01>                                                |         | Row 1 of 3     |                                                     |
| FAST AUDIT FIX PLAN MEMBER LIST                           |         |                |                                                     |
| OPTION ==>                                                |         | SCROLL==> PAGE |                                                     |
| YOU MAY BROWSE (B), EDIT (E), FIX (F) OR VIEW (V) RESULTS |         |                |                                                     |
| COMMAND                                                   | MEMBER  | ACTION         | TITLE                                               |
|                                                           | F1013WA |                | TSO LISTC FOR USER CAT ON DIFFERENT VOL             |
|                                                           | F1013WB |                | TSO DISCONNECT/CONNECT TO CONFLICT VOL ON MASTER    |
|                                                           | F1013WC |                | TSO DISCONNECT/CONNECT TO PRIMARY VOL ON ALT MASTER |

Figure 7-11 Fast Audit Fix Plan Member List for F1013W

You need to run one of the following fixes depending on the catalog in error:

F1013WB TSO DISCONNECT/CONNECT TO CONFLICT VOL ON MASTER  
F1013WC TSO DISCONNECT/CONNECT TO CONFLICT VOL ON ALT MASTER

Select the appropriate fix and generate the fix statements by typing F next to the fix and pressing Enter. The JCL and FIX statements generated for all of the catalogs in error are shown in Figure 7-12. We used fix F1013WB in our example.

```

EDIT          CG.H122810                      Columns 00001 00072
Command ==>                                     Scroll ==> CSR
***** ***** Top of Data *****
000001 //CGA   JOB (0),COLGOR,
000002 //          CLASS=A,MSGCLASS=X,NOTIFY=COLGOR
000003 //*
000004 //STEP1 EXEC PGM=IKJEFT01,
000005 // DYNAMNBR=25,REGION=4096K
000009 //*
000010 //SYSPROC DD DSN=FAUDITX.QABETA.CLIST,DISP=SHR
000011 //*          ****
000012 //SYSOUT DD SYSOUT=*
000013 //SYSTSPRT DD DSN=CG.FIXAUDIT.F1013WB,DISP=SHR
000014 //SYSTSIN DD *
000015 EXPORT 'FACUCAT.EH.VSPCAT' DISCONNECT +
000016 CATALOG('SYSD80.MASTER.CATALOG')
000017 IMPORT OBJECTS(('FACUCAT.EH.VSPCAT' VOLUME(EHT001) +

```

Figure 7-12 JCL and FIX statements generated for F1013WB

Execute the fix by typing **SUBMIT** on the command line and pressing Enter. The results of the execution can be viewed in the product output data set pfx.FA6101.FAUDIT.F1013WB and in the sysout of the job. After the fix is executed, re-run the audit. Keep in mind that the primary catalog is used as a base line against all other master catalogs in the environment.

## 7.2.4 Master catalog audit error 1015W

The next error message is Master Catalog Audit error 1015W UNABLE TO ACCESS USER CAT DEFINED IN PRIMARY.

When *FastAudit/390* executes, it attempts to access each user catalog. If, for some reason, the user catalogs listed are not available to the *FastAudit/390* program and cannot be accessed, error message 1015W is generated.

### Diagnosing the error

Looking up this error in the *FastAudit/390* User Guide, you find the following explanation.

*FastAudit/390* has examined the contents of the VTOC that a user catalog was defined to and didn't find it or the volume was unavailable. Any access to the user catalog would fail. This same catalog might be reported in error 1016W from the Secondary Master.

To view the error detail for this error, type B next to the error in the FastAudit MASTCAT Summary panel and press Enter. The error detail is shown in

Figure 7-13. Our example shows the first catalog in the list, and we have suppressed the remaining entries.

| f1    | f2     | f3   | f4                    | f5                   | f6                    |
|-------|--------|------|-----------------------|----------------------|-----------------------|
| 1015W | SC0003 | 3390 | EMTST3.FAC.UCAT.IMBED | CATALOG.OS390.MASTER | SYST80.MASTER.CATALOG |

Figure 7-13 Error detail for 1015W

In the *FastAudit/390* User Guide, the Field Description Box provided in the documentation and shown in Table 7-4, provides the explanations of each field.

Table 7-4 Diagnostic 1015W Field descriptions

| Dictionary symbol | Example               |
|-------------------|-----------------------|
| Field 1 /ERRNUM   | 1015W                 |
| Field 2 /VOLSER   | SC0003                |
| Field 3 /DEVType  | 3390                  |
| Field 4 /UCAT     | EMTST3.FAC.UCAT.IMBED |
| Field 5 /ALTMAS   | CATALOG.OS390.MASTER  |
| Field 6 /MASTCAT  | SYST80.MASTER.CATALOG |

Possible explanations are that these catalogs could be empty or not on the volume specified. Remember that IMPORT CONNECT does not verify that the catalogs exist when it is executed.

A good starting point is to determine why the catalog cannot be accessed. Perhaps the volume is offline or the catalogs no longer exist. If you find that the catalogs need to be EXPORT DISCONNECTed from the master, you can use *FastAudit/390* fix F1015WB.

### Resolving the error

To view the available fixes, type **F** next to the error message and press Enter. The Fast Audit Fix Plan Member List is shown in Figure 7-14.

```

FAST AUDIT FIX PLAN MEMBER LIST
OPTION ==>                                SCROLL==>
PAGE
YOU MAY BROWSE (B), EDIT (E), FIX (F) OR VIEW (V) RESULTS
COMMAND MEMBER ACTION TITLE
          F1015WA      TSO LISTC FOR USER CAT NOT AVAILABLE ON MASTER
          F1015WB      TSO EXPORT DISCONNECT USERCAT FROM MASTER CAT

```

Figure 7-14 Fast Audit Fix Plan Member List for error 1015W

To generate the JCL and FIX statements, type **F** next to the F1015WB and press Enter. The JCL and fix statements generated are shown in Figure 7-15.

```

Command ==>                                Scroll ==> CSR
***** ***** Top of Data *****
000001 //CGA JOB (0),COLGOR,
000002 //          CLASS=A,MSGCLASS=X,NOTIFY=COLGOR
000003 //*
000004 //STEP1 EXEC PGM=IKJEFT01,
000005 // DYNAMNBR=25,REGION=4096K
000009 //*
000010 //SYSPROC DD DSN=FAUDITX.QABETA.CLIST,DISP=SHR
000011 //*          ****
000012 //SYSOUT DD SYSOUT=*
000013 //SYSTSPRT DD DSN=CG.FA6101.FAUDIT.FIXAUDIT.F1015WB,DISP=SHR
000014 //SYSTSIN DD *
000015 EXPORT 'FACUCAT.EH.VSPCAT' DISCONNECT +
000016 CATALOG('SYSD80.MASTER.CATALOG')

```

Figure 7-15 EXPORT DISCONNECT commands for errant catalogs

Execute the fix by typing **SUBMIT** on the command line and pressing Enter.

The results of the EXPORT DISCONNECT commands can be viewed in the product output data set pfx.FA6101.FAUDIT.F1015WB and in the sysout of the job. After this fix completes, rerun the audit before proceeding with correcting the remaining errors.

## 7.2.5 Master catalog audit error 1016W

The next error message we will look at is Master Catalog Audit error 1016W  
UNABLE TO ACCESS USER CAT DEFINED IN SECONDARY.

### Diagnosing the error

Looking up this error in the *FastAudit/390* User Guide, you find see the following explanation:

*FastAudit/390* has examined the contents of the VTOC that a user catalog was defined to and didn't find it or the volume was unavailable. Any access to the user catalog would fail. This same catalog might be reported in error 1015W from the Primary Master.

The same condition exists for the alternate master catalog as for the primary master catalog. The difference here is that it may be by design. The volume may not be available to the MVS image the user catalog is defined to. A good approach would be to discuss this with the system programmers to determine if this condition exists by design before proceeding with applying fix F1016WB.

To view the error detail for this error, type B next to the error in the FastAudit MASTCAT Summary panel and press Enter. The error detail is shown in Figure 7-16.

|       |        |      |       |                      |
|-------|--------|------|-------|----------------------|
| f1    | f2     | f3   | f4    | f5                   |
| 1016W | EH0001 | 3390 | CAT.A | CATALOG.OS390.MASTER |

Figure 7-16 Error detail for error 1016W

In the *FastAudit/390* User Guide, the Field Description Box provided in the documentation and shown in Table 7-5 provides the explanations of each field.

Table 7-5 Diagnostic 1016W field description

| Dictionary symbol | Example              |
|-------------------|----------------------|
| Field 1 /ERRNUM   | 1016W                |
| Field 2 /VOLSER   | EH0001               |
| Field 3 /DEVType  | 3390                 |
| Field 4 /UCAT     | CAT.A                |
| Field 5 /ALTMAS   | CATALOG.OS390.MASTER |

### Resolving the error

To view the available fixes, type F next to the error message and press Enter. The Fast Audit Fix Plan Member List is shown in Figure 7-17.

```

          FAST AUDIT FIX PLAN MEMBER LIST
OPTION ===>                                SCROLL==> PAGE
YOU MAY BROWSE (B), EDIT (E), FIX (F) OR VIEW (V) RESULTS
OMMAND MEMBER ACTION TITLE
      F1016WA      TSO LISTC FOR USER CAT NOT AVAILABLE ON ALTERNATE
      F1016WB      TSO EXPORT DISCONNECT USERCAT FROM ALTERNATE MASTER

```

Figure 7-17 Fast Audit Fix Plan Member List for 1016W

To generate the JCL and FIX statements, type **F** next to the F1016WB and press Enter. The JCL and FIX statements generated are shown in Figure 7-18.

```

EDIT      CG.H131340                                Columns 00001 00072
Command ===>                                Scroll ==> CSR
***** ***** Top of Data *****
000001 //CGA JOB (0),COLGOR,
000002 //          CLASS=A,MSGCLASS=X,NOTIFY=COLGOR
000003 //*
000004 //STEP1 EXEC PGM=IKJEFT01,
000005 // DYNAMNBR=25,REGION=4096K
000009 //*
000010 //SYSPROC DD DSN=FAUDITX.QABETA.CLIST,DISP=SHR
000011 //*          ****
000012 //SYSOUT DD SYSOUT=*
000013 //SYSTSPRT DD DSN=CG.FIXAUDIT.F1016WB,DISP=SHR
000014 //SYSTSIN DD *
000015 EXPORT 'DANSCAT' DISCONNECT +
000016 CATALOG('SYST80.MASTER.CATALOG')
000017 EXPORT 'DAWNS.CAT' DISCONNECT +
000018 CATALOG('SYST80.MASTER.CATALOG')
000019 EXPORT 'EMTST3.FAC.UCAT.IMBED' DISCONNECT +

```

Figure 7-18 EXPORT DISCONNECT statements generated for Fix F1016WB

Execute the fix by typing **SUBMIT** on the command line and pressing Enter.

The results of the EXPORT DISCONNECT commands can be viewed in the product output data set pfx.FIXAUDIT.F1016WB and in the sysout of the job. After this fix completes, rerun the audit before proceeding with correcting the remaining errors.

## 7.2.6 Master catalog audit error 1017W

The next error message is Master Catalog Audit Error 1017W HLQ IN PRIMARY MASTER NOT DEFINED TO SECONDARY.



# Diagnosing the error

Looking up this error in the *FastAudit/390* User Guide, you find the following explanation:

*FastAudit/390* has found a High Level Qualifier in the primary master catalog defined to a user catalog but the same HLQ is not found in the secondary master catalog.

HLQ is our abbreviation for User Catalog Alias, hereafter known as High Level Qualifier. Errors 1017W through 1020W should be examined only after the errors concerning user catalog definitions are finished. This is because the EXPORT DISCONNECT fixes removes the defined aliases. Furthermore, if DEFINE ALIAS is run as a fix and the catalogs are then disconnected, the fix disappears.

*FastAudit/390* has found a HLQ that is defined for a user catalog in the primary Master Catalog, but that same HLQ is not defined to the user catalog from secondary Master Catalog. This can cause serious problems because an application running on the secondary system may be able to catalog data sets in the master catalog. The same application running on the primary system catalogs the same data set in a user catalog. Not a good situation to be in.

To view the error detail for this error, type B next to the error in the FastAudit MASTCAT Summary panel and press Enter. The error detail is shown in Figure 7-19.

|          |           |                      |                       |
|----------|-----------|----------------------|-----------------------|
| f1       | f2        | f3                   | f4                    |
| 1017W R1 | DAWNS.CAT | CATALOG.OS390.MASTER | SYST80.MASTER.CATALOG |

Figure 7-19 Error detail for 1017W

In the *FastAudit/390* User Guide, the Field Description Box provided in the documentation and shown in Table 7-6 provides the explanations of each field.

Table 7-6 .Diagnostic 1017W field descriptions

| Dictionary symbol | Example               |
|-------------------|-----------------------|
| Field 1 /ERRNUM   | 1017W                 |
| Field 2 /hlq      | R1                    |
| Field 3 /UCAT     | DAWNS.CAT             |
| Field 4 /ALTMAST  | CATALOG.OS390.MASTER  |
| Field 5 /MASTCAT  | SYST80.MASTER.CATALOG |

*FastAudit/390* provides two choices to fix the problem:

F1017WB      TSO DEFINE ALIAS TO UCAT FOR ALTERNATE MASTER

F1017WC      TSO DELETE ALIAS FROM THE MASTER CATALOG

The F1017WB may be the preferred fix to re-define the alias records to the alternate master catalog. However, your environment may be setup so that the alias definitions are not in the alternate master catalog by design. It is important to speak with the system programmers and understand the design of the system before proceeding with F1017WB.

## Resolving the error

To view the available fixes, type **F** next to the error message and press Enter. The Fast Audit Fix Plan Member List is shown in Figure 7-20.

|                                                           |         |                                               |       |
|-----------------------------------------------------------|---------|-----------------------------------------------|-------|
| FAST AUDIT FIX PLAN MEMBER LIST                           |         |                                               |       |
| OPTION ==>                                                |         | SCROLL==>                                     |       |
| PAGE                                                      |         |                                               |       |
| YOU MAY BROWSE (B), EDIT (E), FIX (F) OR VIEW (V) RESULTS |         |                                               |       |
| COMMAND                                                   | MEMBER  | ACTION                                        | TITLE |
|                                                           | F1017WA | TSO LISTC FOR HLQ IN MASTER AND ALTMAS        |       |
|                                                           | F1017WB | TSO DEFINE ALIAS TO UCAT FOR ALTERNATE MASTER |       |
|                                                           | F1017WC | TSO DELETE ALIAS FROM THE MASTER CATALOG      |       |

*Figure 7-20 Fast Audit Fix Plan Member List for error 1017W*

To generate the JCL and FIX statements, type **F** next to the F1017WB and press Enter. The JCL and FIX statements generated are shown in Figure 7-21.

```

EDIT          CG.H132930                      Columns 00001 00072
Command ==>                                     Scroll ==> CSR
***** ***** Top of Data *****
000001 //CGA   JOB (0),COLGOR,
000002 //          CLASS=A,MSGCLASS=X,NOTIFY=COLGOR
000003 //*
000004 //STEP1 EXEC PGM=IKJEFT01,
000005 //  DYNAMNBR=25,REGION=4096K
000006 //* CHANGE THE FOLLOWING STATEMENT TO
000007 //* DSN=PPFX.CLIST WHERE PPFX IS THE PREFIX ASSIGNED TO
000008 //* THE FASTAUDIT PRODUCT
000009 //*
000010 //SYSPROC DD DSN=FAUDITX.QABETA.CLIST,DISP=SHR
000011 //*          ****
000012 //SYSOUT DD SYSOUT=*
000013 //SYSTSPRT DD DSN=CG.FIXAUDIT.F1017WB,DISP=SHR
000014 //SYSTSIN DD *
000015 DEFINE ALIAS (NAME('B9') +
000016 RELATE('USERCAT.UCAT2')) +
000017 CATALOG('SYST80.MASTER.CATALOG')
000018 DEFINE ALIAS (NAME('HFSTST') +
000019 RELATE('SYST80.MASTER.CATALOG')) +

```

Figure 7-21 *DEFINE ALIAS commands for F1017WB*

Execute the fix by typing **SUBMIT** on the command line and pressing Enter. The results of the DEFINE ALIAS commands can be viewed in the product output data set pfx.FIXAUDIT.F1017WB or in the sysout of the job. After the fix completes, rerun the audit before proceeding with correcting the remaining errors.

## 7.2.7 Master catalog audit error 1018W

The next error we will look at is the Master Catalog Audit Error 1018W HLQ IN SECONDARY MASTER NOT DEFINED TO PRIMARY.

### Diagnosing the error

Looking up this error in the *FastAudit/390 User Guide*, you find the following explanation.

*FastAudit/390* has found a High Level Qualifier in the secondary master catalog defined to a user catalog but the same HLQ is not found in the primary master catalog.

This is a similar error to 1017W, but is reporting on the alternate master catalog. To view the error detail for this error, type B next to the error in the FastAudit

MASTCAT Summary panel and press Enter. The error detail is shown in Figure 7-22.

|       |        |                 |                      |                       |
|-------|--------|-----------------|----------------------|-----------------------|
| f1    | f2     | f3              | f4                   | f5                    |
| 1018W | ADB210 | USERCAT.VZ1DB21 | CATALOG.OS390.MASTER | SYST80.MASTER.CATALOG |

Figure 7-22 Display of detail for error 1018W

In the *FastAudit/390* User Guide, the Field Description Box provided in the documentation and shown in Table 7-7 provides an explanation of each field.

Table 7-7 Diagnostic 1018W Field descriptions

| Dictionary symbol | Example               |
|-------------------|-----------------------|
| Field 1 /ERRNUM   | 1018W                 |
| Field 2 /hlq      | ADB210                |
| Field 3 /UCAT     | USERCAT.VZ1DB21       |
| Field 4 /ALTMAS   | CATALOG.OS390.MASTER  |
| Field 5 /MASTCAT  | SYST80.MASTER.CATALOG |

### Resolving the error

To view the available fixes, type **F** next to the error message and press Enter. The Fast Audit Fix Plan Member List is shown in Figure 7-23.

|                                                           |         |                                             |       |
|-----------------------------------------------------------|---------|---------------------------------------------|-------|
| FAST AUDIT FIX PLAN MEMBER LIST                           |         |                                             |       |
| OPTION ===>                                               |         | SCROLL==>                                   |       |
| PAGE                                                      |         |                                             |       |
| YOU MAY BROWSE (B), EDIT (E), FIX (F) OR VIEW (V) RESULTS |         |                                             |       |
| COMMAND                                                   | MEMBER  | ACTION                                      | TITLE |
|                                                           | F1018WA | TSO LISTC FOR HLQ IN MASTER AND ALTMAS      |       |
|                                                           | F1018WB | TSO DEFINE ALIAS TO UCAT FOR MASTER CATALOG |       |
|                                                           | F1018WC | TSO DELETE ALIAS FROM THE ALTERNATE MASTER  |       |

Figure 7-23 FAST AUDIT Fix Plan Member List for error 1018W

To generate the JCL and FIX statements, type **F** next to the F1018WB and press Enter. The JCL and fix statements generated are shown in Figure 7-24.

```

EDIT          CG.H133959                      Columns 00001 00072
Command ==>                               Scroll ==> CSR
***** ***** Top of Data *****
000001 //CGA   JOB (0),COLGOR,
000002 //          CLASS=A,MSGCLASS=X,NOTIFY=COLGOR
000003 //*
000004 //STEP1 EXEC PGM=IKJEFT01,
000005 //  DYNAMNBR=25,REGION=4096K
000006 //* CHANGE THE FOLLOWING STATEMENT TO
000007 //* DSN=PPFX.CLIST WHERE PPFX IS THE PREFIX ASSIGNED TO
000008 //* THE FASTAUDIT PRODUCT
000009 //*
000010 //SYSPROC  DD  DSN=FAUDITX.QABETA.CLIST,DISP=SHR
000011 //*          ****
000012 //SYSOUT   DD  SYSOUT=*
000013 //SYSTSPRT DD  DSN=CG.FIXAUDIT.F1018WB,DISP=SHR
000014 //SYSTSIN DD  *
000015 DEFINE ALIAS (NAME('$10313') +
000016   RELATE('FACUCAT.EH.MANY.ALIAII2')) +
000017   CATALOG('SYSD80.MASTER.CATALOG')
000018 DEFINE ALIAS (NAME('$10421') +
000019   RELATE('FACUCAT.EH.MANY.ALIAII2')) +

```

Figure 7-24 *DEFINE ALIAS statements generated for fix F1018WB*

Execute the fix by typing **SUBMIT** on the command line and pressing Enter.

The results of the DEFINE ALIAS commands can be viewed in the product output data set pfx.FIXAUDIT.F1018WB and in the sysout of the job. After the fix completes, rerun the audit before proceeding with correcting the remaining errors.

## 7.2.8 Master catalog audit error 1019W

The next error message we will look at is Master Catalog Audit Error 1019W  
HLQ IN PRIMARY MASTER CONFLICTS WITH SECONDARY.

### Diagnosing the error

Looking up this error in the *FastAudit/390* User Guide, you find the following explanation.

*FastAudit/390* has found a High Level Qualifier in the primary and secondary master catalogs, but they are connected to two different user catalogs. This error will be repeated in error 1020W for the secondary master catalog.

To view the error detail for this error, type B next to the error in the FastAudit MASTCAT Summary panel and press Enter. The error detail is shown in Figure 7-25.

|       |     |                 |                 |                       |
|-------|-----|-----------------|-----------------|-----------------------|
| f1    | f2  | f3              | f4              | f5                    |
| 1019W | AOP | USERCAT.VIPLT80 | USERCAT.VZ1RES1 | SYST80.MASTER.CATALOG |

Figure 7-25 Error detail for error 1019W

In the *FastAudit/390* User Guide, the Field Description Box provided in the documentation and shown in Table 7-8 provides the explanations of each field.

Table 7-8 Diagnostic 1019W Field descriptions

| Dictionary symbol | Example               |
|-------------------|-----------------------|
| Field 1 /ERRNUM   | 1019W                 |
| Field 2 /HLQ      | AOP                   |
| Field 3 /UCAT     | USERCAT.VIPLT80       |
| Field 4 /CUCAT    | USERCAT.VZ1RES1       |
| Field 5 /MASTCAT  | SYST80.MASTER.CATALOG |

You will need to determine which connection is in error and select the *FastAudit/390* fix to correct the condition. Fix F1019WB deletes the alias and defines it to the conflicting User Catalog.

**Resolving the error**

To view the available fixes, type F next to the error message and press Enter. The Fast Audit Fix Plan Member List is shown in Figure 7-26.

|                                                           |         |                                               |       |
|-----------------------------------------------------------|---------|-----------------------------------------------|-------|
| FAST AUDIT FIX PLAN MEMBER LIST                           |         |                                               |       |
| OPTION ==>                                                |         | SCROLL==>                                     |       |
| PAGE                                                      |         |                                               |       |
| YOU MAY BROWSE (B), EDIT (E), FIX (F) OR VIEW (V) RESULTS |         |                                               |       |
| COMMAND                                                   | MEMBER  | ACTION                                        | TITLE |
|                                                           | F1019WA | TSO LISTC FOR HLQ IN MASTER                   |       |
|                                                           | F1019WB | TSO DELETE ALIAS AND DEFINE ALIAS TO NEW UCAT |       |

Figure 7-26 Fast Audit Fix Plan Member List for error 1019W

To generate the JCL and FIX statements, type F next to the F1019WB member and press Enter. The JCL and FIX statements generated are shown in Figure 7-27.

```

EDIT          CG.H13531                      Columns 00001 00072
Command ==>                                Scroll ==> CSR
***** ***** Top of Data *****
000001 //CGA   JOB (0),COLGOR,
000002 //              CLASS=A,MSGCLASS=X,NOTIFY=COLGOR
000003 /*
000004 //STEP1 EXEC PGM=IKJEFT01,
000005 // DYNAMNBR=25,REGION=4096K
000006 /* CHANGE THE FOLLOWING STATEMENT TO
000007 /* DSN=PPFX.CLIST WHERE PPFX IS THE PREFIX ASSIGNED TO
000008 /* THE FASTAUDIT PRODUCT
000009 /*
000010 //SYSPROC DD DSN=FAUDITX.QABETA.CLIST,DISP=SHR
000011 /*              ****
000012 //SYSOUT DD SYSOUT=*
000013 //SYSTSPRT DD DSN=CG.FIXAUDIT.F1019WB,DISP=SHR
000014 //SYSTSIN DD *
000015 /* DELETE FROM USERCAT.VIPLD80 /*
000016 DELETE 'AOP' ALIAS +
000017 CATALOG('SYSD80.MASTER.CATALOG')
000018 DEFINE ALIAS (NAME('AOP') +
000019 RELATE('USERCAT.VIPLT80')) +

```

Figure 7-27 DELETE/DEFINE ALIAS records for alias records in error

Execute the fix by typing **SUBMIT** on the command line and pressing Enter.

The results of the DELETE/DEFINE Alias commands can be viewed in the product output data set pfx.FIXAUDIT.F1019WB or in the sysout of the job.

After this fix completes, rerun the audit before proceeding with correcting the remaining errors.

## 7.2.9 Master catalog audit error 1020W

The next error message we will look at is Master Catalog Audit Error 1020WHLQ IN SECONDARY MASTER CONFLICTS WITH PRIMARY.

### Diagnosing the error

Looking up this error in the *FastAudit/390* User Guide, you find the following explanation.

*FastAudit/390* has found a High Level Qualifier in the primary and secondary master catalogs, but they are connected to two different user catalogs. This error will be repeated in error 1019W for the primary master catalog. Determine which connection is in error and format and fix errors as appropriate

This error is identical to error 1019W except the master catalog field will be repeated in error 1020W for the secondary master catalog. If you received this, determine which connection is in error and format and fix errors as appropriate. Fix F1020WB will delete the alias and define it to the conflicting User Catalog.

To view the error detail, type B next to the error in the FastAudit MASTCAT Summary panel and press Enter. The error detail is shown in Figure 7-28.

|       |     |                 |                 |                      |
|-------|-----|-----------------|-----------------|----------------------|
| f1    | f2  | f3              | f4              | f5                   |
| 1020W | AOP | USERCAT.VIPLT80 | USERCAT.VZ1RES1 | CATALOG.OS390.MASTER |

Figure 7-28 Error detail for error 1020W

In the *FastAudit/390 User Guide*, the Field Description Box provided in the documentation and shown in Table 7-9 provides an explanation of each field.

Table 7-9 Diagnostic 1020W field descriptions

| Dictionary symbol | Example              |
|-------------------|----------------------|
| Field 1 /ERRNUM   | 1020W                |
| Field 2 /HLQ      | AOP                  |
| Field 3 /UCAT     | USERCAT.VIPLT80      |
| Field 4 /CUCAT    | USERCAT.VZ1RES1      |
| Field 5 /MASTCAT  | CATALOG.OS390.MASTER |

### Resolving the error

To view the available fixes, type F next to the error message and press Enter. The Fast Audit Fix Plan Member List is shown in Figure 7-29.

|                                                           |         |                                               |       |
|-----------------------------------------------------------|---------|-----------------------------------------------|-------|
| FAST AUDIT FIX PLAN MEMBER LIST                           |         |                                               |       |
| OPTION ==>                                                |         | SCROLL==>                                     |       |
| PAGE                                                      |         |                                               |       |
| YOU MAY BROWSE (B), EDIT (E), FIX (F) OR VIEW (V) RESULTS |         |                                               |       |
| OMMAND                                                    | MEMBER  | ACTION                                        | TITLE |
|                                                           | F1020WA | TSO LISTC FOR HLQ IN MASTER                   |       |
|                                                           | F1020WB | TSO DELETE ALIAS AND DEFINE ALIAS TO NEW UCAT |       |

Figure 7-29 Fast Audit Fix Plan Member List for error 1020W

To generate the JCL and FIX statements, type F next to the F1020WB member and press Enter. The JCL and fix statements generated are shown in Figure 7-30.



```

EDIT          CG.H14159                      Columns 00001 00072
Command ==>                                     Scroll ==> CSR
***** ***** Top of Data *****
000001 //CGA   JOB (0),COLGOR,
000002 //          CLASS=A,MSGCLASS=X,NOTIFY=COLGOR
000003 //*
000004 //STEP1 EXEC PGM=IKJEFT01,
000005 // DYNAMNBR=25,REGION=4096K
000009 //*
000010 //SYSPROC DD DSN=FAUDITX.QABETA.CLIST,DISP=SHR
000011 //*          ****
000012 //SYSOUT DD SYSOUT=*
000013 //SYSTSPRT DD DSN=CG.FIXAUDIT.F1020WB,DISP=SHR
000014 //SYSTSIN DD *
000015 DELETE 'AOP' ALIAS + /* FROM USERCAT.VIPLT80 *
000016 LOG') LOG('SYST80.MASTER.CATALOG')
000017 DEFINE ALIAS (NAME('AOP') +
000018 RELATE('USERCAT.VIPLD80')) +
000019 CATALOG('SYST80.MASTER.CATALOG')

```

Figure 7-30 *DEFINE and DELETE ALIAS statements for fix F1020WB*

Execute the fix by typing **SUBMIT** on the command line and pressing Enter.

The results of the DELETE/DEFINE ALIAS commands can be viewed in the product output data set pfx.FIXAUDIT.F1020WB and in the sysout of the job.

After the fix completes, rerun the audit before proceeding with correcting the remaining errors or before running the User Catalog Audits described in the next chapter.

## 7.3 User catalog audits

The User Catalog Audits are provided in the *FastAudit/390* selectable unit, Catalog & DASD Audit. The purpose of the User Catalog Audits is to determine the general health of each User Catalog. *FastAudit/390* reads the contents of the user catalogs for many different audits. As a by-product of the audits between DASD, DFSMSHsm and Tape, the contents of the catalogs have been scanned and questionable information is reported. Among these reports are entries that have no alias defined to the master catalog, empty GDG sphere records, and entries defined to DASD volumes that are not in the system. Since the contents of the User Catalogs are audited against HSM, DASD, and Tape Catalogs, it is important to make sure they are healthy.

Before executing the FAUDITCO procedure, we recommend that the Master Catalog audits are executed and any errors resolved to your satisfaction.

Prior to executing the audit JCL, the following symbolic parameters need to be replaced with your customized values, as shown in our JCL example in Figure 7-31. Instructions on where to find the audit JCL and how to customize the product are documented in the *FastAudit/390 Installation and Maintenance Guide*. The following symbolic values are replaced by your values in the audit JCL:

|                |                                                                                                                                                                                                                                                                                                                                                                                    |
|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>PPFX</b>    | Product Prefix as installed. For example, if the product has been installed with MSC.FAUDIT as a prefix to the libraries, then PPFX=MSC.FAUDIT must be coded in the EXEC PROC statement.                                                                                                                                                                                           |
| <b>PFX</b>     | Prefix represents the Runtime Prefix the user has chosen for this cataloged procedure to allocate work files that contain the audit results. Depending on the SMS treatment of new allocations, the work data sets may be deleted within a day or so. Choose a prefix that is retained for an appropriate amount of time so the results of the audit may be reviewed and repaired. |
| <b>PUNIT</b>   | Permanent unit for allocating data sets containing the audit results. The default is SYSDA. If this unit allocates on a work volume or another volume that scratches data in a short period of time, you may want to change it to a different unit name to allocate on a volume with a longer retention period.                                                                    |
| <b>SOUT</b>    | SYSOUT class is desirable for your installation.                                                                                                                                                                                                                                                                                                                                   |
| <b>ATP</b>     | ATP is allocation type TRK or CYL. Larger shops should use CYL.                                                                                                                                                                                                                                                                                                                    |
| <b>WRKUNIT</b> | Work unit for allocating temporary data sets that are deleted at the end of the job. The default is SYSDA. If your installation has a different standard, code the different name.                                                                                                                                                                                                 |
| <b>LOAD</b>    | Load library as installed for <i>FastAudit/390</i> . This library must be APF authorized.                                                                                                                                                                                                                                                                                          |

Figure 7-31 shows the symbolic parameters in the JCL used to execute procedure FAUDITCO. Our sample shows load library FAUDITX.QABETA.LOAD, product prefix FAUDITX.QABETA and run time prefix DES2.

```

000007 //FAUDITCO PROC PFX=PFX,          <==RUN TIME PREFIX
000008 //* FASTAUDIT <V6 R1.01> SEPTEMBER, 2001
000009 //          PPFX=PPFX,              <==PRODUCT PREFIX
000010 //          PUNIT=SYSDA,              <==UNIT FOR PERMANENT ALLOCATION
000011 //          SOUT=*,                   <==PREFERED SYSOUT CLASS
000012 //          WRKUNIT=SYSDA,           <==UNIT FOR TEMPORARY ALLOCATION
000013 //          ATP=TRK,                   <==TYPE OF ALLOCATION TRK/CYL
000014 //          PRI=100,                  <==PRIMARY ALLOCATION FOR FAUDIT
000015 //          SEC=100,                  <==SECONDARY ALLOCATION FAUDIT
000016 //          LOAD=PPFX.LOAD           <==LOAD LIBRARY AS INSTALLED
000017 //*
- - - - -
000119 //PROC1      EXEC FAUDITCO,
000120 //              LOAD=FAUDITX.QABETA.LOAD,
000121 //              PPFX=FAUDITX.QABETA,
000122 //              PFX=DES2

```

Figure 7-31 Cataloged procedure FAUDITCO showing symbolic substitutions

In the product CNTL library, member XCLUDCO (Figure 7-32) provides a means of excluding catalogs from the audit. To exclude a catalog, type the data set name of the catalog immediately following the XCLUDES FOLLOW statement. If no catalogs are to be excluded, no entry is needed, exit this member (PF3) without changing it.

```

BROWSE    FAUDITX.QABETA.CNTL(XCLUDCO) - 01.00      Line 00000000 Col 001 080
Command ==>                                         Scroll ==> CSR
***** Top of Data *****
XCLUDES FOLLOW
***** Bottom of Data*****

```

Figure 7-32 Member XCLUDCO used to exclude user catalogs from the audit

Figure 7-33 shows the same member with user catalogs PROD.UCAT1 and PROD.UCAT2 being excluded from the audit. All catalogs except PROD.UCAT1 and PROD.UCAT2 are included in the audit.

```

BROWSE    FAUDITX.QABETA.CNTL(XCLUDCO) - 01.00      Line 00000000 Col 001 080
Command ==>                                         Scroll ==> CSR
***** Top of Data *****
XCLUDES FOLLOW
PROD.UCAT1
PROD.UCAT2
***** Bottom of Data*****

```

Figure 7-33 Member XCLUDCO with user catalogs to exclude from audit

Conversely, if the user wants to audit a specific catalog or a list of catalogs, the XCLUDES FOLLOW statement can be changed to ICLUDES FOLLOW and the list of catalog names following the statement are included in the audit. Figure 7-34 shows the XCLUDCO member with catalogs PROD.UCAT1 and PROD.UCAT2 included.

```

BROWSE    FAUDITX.QABETA.CNTL(XCLUDCO) - 01.00      Line 00000000 Col 001 080
Command ==>                                         Scroll ==> CSR
***** Top of Data *****
INCLUDES FOLLOW
PROD.UCAT1
PROD.UCAT2
***** Bottom of Data*****

```

Figure 7-34 Member XCLUDCO used to include user catalogs in the audit

After the audit has run, execute the CLIST FIXAUDIT. This clist displays the main panel, Fast Audit Choose Which Audit To Fix, the various audits available. In this case it is option 5, CAT as shown in Figure 7-35.

```

<V6 R1.01> FAST AUDIT. CHOOSE WHICH AUDIT TO FIX
COMMAND==> 5

1  MCDS   CREATED BY PROCS FAUDITMD, FAUDITCT OR FAUDITAL
2  BCDS   CREATED BY PROC FAUDITBD
3  OCDS   CREATED BY PROC FAUDITOD
4  TAPE   CREATED BY PROCS FAUDITTP, FAUDITZR OR FAUDITRM
5  CAT     CREATED BY PROC FAUDITCO
6  DASD   CREATED BY PROC FAUDITDA OR FAUDITDE
7  MASTCAT CREATED BY PROC FAUDITMC
8  CATRMM  CREATED BY PROC FAUDITCR
9  CATTMS  CREATED BY PROC FAUDITTC

ENTER  END  TO TERMINATE ONLINE FUNCTION

```

Figure 7-35 Primary Fix panel

To view the results of the FAUDITCO audit procedure, select option 5 on this panel and press Enter to display the FastAudit CAT Summary panel as shown in Figure 7-36.

```

<V6 R1.01> Row 1 of 11
FASTAUDIT CAT SUMMARY
OPTION ==> SCROLL==> CSR
ENTER F TO DISPLAY FIXES OR B TO BROWSE ERRORS
S NUMBER  COUNT  MESSAGE
1001W    50082  DATA SET IS IN CATALOG THAT HAS NO CONNECTOR
1002W    27319  GENERATION DATA SET HAS NO ENTRIES
1003W         0  DATA SET IS ON DASD VOLUME WHICH IS VARIED OFF LINE
1004W     5346  DATA SET IS ON DASD VOLUME WHICH IS NOT ON SYSTEM
1005W     7381  NODE IS CONNECTED BUT HAS NO CATALOGED data sets
1005V         4  NODE IS CONNECTED TO CATALOG THAT CANNOT BE ACCESSED
1006W         0  DASD VOLUME IS ONLINE BUT HAS NO CATALOGED DATA SETS
1007W         70  CATALOG EXISTS BUT HAS NO NODES CONNECTED TO IT
1008W         19  CATALOG IS DEFINED BUT CANNOT BE PROCESSED
1009W        754  GDG SPHERE RECORD HAS NOSCRATCH PARAMETER
1010W         1  GDG SPHERE RECORD HAS EMPTY PARAMETER

```

Figure 7-36 Catalog error summary panel

### 7.3.1 Catalog audit error 1001W

The first error we will look at is Catalog Audit Error 1001W DATA SET IS IN CATALOG THAT HAS NO CONNECTOR.

### Diagnosing the error

Looking up this error in the *FastAudit/390* User Guide, you find the following explanation.

The catalog extraction has detected a data set in a catalog whose entry is not connected to the master catalog on this system with a hi-node alias. Access to this data set would require a STEPCAT or JOBCAT. It is also possible in a shared DASD environment that all hi-level nodes are not connected to all master catalogs. The data set cannot be accessed from the system on which *FastAudit/390* ran. Check all connections from the master catalogs for this data set.

Ensure that the audit has executed on the correct system and check with the system programmers to determine if the missing alias entries were by design or are truly in error.

To view the error detail, type B next to the error in the FastAudit MASTCAT Summary panel and press Enter. The error detail is displayed in Figure 7-37.

|       |                    |        |            |
|-------|--------------------|--------|------------|
| f1    | f2                 | f3     | f4         |
| 1001W | APPL1.GDG.G0002V00 | MVS001 | UCAT.APPL1 |

Figure 7-37 Error detail display for audit error 1011W

In the *FastAudit* User Guide, the Field Description Box provided in the documentation and shown in Table 7-10 provides an explanation of the fields.

Table 7-10 Diagnostic 1001W field descriptions

| Dictionary symbol | Example            |
|-------------------|--------------------|
| FIELD 1 /ERRNUM   | 1001W              |
| FIELD 2 /TDSN     | APPL1.GDG.G0002V00 |
| FIELD 3 /VOLSER   | MVS001             |
| FIELD 4 /CATALOG  | UCAT.APPL1         |

These error conditions can be common in sites where systems have been cloned for testing, or workload has been farmed out to an outsourcer and the alias nodes were deleted from the catalogs before the data set entries were cleaned up.

### Resolving the error

You can view all of the available fixes by typing an F next to the error message 1001W and pressing Enter. The Fast Audit Fix Plan Member List is displayed in Figure 7-38.

If you determine that the data set records are no longer valid, you can delete the list of data sets using the fix F1001WB.

```
FAST AUDIT FIX PLAN MEMBER LIST
OPTION ==>                                SCROLL==> PAGE
YOU MAY BROWSE (B), EDIT (E), FIX (F) OR VIEW (V) RESULTS
COMMAND MEMBER ACTION TITLE
      F1001WA      TSO LISTC OF DATA HAVING NO CATALOG CONNECTOR
      F1001WB      TSO DELETE NOSCRATCH FOR DSN MISSING CONNECTOR
```

Figure 7-38 Fast Audit Fix Plan Member List for error 1001W

To generate the JCL and fix statements, type F next to the F1001WB member in this panel and press Enter. The JCL and the fix statements generated for the data sets in error are shown in Figure 7-39.

```
EDIT      CG.H160333                      Columns 00001 00072
Command ==>                                Scroll ==> CSR
***** ***** Top of Data *****
000001 //CGA JOB (0),COLGOR,
000002 //          CLASS=A,MSGCLASS=X,NOTIFY=COLGOR
000003 //*
000004 //STEP1 EXEC PGM=IKJEFT01,
000005 // DYNAMNBR=25,REGION=4096K
000009 //*
000010 //SYSPROC DD DSN=FAUDITX.QABETA.CLIST,DISP=SHR
000011 //*          ****
000012 //SYSOUT DD SYSOUT=*
000013 //SYSTSPRT DD DSN=CG.FIXAUDIT.F1001WB,DISP=SHR
000014 //SYSTSIN DD *
000015 DEL 'ALU.BKP.MFST' +
000016 NSCR +
000017 CATALOG('CAT.ICF.USER2')
000018 DEL 'ALU.TEST.BACKUP' +
000019 NSCR +
```

Figure 7-39 DELETE NOSCRATCH commands generated for F1001WB

Execute the fix by typing **SUBMIT** on the command line and pressing Enter.

The results of the DEL NOSCRATCH commands can be viewed in the product output data set pfx.FIXAUDIT.F1011WB and in the sysout of the job.

### 7.3.2 Catalog audit error 1002W

The next error message we will look at is Catalog Audit Error 1002W  
GENERATION DATA SET HAS NO ENTRIES.

#### Diagnosing the error

Looking up this error in the *FastAudit* User Guide, you find the following explanation.

The catalog extraction has detected an empty generation sphere record. These records may be deleted to 'clean up' the catalog. However exercise caution because many applications may 'empty' a GDG when it reaches its limit and the GDG's existence may be necessary for successful completion of an application.

To view the error detail, type B on the FastAudit CAT Summary panel next to the error and press Enter. The error detail is displayed in Figure 7-40.

|       |                 |            |     |         |           |         |
|-------|-----------------|------------|-----|---------|-----------|---------|
| f1    | f2              | f3         | f4  | f5      | f6        | f7      |
| 1002W | USERID1.EXTRACT | ICFCAT.TSO | 005 | NOEMPTY | NOSCRATCH | 1996034 |

Figure 7-40 Error detail display for audit error 1002W

The Field Description Box provided in the documentation and shown in Table 7-11 provides an explanation of each field.

Table 7-11 Diagnostic 1002W field descriptions

| Dictionary symbol          | Example         |
|----------------------------|-----------------|
| FIELD 1 /ERRNUM            | 1002W           |
| FIELD 2 /TDSN              | USERID1.EXTRACT |
| FIELD 3 /CATALOG           | ICFCAT.TSO      |
| FIELD 4 /LIMIT             | 005             |
| FIELD 5 /EMPTY             | NOEMPTY         |
| FIELD 6 /SCRATCH           | NOSCRATCH       |
| FIELD 7 NONE (create date) | 1996034         |



This condition may not be an error. It may be common for applications to delete all of the GDG's in the base and create new entries the following cycle. Before attempting to delete empty GDG bases, you will need to determine which bases are truly obsolete and which bases may continue to be used.

## Resolving the error

If you find that some or all of the empty GDG bases need to be deleted, you can use the fix provided by *FastAudit/390*. To view the list of available fixes, type **F** next to the error message and press Enter. The Fast Audit Fix Plan Member List is displayed in Figure 7-41.

```
V6 R1.01>                                     Row 1 of 1
      FAST AUDIT FIX PLAN MEMBER LIST
OPTION ==>                                     SCROLL==> PAGE
YOU MAY BROWSE (B), EDIT (E), FIX (F) OR VIEW (V) RESULTS
OMMAND MEMBER  ACTION          TITLE
      F1002WA          TSO DELETE EMPTY GENERATION DATA GROUPS
***** Bottom of data *****
```

Figure 7-41 Fast Audit Fix Plan Member List for error F1002Wa

To generate the JCL and FIX statements, type **F** next to F1002WA in this panel and press Enter. The JCL and fix statements generated for all of the catalogs in error are shown in Figure 7-42.

```

EDIT          CG.H161815                      Columns 00001 00072
Command ==>                                     Scroll ==> CSR
***** ***** Top of Data *****
000001 //CGA   JOB (0),COLGOR,
000002 //          CLASS=A,MSGCLASS=X,NOTIFY=COLGOR
000003 //*
000004 //STEP1 EXEC PGM=IKJEFT01,
000005 //  DYNAMNBR=25,REGION=4096K
000006 //* CHANGE THE FOLLOWING STATEMENT TO
000007 //* DSN=PPFX.CLIST WHERE PPFX IS THE PREFIX ASSIGNED TO
000008 //* THE FASTAUDIT PRODUCT
000009 //*
000010 //SYSPROC DD DSN=FAUDITX.QABETA.CLIST,DISP=SHR
000011 //*          ****
000012 //SYSOUT DD SYSOUT=*
000013 //SYSTSPRT DD DSN=CG.FIXAUDIT.F1002WA,DISP=SHR
000014 //SYSTSIN DD *
000015 DEL 'ALU.BKP.MFST' GDG +
000016 PURGE CATALOG('CAT.ICF.USER2')
000017 DEL 'ALU.TEST.BACKUP' GDG +
000018 PURGE CATALOG('CAT.ICF.USER2')
000019 DEL 'ALU.TRN.GL' GDG +

```

Figure 7-42 DELETE GDG base record commands generated by fix F1002WA

To execute the fix, type **SUBMIT** on the command line and press Enter.

View the results of the **DEL GDG** commands in the product output data set pfx.FIXAUDIT.F1002WA or in the sysout for the job.

### 7.3.3 Catalog audit error 1003W

The next error message we will look at is Catalog Audit Error 1003W DATA SET IS ON DASD VOLUME WHICH IS VARIED OFF LINE.

#### Diagnosing the error

Looking up this error in the *FastAudit/390* User Guide, you find the following explanation.

The catalog extraction has detected an entry that is cataloged to a DASD volume. That volume exists but is varied offline. This data set may not be accessed from the system the catalog extraction process was run on.

To view the error detail, type B next to the error in the *FastAudit* CAT Summary panel and press Enter. The error detail is displayed in Figure 7-43.

|       |              |        |            |
|-------|--------------|--------|------------|
| f1    | f2           | f3     | f4         |
| 1003W | APPL1.NO.VOL | OFF001 | UCAT.APPL1 |

Figure 7-43 Error detail for error 1003W

In the *FastAudit User Guide*, the Field Description Box provided in the documentation and shown in Table 7-12 provides the explanations of each field.

Table 7-12 Diagnostic 1003W field descriptions

| Dictionary symbol | Example      |
|-------------------|--------------|
| FIELD 1 /ERRNUM   | 1003W        |
| FIELD 2 /TDSN     | APPL1.NO.VOL |
| FIELD 3 /VOLSER   | OFF001       |
| FIELD 4 /CATALOG  | UCAT.APPL1   |

Check with your system programmers and storage administrators to determine the reason why these volumes are varied offline. A simple solution may be to vary the volume back online. If this is the case, rerun the audit after the volume(s) has been varied online to verify that the condition has been eliminated.

### Resolving the error

To view the available fixes, type **F** next to the 1003W error and press Enter. The Fast Audit Fix Plan Member List is displayed in Figure 7-44.

```
FAST AUDIT FIX PLAN MEMBER LIST
OPTION ==>                                SCROLL==> PAGE
YOU MAY BROWSE (B), EDIT (E), FIX (F) OR VIEW (V) RESULTS
OMMAND  MEMBER  ACTION                    TITLE
      F1003WA      TSO LISTC OF DATA TO DASD NOT ONLINE
      F1003WB      TSO DELETE NOSCRATCH FOR DSN ON MISSING VOLUME
***** Bottom of data *****
```

Figure 7-44 Fast Audit Fix Plan Member List for error 10013W

To generate the JCL and FIX statements, type **F** next to the error and press Enter. The JCL and FIX statements generated for all of the data sets in error are shown in Figure 7-45.

```

EDIT          CG.H160333                      Columns 00001 00072
Command ==>                                     Scroll ==> CSR
***** ***** Top of Data *****
000001 //CGA   JOB (0),COLGOR,
000002 //          CLASS=A,MSGCLASS=X,NOTIFY=COLGOR
000003 //*
000004 //STEP1 EXEC PGM=IKJEFT01,
000005 // DYNAMNBR=25,REGION=4096K
000009 //*
000010 //SYSPROC DD DSN=FAUDITX.QABETA.CLIST,DISP=SHR
000011 //*          ****
000012 //SYSOUT DD SYSOUT=*
000013 //SYSTSPRT DD DSN=CG.FIXAUDIT.F1001WB,DISP=SHR
000014 //SYSTSIN DD *
000015 DEL 'ALU.BKP.MFST' +
000016 NSCR +
000017 CATALOG('CAT.ICF.USER2')
000018 DEL 'ALU.TEST.BACKUP' +
000019 NSCR +

```

Figure 7-45 DELETE NOSCRATCH commands generated for error 1003W

Execute the fix by typing **SUBMIT** on the command line and pressing Enter. The results of the execution can be viewed in the product output data set pfxFIXAUDIT.F1003WB and in the sysout of the job.

## Catalog audit error 1004W

The next error we will look at is Catalog Audit Error 1004W DATA SET IS ON DASD VOLUME WHICH IS NOT ON SYSTEM.

## Diagnosing the error

Looking up this error in the *FastAudit* User Guide, you find the following explanation.

The catalog extraction has detected an entry that is cataloged to a DASD volume. That volume is not on the system the catalog extraction ran on. This data set cannot be accessed.

Ensure that the audit has executed on the correct system and check with the system programmer to determine if the missing volumes are by design or are truly in error. This error is common when **VOLUMES** are removed from the system but the catalog entries were not cleaned up.

To view the error detail, type **B** next to the error in the FastAudit MASTCAT Summary panel and press Enter. The error detail is displayed in Figure 7-46.

|       |                      |        |               |
|-------|----------------------|--------|---------------|
| f1    | f2                   | f3     | f4            |
| 1004W | CCCB028.ISPF.ISPPROF | PDS005 | CAT.ICF.USER2 |

Figure 7-46 Error detail for error 1004W

In the *FastAudit* User Guide, the Field Description Box provided in the documentation and shown in Table 7-13 provides an explanation of the fields.

Table 7-13 Diagnostic 1004W field descriptions

| Dictionary symbol | Example              |
|-------------------|----------------------|
| FIELD 1 /ERRNUM   | 1004W                |
| FIELD 2 /TDSN     | CCCB028.ISPF.ISPPROF |
| FIELD 3 /VOLSER   | PDS005               |
| FIELD 4 /CATALOG  | CAT.ICF.USER2        |

### Resolving the error

In most cases, data sets cataloged on volumes that no longer exist can be deleted from your user catalogs.

To view the available fixes, type **F** next to the error name and press Enter. The Fast Audit Fix Plan Member List is displayed in Figure 7-47.

|                                                           |         |                |                                                |
|-----------------------------------------------------------|---------|----------------|------------------------------------------------|
| FAST AUDIT FIX PLAN MEMBER LIST                           |         |                |                                                |
| OPTION ==>                                                |         | SCROLL==> PAGE |                                                |
| YOU MAY BROWSE (B), EDIT (E), FIX (F) OR VIEW (V) RESULTS |         |                |                                                |
| COMMAND                                                   | MEMBER  | ACTION         | TITLE                                          |
|                                                           | F1004WA |                | TSO LISTC OF DATA TO DASD NOT ONLINE           |
|                                                           | F1004WB |                | TSO DELETE NOSCRATCH FOR DSN ON MISSING VOLUME |
| ***** Bottom of data *****                                |         |                |                                                |

Figure 7-47 Fast Audit Fix Plan Member List for error 1004W

To delete NOSCRATCH the data sets on the missing VOLUMES, type **F** next to the F1004WB member and press Enter. The JCL and fix statements generated for all of the data sets in error are shown in Figure 7-48.

```

EDIT          CG.H164250                      Columns 00001 00072
Command ==>                                     Scroll ==> CSR
***** ***** Top of Data *****
000001 //CGA   JOB (0),COLGOR,
000002 //          CLASS=A,MSGCLASS=X,NOTIFY=COLGOR
000003 //*
000004 //STEP1 EXEC PGM=IKJEFT01,
000005 //  DYNAMNBR=25,REGION=4096K
000006 //* CHANGE THE FOLLOWING STATEMENT TO
000007 //* DSN=PPFX.CLIST WHERE PPFX IS THE PREFIX ASSIGNED TO
000008 //* THE FASTAUDIT PRODUCT
000009 //*
000010 //SYSPROC DD  DSN=FAUDITX.QABETA.CLIST,DISP=SHR
000011 //*          ****
000012 //SYSOUT  DD  SYSOUT=*
000013 //SYSTSPRT DD  DSN=CG.FIXAUDIT.F1004WB,DISP=SHR
000014 //SYSTSIN DD  *
000015 DEL 'DANSCAT' +
000016   NSCR +
000017   CATALOG('SYST80.MASTER.CATALOG')
000018 DEL 'DAWNS.CAT' +
000019   NSCR +

```

Figure 7-48 *DELETE NOSCRATCH statements for fix F1004WB*

Execute the fix by typing **SUBMIT** on the command line and pressing Enter.

The results of the execution can be viewed in the product output data set pfx.FIXAUDIT.F1004WB. or in the sysout for the job.

### 7.3.4 Catalog audit error 1005W

The next error message we will look at is Catalog Audit Error 1005W NODE IS CONNECTED BUT HAS NO CATALOGED data sets.

#### Diagnosing the error

Looking up this error in the *FastAudit/390 User Guide*, you find the following explanation.

The catalog extraction investigates high-level nodes connected to a user catalog. Those nodes that have no references to them are summarized in this error. This may not strictly be an error but is indicative of the general health of the catalogs.

To view the error detail, type B next to the error in the FastAudit CAT Summary panel and press Enter. The error detail is displayed in Figure 7-49.

|       |      |          |               |
|-------|------|----------|---------------|
| f1    | f2   | f3       | f4            |
| 1005W | DCDV | 00000000 | USERCAT.UCAT1 |

Figure 7-49 Error detail for error 1005W

In the *FastAudit/390 User Guide*, the Field Description Box provided in the documentation and shown in Table 7-14 provides the explanations of each field.

Table 7-14 Diagnostic 1005W field descriptions

| Dictionary symbol      | Example                     |
|------------------------|-----------------------------|
| FIELD 1 /ERRNUM        | 1005W                       |
| FIELD 2 /TDSN          | DCDV                        |
| FIELD 3 DATA SET COUNT | 00000000 (should be zeroes) |
| FIELD 4 /catalog       | USERCAT.UCAT1               |

Check with system programmer, storage administrators, and application owners to determine if the alias entries are obsolete. In some cases, they could be newly defined alias entries that have not been utilized as of yet.

Resolving the error

To view the available fixes, type **F** next to the error message and press Enter. The Fast Audit Fix Plan Member List is shown in Figure 7-50.

```

FAST AUDIT FIX PLAN MEMBER LIST
OPTION ==>                                SCROLL==> PAGE
YOU MAY BROWSE (B), EDIT (E), FIX (F) OR VIEW (V) RESULTS
COMMAND MEMBER ACTION TITLE
      F1005WA      TSO LISTC OF HINODE WITH NO DATA SETS
      F1005WB      TSO DELETE ALIAS HAVING NO data sets
***** Bottom of data *****

```

Figure 7-50 Fast Audit Fix Plan Member List for error 1005W

Generate the JCL and FIX statements by typing **F** next to the **F1005WB** member and press Enter. The JCL and fix statements generated are shown in Figure 7-51.

```

EDIT          CG.H170458                      Columns 00001
00072
Command ==>                                     Scroll ==> CSR
***** Top of Data*****
000001 //CGA   JOB (0),COLGOR,
000002 //          CLASS=A,MSGCLASS=X,NOTIFY=COLGOR
000003 //*
000004 //STEP1 EXEC PGM=IKJEFT01,
000005 // DYNAMNBR=25,REGION=4096K
000006 //* CHANGE THE FOLLOWING STATEMENT TO
000007 //* DSN=PPFX.CLIST WHERE PPFX IS THE PREFIX ASSIGNED TO
000008 //* THE FASTAUDIT PRODUCT
000009 //*
000010 //SYSPROC DD DSN=FAUDITX.QABETA.CLIST,DISP=SHR
000011 //*          ****
000012 //SYSOUT DD SYSOUT=*
000013 //SYSTSPRT DD DSN=CG.FIXAUDIT.F1005WB,DISP=SHR
000014 //SYSTSIN DD *
000015 DEL '$10313' ALIAS
000016 DEL '$10421' ALIAS
000017 DEL '$1044' ALIAS
000018 DEL '$11031' ALIAS
000019 DEL '$11706' ALIAS

```

Figure 7-51 Delete Alias statements generated in fix F1005WB

You may want to remove any alias names that are new or for whatever reason should not be deleted. Execute the fix by typing **SUBMIT** on the command line and pressing Enter.

The results of the DELETE ALIAS commands can be viewed in the product output data set pfx.FIXAUDIT.F1005WB or in the sysout for the job.

### 7.3.5 Catalog audit error 1005V

The next error we will look at is Catalog Audit Error 1005V Node Is Connected To Catalog That Cannot Be Accessed.

#### Diagnosing the error

Looking up this error in the *FastAudit/390* User Guide, you find the following explanation.

An attempt to allocate, read, or otherwise process a catalog has failed. An alias for the catalog has been defined but no verification of the entries in that catalog can be processed.



To view the error detail, type B next to the error in the FastAudit CAT Summary panel and press Enter. The error detail is displayed in Figure 7-52.

|       |    |          |           |
|-------|----|----------|-----------|
| f1    | f2 | f3       | f4        |
| 1005V | A1 | 00000000 | DAWNS.CAT |

Figure 7-52 Error detail for error 1005V

In the *FastAudit User Guide*, the Field Description Box provided in the documentation and shown in Table 7-15 provides the explanations of each field.

Table 7-15 Diagnostic 1005V field descriptions

| Dictionary symbol      | Example                     |
|------------------------|-----------------------------|
| FIELD 1 /ERRNUM        | 1005V                       |
| FIELD 2 /TDSN          | A1                          |
| FIELD 3 DATA SET COUNT | 00000000 (should be zeroes) |
| FIELD 4 /catalog       | DAWNS.CAT                   |

This diagnostic is unique from 1005W, because the referenced catalog could not be read. In this case, it is recommended to delete these aliases.

### Resolving the error

To view the available fixes, type F next to the error message and press Enter. The Fast Audit Fix Plan Member List is shown in Figure 7-53.

```
FAST AUDIT FIX PLAN MEMBER LIST
OPTION ==>                                SCROLL==> PAGE
  YOU MAY BROWSE (B), EDIT (E), FIX (F) OR VIEW (V) RESULTS
COMMAND  MEMBER  ACTION                    TITLE
      F1005VA      TSO LISTC OF HINODE ON UNAVAILABLE CATALOG
      F1005VB      TSO DELETE ALIAS FROM UNAVAIABLE CATALOG
***** Bottom of data *****
```

Figure 7-53 Fast Audit Fix Plan Member for error 1005V

To generate the JCL and FIX statements, type F next to F1005VB and press Enter. The JCL and fix statements generated are shown in Figure 7-54.

```

EDIT          CG.H171438                      Columns 00001 00072
Command ==>                                     Scroll ==> CSR
***** ***** Top of Data *****
000001 //CGA   JOB (0),COLGOR,
000002 //          CLASS=A,MSGCLASS=X,NOTIFY=COLGOR
000003 //*
000004 //STEP1 EXEC PGM=IKJEFT01,
000005 //  DYNAMNBR=25,REGION=4096K
000006 //* CHANGE THE FOLLOWING STATEMENT TO
000007 //* DSN=PPFX.CLIST WHERE PPFX IS THE PREFIX ASSIGNED TO
000008 //* THE FASTAUDIT PRODUCT
000009 //*
000010 //SYSPROC DD DSN=FAUDITX.QABETA.CLIST,DISP=SHR
000011 //*          ****
000012 //SYSOUT DD SYSOUT=*
000013 //SYSTSPRT DD DSN=CG.FIXAUDIT.F1005VB,DISP=SHR
000014 //SYSTSIN DD *
000015 DEL '$VALUE.NOVOLSER' ALIAS
000016 DEL 'A1' ALIAS
000017 DEL 'DANSCAT' ALIAS
000018 DEL 'R1' ALIAS
000019 DEL 'Z1' ALIAS

```

Figure 7-54 DELETE ALIAS commands generated in fix F1005VB

Execute the fix by typing **SUBMIT** on the command line and pressing Enter. The results of the DELETE ALIAS commands can be viewed in the product output data set pfx.FIXAUDIT.F1005VB or in the sysout for the job.

### 7.3.6 Catalog audit error 1006W

The next error we will look at is Catalog Audit Error 1006W DASD VOLUME IS ONLINE BUT HAS NO CATALOGED DATA SETS.

#### Diagnosing the error

Looking up this error in the *FastAudit* User Guide, you find the following explanation.

The catalog extraction has found a DASD volume that is on-line but has no data sets cataloged to it.

To view the error detail, type B next to the error in the FastAudit CAT Summary panel and press Enter. The error detail is displayed in Figure 7-55.

|       |        |
|-------|--------|
| f1    | f2     |
| 1006W | EMPTYV |

Figure 7-55 Error detail for error 1006W

In the *FastAudit/390* User Guide, the Field Description Box provided in the documentation and shown in Table 7-16 provides the explanations of each field.

Table 7-16 Diagnostic 1006W Field descriptions

| Dictionary symbol | Example |
|-------------------|---------|
| FIELD 1 /ERRNUM   | 1006W   |
| FIELD 2 /VOLSER   | EMPTYV  |

After scanning all the catalogs, the volumes in this list were not referenced by any catalog entries. It could be this is desired, however, its good information to know. Furthermore, it may be the volumes do have data sets on them but they are not cataloged.

There is no recommended fix for this message. You may want to access the volumes via the VTOC to ensure that data sets on the volume are not uncataloged.

### 7.3.7 Catalog audit error 1007W

The next error we will look at is Catalog Audit Error 1007W CATALOG EXISTS BUT HAS NO NODES CONNECTED TO IT.

#### Diagnosing the error

Looking up this error in the *FastAudit/390* User Guide, you find the following explanation.

The catalog extraction has found a catalog with entries. None of those entries are connected to the master catalog.

To view the error detail, type B next to the error in the FastAudit CAT Summary panel and press Enter. The error detail is displayed in Figure 7-56.

|       |                       |        |     |
|-------|-----------------------|--------|-----|
| f1    | f2                    | f3     | f4  |
| 1007W | EMTST3.FAC.UCAT.IMBED | SC0003 | 007 |

Figure 7-56 Error detail for error 1007W

In the *FastAudit/390* User Guide, the Field Description Box provided in the documentation and shown in Table 7-17 provides the explanations of each field.

Table 7-17 Diagnostic 1007W field descriptions

| Dictionary symbol                | Example                |
|----------------------------------|------------------------|
| Field 1 /errnum                  | 1007W                  |
| Field 2 /CATALOG                 | EMTST3..FAC.UCAT.IMBED |
| Field 3 /volser                  | SC0003                 |
| FIELD 4 CATALOG NUMBER (no dict) | 007                    |

This error may be a result of a newly defined catalog or one that is planned for deletion. Check with your system programmers and others who delete and define catalogs in the environment to determine the correct status of the catalogs. There is no recommended fix for this error message.

### 7.3.8 Catalog audit error 1008W

The next error message we will look at is Catalog Audit Error 1008W CATALOG EXISTS BUT CANNOT BE ALLOCATED.

#### Diagnosing the error

Looking up this error in the *FastAudit/390* User Guide, you find the following explanation.

The catalog extraction has found a catalog name in the master catalog but was unable to allocate the catalog to list its contents

To view the error detail, type B next to the error in the FastAudit CAT Summary panel and press Enter. The error detail is displayed in Figure 7-57.

|       |                  |        |     |
|-------|------------------|--------|-----|
| f1    | f2               | f3     | f4  |
| 1008W | FACUCAT.EH.MCAT1 | EHT004 | 056 |

Figure 7-57 Error detail for error 1008W

In the *FastAudit/390* User Guide, the Field Description Box provided in the documentation and shown in Table 7-18 provides the explanations of each field.

Table 7-18 Diagnostic 1008W field descriptions

| Dictionary symbol                | Example          |
|----------------------------------|------------------|
| Field 1 /errnum                  | 1008W            |
| Field 2 /CATALOG                 | FACUCAT.EH.MCAT1 |
| Field 3 /volser                  | EHT004           |
| FIELD 4 CATALOG NUMBER (no dict) | 056              |

This might be caused by an improper IMPORT CONNECT command. No verification is made by *FastAudit/390* to determine if the VOLSER is incorrect.

There is no fix provided for this error message, however, there is a fix provided that executes TSO LISTC for the user catalog that cannot be accessed.

7.3.9 Catalog audit error 1009W

The next error we will look at is Catalog Audit Error 1009W GDG SPHERE RECORD HAS NOSCRATCH PARAMETER.

Diagnosing the error

Looking up this error in the *FastAudit/390* User Guide, you find the following explanation.

The catalog extraction has found a GDG sphere record defined with the NOSCRATCH parameter. Prior to SMS this would leave uncataloged data sets on DASD and cause an error. With SMS it probably is no longer an error but the catalog manager needs to know about it.

To view the error detail, type B next to the error in the FastAudit CAT Summary panel and press Enter. The error detail is displayed in Figure 7-58.

|       |                 |     |               |     |         |           |         |
|-------|-----------------|-----|---------------|-----|---------|-----------|---------|
| F1    | f2              | f3  | f4            | f5  | f6      | f7        | f8      |
| 1009W | USERID1.EXTRACT | GDG | CAT.ICF.USER2 | 005 | NOEMPTY | NOSCRATCH | 1991352 |

Figure 7-58 Error detail for error 1009W

In the *FastAudit/390* User Guide, the Field Description Box provided in the documentation and shown in Table 7-19 provides the explanations of each field.

Table 7-19 Diagnostic 1009W field descriptions

| Dictionary symbol               | Example         |
|---------------------------------|-----------------|
| Field 1 /ERRNUM                 | 1009W           |
| Field 2 /TDSN                   | USERID1.EXTRACT |
| Field 3 GDG INDICATOR (NO DICT) | GDG             |
| FIELD 4 /CATALOG                | CAT.ICF.USER2   |
| Field 5 /LIMIT                  | 005             |
| Field 6 /EMPTY                  | NOEMPTY         |
| Field 7 /SCRATCH                | NOSCRATCH       |
| Field 8 Creation Date (NO DICT) | 1991352         |

GDG data sets with the noscratch parameter coded in the define statements can cause data sets to become uncataloged on the volumes. In one example, thousands of GDG sphere records were created using the noscratch parameter causing data sets that had rolled off the GDG base to become uncataloged. Some of these data sets were resident on migration level 1 (ML1) volumes. The customer's ML1 environment had been growing rapidly over a period of time with no explanation. After running this audit, it was determined that the cause was in fact GDG base records inadvertently created with the NOSCRATCH parameter. After deleting the uncataloged data sets and altering the GDG base records to SCRATCH, the customer was able to re-deploy almost half of their ML1 DASD resources.

## Resolving the error

The resolution to this error is two fold. The first resolution is to alter the GDG base records from NOSCRATCH to SCRATCH. Prior to running this fix, you may want to discuss the change with the system programmers and application owners to ensure that you may alter all of the GDG base records in question. After clearing this change, proceed with fix F1009WB to alter the GDG base records.

To view the available fixes, type **F** next to the error message and press Enter. The Fast Audit Fix Plan Member List is shown in Figure 7-59.

```

          FAST AUDIT FIX PLAN MEMBER LIST
OPTION ===>                                SCROLL==> PAGE
YOU MAY BROWSE (B), EDIT (E), FIX (F) OR VIEW (V) RESULTS
COMMAND MEMBER ACTION TITLE
          F1009WA      TSO LISTC GDG BASE INFORMATION
          F1009WB      TSO ALTER GDG BASE TO SCRATCH
***** Bottom of data *****

```

Figure 7-59 Fast Audit Fix Plan Member List for fix F1009WA

Generate the JCL and FIX statements by typing **F** next to F1009WB and press Enter. The JCL and FIX statements generated are shown in Figure 7-60.

```

EDIT      CG.H174543                        Columns 00001 00072
Command ===>                                Scroll ==> CSR
***** ***** Top of Data *****
000001 //CGA JOB (0),COLGOR,
000002 //          CLASS=A,MSGCLASS=X,NOTIFY=COLGOR
000003 //*
000004 //STEP1 EXEC PGM=IKJEFT01,
000005 // DYNAMNBR=25,REGION=4096K
000006 //* CHANGE THE FOLLOWING STATEMENT TO
000007 //* DSN=PPFX.CLIST WHERE PPFX IS THE PREFIX ASSIGNED TO
000008 //* THE FASTAUDIT PRODUCT
000009 //*
000010 //SYSPROC DD DSN=FAUDITX.QABETA.CLIST,DISP=SHR
000011 //*          ****
000012 //SYSOUT DD SYSOUT=*
000013 //SYSTSPRT DD DSN=CG.FIXAUDIT.F1009WB,DISP=SHR
000014 //SYSTSIN DD *
000015 ALTER 'ARX.ARP108.CUSTOMER.ISSUE' SCRATCH +
000016 CATALOG('CAT.ICF.USER2')
000017 ALTER 'ARX.DRP.ARPINQ' SCRATCH +
000018 CATALOG('CAT.ICF.USER2')
000019 ALTER 'ARX.DRP.ARSRTCUS' SCRATCH +

```

Figure 7-60 ALTER statements for FIX F1009WB

Execute the fix by typing **SUBMIT** on the command line and pressing Enter.

The results of the ALTER commands can be viewed in the product output data set pfx.FIXAUDIT.F1009WB or in the sysout for the job. The second step is to delete the uncataloged data sets.

### 7.3.10 Catalog audit error 1010W

The next error message we will look at is Catalog Audit Error 1010W GDG SPHERE RECORD HAS EMPTY PARAMETER.

#### Diagnosing the error

Looking up this error in the *FastAudit/390* User Guide, you find the following explanation.

The catalog extraction has found a GDG sphere record defined with the EMPTY parameter. Prior to SMS this would leave uncataloged data sets on DASD and cause an error. With SMS it probably is no longer an error but the catalog manager needs to know about it.

To view the error detail, type B next to the error in the FastAudit CAT Summary panel and press Enter. The error detail is displayed in Figure 7-61.

|       |                    |                 |     |       |         |         |
|-------|--------------------|-----------------|-----|-------|---------|---------|
| f1    | f2                 | f3              | f4  | f5    | f6      | f7      |
| 1010W | EDPT9.CAT.LAB.GDGX | USERCAT.EDPTST2 | 010 | EMPTY | SCRATCH | 2001107 |

Figure 7-61 Error detail for error 1010W

In the *FastAudit/390* User Guide, the Field Description Box provided in the documentation and shown in Table 7-20 provides the explanations of each field.

Table 7-20 Diagnostic 1010W field descriptions

| Dictionary symbol               | Example            |
|---------------------------------|--------------------|
| FIELD 1 /ERRNUM                 | 1010W              |
| FIELD 2 /TDSN                   | EDPT9.CAT.LAB.GDGX |
| FIELD 3 /CATALOG                | USERCAT.EDPTST2    |
| FIELD 4 /LIMIT                  | 010                |
| FIELD 5 /EMPTY                  | EMPTY              |
| FIELD 6 /SCRATCH                | SCRATCH            |
| FIELD 7 CREATION DATE (NO DICT) | 2001107            |

Although this is a legitimate parameter, frequently this is not the desired purpose of GDG(s). This diagnostic serves as a reminder to review this unusual GDG definition with the system programmers or application owners.



### Resolving the error

If you determine that the GDG base was incorrectly defined, you can alter the entry using the *FastAudit/390* fix F1010WB. To view the available fixes, type F next to the error message and press Enter. The Fast Audit Fix Plan Member List is shown in Table 7-62.

```
FAST AUDIT FIX PLAN MEMBER LIST
OPTION ==>                                SCROLL==> PAGE
YOU MAY BROWSE (B), EDIT (E), FIX (F) OR VIEW (V) RESULTS
COMMAND MEMBER ACTION TITLE
      F1010WA      TSO LISTC GDG BASE INFORMATION
      F1010WB      TSO ALTER GDG BASE TO NOEMPTY
***** Bottom of data *****
```

Figure 7-62 Fast Audit Fix Plan Member List for F1010W

To generate the JCL and FIX statements, type F next to F101WB and press Enter. The JCL and FIX statements generated are shown in Figure 7-63.

```
EDIT      CG.H180841                        Columns 00001 00072
Command ==>                                Scroll ==> CSR
***** ***** Top of Data *****
000001 //CGA JOB (0),COLGOR,
000002 //          CLASS=A,MSGCLASS=X,NOTIFY=COLGOR
000003 //*
000004 //STEP1 EXEC PGM=IKJEFT01,
000005 // DYNAMNBR=25,REGION=4096K
000006 //* CHANGE THE FOLLOWING STATEMENT TO
000007 //* DSN=PPFX.CLIST WHERE PPFX IS THE PREFIX ASSIGNED TO
000008 //* THE FASTAUDIT PRODUCT
000009 //*
000010 //SYSPROC DD DSN=FAUDITX.QABETA.CLIST,DISP=SHR
000011 //*          ****
000012 //SYSOUT DD SYSOUT=*
000013 //SYSTSPRT DD DSN=CG.FIXAUDIT.F1010WB,DISP=SHR
000014 //SYSTSIN DD *
000015 ALTER 'EDPT9.CAT.LAB.GDGX' NOEMPTY +
000016 CATALOG('USERCAT.EDPTST2')
```

Figure 7-63 ALTER GDG statements for fix F1010WB

Execute the fix by typing **SUBMIT** on the command line and pressing Enter.

The results of the ALTER commands can be viewed in the product output data set pfx.FA6101.FAUDIT.F1010WB or in the sysout for the job.





## **Auditing your tape environment using Tape Audit**

Mainstar created Tape Audit to audit the ICF catalog entries against the CA-1, ZARA, or RMM tape databases. This audit is a valuable tool in determining whether tapes are being returned to the scratch pool, or, due to procedural or system errors, are lost for future use. Using Tape Audit, many customers have recovered hundreds or thousands of tapes representing a cost savings to their organization.

## 8.1 DFSMShsm tape handling

DFSMShsm has a method of managing scratch tapes. Once a tape, or more appropriately, a cartridge, has been assigned to DFSMShsm, it enters the unassigned pool in the Offline Control Data Set (OCDS). Whenever DFSMShsm needs a new tape for migration level 2 (L2), spill (SP), or cycle (01..07), it locates a tape in the unassigned pool and does an ADDVOL for the appropriate usage.

When a tape is recycled it is returned to the unassigned pool. Subsequently, the tape may be reused for other DFSMShsm tape functions. Whenever additional tapes are needed, ADDVOL functions must be performed and marked as in use by "other" tape management systems, such as RMM or CA-1.

DFSMShsm provides an exit (ARCTVEXT) to interface between DFSMShsm and CA-1 or ZARA to automate the management of tapes. When this exit is implemented, the DFSMShsm unassigned pool is no longer used. When a new L2, SP or CYCLE tape is needed, an ADDVOL is automatically performed for DFSMShsm and the tape is marked as "in use" in the tape database. When a tape is recycled or otherwise returned to the scratch pool, the ARCTVEXT exit routine removes the tape from the DFSMShsm HSM inventory (DELVOL) and the tape management database is updated to indicate the tape is returned to scratch. To ensure this interface is working correctly, Mainstar provides the HSM TAPE inventory to CA-1 (FAUDITTP) and ZARA (FAUDITZP) audit.

Tape Audit gathers the following information from DFSMShsm for the audit:

|                    |                                                 |
|--------------------|-------------------------------------------------|
| Level 2            | DFSMShsm MIGRATION LEVEL 2 TAPES                |
| Level 2 copy tapes | If TAPECOPY is done for level 2 tapes.          |
| Cycle Tapes        | Backup tapes assigned to a cycle day (01... 07) |
| Spill Tapes        | Recycled Backup tapes                           |
| Backup Copy tapes  | If TAPECOPY is done for backup tapes            |
| Dump Volume tapes  | If HSM DUMP VOL is done                         |

DFSMShsm may be assigned one or more "prefixes" such as HSM, DFHSM, or SY2HSM. Subsequently, the data set name written on the tape uses this prefix as the first node on each tape. HSM Tape Audit detects each new prefix and generates control statements to extract information from the tape management database.

The data set naming conventions are:

- ▶ &HSMPFX.HMIG.\*\*
- ▶ &HSMPFX.HMIGTAPE.\*\*
- ▶ &HSMPFX.BACK.\*\*

- ▶ &HSMPFX.BACKTAPE.\*\*
- ▶ &HSMPFX.COPY.HMIG.\*\*
- ▶ &HSMPFX.TAPECOPY.HMIG
- ▶ &HSMPFX.COPY.BACKTAPE.\*\*
- ▶ &HSMPFX.TAPECOPY.BACKTAPE
- ▶ &HSMPFX.DMP.\*\*

In addition to auditing the tapes found in DFSMSHsm to the tape management database, this audit does some basic health checks in DFSMSHsm. For example, a tape having a TTOC entry must also have an MCT (backup volume) or an MCV (migration volume) and vice versa. Furthermore, its possible there may be migration or backup records that reference a tape that is not on the system. These data may be important and this audit provides a method of researching and correcting the problem.

The cataloged procedures available to audit the DFSMSHsm tape information against the tape management database are:

- ▶ *FAUDITRM* procedure is used to audit the DFSMSHsm information about tapes against information in the Removable Media Manager (RMM) database.
- ▶ *FAUDITTP* procedure is used to audit the DFSMSHsm information about tapes against information in the CA-1 TMS database.
- ▶ *FAUDITZR* procedure is used to audit the DFSMSHsm information about tapes against information in the CA-1 TMS database.

### 8.1.1 FAUDITRM procedure setup

The parameters in the FAUDITRM procedure must be customized for your environment.

- |              |                                                                                                                                                                                                                                                                                                                                                                        |
|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>PPFX</b>  | Product Prefix as installed. For example, if the product has been installed with MSC.FAUDIT as a prefix to the libraries, then PPFX=MSC.FAUDIT must be coded in the EXEC PROC statement.                                                                                                                                                                               |
| <b>PFX</b>   | Prefix represents the Runtime Prefix the user has chosen for this cataloged procedure to allocate work files that contain the audit results. Depending on the SMS treatment of new allocations, the work data sets may be deleted within a day or so. Choose a prefix that will not be scratched for a while so the results of the audit may be reviewed and repaired. |
| <b>PUNIT</b> | Permanent unit for allocating data sets containing the audit results. The default is SYSDA. This unit should allocate data sets to a device that retains data for several days or weeks.                                                                                                                                                                               |

|                |                                                                                                                                                                                        |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>WRKUNIT</b> | Work Unit for allocating temporary data sets that will be deleted at the end of the job. The default is SYSDA. If your installation has a different standard, code the different name. |
| <b>ATP</b>     | Allocation Type. Allocation is given in units such as PRI and SEC. For the larger shops the unit TRKS sometimes isn't enough. If your installation is large, change the ATP to CYL.    |
| <b>PRI</b>     | The number of primary allocation units. The default is 100. This value must be adjusted to accommodate the size of your HSM MCDS and the number of records.                            |
| <b>SEC</b>     | The number of secondary allocation units. The default is 200. This parameter is important to adjust for the size of your MCDS.                                                         |
| <b>MSG</b>     | MSG is a requirement of the RMM program EDGHSPK. It is the dsname of the message data set.                                                                                             |
| <b>RPT</b>     | RPT is a requirement of the RMM program EDGHSPK. It is the dsname of the report data set.                                                                                              |
| <b>MCDS</b>    | MCDS is required. It is the first split of the DFSMSHsm MCDS dataset or the only split if the MCDS is not split.                                                                       |
| <b>MCDS1</b>   | MCDS1 is the second split of the MCDS. The default is NULLFILE.                                                                                                                        |
| <b>MCDS2</b>   | MCDS2 is the third split of the MCDS. The default is NULLFILE.                                                                                                                         |
| <b>MCDS3</b>   | MCDS3 is the fourth split of the MCDS. The default is NULLFILE.                                                                                                                        |
| <b>BCDS</b>    | BCDS is required. It is the first split of the DFSMSHsm BCDS data set or the only split if the BCDS is not split.                                                                      |
| <b>BCDS1</b>   | BCDS1 is the second split of the BCDS. The default is NULLFILE.                                                                                                                        |
| <b>BCDS2</b>   | BCDS2 is the third split of the BCDS. The default is NULLFILE.                                                                                                                         |
| <b>BCDS3</b>   | BCDS3 is the fourth split of the BCDS. The default is NULLFILE.                                                                                                                        |
| <b>OCDS</b>    | OCDS is the name of the Offline Control Data Set.                                                                                                                                      |
| <b>LOAD</b>    | LOAD is the name of the load library as installed with <i>FastAudit/390</i> . It must be authorized.                                                                                   |

Figure 8-1 shows the symbolic parameters in the FAUDITRM JCL used to execute the procedure for the DFSMSHsm Tape to RMM Audit. The figure shows a multi-cluster MCDS with two splits and BCDS with two splits. The OCDS is named SIS.HSM.OCDS. The product prefix is FAUDITX.QABETA and the runtime prefix is DES2.

```

000010 //FAUDITRM PROC PFX=PFX,          <==RUN TIME PREFIX
000011 //* FASTAUDIT <V6 R1.01> SEPTEMBER, 2001
000012 //      PPFX=PPFX,                <==PRODUCT PREFIX
000013 //      PUNIT=SYSDA,                <===UNIT FOR PERMANENT ALLOCATION
000014 //      SOUT=*,                    <===PREFERED SYSOUT CLASS
000015 //      WRKUNIT=SYSDA,              <===UNIT FOR TEMPORARY ALLOCATION
000016 //      ATP=TRK,                   <===TYPE OF ALLOCATION TRK/CYL
000017 //      PRI=100,                   <===PRIMARY ALLOCATION FOR FAUDIT
000018 //      SEC=200,                   <===SECONDARY SPACE ALLOCATION
000019 //      MSG=RMM.SMS.MESSAGE,        <==RMM MESSAGE LIBRARY
000020 //      RPT=RMM.SMS.REPORT,          <==RMM REPORT LIBRARY
000021 //      MCDS=HSM.MCDS,              <===INSTALLATION MCDS DSNAME
000022 //      MCDS1=NULLFILE,            <==1ST MCDS SPLIT
000023 //      MCDS2=NULLFILE,            <==2ND MCDS SPLIT
000024 //      MCDS3=NULLFILE,            <==3RD MCDS SPLIT
000025 //      BCDS=HSM.BCDS,              <===INSTALLATION BCDS DSNAME
000026 //      BCDS1=NULLFILE,            <==1ST BCDS SPLIT
000027 //      BCDS2=NULLFILE,            <==2ND BCDS SPLIT
000028 //      BCDS3=NULLFILE,            <==3RD BCDS SPLIT
000029 //      OCDS=HSM.OCDS,              <===INSTALLATION OCDS DSNAME
000030 //      OCDS1=NULLFILE,            <==1ST OCDS SPLIT
000031 //      OCDS2=NULLFILE,            <==2ND OCDS SPLIT
000032 //      OCDS3=NULLFILE,            <==3RD OCDS SPLIT
000033 //      LOAD=PPFX.LOAD               <==LOAD LIBRARY AS INSTALLED
000034 //*
-----
000265 //TTCVRFY DD DSN=&PFX..FAUDIT.TTCVRF,DISP=SHR,
000266 //      DCB=BUFNO=10
000267 //      PEND
000268 //PROC1 EXEC FAUDITRM,
000269 //      OCDS=SIS.HSM.OCDS,
000270 //      BCDS=SIS.HSM.SPLIT.BCDS1,
000271 //      MCDS=SIS.HSM.SPLIT.MCDS1,
000272 //      MCDS1=SIS.HSM.SPLIT.MCDS2,
000273 //      MSG=SIS.RMM.SMS.MESSAGE,
000274 //      RPT=SIS.RMM.SMS.REPORT,
000275 //      LOAD=FAUDITX.QABETA.LOAD,
000276 //      PPFX=FAUDITX.QABETA,
000277 //      PFX=DES2

```

Figure 8-1 Cataloged procedure FAUDITRM showing symbolic substitutions

## 8.1.2 FAUDITTP procedure setup

The parameters in the FAUDITTP procedure must be customized for your environment.

|                |                                                                                                                                                                                                                                                                                                                                                                                       |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>PPFX</b>    | Product Prefix as installed. For example, if the product has been installed with MSC.FAUDIT as a prefix to the libraries, then PPFX=MSC.FAUDIT must be coded in the EXEC PROC statement.                                                                                                                                                                                              |
| <b>PFX</b>     | Prefix represents the Runtime Prefix the user has chosen for this cataloged procedure to allocate work files that contain the audit results. Depending on the SMS treatment of new allocations, the work data sets may be deleted within a day or so. Choose a prefix that will not be scratched for a while so the results of the audit may be <i>reviewed</i> and <i>repaired</i> . |
| <b>PUNIT</b>   | Permanent unit for allocating data sets containing the audit results. The default is SYSDA. This unit should allocate data sets to a device that retains data for several days or weeks.                                                                                                                                                                                              |
| <b>WRKUNIT</b> | Work Unit for allocating temporary data sets that will be deleted at the end of the job. The default is SYSDA. If your installation has a different standard, code the different name.                                                                                                                                                                                                |
| <b>ATP</b>     | Allocation Type. Allocation is given in units, such as PRI and SEC. For the larger shops the unit TRKS sometimes isn't enough. If your installation is large, change the ATP to CYL.                                                                                                                                                                                                  |
| <b>PRI</b>     | The number of primary allocation units. The default is 100. This value must be adjusted to accommodate the size of your HSM MCDS and the number of records.                                                                                                                                                                                                                           |
| <b>SEC</b>     | The number of secondary allocation units. The default is 200. This parameter is important to adjust for the size of your MCDS.                                                                                                                                                                                                                                                        |
| <b>TMC</b>     | TMC is the name of the CA-1 Tape Management Catalog and is similar to the RMM database in contents.                                                                                                                                                                                                                                                                                   |
| <b>MCDS</b>    | MCDS is required. It is the first split of the DFSMSHsm MCDS data set or the only split if the MCDS is not split.                                                                                                                                                                                                                                                                     |
| <b>MCDS1</b>   | MCDS1 is the second split of the MCDS. The default is NULLFILE.                                                                                                                                                                                                                                                                                                                       |
| <b>MCDS2</b>   | MCDS2 is the third split of the MCDS. The default is NULLFILE.                                                                                                                                                                                                                                                                                                                        |
| <b>MCDS3</b>   | MCDS3 is the fourth split of the MCDS. The default is NULLFILE.                                                                                                                                                                                                                                                                                                                       |
| <b>BCDS</b>    | BCDS is required. It is the first split of the DFSMSHsm BCDS data set or the only split if the BCDS is not split.                                                                                                                                                                                                                                                                     |
| <b>BCDS1</b>   | BCDS1 is the second split of the BCDS. The default is NULLFILE.                                                                                                                                                                                                                                                                                                                       |
| <b>BCDS2</b>   | BCDS2 is the third split of the BCDS. The default is NULLFILE.                                                                                                                                                                                                                                                                                                                        |
| <b>BCDS3</b>   | BCDS3 is the fourth split of the BCDS. The default is NULLFILE.                                                                                                                                                                                                                                                                                                                       |
| <b>OCDS</b>    | OCDS is the name of the Offline Control Data Set.                                                                                                                                                                                                                                                                                                                                     |
| <b>LOAD</b>    | LOAD is the name of the load library as installed with <i>FastAudit/390</i> . It must be authorized.                                                                                                                                                                                                                                                                                  |



Figure 8-2 shows the symbolic parameters in the FAUDITTP JCL used to execute the procedure for the DFSMSHsm Tape to CA-1 Audit. The figure shows a single MCDS named HSM.HSM.MCDS, a single BCDS named HSM.HSM.BCDS, and the OCDS is named HSM.HSM.OCDS. The product prefix is ZGST66.DAWN.PS6101, the run time prefix is ZGST66. The load library for *FastAudit/390* in this case is ZGST66.HFA6101.LOAD.

```

000010 //FAUDITTP PROC PFX=PFX,          <==RUN TIME PREFIX
000011 //* FASTAUDIT <V6 R1.01> SEPTEMBER, 2001
000012 //      PFX=PFX,          <==PRODUCT PREFIX
000013 //      PUNIT=SYSDA,      <==UNIT FOR PERMANENT ALLOCATION
000014 //      SOUT=*,          <==PREFERED SYSOUT CLASS
000015 //      WRKUNIT=SYSDA,    <==UNIT FOR TEMPORARY ALLOCATION
000016 //      ATP=TRK,          <==TYPE OF ALLOCATION TRK/CYL
000017 //      PRI=100,          <==PRIMARY ALLOCATION FOR FAUDIT
000018 //      SEC=200,          <==SECONDARY SPACE ALLOCATION
000019 //      MSG=RMM.SMS.MESSAGE, <==RMM MESSAGE LIBRARY
000020 //      RPT=RMM.SMS.REPORT, <==RMM REPORT LIBRARY
000021 //      MCDS=HSM.MCDS,     <==INSTALLATION MCDS DSNAME
000022 //      MCDS1=NULLFILE,    <==1ST MCDS SPLIT
000023 //      MCDS2=NULLFILE,    <==2ND MCDS SPLIT
000024 //      MCDS3=NULLFILE,    <==3RD MCDS SPLIT
000025 //      BCDS=HSM.BCDS,     <==INSTALLATION BCDS DSNAME
000026 //      BCDS1=NULLFILE,    <==1ST BCDS SPLIT
000027 //      BCDS2=NULLFILE,    <==2ND BCDS SPLIT
000028 //      BCDS3=NULLFILE,    <==3RD BCDS SPLIT
000029 //      OCDS=HSM.OCDS,     <==INSTALLATION OCDS DSNAME
000030 //      OCDS1=NULLFILE,    <==1ST OCDS SPLIT
000031 //      OCDS2=NULLFILE,    <==2ND OCDS SPLIT
000032 //      OCDS3=NULLFILE,    <==3RD OCDS SPLIT
000033 //      LOAD=PPFX.LOAD      <==LOAD LIBRARY AS INSTALLED
000034 //*
-----
000253 //TTCVRFY DD DSN=&PFX..FAUDIT.TTCVRF,DISP=SHR,DCB=BUFNO=255
000254 //*
000255 //      PEND
000256 //PROC1 EXEC FAUDITTP,
000257 //      MCDS=HSM.HSM.MCDS,
000258 //      TMC=TAPES.TMS.TMC,
000259 //      BCDS=HSM.HSM.BCDS,
000260 //      OCDS=HSM.HSM.OCDS,
000261 //      LOAD=ZGST66.HFA6101.LOAD,
000262 //      PFX=ZGST66.DAWN.PS6101,
000263 //      PFX=ZGST66

```

Figure 8-2 Cataloged procedure FAUDITTP showing symbolic substitutions

### 8.1.3 FAUDITZR procedure setup

The parameters in the FAUDITZR procedure must be customized for your environment.

|                |                                                                                                                                                                                                                                                                                                                                                                        |
|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>PPFX</b>    | Product Prefix as installed. For example, if the product has been installed with MSC.FAUDIT as a prefix to the libraries, then PPFX=MSC.FAUDIT must be coded in the EXEC PROC statement.                                                                                                                                                                               |
| <b>PFX</b>     | Prefix represents the Runtime Prefix the user has chosen for this cataloged procedure to allocate work files that contain the audit results. Depending on the SMS treatment of new allocations, the work data sets may be deleted within a day or so. Choose a prefix that will not be scratched for a while so the results of the audit may be reviewed and repaired. |
| <b>PUNIT</b>   | Permanent unit for allocating data sets containing the audit results. The default is SYSDA. This unit should allocate data sets to a device that retains data for several days or weeks.                                                                                                                                                                               |
| <b>WRKUNIT</b> | Work Unit for allocating temporary data sets that will be deleted at the end of the job. The default is SYSDA. If your installation has a different standard, code the different name.                                                                                                                                                                                 |
| <b>ATP</b>     | Allocation Type. Allocation is given in units, such as PRI and SEC. For the larger shops the unit TRKS sometimes isn't enough. If your installation is large, change the ATP to CYL.                                                                                                                                                                                   |
| <b>PRI</b>     | The number of primary allocation units. The default is 100. This value must be adjusted to accommodate the size of your HSM MCDS and the number of records.                                                                                                                                                                                                            |
| <b>SEC</b>     | The number of secondary allocation units. The default is 200. This parameter is important to adjust for the size of your MCDS.                                                                                                                                                                                                                                         |
| <b>TMC</b>     | TMC is the name of the Zara Tape Management Catalog and is similar to the RMM database in contents.                                                                                                                                                                                                                                                                    |
| <b>VAULT</b>   | The name of the vault data set used by the ZARA tape management product.                                                                                                                                                                                                                                                                                               |
| <b>MCDS</b>    | MCDS is required. It is the first split of the DFSMSHsm MCDS data set or the only split if the MCDS is not split.                                                                                                                                                                                                                                                      |
| <b>MCDS1</b>   | MCDS1 is the second split of the MCDS. The default is NULLFILE.                                                                                                                                                                                                                                                                                                        |
| <b>MCDS2</b>   | MCDS2 is the third split of the MCDS. The default is NULLFILE.                                                                                                                                                                                                                                                                                                         |
| <b>MCDS3</b>   | MCDS3 is the fourth split of the MCDS. The default is NULLFILE.                                                                                                                                                                                                                                                                                                        |
| <b>BCDS</b>    | BCDS is required. It is the first split of the DFSMSHsm BCDS data set or the only split if the BCDS is not split.                                                                                                                                                                                                                                                      |
| <b>BCDS1</b>   | BCDS1 is the second split of the BCDS. The default is NULLFILE.                                                                                                                                                                                                                                                                                                        |

|              |                                                                                                      |
|--------------|------------------------------------------------------------------------------------------------------|
| <b>BCDS2</b> | BCDS2 is the third split of the BCDS. The default is NULLFILE.                                       |
| <b>BCDS3</b> | BCDS3 is the fourth split of the BCDS. The default is NULLFILE.                                      |
| <b>OCDS</b>  | OCDS is the name of the Offline Control Data Set.                                                    |
| <b>LOAD</b>  | LOAD is the name of the load library as installed with <i>FastAudit/390</i> . It must be authorized. |

Figure 8-3 shows the symbolic parameters in the FAUDITZR JCL used to execute the procedure for the DFSMSHsm Tape to CA-1 Audit. The figure shows a single MCDS named HSM.MCDS, a single BCDS named HSM.HSM.BCDS, and the OCDS is named HSM.OCDS. The product prefix is ZGST66.DAWN.PS6101, the run time prefix is ZGST66. The load library for *FastAudit/390* in this case is ZGST66.HFA6101.LOAD. The tape management catalog is TMS.PR.ZARA.DATABASE and the vault name is TMS.PR.ZARA.DATABASE.DUPLEX.

```

//USERID1 JOB      ,'HSM FASTAUDIT07S',CLASS=B,MSGLEVEL=(1,1),
//              MSGCLASS=X,NOTIFY=USERID,TIME=1
//*
//FAUDITZR PROC PFX=PFX,                <==RUN TIME PREFIX
//* <V6 R1.xx> OCT 15, 2001
//      PPFX=PPFX,                      <==PRODUCT PREFIX
//      PUNIT=SYSDA,                    <==UNIT FOR PERMANENT ALLOCATION
//      SOUT=*,                          <==PREFERED SYSOUT CLASS
//      WRKUNIT=SYSDA,                  <==UNIT FOR TEMPORARY ALLOCATION
//      PRI=100,                        <==PRIMARY ALLOCATION FOR FAUDIT
//      SEC=200,                        <==SECONDARY SPACE ALLOCATION
//      TMC=TMS.TMC,                    <==ZARA TAPE MANAGEMENT CATALOG
//      VAULT=VAULT,                    <==ZARA TAPE MANAGEMENT CATALOG
//      MCDS=HSM.MCDS,                  <==INSTALLATION MCDS DSNAME
//      MCDS1=NULLFILE,                 <==1ST MCDS SPLIT
//      MCDS2=NULLFILE,                 <==2ND MCDS SPLIT
//      MCDS3=NULLFILE,                 <==3RD MCDS SPLIT
//      BCDS=HSM.BCDS,                  <==INSTALLATION BCDS DSNAME
//      BCDS1=NULLFILE,                 <==1ST BCDS SPLIT
--
//*
//PROC1      EXEC FAUDITZR,
//            MCDS=HSM.MCDS,              <==change to HSM MCDS name
//            TMC=TMS.PR.ZARA.DATABASE,   <==change to zara data base
//            VAULT=TMS.PR.ZARA.DATABASE.DUPLEX, <==change to zara vault name
//            BCDS=HSM.BCDS,              <==change to hsm BCDS
//            OCDS=HSM.OCDS,              <==change to hsm OCDS
//            LOAD=ZGST66.HFA6101.LOAD,   <==change to product load
library
//            PFX=ZGST66                  <==change to run time prefix
//            PPFX=ZGSTGG.DAWN.PS6101    <==change to product prefix

```

Figure 8-3 Cataloged procedure FAUDITZP showing symbolic substitutions

## 8.2 Viewing the results of the audit

After the audit job completes, execute the CLIST FIXAUDIT. This CLIST displays the main panel, Fast Audit, Choose Which Audit To Fix, shown in Figure 8-4, and the various audit results options available.

```

<V6 R1.01> FAST AUDIT. CHOOSE WHICH AUDIT TO FIX
COMMAND==> 4
 1 MCDS   CREATED BY PROCS FAUDITMD, FAUDITCT OR FAUDITAL
 2 BCDS   CREATED BY PROC FAUDITBD
 3 OCDS   CREATED BY PROC FAUDITOD
 4 TAPE   CREATED BY PROCS FAUDITTP, FAUDITZR OR FAUDITRM
 5 CAT    CREATED BY PROC FAUDITCO
 6 DASD   CREATED BY PROC FAUDITDA OR FAUDITDE
 7 MASTCAT CREATED BY PROC FAUDITMC
 8 CATRMM CREATED BY PROC FAUDITCR
 9 CATTMS  CREATED BY PROC FAUDITTC

ENTER  END  TO TERMINATE ONLINE FUNCTION

```

Figure 8-4 Primary Choose Which Audit To Fix panel

Select option 4 TAPE, by typing the number 4 on the command line and pressing Enter. The Tape Error Summary Panel is displayed in Figure 8-5.

```

<V6 R1.01> Row 1 of 10
FASTAUDIT TAPE SUMMARY
OPTION ==>
ENTER F TO DISPLAY FIXES OR B TO BROWSE ERRORS
SCROLL==> CSR
S NUMBER  COUNT  MESSAGE
3101W      0    TAPE IS IN MCV/MCT, NOT IN TTOC
3102C      0    TAPE IS IN TTOC, NOT IN MCV/MCT
3103C      5    TAPE IS NOT IN TTOC OR MCV/MCT BUT HAS ENTRIES
3104W      0    TAPE IS IN TTOC & MCV/MCT, NOT IN TMS
3110W      0    TAPE IS PERMANENT IN TMS, NOT IN TTOC
3111W      1    TAPE IS IN TMS, VALID IN HSM, HAS NO ENTRIES
3112W      0    TAPE IS VALID IN HSM BUT EXPIRED IN TMS
3114I     29    TAPE IS IN TMS SCRATCH STATUS
3115I    4072   TAPE IS IN TMS DELETE STATUS
3116W      0    TAPE DOES NOT INDICATE EXTERNAL DATA MANAGEMENT

```

Figure 8-5 Tape error summary panel

There are subsequent panels behind each error number listed. These panels provide the list of FIXES to either diagnose or resolve the various errors. In all cases, you can view the list of available fixes by keying the letter F next to the error number and pressing Enter. The Fast Audit Fix Plan Member List for that error number is displayed and lists the available fixes for the error as shown in Figure 8-6.

```

<V6 R1.01>
FAST AUDIT FIX PLAN MEMBER LIST
OPTION ==>
YOU MAY BROWSE (B), EDIT (E), FIX (F) OR VIEW (V) RESULTS
COMMAND MEMBER ACTION TITLE
F3102CA HSM FIXCDS DISPLAY V RECORD FOR MISSING MCV
F3102CB RMM INTERROGATE RMM FOR HSM TAPES
F3102CC HSM FIXCDS TO ADD NEW MCV LEVEL 2 RECORD
F3102CD HSM FIXCDS TO ADD NEW MCT SPILL TAPE
F3102CE HSM FIXCDS DISPLAY X RECORD FOR MISSING MCT
F3102CF HSM FIXCDS PATCH TIOC FOR COPY VOLUME
F3102CG HSM FIXCDS TO ADD NEW MCT CYCLE TAPE
F3102CH HSM FIXCDS DELETE X RECORD
***** Bottom of data *****

```

Figure 8-6 Tape Audit Fix Plan Member List

FastAudit/390 automatically generates the required JCL and FIX statements for each FIX listed when you type the letter F next to the FIX member name and press Enter. Figure 8-6 is an example of the JCL and FIX statements generated for fix F4012CR2.

```

File Edit Edit_Settings Menu Utilities Compilers Test Help

EDIT CG.H074639 Columns 00001 00072
Command ==> Scroll ==> CSR
***** Top of Data *****
000001 //CGA JOB (0),COLGOR,
000002 // CLASS=A,MSGCLASS=X,NOTIFY=COLGOR
000003 //*
000004 //STEP1 EXEC PGM=IKJEFT01,
000005 // DYNAMNBR=25,REGION=4096K
000006 //* CHANGE THE FOLLOWING STATEMENT TO
000007 //* DSN=PPFX.CLIST WHERE PPFX IS THE PREFIX ASSIGNED TO
000008 //* THE FASTAUDIT PRODUCT
000009 //*
000010 //SYSPROC DD DSN=FAUDITX.QABETA.CLIST,DISP=SHR
000011 //* *****
000012 //SYSOUT DD SYSOUT=*
000013 //SYSTSPRT DD SYSOUT=*
000014 //SYSTSIN DD *
000015 HSEND WAIT FIXCDS V 003625 CREATE(X'00' '3480 ') +
000016 ODS('CG.FIXAUDIT.F3102CC') LOGONLY
000017 HSEND FIXCDS V 003625 PATCH(X'08' X'7140') +
000018 ODS('CG.FIXAUDIT.F3102CC') LOGONLY
000019 HSEND WAIT FIXCDS V 111111 CREATE(X'00' '3480 ') +

```

Figure 8-7 JCL statements automatically generated for fix F3102CC

You can use ISPF edit commands to edit the member and remove any records that you do not want to FIX or diagnose. Once your editing is complete or if no editing is desired, execute the FIX JCL by typing the word SUBMIT or the short version SUB on the command line in the edit entry panel and pressing Enter. A batch job is submitted to execute the FIX commands.

When the FIX batch job has completed you can view the results by keying the letter V next to the FIX member name and pressing Enter. This action results in an ISPF view of the FIX results.

The error messages are identified as:

- ▶ (C) - Critical means that a data set cannot be recovered using IBM provided Recover facilities.
- ▶ (V) - Volume is summaries of error detail volumes.
- ▶ (I) - Information only.
- ▶ (W) - Warning means that the data set might not be recoverable and further investigation is required to determine the status of the data set.

The following shows all of the error messages and their description for Tape Audit.

|       |                                                |
|-------|------------------------------------------------|
| 3101W | TAPE IS IN MCV/MCT, NOT IN TTOC                |
| 3102C | TAPE IS IN TTOC, NOT IN MCV/MCT                |
| 3103C | TAPE IS NOT IN TTOC OR MCV/MCT BUT HAS ENTRIES |
| 3104W | TAPE IS IN TTOC & MCV/MCT, NOT IN RMM          |
| 3110W | TAPE IS PERMANENT IN RMM, NOT IN TTOC          |
| 3111W | TAPE IS IN RMM, VALID IN HSM, HAS NO ENTRIES   |
| 3112W | TAPE IS VALID IN HSM BUT EXPIRED IN RMM        |
| 3114I | TAPE IS IN RMM SCRATCH STATUS                  |

### 8.2.1 *FastAudit/390* audit error 3101W

Now we will look at error messages produced by Tape Audit. The first error message is 3101W TAPE IS IN MCV/MCT, NOT IN TTOC.

Looking up the error in the *FastAudit/390* User Guide, we see the following explanation:

When *FastAudit/390* extracts the tape information it examines the elements in DFSMSHsm that control the tapes. The MCV record is for Level 2 tapes, and the MCT record is for cycle and spill tapes (backup). Recall or recover may be successful even though a TTOC record for the tape is missing. If a TTOC

record is missing DFSMSHsm cannot recycle the tapes. If the tapes are under CA-1 control then there may be many tapes that are empty and should be scratched. The information in this error record may be used to interrogate CA-1.

To view the error detail, type **B** on the FastAudit Tape Summary panel next to the error and press Enter. The Field Description Box provided in the documentation and shown in Table 8-1 provides an explanation of each field.

Table 8-1 Diagnostic 3101W field descriptions

| Field # / Description | Example                 |
|-----------------------|-------------------------|
| FIELD 1 ERROR NUMBER  | 3101W                   |
| FIELD 2 VOLSER        | 010018                  |
| FIELD 3 TAPE TYPE     | L2 (SP CYCLE DAY OR DP) |
| FIELD 4 DSN COUNT     | 000034                  |

Resolving the error

To view the list of available fixes, type **F** next to the error message and press Enter. The Fast Audit Fix Plan Member List is displayed in Figure 8-8.

```
<V6 R1.01> Row 1 of 6
      FAST AUDIT FIX PLAN MEMBER LIST
OPTION ===> SCROLL==> PAGE
YOU MAY BROWSE (B), EDIT (E), FIX (F) OR VIEW (V) RESULTS
COMMAND MEMBER ACTION TITLE
      F3101WA TMS INTERROGATE TMS FOR HSM TAPES SANS TTOC
      F3101WB HSM VERIFY TTOC VOLUME MISSING FROM OCDS
      F3101WC ZRA INTERROGATE ZARA FOR HSM TAPES SANS TTOC
      F3101WD RMM INTERROGATE RMM FOR HSM TAPES
      F3101WE HSM DELETE MCV RECORD FROM MCDS
      F3101WF HSM FIXCDS DISPLAY THE TTOC RECORD FROM OCDS
***** Bottom of data *****
```

Figure 8-8 Fix Plan Member List for error 3101W

The fix F3101WB can be executed to verify that the TTOC records are missing for the tapes. If the TTOC records are missing DFSMSHsm cannot recycle the tape even though valid data sets can be recalled or recovered. If the tapes are empty (no valid entries), the fix F3101WE can be executed to delete the MCV record and the tapes can be scratched. If there is valid data on the tapes, execute the DFSMSHsm Audit Command AUDITMEDIACONTROLS for the tapes. An alternative to running AUDITMEDIACONTROLS is to recall the data sets on the tape (if there are few), run the fix F3101WE and scratch the tape.



## 8.2.2 *FastAudit/390* audit error 3102C

The next error message we will look at is 3102C TAPE IS IN TTOC, NOT IN MCV/MCT.

Looking up the error in the *FastAudit/390* User Guide, we see the following explanation:

When *FastAudit/390* extracts the tape information it examines the elements in DFSMSHsm that control the tapes. It has found a TTOC entry that is not being managed by DFSMSHsm. The valid data sets on these tapes cannot be recalled. In addition they may have been scratched and/or written over. Use this information to interrogate the catalog database. If the tape still has DFSMSHsm data set names on it, the data might be reconstructed.

To view the error detail, type **B** on the FastAudit Tape Summary panel next to the error and press Enter. The Field Description Box provided in the documentation and shown in Table 8-2 provides an explanation of each field.

Table 8-2 *Diagnostic 3102C field descriptions*

| Field # / Description | Example                 |
|-----------------------|-------------------------|
| FIELD 1 ERROR NUMBER  | 3102C                   |
| FIELD 2 VOLSER        | 010020                  |
| FIELD 3 TAPE TYPE     | L2 (SP CYCLE DAY OR DP) |
| FIELD 4 DSN COUNT     | 000004                  |

### Resolving the error

To view the list of available fixes, type **F** next to the error message and press Enter. The Fast Audit Fix Plan Member List is displayed in Figure 8-9.

```
<V6 R1.01>                                     Row 1 of 8
FAST AUDIT FIX PLAN MEMBER LIST
OPTION ==>                                     SCROLL==> PAGE
YOU MAY BROWSE (B), EDIT (E), FIX (F) OR VIEW (V) RESULTS
COMMAND  MEMBER  ACTION  TITLE
F3102CA  HSM FIXCDS DISPLAY V RECORD FOR MISSING MCV
F3102CB  RMM INTERROGATE RMM FOR HSM TAPES
F3102CC  HSM FIXCDS TO ADD NEW MCV LEVEL 2 RECORD
F3102CD  HSM FIXCDS TO ADD NEW MCT SPILL TAPE
F3102CE  HSM FIXCDS DISPLAY X RECORD FOR MISSING MCT
F3102CF  HSM FIXCDS PATCH TTOC FOR COPY VOLUME
F3102CG  HSM FIXCDS TO ADD NEW MCT CYCLE TAPE
F3102CH  HSM FIXCDS DELETE X RECORD
***** Bottom of data *****
```

Figure 8-9 Fix Plan Member List for error 3102C

If the tapes have been scratched and written over, the data cannot be recovered. The valid data sets on the tape must be deleted using the DFSMSHsm HDELETE command and recovered using the DFSMSHsm HRECOVER command. If the data has not been written over, you can execute fixes F3102CC (if the tape was an ML2 tape), fix F3102CD (if the tape was a spill tape), or fix F3102CG (if the tape was a cycle tape), to create the new records for the tape.

### 8.2.3 FastAudit/390 audit error 3103C

The next error we will look at is 3103C TAPE IS NOT IN TTOC OR MCV/MCT BUT HAS ENTRIES.

#### Diagnosing the error

Looking up the error in the *FastAudit/390* User Guide, we see the following explanation:

*FastAudit/390* has discovered tapes that have Level 2 dump volumes or backup data sets, but the tape is not under DFSMSHsm control. The tapes might still be managed by CA-1. If this data is critical, use CA-1 to interrogate the status of these tapes. If no record of them is in CA-1, the data sets referencing these tapes may be deleted from DFSMSHsm.

To view the error detail, type **B** on the FastAudit Tape Summary panel next to the error and press Enter. The Field Description Box provided in the documentation and shown in Table 8-3 provides an explanation of each field.

Table 8-3 Diagnostic 3103C field descriptions

| Field # / Description | Example                   |
|-----------------------|---------------------------|
| FIELD 1 ERROR NUMBER  | 3103C                     |
| FIELD 2 VOLSER        | 010020                    |
| FIELD 3 TAPE TYPE     | L2 (SP, CYCLE DAY, OR DP) |
| FIELD 4 DSN COUNT     | 000004                    |

### Resolving the error

To view the list of available fixes, type **F** next to the error message and press Enter. The Fast Audit Fix Plan Member List is displayed in Figure 8-10.

```
<V6 R1.01>                                     Row 1 of 4
      FAST AUDIT FIX PLAN MEMBER LIST
OPTION ==>                                     SCROLL==> PAGE
YOU MAY BROWSE (B), EDIT (E), FIX (F) OR VIEW (V) RESULTS
COMMAND MEMBER ACTION TITLE
      F3103CA      TMS INTERROGATE TMS FOR TAPE WITH ENTRIES
      F3103CB      ZRA INTERROGATE ZARA FOR HSM TAPES WITH ENTRIES
      F3103CC      RMM INTERROGATE RMM FOR HSM TAPES
      F3103CR1     HSM BUILD LIST TTOC FOR TAPES W/ENTRIES
***** Bottom of data *****
```

Figure 8-10 Fix Plan Member List for error 3103C

These tapes are most often old or obsolete and can be scratched. In some cases, they are tapes that were owned by another DFSMSHsm or CA-1 environment and the two environments have been merged or the errors can result from a split apart of your DFSMSHsm environment. If they are valid tapes they are currently not being managed by DFSMSHsm. Execute fix F3103CA to interrogate CA-1 or RMM and if they are being managed by the tape management system and what their status is. Execute fix F3103CR1 to LIST TTOC on the volumes and get a listing that shows all of the valid entries on the tape. If the entries are invalid (from another DFSMSHsm, or no longer needed), the data set records should be deleted from DFSMSHsm.

### 8.2.4 FastAudit/390 audit error 3104W

The next error we will look at is 3104W TAPE IS IN TTOC & MCV/MCT, NOT IN TMS (RMM).

### Diagnosing the error

Looking up the error in the *FastAudit/390* User Guide, we see the following explanation:

*FastAudit/390* has compared the DFSMSHsm tape extraction against the tape Inventory extraction and has found DFSMSHsm managed tapes that are not under tape management control. It's possible the tape has been written over. Use the Tape Inventory interrogation system to find the current name of the tape. If the tape shows 'invalid' then it is not managed and the data is probably safe. If Tape Inventory shows a DSNName other than the appropriate DFSMSHsm data set name then the tape has probably been written on and the DFSMSHsm data has been lost.

To view the error detail, type **B** on the FastAudit Tape Summary panel next to the error and press Enter. The field description box provided in the documentation and shown in Table 8-4 provides an explanation of each field.

Table 8-4 Diagnostic 3104W field descriptions

| Field # / Description | Example                 |
|-----------------------|-------------------------|
| FIELD 1 ERROR NUMBER  | 3104W                   |
| FIELD 2 VOLSER        | 010021                  |
| FIELD 3 TAPE TYPE     | L2 (SP CYCLE DAY OR DP) |
| FIELD 4 DSN COUNT     | 000004                  |

### Resolving the error

To view the list of available fixes, type **F** next to the error message and press Enter. The Fast Audit Fix Plan Member List is displayed in Figure 8-11.

```
<V6 R1.01>                                     Row 1 of 11
      FAST AUDIT FIX PLAN MEMBER LIST
OPTION ==>                                     SCROLL==> PAGE
YOU MAY BROWSE (B), EDIT (E), FIX (F) OR VIEW (V) RESULTS
COMMAND  MEMBER  ACTION  TITLE
F3104WA  TMS INTERROGATE EXPIRED TAPE IN CA-1 TMS
F3104WB  ZRA INTERROGATE EXPIRED TAPE IN ZARA
F3104WC  HSM LIST CONTENTS OF TAPE MISSING IN ZARA
F3104WD  RMM INTERROGATE RMM FOR HSM TAPES
F3104WE  HSM FIXCDS DISPLAY OF TT0C RECORD
F3104WF  HSM FIXCDS DISPLAY OF DVL RECORD
F3104WG  HSM FIXCDS T PATCH COPY VOLUME GONE
F3104WR1 HSM LIST ACTIVE DATA FROM TAPE
F3104WR2 HSM DELVOL PURGE FOR BACKUP VOLUMES
F3104WR3 HSM DELVOL PURGE FOR ML2 VOLUMES
F3104WR4 HSM DELVOL PURGE FOR DUMP VOLUMES
***** Bottom of data *****
```

Figure 8-11 Fix Plan Member List for error 3104W

Tape Audit provides several fixes to interrogate TMS, ZARA or RMM. These fixes list the status of the tape in the tape management database. The report also shows the data set name of the tape. From this you can determine if the tape has been written over or not. If the data set name of the tape is the valid DFSMSHsm data set name, the problem could be that the tape management system does not show the tapes being managed by DFSMSHsm. This could indicate that some DFSMSHsm tapes were created at a time when the ARCTVEXT was not activated. If the tape is not valid, use fixes F3104WR2, F3104WR3 or F4104WR4 to DELVOL the tape from DFSMSHsm management.

## 8.2.5 FastAudit/390 audit error 3110W

The next error we will look at is 3110W TAPE IS PERMANENT IN TMS (RMM), NOT IN TTOC.

### Diagnosing the error

Looking up the error in the *FastAudit/390 User Guide*, we see the following explanation:

*FastAudit/390* has compared the DFSMSHsm tape extraction against the Tape management database extraction and has found managed tapes marked as 'permanent'. DFSMSHsm has no record of the tapes. It's possible that these tapes have been orphaned and may be scratched.

To view the error detail, type **B** on the FastAudit Tape Summary panel next to the error and press Enter. The Field Description Box provided in the documentation and shown in Table 8-5 provides an explanation of each field.

Table 8-5 Diagnostic 3110W field descriptions

| Field # / Description   | Example                 |
|-------------------------|-------------------------|
| FIELD 1 ERROR NUMBER    | 3110W                   |
| FIELD 2 VOLSER          | 010021                  |
| FIELD 3 TAPE TYPE       | L2 (SP CYCLE DAY OR DP) |
| FIELD 4 DSN COUNT       | 000004                  |
| FIELD 5 EXPIRATION DATE | PERMANENT (1999/365)    |
| FIELD 6 TAPE DSNAME     | HSM.HMIGTAPE.DATASET    |
| FIELD 7 CREATING JOB    | HSM                     |

Resolving the error

To view the list of available fixes, type **F** next to the error message and press Enter. The Fast Audit Fix Plan Member List is displayed in Figure 8-12.

```
<V6 R1.01> Row 1 of 8
FAST AUDIT FIX PLAN MEMBER LIST
OPTION ===> SCROLL==> PAGE
YOU MAY BROWSE (B), EDIT (E), FIX (F) OR VIEW (V) RESULTS
COMMAND MEMBER ACTION TITLE
F3110WA HSM FIXCDS DISPLAY MCV FOR TMS MISSING TTOC
F3110WB TMS INTERROGATE TAPE IN TMS, NOT IN TTOC
F3110WC RMM INTERROGATE RMM FOR HSM TAPES
F3110WD HSM FIXCDS DISPLAY OF DVL RECORD
F3110WR1 HSM ADDVOL/DELVOL FOR BACKUP TMS MISSING TTOC
F3110WR2 HSM ADDVOL/DELVOL FOR LEVEL2 TMS MISSING TTOC
F3110WU TMC EXPIRE TMS TAPES
F3110WV ZRA EXPIRE ZARA TAPES
***** Bottom of data *****
```

Figure 8-12 Fix Plan Member List for error 3110W

These tapes are being held permanently by the tape management system as DFSMSHsm owned tapes, but DFSMSHsm does not own them. If the condition is valid, the tapes can be scratched and reclaimed by executing fixes F3110WU or F3110WV. If the condition is not valid DFSMSHsm should own these tapes. It is possible that the ARCTVEXT was not in place when these tapes were created.

Execute fix F3110WR1 or F3110WR2 to ADDVOL the tapes so that DFSMSHsm can manage them.

### 8.2.6 FastAudit/390 audit error 3111W

The next error we will look at is 3111W TAPE IS IN TMS (RMM), VALID IN HSM, HAS NO ENTRIES.

#### Diagnosing the error

Looking up the error in the *FastAudit/390 User Guide*, we see the following explanation:

*FastAudit/390* has compared the DFSMSHsm tape extraction against the tape management database extraction and has found managed tapes marked as 'permanent'. DFSMSHsm has all the appropriate records but no data is written on the tape. This condition could be valid. However, if there are a very large number of these tapes, the analyst should investigate why. *FastAudit/390* has extracted all data sets on ML2 all backup versions on tape, and all volume dump tapes. When any tape volume is referenced it's marked as occupied. This warning message displays the tape volumes that have not been referenced, however, all the appropriate structures for tapes in the DFSMSHsm environment are correct and tape management database indicates that DFSMSHsm has control.

To view the error detail, type **B** on the FastAudit Tape Summary panel next to the error and press Enter. The Field Description Box provided in the documentation and shown in Table 8-6 provides an explanation of each field.

Table 8-6 Diagnostic 311W field descriptions

| Field # / Description   | Example                   |
|-------------------------|---------------------------|
| FIELD 1 ERROR NUMBER    | 3111W                     |
| FIELD 2 VOLSER          | 010022                    |
| FIELD 3 TAPE TYPE       | L2 (SP, CYCLE DAY, OR DP) |
| FIELD 4 DSN COUNT       | 000004                    |
| FIELD 5 EXPIRATION DATE | PERMANENT (1999/365)      |
| FIELD 6 TAPE DSNAME     | HSM.HMIGTAPE.DATASET      |
| FIELD 7 CREATING JOB    | HSM                       |

### Resolving the error

To view the list of available fixes, type **F** next to the error message and press Enter. The Fast Audit Fix Plan Member List is displayed in Figure 8-13.

```
<V6 R1.01>                                     Row 1 of 11
      FAST AUDIT FIX PLAN MEMBER LIST
OPTION ===>                                     SCROLL==> PAGE
YOU MAY BROWSE (B), EDIT (E), FIX (F) OR VIEW (V) RESULTS
COMMAND  MEMBER  ACTION  TITLE
F3111WA  TMS INTERROGATE TMS FOR EMPTY HSM TAPES
F3111WB  HSM LIST CONTENTS OF TAPE WITH NO ENTRIES
F3111WC  ZRA INTERROGATE ACTIVE TAPE IN ZARA
F3111WD  RMM INTERROGATE RMM FOR HSM TAPES
F3111WE  HSM FIXCDS TO ADD NEW MCV LEVEL 2 RECORD
F3111WF  HSM FIXCDS DISPLAY OF MCV RECORD
F3111WR1 HSM DELVOL PURGE FOR ML2 VOLUMES
F3111WR2 HSM RECYCLE VOLUMES THAT HAD VALID DATA
F3111WR3 HSM DELVOL PURGE FOR BACKUP VOLUMES
F3111WR4 HSM DELVOL PURGE FOR DUMP VOLUMES
F3111WR5 HSM AUDIT MEDIACONTROLS FOR RECYCLE
***** Bottom of data *****
```

Figure 8-13 Fix Plan Member List for error 3111W

If there are a large number of these tapes, you should investigate why. These tapes are managed by DFSMSHsm, but they have not been referenced by DFSMSHsm. However, all of the appropriate structures for tapes in the DFSMSHsm environment are correct and the tape management database indicates that DFSMSHsm owns the tape. Tape Audit provides a number of fixes to help you interrogate the tape management database and DFSMSHsm to ensure that this condition is valid.

### 8.2.7 FastAudit/390 audit error 3112W

The next error we will look at is 3112W TAPE IS VALID IN HSM BUT EXPIRED IN TMS (RMM).

### Diagnosing the error

Looking up the error in the *FastAudit/390* User Guide, we see the following explanation:

*FastAudit/390* has compared the DFSMSHsm tape extraction against the tape management database and has found managed tapes that are due to expire. These tapes could be written over and valid DFSMSHsm data lost. It is possible that these tapes were released to Tape Inventory if recycle was running during the audit. If in doubt, rerun audit when recycle is not executing.



Examine the expiration date for the current Julian date to verify the scratch status. Other audit options may be necessary to determine the status of these tapes.

To view the error detail, type B on the FastAudit Tape Summary panel next to the error and press Enter. The Field Description Box provided in the documentation and shown in Table 8-7 provides an explanation of each field.

Table 8-7 Diagnostic 3112W field descriptions

| Field # / Description   | Example                 |
|-------------------------|-------------------------|
| FIELD 1 ERROR NUMBER    | 3112W                   |
| FIELD 2 VOLSER          | 010024                  |
| FIELD 3 TAPE TYPE       | SP (SP CYCLE DAY OR DP) |
| FIELD 4 DSN COUNT       | 000004                  |
| FIELD 5 EXPIRATION DATE | 15JUN1996               |
| FIELD 6 TAPE DSNAME     | HSM.HMIGTAPE.DATASET    |
| FIELD 7 CREATING JOB    | HSM                     |

## Resolving the error

To view the list of available fixes, type F next to the error message and press Enter. The Fast Audit Fix Plan Member List is displayed in Figure 8-7.

|                                                           |         |                                          |
|-----------------------------------------------------------|---------|------------------------------------------|
| <V6 R1.01>                                                |         | Row 1 of 5                               |
| FAST AUDIT FIX PLAN MEMBER LIST                           |         |                                          |
| OPTION ==>                                                |         | SCROLL==> PAGE                           |
| YOU MAY BROWSE (B), EDIT (E), FIX (F) OR VIEW (V) RESULTS |         |                                          |
| COMMAND                                                   | MEMBER  | ACTION TITLE                             |
|                                                           | F3112WA | TMS INTERROGATE EXPIRED TAPE IN CA-1 TMS |
|                                                           | F3112WB | HSM LIST CONTENTS OF TAPE EXPIRED IN TMS |
|                                                           | F3112WC | ZRA INTERROGATE EXPIRED TAPE IN ZARA     |
|                                                           | F3112WD | RMM INTERROGATE RMM FOR HSM TAPES        |
|                                                           | F3112WE | HSM GENERATE FIXCDS DELETE FOR DUMP TAPE |
| ***** Bottom of data *****                                |         |                                          |

Figure 8-14 Fix plan member list for error 3112W

It is possible that this error occurred because the audit was executed at the same time that recycle was running. If this is the case, rerun the audit when recycle is not running and check the results. These tapes have DFSMSHsm data on them but are not managed by DFSMSHsm. They are owned by the tape management

database and either expired or due to expire. If you find the condition to be true, check to ensure that the ARCTVEXT is activated.

### 8.2.8 FastAudit/390 audit error 3114I

The next error we will look at is 3114I TAPE IS SCHEDULED FOR SCRATCH.

#### Diagnosing the error

Looking up the error in the *FastAudit/390 User Guide*, we see the following explanation:

*FastAudit/390* has compared the DFSMSHsm tape extraction against the tape management database extraction and has found managed tapes that have expired. There are no DFSMSHsm references to these tapes and everything is working, as it should. This message is for information only to review the pool of scratch tapes released by DFSMSHsm but not yet selected for use by other applications.

To view the error detail, type **B** on the FastAudit Tape Summary panel next to the error and press Enter. The Field Description Box provided in the documentation and shown in Table 8-8 provides an explanation of each field.

Table 8-8 Diagnostic 3114I field descriptions

| Field # / Description   | Example              |
|-------------------------|----------------------|
| FIELD 1 ERROR NUMBER    | 3114I                |
| FIELD 2 VOLSER          | 010024               |
| FIELD 3 DSN COUNT       | 000000               |
| FIELD 4 EXPIRATION DATE | 15JUN1996            |
| FIELD 5 TAPE DSNAME     | HSM.HMIGTAPE.DATASET |
| FIELD 6 CREATING JOB    | HSM                  |

#### Resolving the error

To view the list of available fixes, type **F** next to the error message and press Enter. The Fast Audit Fix Plan Member List is displayed in Figure 8-15.

```
<V6 R1.01>                                     Row 1 of 1
      FAST AUDIT FIX PLAN MEMBER LIST
OPTION ===>                                     SCROLL==> PAGE
  YOU MAY BROWSE (B), EDIT (E), FIX (F) OR VIEW (V) RESULTS
COMMAND  MEMBER  ACTION      TITLE
      F3114IA      RMM INTERROGATE RMM FOR HSM TAPES
***** Bottom of data *****
```

*Figure 8-15 Fix Plan Member List for error 3114I*

This is an informational message only and does not indicate an error. You can execute fix F3114IA to list the tapes in the tape management database.





## **Auditing DFSMShsm CDSs using command audit**

DFSMShsm provides audit and enhanced audit commands to detect, report and diagnose errors in the control data sets. In some cases, an automatic fix is also provided. Chapter 4, “Why audit the DFSMShsm control data sets?” on page 33 provides information about why it is so important to audit and keep your control data sets free of errors. This chapter describes the enhanced audit commands and what the audits provide.

## 9.1 Using the AUDIT command

As we discussed in the introduction chapter of this book, catalogs, control data sets and the tape management catalogs are all metadata structures. Audit is used to ensure the integrity of these structures so that the data they manage is accessible to the organization.

A word of caution when using the DFSMSHsm provided audit and enhanced audit commands is that they should be issued during periods of low system activity. During an audit, DFSMSHsm serializes control records while it is comparing it to several sources of information, such as the ICF catalogs. While the control records are static, the comparisons are valid. If DFSMSHsm is active while the audit is run, some of these records may no longer be valid after the audit has completed. An example is a migrated data set that is being recalled while the audit is executing. The audit may report that the migrated data set's catalog entry is invalid because the VOLSER is not MIGRAT. Because the data set was being recalled while the audit was executing the catalog information is incorrect. In fact, the data set resides on a DASD volume that is properly recorded in the ICF catalog entry.

Another consideration is that the audit of each record may require several I/Os to one or more of the control data sets. In a large DFSMSHsm environment, the sheer quantity of records can cause an enormous amount of I/O activity to the control data set, impacting the overall performance of DFSMSHsm.

If your DFSMSHsm environment is always active or there is a large quantity of records in the control data sets causing the DFSMSHsm audit command to run for several hours, we recommend that you use Mainstar: *FastAudit/390* to audit your control data set structures.

## 9.2 Enhanced audit commands

The enhanced audit commands audit and report errors within the DFSMSHsm control data sets. The following enhanced audit commands are available with DFSMSHsm:

- ▶ AUDIT ABARSCONTROLS(agency)
- ▶ AUDIT DATASETCONTROLS(BACKUP)
- ▶ AUDIT DATASETCONTROLS(MIGRATION)
- ▶ AUDIT DIRECTORYCONTROLS VOLUMES
- ▶ AUDIT MEDIACONTROLS(SDSP) VOLUMES
- ▶ AUDIT MEDIACONTROLS VOLUMES

- ▶ AUDIT VOLUMECONTROLS(BACKUP)
- ▶ AUDIT VOLUMECONTROLS(MIGRATION)
- ▶ AUDIT VOLUMECONTROLS(RECOVERABLE)
- ▶ AUDIT COMMONQUEUE(RECALL)

When the FIX command is specified on the enhanced audit command, the enhanced audit command can repair many of the errors. We recommend that the first execution of the enhanced audit be run with the NOFIX command. This allows you to analyze the output and see what fix DFSMSHsm would have chosen for the error condition. If you agree with the fix chosen, run the enhanced audit command again with FIX specified. In some cases, the FIX parameter may cause valid migrated data to be deleted, because it is missing the ICF catalog record. The missing catalog record could have been a result of a restored catalog, human error, or another condition causing the catalog entry to be deleted. The migrated data could be valid data and the proper fix would be to re-catalog the data set to VOLSER MIGRAT.

## 9.3 The audit report

The report generated from the enhanced audit commands contains a header and a body. The header of the report confirms the performed audit and shows the time, the date, and the system identification. The second line of the report header shows the enhanced audit command and parameters you entered.

The body of the report normally consists of two lines for each error detected. The first line provides the error code and a brief error description. The lines immediately following the error list the repair commands when an automatic correction for the error is provided.

### 9.3.1 Running audits and diagnosing the errors

To provide an example of running an enhanced audit command, we will run an enhanced audit by executing the **AUDIT ABARSCONTROL(agname)** command. We will then look at the errors reported, analyze them, and correct the errors using either the automatic FIX command or FIXCDS commands we create.

The **AUDIT ABARSCONTROL** command requires the name of an aggregate to audit. Aggregates are groups of related data sets that are backed up using DFSMSHsm ABARS. Aggregates are defined to DFSMS in the Aggregate Group panel, option 9, on the ISMF menu.

We issued the enhanced audit command for aggregate BMPTEST. Executing this enhanced audit causes DFSMSShsm to check each ABR (Q) record in the BCDS. The output from the enhanced audit execution is shown in Figure 9-1.

```
-DFSMSHSM AUDIT-      ENHANCED AUDIT -- LISTING - AT 15:42:05 ON 02/04/15
FOR SYSTEM=DVLP
COMMAND ENTERED:
AUDIT ABARSCONTROLS(BMPTEST) ODS('CG.AUDIT.BMPTEST')

/* ERR 168 - CONTROL FILE NAME = EDP.BRM.BMPTEST1.TODISK.C.C01V0001 , +
  INDICATED IN THE ABR RECORD, BUT IS NOT CATALOGED.
ABR RECORD KEY = RECORD KEY = BMPTEST1.2001059000101
- END OF -      ENHANCED AUDIT - LISTING -
```

Figure 9-1 Enhanced audit command report

The output shows that we issued the enhanced audit command **AUDIT ABARSCONTROLS(BMPTEST)** and wrote the output to a data set called CG.AUDIT.BMPTEST. The output data set is automatically created; it does not have to be pre-allocated.

The body of the report shows the error that was generated from the enhanced audit, ERR 168 and a brief description of the error. This information is followed by the ABR Record Key that we will use later in the fix command.

The *DFSMSShsm Storage Administration Reference*, SC35-0422 provides error codes and a diagnosis table that lists the error description, audit repair action, and troubleshooting hints for each audit error. The following are the possible error conditions returned by the **AUDIT ABARSCONTROLS** command.

### 9.3.2 Audit error 165

If no data file is present for an ABR record, AUDIT reports ERR 165.

#### Error 165 description

The ABR record with key *abrrecordkey* indicates that neither a DFSMSdss data file nor an internal data file is included with the aggregate backup.

#### Troubleshooting

The ABR record specified is invalid and should be deleted.



### 9.3.3 Audit error 166

If no ABR records are found for a specified aggregate group, AUDIT reports ERR 166.

#### Error 166 description

The aggregate group name specified with the AUDIT command has no ABR records (has not been backed up).

#### Troubleshooting

The aggregate group name may have been spelled incorrectly.

### 9.3.4 Audit error 167

If no ABR record is found at all, AUDIT reports ERR 167.

#### Error 167 description

The BCDS has no record of aggregate backup activity.

#### Troubleshooting

No ABR record exists, therefore no aggregate backups have completed successfully.

### 9.3.5 Audit error 168

If the control file indicated in an ABR record is not cataloged, AUDIT reports ERR 168.

#### Error 168 description

The aggregate group backup refers to a particular file, but the file cannot be found in the catalog.

#### Troubleshooting

You should catalog the file indicated by filename.

In our example (shown in Figure 9-1), we received error condition 168. The description of this error is *the aggregate group backup refers to a particular file, but the file cannot be found in the catalog*.

The report shows us that control file name  
EDP.BRM.BMPTEST1.TODISK.C.C01V0001 is the control file for aggregate

BMPTEST. A catalog check of this data set name reveals that the file is no longer cataloged. The troubleshooting hints state that the file indicated by the file name should be cataloged.

Because we ran this enhanced audit in our environment, and we are familiar with the aggregates that we run, we determined that the file no longer exists and that the ABR record is no longer valid. In this case, the ABR record should be deleted. In your environment, you may find that the aggregate is valid and that the control file has simply become uncataloged for some other reason.

To delete this control data set record, a FIXCDS DELETE command is used. This is documented in Appendix A. “DFSMSHsm Maintenance Commands” in the *DFSMSHsm Diagnosis Guide*, LY35-0114. The command we are looking for is the FIXCDS command.

FIXCDS commands display, modify, create or delete records in the MCDS, BCDS, or OCDS. A record in the CDS consists of a key field (44 bytes), followed by a header field (20 bytes), followed by variable data. The formats of the records are shown in the *DFSMSHsm Storage Administration Reference*, SC35-0422.

The format of the FIXCDS command is **FIXCDS TYPE KEY** followed by a variety of optional parameters. The optional parameters of the FIXCDS command are shown in Figure 9-2.

```

ADDMIGRATEDDATASET(volser)
ASSIGNEDBIT (ON) (OFF)
CREATE (offset data)
DELETE
DISPLAY
DISPLAY(offset) GEN(nn)
ENTRY(volser) (dsname)
EXPAND (bytes)
NEWKEY(keyname)
VERIFY(offset data) BITS(bits)
PATCH (offset data) BITS(bits) GEN(nn)
LOGONLY
OUTDATASET(dsname)
LENGTH(bytes)

```

Notes:

- ▶ You can specify one of the ADDMIGRATEDDATASET, ASSIGNEDBIT, CREATE, DELETE, DISPLAY, ENTRY, EXPAND, NEWKEY, VERIFY, or PATCH with each FIXCDS command
- ▶ If you specify GEN(nn) with the DISPLAY or PATCH parameters, the required parameter type must be 'C'.

Figure 9-2 FIXCDS command parameters

We want to create a FIXCDS command for an ABR Q record. The positional parameter *type* is Q in our command:

### **FIXCDS Q**

The third positional parameter, *key*, is required. The *key* specifies the control record key for the control data set record you want to fix or display. The key can be specified in EBCDIC or hexadecimal characters. To determine what the key is for a type Q record, reference the *DFSMSHsm Diagnosis Guide*, LY35-0114. The guide lists the key information for each record type. For type Q aggregate group version records, the key corresponds to the aggregate group name, version and local copy number. The next available version number for the ABACKUP output files and ABR key is determined by DFSMSHsm. If all available version numbers (1 – 9999) have been used, a message is issued and the ABACKUP command is failed. An example of the key used with Q aggregate backup and recovery record is:

**FIXCDS Q AGNAME.yyyyddnnnncc**

Where:

**Agname** Is the name of the aggregate group

**Yyyyddd** Indicates the date of the aggregate backup. YYYY is the 4 digit year, and ddd is the relative (julian) day of the year.

**Nnnn** Indicates the version number of the aggregate group version. This maintains a one-to-one correspondence between an ABR record and an aggregate backup version

**Cc** Indicates the local copy number described by this record.

From our report in Figure 9-1 on page 240 the record key is RECORD KEY = BMPTEST1.2001059000101

So far, our FIXCDS command is **FIXCDS Q BMPTEST1.2001059000101**

Where:

**FIXCDS** Is the command

**Q** Is the second positional parameter of type

**BMPTEST1.2001059000101** Is the third positional parameter of key

The next field is for one of the many optional parameters provided with the FIXCDS command. See Figure 9-2 for a list of optional parameters. We know that the record is no longer valid; our intent is to DELETE the record so we use the optional parameter DELETE:

**FIXCDS Q BMPTEST1.2001059000101 DELETE**

To issue this command, we typed the command into the TSO interface of ISPF, option 6, prefixed with the HSEND command required for all DFSMSHsm commands. It is also necessary that the submitter's USERID is authorized in DFSMSHsm to issue the command:

**HSEND FIXCDS Q 'BMPTEST1.2001059000101' DELETE**

DFSMSHsm returns the results of the command to the terminal as shown in Figure 9-3.

```
ARC0197I TYPE Q, KEY BMPTEST1.2001059000101, FIXCDS DELETE SUCCESSFUL
ARC1007I COMMAND REQUEST 00000007 SENT TO DFSMSHSM
***
```

*Figure 9-3 Result of FIXCDS command*

Optionally, we could use the OUTDATASET('output data set name') command on the FIXCDS command to write the output to a data set like we did with the enhanced audit command.

To review, the *DFSMSHsm Storage Administration Reference*, SC35-0422, provides the reference for the enhanced audit commands, the possible error codes, and a table that provides error codes, Audit Repair Action (if any), and Troubleshooting Hints. After our analysis of the error code, we determined that a FIXCDS command was necessary to DELETE the errant ABR record. We used the *DFSMSHsm Diagnosis Guide*, LY35-0114 as a reference to construct the FIXCDS command.

The guide provided the information on what positional parameters are required for the FIXCDS command, how to determine the key for type Q records and what optional parameters are available. From this, we constructed our FIXCDS Q BMPTEST1.2001059000101 DELETE command that we submitted to DFSMSHsm, and successfully deleted the ABR record.

## 9.4 Auditing the MCDS

The **AUDIT DATASETCONTROLS(MIGRATION)** command checks all of the records in the MCDS and creates a list of errors. To audit the MCDS, we used the following command:

```
HSEND AUDIT DATASETCONTROLS(MIGRATION) ODS('CG.MIG.AUDIT')
```

The following messages in Figure 9-4 were sent to the terminal by DFSMSHsm.

```
ARC0801I DFSMSHSM AUDIT STARTING
ARC1007I COMMAND REQUEST 00000028 SENT TO DFSMSHSM
```

*Figure 9-4 Result of AUDIT command for the MCDS*

To check the progress of our enhanced audit, we issued the **HSEND QUERY ACTIVE** command to DFSMSHsm. The query command showed us that enhanced audit is still active (Figure 9-5).

```
ARC0101I QUERY ACTIVE COMMAND STARTING ON HOST=1
ARC0144I AUDIT=NOT HELD AND ACTIVE, LIST=NOT HELD AND INACTIVE, RECYCLE=NOT
```

*Figure 9-5 Result of QUERY ACTIVE command with active audit*

Later, we checked the status again. The query command showed us that the enhanced audit is now inactive (Figure 9-6). This means that the enhanced audit is complete and we can check the output listing for errors.

```

ARCO101I QUERY ACTIVE COMMAND STARTING ON HOST=1
ARCO144I AUDIT=NOT HELD AND INACTIVE, LIST=NOT HELD AND INACTIVE,
RECYCLE=NOT
ARCO144I (CONT.) HELD AND INACTIVE, REPORT=NOT HELD AND INACTIVE

```

Figure 9-6 Result of QUERY ACTIVE command with completed audit

## 9.5 The enhanced audit report

We can now browse the output listing for a list of errors. Example 9-1 shows the listing and the errors reported.

Example 9-1 Enhanced audit report listing -Part 1

---

```

-DFSMSHSM AUDIT-          ENHANCED AUDIT -- LISTING - AT 12:27:31 ON 02/04/16 FOR
COMMAND ENTERED:
AUDIT DATASETCONTROLS(MIGRATION) ODS('CG.MIG.AUDIT')

```

```

/* ERR 28 JJJ.ABARSMGR.IVP.GDG.G0002V00 - EXTRANEIOUS ALIAS RECORD
/* FIXCDS A HSM.HMIG.T023114.JJJ.ABARSMGR.J0235 DELETE
/* ERR 28 KAS.GDG2.G0002V00 - EXTRANEIOUS ALIAS RECORD
/* FIXCDS A HSM.HMIG.T033114.KAS.GDG2.J0235 DELETE
/* ERR 28 RMP.GDG2.G0500V00 - EXTRANEIOUS ALIAS RECORD
/* FIXCDS A HSM.HMIG.T043214.RMP.GDG2.J0235 DELETE
/* ERR 28 RMP.GDG3.G0001V00 - EXTRANEIOUS ALIAS RECORD
/* FIXCDS A HSM.HMIG.T043214.RMP.GDG3.J0235 DELETE
/* ERR 28 KAS.GDG2.G0003V00 - EXTRANEIOUS ALIAS RECORD
/* FIXCDS A HSM.HMIG.T053114.KAS.GDG2.J0235 DELETE
/* ERR 28 TL01.DSROUT.G0004V00 - EXTRANEIOUS ALIAS RECORD
/* FIXCDS A HSM.HMIG.T083214.TL01.DSROUT.J0235 DELETE
/* ERR 28 TL01.DSROUTC.G0003V00 - EXTRANEIOUS ALIAS RECORD
/* FIXCDS A HSM.HMIG.T113214.TL01.DSROUTC.J0235 DELETE
/* ERR 28 TL01.DSROUTC2.G0003V00 - EXTRANEIOUS ALIAS RECORD
/* FIXCDS A HSM.HMIG.T113214.TL01.DSROUTC2.J0235 DELETE
/* ERR 28 TL01.DSROUTEX.G0039V00 - EXTRANEIOUS ALIAS RECORD
/* FIXCDS A HSM.HMIG.T113214.TL01.DSROUTEX.J0235 DELETE
/* ERR 28 TL01.DSROUTEX.G0040V00 - EXTRANEIOUS ALIAS RECORD
/* FIXCDS A HSM.HMIG.T123214.TL01.DSROUTEX.J0235 DELETE
/* ERR 28 TL01.EXTRACT.G0003V00 - EXTRANEIOUS ALIAS RECORD
/* FIXCDS A HSM.HMIG.T123214.TL01.EXTRACT.J0235 DELETE
/* ERR 28 EM.Z.PRESTAGE.INPUTJCL - EXTRANEIOUS ALIAS RECORD
/* FIXCDS A HSM.HMIG.T210408.EM.Z.J0230 DELETE
/* ERR 28 KAS.AMPASAP.LAB1B.SELECT - EXTRANEIOUS ALIAS RECORD
/* FIXCDS A HSM.HMIG.T224019.KAS.AMPASAP.J0325 DELETE
/* ERR 28 JJJ.ABARSMGR.IVP.GDG.G0001V00 - EXTRANEIOUS ALIAS RECORD
/* FIXCDS A HSM.HMIG.T260419.JJJ.ABARSMGR.J0235 DELETE

```

```

/* ERR 28 KAS.GDG2.G0002V00 - EXTRANEIOUS ALIAS RECORD
/* FIXCDS A HSM.HMIG.T260419.KAS.GDG2.J0235 DELETE
/* ERR 28 JJJ.ABARSMGR.IVP.GDG.G0001V00 - EXTRANEIOUS ALIAS RECORD
/* FIXCDS A HSM.HMIG.T433014.JJJ.ABARSMGR.J0235 DELETE
/* ERR 28 RMP.GDG2.G1001V00 - EXTRANEIOUS ALIAS RECORD
/* FIXCDS A HSM.HMIG.U043214.RMP.GDG2.J0235 DELETE
/* ERR 28 RMP.GDG3.G0003V00 - EXTRANEIOUS ALIAS RECORD
/* FIXCDS A HSM.HMIG.U043214.RMP.GDG3.J0235 DELETE
/* ERR 28 TL01.DSROUTEX.G0042V00 - EXTRANEIOUS ALIAS RECORD
/* FIXCDS A HSM.HMIG.U123214.TL01.DSROUTEX.J0235 DELETE
/* ERR 28 EM.Z.PRESTAGE.JOBS - EXTRANEIOUS ALIAS RECORD
/* FIXCDS A HSM.HMIG.U210408.EM.Z.J0230 DELETE
/* ERR 28 JJJ.ABARSMGR.IVP.GDG.G0002V00 - EXTRANEIOUS ALIAS RECORD
/* FIXCDS A HSM.HMIG.U260419.JJJ.ABARSMGR.J0235 DELETE
/* ERR 28 KAS.GDG2.G0003V00 - EXTRANEIOUS ALIAS RECORD
/* FIXCDS A HSM.HMIG.U260419.KAS.GDG2.J0235 DELETE
/* ERR 28 RMP.GDG2.G9000V00 - EXTRANEIOUS ALIAS RECORD
/* FIXCDS A HSM.HMIG.V043214.RMP.GDG2.J0235 DELETE
/* ERR 28 KAS.GDG2.G0005V00 - EXTRANEIOUS ALIAS RECORD
/* FIXCDS A HSM.HMIG.V053114.KAS.GDG2.J0235 DELETE
/* ERR 28 EM.Z.USERDSN.JOBS - EXTRANEIOUS ALIAS RECORD
/* FIXCDS A HSM.HMIG.V210408.EM.Z.J0230 DELETE
/* ERR 28 KAS.GDG2.G0005V00 - EXTRANEIOUS ALIAS RECORD
/* FIXCDS A HSM.HMIG.W260419.KAS.GDG2.J0235 DELETE
/* ERR 24 VSPA-1.KSDS1 - MISSING VSAM INTERCEPT OBJECT
/* FIXCDS O HSM.HMIG.T521107.VSPA-1.KSDS1.J1297 VERIFY(X'0000002C' X'E060')
/*PATCH(X'0000002C' X'8060')/* ERR 03 VSPA-1.KSDS1 NOT CATALOGED, HAS MIGRATION
/*COPY +
/* VSPA-1.KSDS1
/* ERR 24 VSPA-1.KSDS1 - MISSING VSAM INTERCEPT OBJECT
/* FIXCDS O HSM.HMIG.T521107.VSPA-1.KSDS1.J1297 VERIFY(X'0000002C' X'E060')
/* PATCH(X'0000002C' X'8060')
- END OF -          ENHANCED AUDIT - LISTING -

```

---

## 9.5.1 Analyzing ERR 28

Most of our errors in Example 9-1 are ERR 28. We have an ERR 24 condition as well. The enhanced audit command built the possible FIXCDS command we can use to resolve the errors.

Looking at the *DFSMSHsm Storage Administration Reference*, SC35-0422, ERR 28 is caused when an alias record refers to a data set record that does not exist. The *DFSMSHsm Diagnosis Guide*, LY35-0114 shows us that an alias record is created when a data set (non-VSAM) is migrated by DFSMSHsm. If the data set is VSAM, the alias record is the name of a VSAM component of a migration data set eligible for automatic recall.

We checked the catalog for several of the data set names listed. We did not find any catalog entries for the data set listed. It appears that these data sets have an MCA record in the MCDS, but no MCD (data set) record, and they are no longer cataloged in an ICF catalog. We determined that we want to delete the MCA records out of the MCDS. To do this we can run the enhanced audit command again with the optional parameter FIX to automatically issue the FIXCDS commands created by the audit execution. But, before we do this, let's look at the second error type we received, ERR 24.

## 9.5.2 Analyzing ERR 24

In the *DFSMSHsm Storage Administration Reference*, SC35-0422, we find that ERR 24 is caused when a VSAM-associations record refers to a migrated VSAM component, but there is no catalog entry for the component, or the catalog entry is something other than a non-VSAM entry referring to a single volume serial MIGRAT.

The MCD record and its catalog entry are the important records used to recall a data set, not the MCO. The data set record or (D) record represents the cluster name. The (O) record is a strange and interesting way to access the data component of a migrated VSAM cluster by the data name, not the cluster name. If someone, through JCL or a recall command, uses the data component name, there is an MCA (alias) record that is spelled the same as the component's name. This MCA record points to an MCO record that has buried in it the cluster name. The cluster is then recalled by the cluster name if it is cataloged and there is a valid migration copy (D) record. Of course after the cluster has been recalled the data component is available for allocation.

The enhanced audit also generated ERR 03 for this error as shown in Example 9-2. ERR 03 is reported when a data set record indicates a valid migrated copy, but the data set is uncataloged.

### *Example 9-2 Enhanced audit report listing - Part 2*

---

```
/*ERR 24 VSPA-1.KSDS1 - MISSING VSAM INTERCEPT OBJECT
/* FIXCDS 0 HSM.HMIG.T521107.VSPA-1.KSDS1.J1297 VERIFY(X'0000002C' X'E060')
/*PATCH(X'0000002C' X'8060')
/* ERR 03 VSPA-1.KSDS1 NOT CATALOGED, HAS MIGRATION COPY +
/* VSPA-1.KSDS1
/* ERR 24 VSPA-1.KSDS1 - MISSING VSAM INTERCEPT OBJECT
/* FIXCDS 0 HSM.HMIG.T521107.VSPA-1.KSDS1.J1297 VERIFY(X'0000002C' X'E060')
/* PATCH(X'0000002C' X'8060')
```

---



### 9.5.3 Correcting ERR 28 and ERR 24

The PATCH command that was generated by the enhanced audit for ERR 24 patches the MCOCTFLG byte in the MCO record to indicate that the data and index portion of the VSAM cluster is not cataloged. This may not be the fix you would want to use to resolve this problem. The (0) record is not required to recall the data set by its cluster name.

To analyze this error, we executed a **FIXCDS DISPLAY** command to validate that a MCD record and an MCA record exists. Figure 9-7 shows the display of the D record and that the record is valid.

```

MCH= 01D40000 B6A20E85 6FB5DE80 B6A16EE5 0766E2C0
* M V S *
+0000 D4D3F1F0 F0F08881 2C348000 0101024C 00000000 0101297C 00000000 0000000F
*ML1000 @ *
+0020 07115332 0101297F 00081000 00000000 00000002 0000059B 00000001 00020000
* *
+0040 C5C8E3F0 F0F20200 C7C5D540 3030200F 07111921 0101297F 3030200F 00000000
*EHT002 GEN *
+0060 00000000 00000000 00000000 00000000 00000000 00000000 00000000 00000000
* *
+0080 00000000 00000000 00000000 00000000 00000000 00000000 00000000 C8E2D44B
* HSM.*
+00A0 C8D4C9C7 4BE3F5F2 F1F1F0F7 4BE5E2D7 C160F14B D2E2C4E2 F14BD1F1 F2F9F740
*HMI G.T521107.VSPA 1.KSDS1.J1297 *
+00C0 40404040 40404040 00000000 00000000 00000000 00000000 00000000 00000000
* *
+00E0 00000000 00004040 40404040 40404040 40404040 40404040 40404040 40404040
* *
+0100 40404040 0006C5D3 D3C9D6E3 40404040 40404040 40404040 40404040 40404040
* ELLIOT *
+0120 40404040 0005D4C3 C1C7C740 40404040 40404040 40404040 40404040 40404040
* MCAGG *
+0140 40404040 00000000 00000000 40800000 00000000 00000000 00000000 00000000
* *
+0160 00000000 00008800 00000000 00000000 00000000 02020A00 00000000 00000000
* *
+0180 00000000 00000000 00000000 00000000 00000000
* *
ARC0197I TYPE D, KEY VSPA-1.KSDS1, FIXCDS DISPLAY SUCCESSFUL
ARC1007I COMMAND REQUEST 00000051 SENT TO DFSMSHSM
***

```

Figure 9-7 Result of the DISPLAY of the data sets (D) record

Figure 9-8 shows the display of the data set's alias (A) record.

```

MCH= 00701100 B6A20E84 A086F740 B6A20E84 A086F740
*      7      7      *
+0000 D4C30000 E5E2D7C1 60F14BD2 E2C4E2F1 40404040 40404040 40404040 40404040
*MC VSPA 1.KSDS1 *
+0020 40404040 40404040 40404040 40404040
*      *
ARC0197I TYPE A, KEY HSM.HMIG.T521107.VSPA-1.KSDS1.J1297, FIXCDS DISPLAY
ARC0197I (CONT.) SUCCESSFUL
***

```

Figure 9-8 Result of the DISPLAY of the data sets (A) record

We checked the catalog and found that the data set is not cataloged. See Figure 9-9 for a display of the catalog entry. We then checked the catalog for a valid alias record for VSPA-1 and found that the alias no longer exists.

We checked with the data set's owner and found that the data set, VSPA-1.KSDS1 is no longer valid. In our case, any CDS records for this data set should be deleted including the errant (0) record.

```

TSO LISTC ENT('VSPA-1.KSDS1') ALL
IDC3012I ENTRY VSPA-1.KSDS1 NOT FOUND+
IDC1566I ** VSPA-1.KSDS1 NOT LISTED
IDC0014I LASTCC=4
***

```

Figure 9-9 Result of the LISTC ENT command for the data set

This is an example of why you may not want to run the enhanced audit with the FIX command before you have an opportunity to view the output and analyze the errors. Running the enhanced audit once with NOFIX (which is the default if not coded) command and analyzing the output first helps to avoid potentially disastrous results.

An example is several data sets that have valid migration records but are no longer cataloged. The automatic FIX generated by the enhanced audit command deletes the migration records in the MCDS causing the data to be no longer accessible. Automatic FIX assumes that if the data sets are not cataloged, they are no longer valid. In fact, the reason that the data sets are no longer cataloged may be because of a catalog recovery or human error.

In this example, we cannot rerun the enhanced audit with the FIX command, because we do not want the enhanced audit execution to PATCH the MCO records found in ERR 24. We want to delete the errant (0) record and the associated (D) record with the following **FIXCDS** commands in Figure 9-10.

```

FIXCDS O 'HSM.HMIG.T521107.VSPA-1.KSDS1.J1297' DELETE
FIXCDS A 'HSM.HMIG.T521107.VSPA-1.KSDS1.J1297' DELETE
FIXCDS D 'VSPA-1.J1297' DELETE

```

*Figure 9-10 FIXCDS command to delete errant records*

To fix ERR 28, we will need to execute the FIXCDS commands in an IKJEFT01 batch job or type them in ISPF option 6. We chose to execute the FIXCDS commands for the ERR 28 by copying the FIXCDS commands to an IKJEFT01 batch utility. We removed extraneous information other than the FIXCDS commands and removed the FIXCDS command for ERR 24 before we submitted it. The IKJEFT01 job we used is shown in Figure 9-11.

```

//STEP01      EXEC  PGM=IKJEFT01
//SYSPRINT DD  SYSOUT=*
//SYSTSPRTDD  SYSOUT=*
//SYSIN       DD  DUMMY
//SYSTSIN DD   *
    HSEND FIXCDS A HSM.HMIG.T023114.JJJ.ABARSMGR.J0235 DELETE
    HSEND FIXCDS A HSM.HMIG.T033114.KAS.GDG2.J0235 DELETE
    HSEND FIXCDS A HSM.HMIG.T043214.RMP.GDG2.J0235 DELETE
    HSEND FIXCDS A HSM.HMIG.T043214.RMP.GDG3.J0235 DELETE
    HSEND FIXCDS A HSM.HMIG.T053114.KAS.GDG2.J0235 DELETE
    HSEND FIXCDS A HSM.HMIG.T083214.TL01.DSR0UT.J0235 DELETE
    HSEND FIXCDS A HSM.HMIG.T113214.TL01.DSR0UTC.J0235 DELETE
    HSEND FIXCDS A HSM.HMIG.T113214.TL01.DSR0UTC2.J0235 DELETE
    HSEND FIXCDS A HSM.HMIG.T113214.TL01.DSR0UTEX.J0235 DELETE
    HSEND FIXCDS A HSM.HMIG.T123214.TL01.DSR0UTEX.J0235 DELETE
    HSEND FIXCDS A HSM.HMIG.T123214.TL01.EXTRACT.J0235 DELETE
    HSEND FIXCDS A HSM.HMIG.T210408.EM.Z.J0230 DELETE
    HSEND FIXCDS A HSM.HMIG.T224019.KAS.AMPASAP.J0325 DELETE
    HSEND FIXCDS A HSM.HMIG.T260419.JJJ.ABARSMGR.J0235 DELETE
    HSEND FIXCDS A HSM.HMIG.T260419.KAS.GDG2.J0235 DELETE
    HSEND FIXCDS A HSM.HMIG.T433014.JJJ.ABARSMGR.J0235 DELETE
    HSEND FIXCDS A HSM.HMIG.U043214.RMP.GDG2.J0235 DELETE
    HSEND FIXCDS A HSM.HMIG.U043214.RMP.GDG3.J0235 DELETE
    HSEND FIXCDS A HSM.HMIG.U123214.TL01.DSR0UTEX.J0235 DELETE
    HSEND FIXCDS A HSM.HMIG.U210408.EM.Z.J0230 DELETE
    HSEND FIXCDS A HSM.HMIG.U260419.JJJ.ABARSMGR.J0235 DELETE
    HSEND FIXCDS A HSM.HMIG.U260419.KAS.GDG2.J0235 DELETE
    HSEND FIXCDS A HSM.HMIG.V043214.RMP.GDG2.J0235 DELETE
    HSEND FIXCDS A HSM.HMIG.V053114.KAS.GDG2.J0235 DELETE
    HSEND FIXCDS A HSM.HMIG.V210408.EM.Z.J0230 DELETE
    HSEND FIXCDS A HSM.HMIG.W260419.KAS.GDG2.J0235 DELETE

```

*Figure 9-11 IKJEFT01 job created from audit report*

## 9.5.4 Auditing ML1 or ML2 volumes

To complete the audit of the MCDS, an enhanced audit of the ML1 volumes is also necessary. The AUDIT DIRECTORYCONTROLS enhanced audit cross checks information in the volume's directory (a VTOC or TTOC) with control records in the MCDS. We can also audit ML2 volumes with the same command. When ML2 volumes are audited, the ML2 volume is mounted so audit can cross check the TTOC records with the control records in the MCDS.

This audit we executed for our ML1 volume, shown in Figure 9-12, returned no errors.

```
***** Top of Data *****
-DFSMSHSM AUDIT-      ENHANCED AUDIT -- LISTING - AT 14:31:53 ON 02/04/16 FOR S
COMMAND ENTERED:
AUDIT DIRECTORYCONTROLS VOLUMES(ML1000) ODS('CG.MIGVOL.AUDIT')

- END OF -      ENHANCED AUDIT - LISTING -
***** Bottom of Data *****
```

Figure 9-12 AUDIT DIRECTORYCONTROLS VOLUMES (ML1001) command output

## 9.6 Auditing the common recall queue

The common recall queue enables you to have all DFSMSHsm address spaces in an HSMplex process recall requests from a single common queue.

It is possible that problems may be encountered with the contents of the common recall queue structure. DFSMSHsm will initially attempt to correct any problems with the contents of the common recall queue structure that it encounters.

If DFSMSHsm is not successful at correcting the structures contents you may be able to do so using the AUDIT command. The *DFSMSHsm Storage Administration Reference*, SC35-0422, recommends using the AUDIT command under the following conditions:

- ▶ After receiving an ARC1506E message
- ▶ After receiving an ARC1187E message
- ▶ When recall requests are unexpectedly not being selected for processing

The structure of the common queues is not externalized. It is possible that losses of connectivity, ABENDS or internal errors may cause the common recall queue to become corrupted. Just like the DFSMSHsm control data sets the common recall queue has interrelated entries that can get out-of-sync.

You can use the **AUDIT COMMONQUEUE(RECALL)** command to analyze the entire structure and reports/fix the errors that it finds. It issues ARC1544I to report the number of errors. It cannot find or correct all errors. It creates PDA entries to record what it found and then fixed.

**AUDIT COMMONQUEUE(RECALL)** does not return FIXCDS commands, as there are no control data set records for the common recall queue these are not applicable. Also **AUDIT COMMONQUEUE(RECALL)** does not create an output data set. You may see differences in the number of errors found between AUDITs specified with FIX and NOFIX. Common recall queue errors are usually interrelated. Fixing one error will fix many errors reported by NOFIX.

## 9.7 Using the original AUDIT command

The original audit is used to detect and to report discrepancies in data set information in DFSMSHsm control data sets, the VTOCs identifying the data sets, and catalogs. The scope of these audits is less extensive than the enhanced audits and there is some overlap between the parameters.

The AUDIT commands include:

- ▶ AUDIT DATASET NAMES and LEVELS
- ▶ AUDIT VOLUMES
- ▶ AUDIT BACKUP VOLUMES and BACKUP TYPE
- ▶ AUDIT MASTER CATALOG and USER CATALOG
- ▶ AUDIT MIGRATION CONTROL DATA SET
- ▶ AUDIT OFFLINE CONTROL DATA SET
- ▶ AUDIT ALL

With these audit commands, you can audit:

- ▶ All cataloged data sets that have the same set of initial characters of the data set name
- ▶ All data sets that reside on a specific non-SMS-managed volume
- ▶ All data sets that reside on backup volumes of a specified backup type
- ▶ All data sets cataloged in a specified catalog
- ▶ All data sets that have a data set record in the MCDS
- ▶ All data sets that have a data set record in the BCDS
- ▶ All data sets that have an entry in an OCDS TTOC record of a specified type

These commands are useful when you have a specific symptom with a data set or volume. A symptom can be a data set that you cannot recall or recover, or a volume that cannot be recycled.

As an example, we can execute an audit using the same data set name, VSPA-1.KSDS1, from our previous example, using the following command:

```
AUDIT DATASETNAMES(VSPA-1.KSDS1) NOFIX ODS('CG.AUDIT.OUTPUT')
```

The report (Figure 9-13) shows the ERR03 but not the ERR24 from the enhanced audit. The other item to note is that the automatically generated FIXCDS commands are not provided in this audit.

|                                                                                   |                     |             |           |           |           |  |
|-----------------------------------------------------------------------------------|---------------------|-------------|-----------|-----------|-----------|--|
| ***** Top of Data *****                                                           |                     |             |           |           |           |  |
| -DFSMSHSM AUDIT- DATASET -- LISTING - AT 08:09:51 ON 02/04/17 FOR FOR SYSTEM=DVLP |                     |             |           |           |           |  |
| COMMAND ENTERED:                                                                  |                     |             |           |           |           |  |
| AUDIT DATASETNAMES(VSPA-1.KSDS1) NOFIX ODS('CG.AUDIT.OUTPUT')                     |                     |             |           |           |           |  |
| ERROR                                                                             | DATA SET NAME       | DATA SET ON | CATALOGED | MIGRATED  | BACKED UP |  |
| TYPE                                                                              |                     | VOLUME      | TO VOLUME | TO VOLUME | TO VOLUME |  |
| *ERR 03                                                                           | VSPA-1.KSDS1        | ??????      | -NONE-    | ML1000    | -NONE-    |  |
| - END OF -                                                                        | DATASET - LISTING - |             |           |           |           |  |
| ***** Bottom of Data *****                                                        |                     |             |           |           |           |  |

Figure 9-13 AUDIT DATASETNAMES output

The *DFSMSHsm Storage Administration Reference*, SC35-0422, provides the following error description for ERR03:

The catalog has no entry for the data set that is being audited, but the MCDS record indicates that the data set is on a migration volume.

The troubleshooting hints provides the following help:

Because there is no catalog entry for the data set with the migration copy, that copy is most likely an old one that has not yet been scratched, probably because of a previous error when the data set was being recalled or deleted. There are two methods to correct the problem: either issue the HSEND DELETE command for the data set, deleting the D and A records in the MCDS immediately; or have the migration copy and its D record removed during the next migration cleanup by using the FIXCDS D dsname PATCH() command to reset the assigned flag (MCDFASN) bit to 0 (off) and to set the needs scratch flag (MCDFNSCR) bit to 1 (on).

```
FIXCDS D dsname PATCH(X'06' BITS(0.....1.))
```

This command resets the assigned bit and sets the needed scratch bit. Also possible is that the catalog entry was inadvertently deleted, and the migrated copy is still valid. If this is the case, rebuild the catalog entry as a non-VSAM data set with a volume serial of "MIGRAT".

Because we know that this data set is no longer valid, the suggested troubleshooting hints would be appropriate in order to delete the MCD and MCA records using the **HSEND DELETE** command. This audit did not detect the errant MCO record that subsequently needs to be deleted using the **FIXCDS DELETE** command.

### 9.7.1 Audit MASTERCATALOG and Audit USERCATALOG parameters

DFSMSHsm provides a way to audit the master catalog or user catalog using the audit command. Audit checks all data sets in the catalog that have an entry that indicates a data set on a primary volume of a device type managed by DFSMSHsm. Audit uses the obtain function to verify that the non-VSAM or VSAM data set is indeed on the indicated volume. Errors are returned when discrepancies are found.

Again, we recommend that you submit this and any other audit commands with the NOFIX parameter first. When FIX is specified with the audit MASTERCATALOG or USERCATALOG command, the audit function enqueues continually on the DFSMSHsm control data sets during the entire audit, to minimize the possibility of other DFSMSHsm functions making changes to them. Because auditing a complete catalog can take considerable time, AUDIT issues the following message in Figure 9-14.

ARC0803A WARNING: AUDIT OF CATALOG MAY DEGRADE PERFORMANCE,  
REPLY 'Y' TO START AUDIT OR 'N' TO CANCEL AUDIT COMMAND.

*Figure 9-14 AUDIT message ARC0803A*

We submitted the **AUDIT USERCATALOG** command for catalog USERCAT.UCAT1 with the NOFIX parameter:

```
HSEND AUDIT USERCATALOG(USERCAT.UCAT1) NOFIX ODS('CG.AUDIT.CTLG')
```

Figure 9-15 shows the report produced by this command that we wrote to the output data set CG.AUDIT.CTLG.

```

- END OF -      CATALOG - LISTING -
-DFSMSHSM AUDIT-      CATALOG -- LISTING - AT 14:10:41 ON 02/04/17 FOR
COMMAND ENTERED:
AUDIT USERCATALOG(USERCAT.UCAT1) NOFIX ODS('CG.AUDIT.CTLG')

AUDITING USERCAT.UCAT1

ERROR          DATA SET NAME      DATA SET ON CATALOGED   MIGRATED   BACKED UP
TYPE                                VOLUME    TO VOLUME   TO VOLUME  TO VOLUME

*ERR 04      BB01.ISPF.CATALOG.ERROR      ??????      NONSMS      -NONE-
*ERR 04      BK.TEST.LIKE                  ??????      TS0004      -NONE-
*ERR 10      DCD.FAK.DSN3                  ??????      MIGRAT      -NONE-
*ERR 04      DCD.NO.DASD                   ??????      TS0003      -NONE-
*ERR 04      DCD.NO.DASD.MISS              ??????      TS0001      -NONE-
*ERR 04      DCD.NO.DASD.NOVOL             ??????      386003      -NONE-
*ERR 04      BB01.ISPF.CATALOG.ERROR      ??????      PEREGO      -NONE-
*ERR 04      BK.TEST.LIKE                  ??????      TS0001      -NONE-
*ERR 10      DCD.FAK.DSN3                  ??????      MIGRAT      -NONE-
*ERR 04      DCD.NO.DASD                   ??????      TS0001      -NONE-
*ERR 04      DCD.NO.DASD.MISS              ??????      TS0001      -NONE-
*ERR 04      DCD.NO.DASD.NOVOL             ??????      NONSMS      -NONE-
*ERR 04      DR03.NVSAM                    ??????      X01         -NONE-
*ERR 04      DR03.NVSAM1                   ??????      TS0001      -NONE-
*ERR 04      DR03.NVSAM2                   ??????      NONSMS      -NONE-
*ERR 04      EH.MANYVOLS                   ??????      X01         -NONE-
*ERR 04      EH.MODEL.DSCB1                ??????      EHN001      -NONE-
*ERR 04      EH.MODEL.DSCB2                ??????      EHN001      -NONE-
*ERR 04      EH.NONVSAM.NONSMS.NONCAT01    ??????      A           -NONE-
*ERR 04      EH.NONVSAM.NONSMS.NONCAT02    ??????      EHN001      -NONE-
*ERR 04      HFS.P390.ETC                  ??????      Z1USS1      -NONE-
*ERR 04      MH.DIAVBV.DSK318              ??????      DSK318      -NONE-
*ERR 04      MH2.CATTEST.AIX1.DATA          ??????      UCAT01      -NONE-
*ERR 04      MH2.CATTEST.AIX1.DATA          ??????      TS0002      -NONE-
*ERR 10      VSP6.PMR11956.LDS1            ??????      MIGRAT      -NONE-
*ERR 10      VSP6.PMR11956.LDS1.DATA        ??????      MIGRAT      -NONE-
*ERR 04      Z999.FLAT.FILE                 ??????      TS0002      -NONE-
- END OF -      CATALOG - LISTING -

```

Figure 9-15 Output listing from the AUDIT USERCATALOG command

Our most common error is ERR 04. This error code is issued when there is no MCDS data set record for the data set. We also see ERR 10 this error is issued when the data set is catalog to VOLSER MIGRAT but there is no MCDS data set record (MCD) in the MCDS.



The first step in correcting these errors is to determine if these catalog entries are valid entries for each data set, and if the VOLUME they are cataloged to exists. Our check revealed that the data sets were indeed cataloged on the volume indicated but that the volume was no longer mounted on our system. To clean up these entries, the delete command can be issued with the NOSCRATCH command.

ERR 10 can be researched by listing the catalog entry for each data set as we did with DCD.FAK.DSN3. The catalog entry shown in Figure 9-16 reveals that the data set is cataloged to VOLSER MIGRAT.

```
NONVSAM ----- DCD.FAK.DSN3
IN-CAT --- USERCAT.UCAT1
HISTORY
  DATASET-OWNER----- (NULL)      CREATION-----2001.057
  RELEASE-----2      EXPIRATION-----0000.000
VOLUMES
  VOLSER-----MIGRAT      DEVTYPE-----X'3010200F'      FSEQN-----
-----0
  ASSOCIATIONS----- (NULL)
  ATTRIBUTES
***
```

Figure 9-16 Results of the LISTC ENT('dcd.fak.dsn3') ALL command

We then check to see if a valid MCD record exist by issuing the command:

```
HLIST DSN('DCD.FAK.DSN3') MCDS
```

DFSMSHsm shows us that no MCDS record exists (Figure 9-17).

```
ARC0138I NO MCDS INFORMATION FOUND FOR DATASET DCD.FAK.DSN3
ARC0140I LIST COMPLETED, 1 LINE(S) OF DATA OUTPUT      ***
```

Figure 9-17 HLIST DSMN command output

Since there is no valid MCD record, the data set is no longer valid. The data set can be recovered from a backup copy if a backup copy exists. We deleted the catalog entry using the DFSMSHsm HDELETE 'DCD.FAK.DSN3' command.

## 9.8 Summary on running audits

In this chapter, we have shown you several examples of running the enhanced audit command and the audit command provided by DFSMSHsm. In each example, we executed the command first without the **FIX** command so we could

analyze the output. We researched each error to determine the proper **FIX** command to use. In some cases, we constructed our own **FIXCDS** commands to resolve the error. In other cases, we simply deleted the records, because our research showed that the data sets were no longer valid. We also showed you some differences in the output between the enhanced audit commands and the audit commands provided by DFSMSHsm.

Chapter 10, “Recreating the DFSMSHsm environment at the DR site” on page 259 details how to execute all of the audits provided by DFSMSHsm using Mainstar’s *FastAudit/390* tool. We also discuss how to analyze and correct the errors using the tool.



## Recreating the DFSMSHsm environment at the DR site

This chapter describes how to prepare your environment for disaster recovery, how to recover and initialize the control data sets at the recovery site and how to reconcile control data sets with ICF user catalog and tape management databases using Mainstar *FastAudit/390* HSM Audit, Tape Audit and Catalog & DASD Audit.

## 10.1 Preparing the environment

An important aspect of DFSMSHsm Control Data Set (CDS) support and maintenance is the requirement to prepare for creating or recreating the DFSMSHsm environment at a test of disaster recovery procedures or at a real disaster site. The DFSMSHsm environment can be initialized empty and re-populated using DFSMSHsm ABARS or it may be copied and restored with entries from the primary site.

When the DFSMSHsm environment is copied and restored at the disaster site, inconsistencies between the records in the control data sets and the ICF user catalog or tape management database can occur. This is due to the timing of the backup, whether the backup is performed while DFSMSHsm is not active or if DFSMSHsm is performing space management functions.

When the backup is taken while DFSMSHsm is performing space management functions, such as migration, recall, or automatic expiration, these “in flight” records may not be captured by the backup. When the control data sets are recovered, they may reflect that a data set is migrated, when in fact, the data set was recalled and is resident on DASD.

### 10.1.1 Use of DFSMSHsm ABARS

A good approach is to determine if your recovery procedures involve the use of DFSMSHsm ABARS. If you are using ABARS, you can copy data from DFSMSHsm migration at the backup site to an ABARS aggregate output tape. At the recovery site, the ABARS ARECOVER procedure by default recovers the data, in its migrated form, and recreates the migration control and other records in the control data sets.

If you are not using ABARS, data that is migrated, either to migration level 1 (ML1) DASD or migration level 2 (ML2) tape must be copied to another set of storage media and removed from the primary site for safekeeping.

DFSMSHsm has a facility for making an alternative copy of data on ML2 for this purpose. Data on ML1 must be full volume dumped or mirrored to the recovery site.

Knowing your recovery methodology is important in designing a recovery scenario for DFSMSHsm. If you plan on initializing your DFSMSHsm environment empty at the recovery site, but you are not using ABARS to recreate the CDS entries, the recovery of your DFSMSHsm environment can fail. To avoid this, get involved with the department responsible for contingency planning, storage administration, technical support, as well as other department's efforts, to create and maintain a viable recovery plan.

## 10.2 Empty CDS recovery

Empty CDS recovery is possible when your site uses DFSMSHsm ABARS to copy migrated data (ML1 or ML2) to an ABARS aggregate output tape that is restored at the recovery site. The ARECOVER process recreates the needed entries in the control data sets, copies the data from the aggregate output tape to either ML1 or ML2, and creates the ICF catalog entry with a VOLSER of MIGRAT. The data can be recalled in the normal manner, by accessing the data set or issuing the DFSMSHsm RECALL command.

Creating the DFSMSHsm environment empty at the disaster site requires batch JCL to be executed to create the following data sets:

- ▶ Migration Control Data Set (MCDS)
- ▶ Backup Control Data Set (BCDS)
- ▶ Offline Control Data Set (OCDS)
- ▶ Journal File
- ▶ HSM LOGX and LOGY data sets

In addition, the empty control data sets must be primed with a record before they can be accessed by DFSMSHsm.

### 10.2.1 Creating a user catalog for the DFSMSHsm entries

The first step in creating your DFSMSHsm environment is to create an ICF user catalog for the DFSMSHsm entries. When DFSMSHsm migrates a data set or creates a backup copy of a data set, DFSMSHsm creates a data set name using HSM (or your customized) high-level qualifier. This new data set name is cataloged in the ICF user catalog you create.

The variables in the sample JCL shown in Figure 10-1 are:

|                |                                                                                                 |
|----------------|-------------------------------------------------------------------------------------------------|
| ?UCATNAM       | The name of the user catalog                                                                    |
| CYLINDERS(? ?) | Replace the first “?” with the primary space amount and the second “?” with the secondary space |
| ?UCATVOL       | The name of the output VOLSER the catalog is allocated on                                       |
| ?MCATNAM       | The name of the master catalog on the system                                                    |
| ?UID           | The high-level qualifier you choose for DFSMSHsm                                                |
| ?UCATNAM       | The name of the user catalog                                                                    |

The sample JCL in Figure 10-1 is used to create an ICF user catalog.

```

//CATALOG          EXEC PGM=IDCAMS
//*
//SYSPRINT      DD  SYSOUT=*
//HSMCAT        DD  UNIT=3390,
//              DISP=SHR,VOL=SER=VOL001
//SYSIN          DD  *
DEFINE UCAT(NAME(?UCATNAM) -
            CYLINDERS(? ?) VOLUME(?UCATVOL) -
            FILE(HSMCAT) FREESPACE(0 0) -
            RECORDSIZE(4086 4086) -
            ICFCATALOG) -
            CATALOG(?MCATNAM)
IF MAXCC = 0 THEN DO
  DEFINE ALIAS(NAME(?UID) RELATE(?UCATNAM))
END

```

Figure 10-1 Sample JCL to create a user catalog

## 10.2.2 Creating the primer record

The second step is to create the primer record for the control data sets. The primer record may not be needed depending on the release of DFSMSHsm you are using. Refer to the *DFSMSHsm Implementation and Customization Guide*, SC35-0418, for the release you are using to verify if the primer record is necessary.

Sample JCL to create the starter record is shipped with DFSMSHsm. It can be found in the DFSMSHsm starter set provided with the product. Refer to the *DFSMSHsm Implementation and Customization Guide*, SC35-0418 for more information and JCL samples to create the control data sets and primer record.

Figure 10-2 is sample JCL used to create an output data set HSM.CDS.PRIMER to be used subsequently as input to create the MCDS, BCDS, and OCDS data sets.



```

//IDCAMS      EXEC  PGM=IDCAMS
//*
SYSPRINT      DD  SYSOUT=*
//*
//PRIMER      DD  DSN=HSM.CDS.PRIMER,DISP=(OLD)
//SYSIN       DD  *
               DEFINE CLUSTER (NAME(?UID.MCDS) VOLUMES(?MCDSVOL) -
                           CYLINDERS(?CDSSIZE) FILE(HSMMCDS) -
                           STORCLAS(?SCCDSNM) -
                           MGMTCLAS(?MCDFHSM) -
                           RECORDSIZE(435 2040) FREESPACE(0 0) -
                           INDEXED KEYS(44 0) SHAREOPTIONS(3 3) -
                           SPEED BUFFERSPACE(530432) -
                           UNIQUE NOWRITECHECK) -
                           DATA(NAME(?UID.MCDS.DATA) -
                           CONTROLINTERVALSIZE(12288)) -
                           INDEX(NAME(?UID.MCDS.INDEX) -
                           CONTROLINTERVALSIZE(2048))
               REPRO ODS('?UID.MCDS') INFILE(PRIMER)
               END

```

Figure 10-3 Sample JCL used to create the MCDS

The variables in the sample JCL shown in Figure 10-3 are:

|                 |                                                       |
|-----------------|-------------------------------------------------------|
| ?UID.MCDS       | The output data set name of the MCDS                  |
| ?MCDSVOL        | The VOLSER that the data set is allocated on          |
| ?CDSSIZE        | The primary and secondary space allocation            |
| ?SCCDSNM        | Optional STORCLAS name if the CDS is DFSMS managed    |
| ?MCDFHSM        | Optional MGMTCLAS name if the CDS is DFSMSHsm managed |
| ?UID.MCDS.DATA  | The name of the data portion of the MCDS              |
| ?UID.MCDS.INDEX | The name of the index portion of the MCDS             |

## 10.2.4 Creating the Backup Control Data Set

The BCDS is required if automatic functions such as AUTOBACKUP are run. It is also necessary to execute ABARS ABACKUP events at the recovery site. ABARS may be used to create aggregates backups of the data at the recovery site used to return to the primary site.



Figure 10-4 is sample JCL used to create a single BCSD. If your control data sets exceed the 4 GB limit, you may need to create a multi-cluster control data set in your environment. Additional sample JCL can be found in the DFSMSHsm starter set JCL provided with the product. Note that the primer record created in Figure 10-4 is used as input in DD statement PRIMER. Refer to the *DFSMSHsm Implementation and Customization Guide*, SC35-0418 for more information and JCL samples used to create both single and multi-cluster control data sets.

```
//IDCAMS      EXEC  PGM=IDCAMS
//*
SYSPRINT      DD  SYSOUT=*
//*
//PRIMER      DD  DSN=HSM.CDS.PRIMER,DISP=(OLD)
//SYSIN       DD  *
               DEFINE CLUSTER (NAME(?UID.BCDS) VOLUMES(?BCDSVOL) -
                           CYLINDERS(?CDSSIZE) FILE(HSMBBCDS) -
                           STORCLAS(?SCCDSNM) -
                           MGMTCLAS(?MCDFHSM) -
                           RECORDSIZE(435 2040) FREESPACE(0 0) -
                           INDEXED KEYS(44 0) SHAREOPTIONS(3 3) -
                           SPEED BUFFERSPACE(530432) -
                           UNIQUE NOWRITECHECK) -
                           DATA(NAME(?UID.BCDS.DATA) -
                           CONTROLINTERVALSIZE(12288)) -
                           INDEX(NAME(?UID.BCDS.INDEX) -
                           CONTROLINTERVALSIZE(2048))
               REPRO  ODS(*?UID.BCDS*) INFILE(PRIMER)
END
```

Figure 10-4 Sample JCL to create the BCDS

The variables in the sample JCL shown in Figure 10-4 are:

|                 |                                                       |
|-----------------|-------------------------------------------------------|
| ?UID.BCDS       | The output data set name of the BCDS                  |
| ?BCDSVOL        | The VOLSER that the data set is allocated on          |
| ?CDSSIZE        | The primary and secondary space allocation            |
| ?SCCDSNM        | Optional STORCLAS name if the CDS is DFSMS managed    |
| ?MCDFHSM        | Optional MGMTCLAS name if the CDS is DFSMSHsm managed |
| ?UID.BCDS.DATA  | The name of the data portion of the BCDS              |
| ?UID.BCDS.INDEX | The name of the index portion of the BCDS             |

## 10.2.5 Creating the Offline Control Data Set

The OCDS is required when tape output is created by DFSMSHsm. Figure 10-5 is sample JCL used to create a single OCSD. If your control data sets exceed the 4 GB limit, you will need to create a multicluster control data set in your environment. Additional sample JCL can be found in the DFSMSHsm starter set JCL provided with the product. Note that the primer record created in Figure 10-5 is used as input in DD statement PRIMER.

Refer to the *DFSMSHsm Implementation and Customization Guide*, SC35-0418 for more information and JCL samples used to create both single and multicluster control data sets.

```
/IDCAMs      EXEC  PGM=IDCAMs
//*
SYSPRINT      DD  SYSOUT=*
//*
//PRIMER DD  DSN=HSM.CDS.PRIMER,DISP=(OLD)
//SYSIN      DD  *
              DEFINE CLUSTER (NAME(?UID.OCDS) VOLUMES(?OCDSVOL) -
                            CYLINDERS(?CDSSIZE) FILE(HSMOCDS) -
                            STORCLAS(?SCCDSNM) -
                            MGMTCLAS(?MCDFHSM) -
                            RECORDSIZE(435 2040) FREESPACE(0 0) -
                            INDEXED KEYS(44 0) SHAREOPTIONS(3 3) -
                            SPEED BUFFERSPACE(530432) -
                            UNIQUE NOWRITECHECK) -
                            DATA(NAME(?UID.OCDS.DATA) -
                            CONTROLINTERVALSIZE(12288)) -
                            INDEX(NAME(?UID.OCDS.INDEX) -
                            CONTROLINTERVALSIZE(2048))
              REPRO ODS('?UID.OCDS') INFILE(PRIMER)
END
```

Figure 10-5 Sample JCL used to create the OCDS

The variables in the sample JCL shown in Figure 10-5 are:

|           |                                                       |
|-----------|-------------------------------------------------------|
| ?UID.OCDS | The output data set name of the OCDS                  |
| ?OCDSVOL  | The VOLSER that the data set is allocated on          |
| ?CDSSIZE  | The primary and secondary space allocation            |
| ?SCCDSNM  | Optional STORCLAS name if the CDS is DFSMS managed    |
| ?MCDFHSM  | Optional MGMTCLAS name if the CDS is DFSMSHsm managed |

|                 |                                           |
|-----------------|-------------------------------------------|
| ?UID.OCDS.DATA  | The name of the data portion of the OCDS  |
| ?UID.OCDS.INDEX | The name of the index portion of the OCDS |

## 10.2.6 Creating the journal and log file data sets

When DFSMSHsm is active, critical changes are logged to the control data set. The journal file is emptied each time the CDS is successfully backed up. The journal file is used to help recover the control data sets by merging the journal with a backed up version of the control data set. Without the journal file, critical updates to the control data sets are lost, should a control data set need to be recovered for any reason.

The journal data set contains a control record at the beginning that points to the last record written. The rest of the data set is processed as a sequential data set, each records a significant change made to one of the control data sets. There is no end-of-file marker at the end of the journal data set. The journal file should only be allocated on DASD.

The DFSMSHsm log data sets record events on all automatic and command functions for each processing unit. The events are logged in chronological order. Many vendor products use the log data sets as input to provide reporting on automatic functions, such as migration and backup, and user issued command functions, such as HRECALL. Figure 10-6 shows sample JCL to create the DFSMSHsm journal and DFSMSHsm LOGX and LOGY data sets.

```

//LOGALC      EXEC  PGM=IEFBE14
//HSMLOGx     DD   DSN=?UID.HSMLOGX,DISP=(,CATLG),
//              UNIT=?LOGUNIT,
//              VOL=SER=?LOGVOL,SPACE=(CYL,(?SPC)),
//              STORCLAS=?SCLOGNM,
//              MGMTCLAS=?MCDFHSM
//HSMLOGY     DD   DSN=?UID.HSMLOGY,DISP=(,CATLG),
//              UNIT=?LOGUNIT,
//              VOL=SER=?LOGVOL,SPACE=(CYL,(?SPC)),
//              STORCLAS=?SCLOGNM,
//              MGMTCLAS=?MCDFHSM
//EDITLOG     DD   DSN=?UID.EDITLOG,DISP=(,CATLG),
//              UNIT=?LOGUNIT,
//              VOL=SER=?LOGVOL,SPACE=(CYL,(?SPC)),
//              STORCLAS=?SCLOGNM,
//              MGMTCLAS=?MCDFHSM
//JOURNAL     DD   DSN=?UID.JOURNAL,DISP=(,CATLG),
//              UNIT=?LOGUNIT,
//              VOL=SER=?LOGVOL,SPACE=(CYL,(?SPC)),
//              STORCLAS=?SCLOGNM,
//              MGMTCLAS=?MCDFHSM
/*

```

Figure 10-6 Sample JCL to create HSM journal and HSM LOGX/LOGY data sets

The variables in the sample JCL shown in Figure 10-6 are:

|              |                                                               |
|--------------|---------------------------------------------------------------|
| ?UID.HSMLOGX | The HSMLOGX data set name                                     |
| ?LOGUNIT     | The unit name of the DASD device the data set is allocated on |
| ?LOGVOL      | The VOLSER of the DASD device the data set is allocated on    |
| ?SPC         | The space in cylinders                                        |
| ?SCLOGNM     | The STORCLAS name if the data set is DFSMS managed            |
| ?MCDFHSM     | The MGMTCLAS name if the data set is DFSMSShsm managed        |
| ?UID.HSMLOGY | The HSMLOGY data set name                                     |

### 10.3 Using the newly created DFSMSHsm environment

Once the catalog, primer record, control data sets, journal file and HSM log data sets have been created, the DFSMSHsm subsystem can be started. ABARS

ARECOVER events can begin to recover data and populate the CDS files. Restored data is easily accessible to the user by referencing the data or by issuing the RECALL DFSMSHsm command.

### 10.3.1 Full CDS recovery

If you are not using DFSMSHsm ABARS, you must recover your DFSMSHsm environment by copying the ICF user catalog, ML1 volumes, and control data sets to tape and restoring them or mirroring them to the recovery site.

The backup of the ICF user catalog, control data sets, and ML1 volumes require careful planning and testing to ensure a synchronized backup. The DFSMSHsm environment should not be active and the ICF user catalog should be the last data set backed up in the series. These steps may help to minimize the number of discrepancies between the ICF user catalog and the control data sets.

Since data on ML1 volumes is highly active, that is, data can be migrated, recalled, or deleted often throughout any 24-hour period, data on ML1 volumes must be backed up as often as the applications using those volumes. As an example, if an application writes data to DASD that is immediately eligible for migration and interval migration is used, the data is not included in a full volume dump or incremental backup of the primary DASD volumes taken later that day or later in the week. The data resides only on the ML1 volumes until and unless it is migrated to ML2 or recalled back to DASD, or deleted. Because of the potential for movement of the data, the backup of the ML1 volumes must be coordinated with the backup of the application volumes.

Data on ML2 volumes can be automatically duplicated using the DUPLEX parameter in DFSMSHsm PARMLIB. Specifying DUPLEX allows DFSMSHsm to create two tapes concurrently, one to be kept onsite and the alternate to be removed and kept offsite or written to a remote tape library. Figure 10-7 shows a sample of the DUPEX command for backup tapes. To duplex MIGRATION volumes, use DUPLEX(MIGRATE(Y)). More information on the DUPLEX command is available in the *DFSMSHsm Storage Administration Reference*, SC35-0422. Figure 10-7 shows sample SETSYS command with the DUPLEX parameter.

```

SETSYS AUTOBACKUPSTART(0030 0400 0730) +
      BACKUP(INUSE(RETRY(Y) DELAY(60) SERIALIZATION(PREFERRED))) +
      BACKUPPREFIX(HSM)      +
      COMPACT(DASDBACKUP)    +
      COMPACTPERCENT(40)     +
      DUPLEX(BACKUP(Y))      +
      EXITON(BD SA)          +
      INCREMENTALBACKUP(CHANGEDONLY) +
      MAXBACKUPTASKS(8)      +
      DSBACKUP(TAPE(DEMOUNTDELAY(MAXIDLETASKS(2)
MINUTES(120))))

```

Figure 10-7 Sample SETSYS command showing the DUPLEX parameter

### 10.3.2 Recovery planning

When full volume dump or full volume dump and incremental backups are used for recovery of the primary volumes at the recovery site, the data center must plan on how often and at what point in the daily or weekly processing cycle the control data sets and the ML1 volume backups occur. DFSMShsm backs up the control data sets each time automatic backup is scheduled to execute. The backup of the control data sets can be made to DASD or to tape. This backup is normally regarded as the on-site backup copy used to recover a CDS, should a failure occur.

It is strongly recommended that you specify DFDSS as the data mover on the **BACKVOL CDS** command. DFDSS provides advantages of validation during backup as well as the use of the concurrent copy function that can reduce serialization of the control data sets during backup. Refer to the *DFSMShsm Storage Administration Reference*, SC35-0422 or *DFSMShsm Storage Administration Guide*, SC35-0421 for more information on the BACKVOL CDS command.

Off-site backup copies of the control data sets are accomplished using DFDSS logical backup. This logical backup can be executed at a specific point-in-time for disaster recovery. The recommended steps for backing up ML1 DASD volumes, the ICF user catalog, and the control data sets are:

- ▶ Stop DFSMShsm activity
- ▶ Dump ML1 DASD volumes
- ▶ Backup the ICF user catalog
- ▶ Backup the control data sets
- ▶ Resume DFSMShsm activity

### 10.3.3 Keeping your DFSMSHsm control data sets free of errors

To ensure a successful and timely recovery at the disaster site, it is necessary to regularly audit your DFSMSHsm environment at the primary site and keep your control data sets free of errors. During a recovery test or in the event of an actual disaster, time is an important issue. You can minimize the time it takes to audit your DFSMSHsm and ICF user catalog environment by performing regular audits at the primary site and resolving any errors reported. Refer to Chapter 5, “Using FastAudit/390” on page 39 for information on how to perform audit and error correction in your DFSMSHsm environment.

## 10.4 Recovery of the DFSMSHsm environment

Once at the disaster site and with the operating system active, the recovery of the ICF user catalog, ML1 volumes, and control data sets can occur. Most likely, the DASD volumes that the control data sets and the ICF user catalog reside on have been recovered as part of the operating system. The logical backup of the control data sets using DFDSS and the backup of the ICF user catalog must be restored to replace the existing control data sets and catalog. The recommended steps for restoring the ICF user catalog, ML1 DASD volumes, and the control data sets are:

1. Ensure DFSMSHsm is not active.
2. Restore the ICF user catalog, replacing the catalog, if present due to the full volume restore of the operating system.
3. Restore the control data sets from the DFDSS logical backup, replacing the control data sets if present due to the full volume restore of the operating system.
4. Restore the ML1 DASD volumes.
5. Start DFSMSHsm issue the HOLD ALL command.
6. Issue the TAPEREP L ALL DAVOLUMES DFSMSHsm command.
7. Reconcile the ICF user catalogs with the DFSMSHsm control data sets.
8. Issue the SETSYS DIASSTERMODE DFSMSHsm command.
9. Release MIGRATION and RECALL.
10. Audit the BCDS using HSM *FastAudit*.
11. Release BACKUP and RECOVER.

For more information about the **TAPEREP L ALL DAVOLUMES** and the **SETSYS DIASSTERMODE** DFSMSHsm commands, see the section titled “Disaster Alternate Volumes” in *DFSMSHsm Storage Administration Guide*, SC35-0421. For more

information about the HOLD and RELEASE operator commands, see the *DFSMSHsm Storage Administration Guide*, SC35-0421.

### 10.4.1 Reconciling ICF catalogs with DFSMSHsm control data sets

It is often necessary to audit the DFSMSHsm control data sets and ICF user catalogs upon recovery to ensure that the ICF user catalogs reflect the correct disposition of data that DFSMSHsm is managing. This is especially true if DFSMSHsm is active at the time the ICF user catalog, the control data sets, and the ML1 volumes were backed up. We recognize that in today's IT environments, it is not always possible to stop DFSMSHsm activity to back up the environment.

Mainstar: *FastAudit/390* provides audits to reconcile out-of-synchronization conditions at the recovery site. At the recovery site, the following steps are recommended to correct these discrepancies:

1. Run FAUDITAL to audit all of the MCDS migration components and the catalog entries that are cataloged to VOLSER MIGRAT. This audit checks the data chain for each of the MCD, MCA, and MCO records and verifies them from the recovered MCDS to the ICF user catalogs, ensuring that these structures are in synchronization with each other. Errors are reported in the HSM *FastAudit* ISPF panels provided by the product. See 6.1, "Migration Control Data Set audits" on page 56 for more information on the FAUDITAL procedure and fixing the errors reported in the audit.
2. Correct the errors found by the audit by executing the FIX commands supplied by the HSM *FastAudit* product. See Chapter 5, "Using *FastAudit/390*" on page 39 for more information about executing the FIX statements provided to correct the errors.
3. Issue the TAPEREPL ALL DAVOLUMES DFSMSHsm command.
4. Run the HSM *FastAudit* procedure, FAUDITBD, to audit the BCDS.
5. Correct the errors found by the audit by executing the FIX commands supplied by the HSM *FastAudit* product. See 6.2, "Backup Control Data Set audits" on page 112 for more information about executing the FIX statements provided to correct the errors.
6. Release the BACKUP and RECOVER DFSMSHsm functions.
7. Start DFSMSHsm and release MIGRATION and RECALL.

### 10.4.2 Auditing the BCDS

Auditing the BCDS is optional. If you plan on using DFSMSHsm backups to recover data during your recovery test or in the event of an actual disaster, auditing the BCDS should be a priority. This audit matches the MCC records to



the MCB records, validates the TTOC records and checks the VTOC entry for each backup tape. Refer to Chapter 5, “Using FastAudit/390” on page 39 for a complete description of auditing the BCDS using HSM *FastAudit*.

### 10.4.3 Auditing the OCDS

If you plan on being at your recovery site for a length of time or plan to process jobs that might require recalling or recovering data sets, it would be advisable to audit the OCDS. Execute the HSM *FastAudit* FAUDITOD procedure to audit the volume pointer chains, forwards and backwards, ensuring that the tapes in the OCDS can be recycled properly. Refer to Chapter 6, “Auditing DFSMSHsm using HSM *FastAudit*” on page 55 for a complete description of auditing the OCDS using HSM *FastAudit*.

### 10.4.4 Auditing the ICF User Catalogs and DASD VTOC

Another type of error condition prominent at the recovery site, is discrepancies between the entries in the ICF user catalogs that have been recovered, and the actual content of the DASD volumes that have been restored. *FastAudit/390* provides a selectable unit, Catalog & DASD Audit, for this purpose.

The audit procedure, FAUDITDE, audits the contents of the DASD VTOCs against the extraction of the ICF user catalogs. It then reverses the process and audits the catalogs to the DASD VTOCs. The difference between FAUDITDE and FAUDITDA is that FAUDITDE audits the entire content of all the DASD VTOCs at the recovery site.

FAUDITDE uses the same control card structures as in FAUDITDA, so there are no new or additional control card formats that need to be learned or documented. Each data set and the associated sequence record is audited and verified. Multi-volume data sets are treated like multiple records and not dependent upon each other. This audit provides the flexibility to audit all your recovered DASD volumes to your ICF user catalogs, audit selected DASD volumes to selected catalogs, or audit specific ICF user catalogs to all DASD volumes. The audit and subsequent FIX commands when executed, ensures access all your data.

### 10.4.5 How to ensure the DFSMSHsm environment is accurate

The final task is to test, to verify a complete and accurate recreation of your ICF user catalogs, DASD volumes, and DFSMSHsm environment at your recovery site. Possibly the best way to verify your recovery site environment is to re-process all of the *FastAudit/390* audit jobs mentioned in this chapter. If the result of the audit is no critical errors reported, you can be assured that your recovered environment is ready for access.

Alternatively, testing of the environment could include migrating and recalling various data sets, backup and recovering data sets, running test jobs and verifying the output.

Table 10-1 shows an example of a Verification and Test Log provided as a checklist to test various functions of the DFSMSHsm environment.

*Table 10-1 Checklist for DFSMSHsm function at recovery site*

| <b>Alternate Site Verification and Testing Tasks:</b> |                   |                     |                         |
|-------------------------------------------------------|-------------------|---------------------|-------------------------|
| <b>Company Name:</b>                                  |                   |                     |                         |
| <b>Date:</b>                                          |                   |                     |                         |
| <b>Task</b>                                           | <b>System/Job</b> | <b>Successful –</b> | <b>Tester ID number</b> |
| <b>Migration</b>                                      | HLQ USERa         | YES NO              |                         |
|                                                       | HLQ-TESTa         | YES NO              |                         |
|                                                       | HLQ-PRODa         | YES NO              |                         |
| <b>Recall</b>                                         | HLQ-USERb         | YES NO              |                         |
|                                                       | HLQ-TESTb         | YES NO              |                         |
|                                                       | HLQ-PTODb         | YES NO              |                         |
| <b>Backup</b>                                         | HLQ-USERc         | YES NO              |                         |
|                                                       | HLQ-TESTc         | YES NO              |                         |
|                                                       | HLQ-PRODc         | YES NO              |                         |
| <b>Recover</b>                                        | HLQ-USERd         | YES NO              |                         |
|                                                       | HLQ-TESTd         | YES NO              |                         |
|                                                       | HLQ-PRODd         | YES NO              |                         |
| <b>Listcat</b>                                        | Master Catalog    | YES NO              |                         |
|                                                       | USER1 Catalog     | YES NO              |                         |
|                                                       | USER2-Catalog     | YES NO              |                         |
| <b>Allocate</b>                                       | USER data set     | YES NO              |                         |
|                                                       | TEST data set     | YES NO              |                         |
| <b>Management SignOff</b>                             |                   | YES NO              |                         |
| <b>Production</b>                                     | Start Production  | YES NO              |                         |

# Related publications

The publications listed in this section are considered particularly suitable for a more detailed discussion of the topics covered in this redbook.

## IBM Redbooks

For information on ordering these publications, see “How to get IBM Redbooks” on page 276.

- ▶ *ICF Catalog Backup and Recovery, A Practical Guide*, SG24-5644
- ▶ *DFSMSHsm Primer*, SG24-5272

## Other publications

These publications are also relevant as further information sources:

- ▶ *DFSMSHsm Diagnosis Reference*, LY35-0111
- ▶ *DFSMSHsm Implementation and Customization Guide*, SC35-0418
- ▶ *DFSMSHsm Storage Administration Guide*, SC35-0421
- ▶ *DFSMSHsm Diagnosis Guide*, LY35-0114
- ▶ *DFSMSHsm Storage Administration Reference*, SC35-0422

## Online resources

These Web sites are also relevant as further information sources:

- ▶ Mainstar Software Corporation  
<http://www.mainstar.com/>
- ▶ IBM TotalStorage™ home  
<http://www.storage.ibm.com/ssg.html>

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## DFSMShm Audit and Mainstar: Fas/Audit/390

(0.5" spine)  
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**Redbooks**

# DFSMSHsm Audit and Mainstar:*FastAudit*/390

**You should audit your HSM hosts. This IBM Redbook shows you how!**

**Learn Mainstar: *FastAudit* and DFSMSHsm AUDIT commands**

**Interpret audit results**

The DFSMSHsm control data sets contain information describing the status of the DFSMSHsm managed data sets and volumes. Each control data set requires regular maintenance (backup and reorganization) and auditing to ensure the integrity of the data managed by DFSMSHsm. The DFSMSHsm product provides the AUDIT command to detect, report, diagnose, and often provide automated correction for discrepancies between the control data set, ICF user catalog, and DFSMSHsm owned volumes.

Mainstar: *FastAudit* family of products provides the same audit capabilities as with the DFSMSHsm AUDIT command, but at faster speeds. Using the *FastAudit* products, the audits are comprehensive (perform multiple audits with one command), flexible, and easy to use. Audits of some, or all of the z/OS and OS/390 metadata environments can now be done frequently, even daily, to identify and resolve errors before they lead to data loss or failed access. This IBM Redbook outlines the tasks to use DFSMSHsm Audit and the use of Mainstar's *FastAudit* program to manage your HSM environment.

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